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Jeeves ERP
Product overview
Updated for Jeeves ERP 2026.Q2.2

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1 About this document

1.1 Introduction

As today's companies strive to maintain their competitive edge, they are constantly looking to improve the way they are doing business. Keeping pace with technological advantages, thriving in a fiercely competitive global marketplace, responding to demands for highly customized products – these are just some of the seemingly overwhelming challenges that must be met to ensure your company's success. At Jeeves, we realize that a powerful ERP system plays a vital role in this success. We have therefore developed Jeeves ERP to empower your organization to quickly and efficiently manage change, today and in the future.

1.2 Product overview

Welcome to Jeeves' Product overview — an overview of the features and functionality in your Jeeves ERP. This document is a high-level overview of the major capabilities contained in the described version, and is intended for anyone who wishes to gain a general understanding of Jeeves ERP's overall features and functionality. It is organized according to functional groups. The document does not assume that the reader has detailed knowledge of the current and overall functionality, but requires general business knowledge relative to the application.

The Product overview presents the Jeeves ERP applications and components and highlights the key features and functions available. This offers you a concise overview of the business process they support and the benefits your company can gain from using Jeeves ERP.

Please note that Jeeves Professional Services can provide additional functionality as part of the packaging for different industries and companies. Your Jeeves Professional Services representative can advise you of suitable additions for your company's processes.

1.3 Document version/revision

Version/revision	Description	Date
Version 1	Updated for Jeeves ERP 6.0	July 2019
Version 2	Updated for Jeeves ERP 6.0 Update 1	May 2020
Version 3	Updated for Jeeves ERP 6.0 Update 2	August 2020
Version 4	Updated for Jeeves ERP 6.0 Update 3 and 4	February 2021
Version 5	Updated for Jeeves ERP 6.0 Update 5 and 6	September 2021
Version 6	Updated for Jeeves ERP 6.0 Update 7 and 8	January 2022
Version 7	Updated for Jeeves ERP 6.0 Update 9 and 10	September 2022
Version 8	Updated for Jeeves ERP 6.0 Update 11 and 12	January 2023
Version 9	Updated for Jeeves ERP 6.0 Update 13	October 2023
Version 10	Updated for Jeeves ERP 6.0 Update 14-16	November 2023
Version 11	Updated for Jeeves ERP 2024.Q2.0	January 2025
Version 12	Updated for Jeeves ERP 2025.Q2.0	February-March 2025
Version 13	Updated for Jeeves ERP 2025.Q2.1	June 2025
Version 14	Updated for Jeeves ERP 2025.Q2.2	September 2025
Version 15	Updated for Jeeves ERP 2025.Q2.3	December 2025
Version 16	Updated for Jeeves ERP 2026.Q2.0	May 2026
Version 17	Updated for Jeeves ERP 2026.Q2.1	June 2026
Version 18	Updated for Jeeves ERP 2026.Q2.2	September 2026

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The following applications are not described in Jeeves' Product overview, nor are any separate packages that are not part of Jeeves ERP standard:

- Jeeves EDI
- Jeeves Workflow

1.4 Audience

This document is intended for anyone who wishes to gain a general understanding of Jeeves ERPs overall features and functionality.

1.5 Document conventions

Use of typographic conventions in documentation helps users locate and interpret information easily. The following guidelines present specific typographic conventions. This document enables you to establish and maintain a common understanding of various conventions used in the document. For simplicity and consistency, this document describes the method of performing tasks. Other conventions used in this document are listed in the following table.

Representation	Description
Bold text	Procedures, references to screens, dialog boxes and their elements, file and folder names, commands and buttons.
Numbered list	Stepwise instructions.
Bulleted list	Sets of features or options used to perform an activity.
Note text	Any exception, extra information, or pitfalls.
Italic text	References to related documents.

1.6 List of acronyms

Acronym	Description
AL	Alternative supplier
API	Application programming interface
ATP	Available to promise
AWS	Amazon web services
BOA	Business operation area
BOE	Bills of exchange
BOM	Bills of material
B2B	Business to business
B2S	Business to salespersons
CBAM	Carbon Border Adjustment Mechanism
CSRD	Corporate Sustainability Reporting Directive
DDS	Due Diligence Statement
DSO	Delivery sales order (in delivering company in ICT)
EDM	Electronic document management
EOQ	Economic order quantity
EPOS	Electronic point of sales
EUDR	EU Deforestation Regulation
ES	EazyStock

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Acronym	Description
FIFO	First in first out
FDA	Food and Drug Administration
GL	General ledger
HMRC	Her majesty’s revenue and customs
ICT	Inter-company transactions
IDEP	Intrastat data entry package
JIS	Jeeves information systems
JMI	Jeeves message integrator
JSON	JavaScript Object Notation
JSB	Jeeves service builder
KSeF	Krajowy System e-Faktur (Poland's National e-Invoice System)
MHA	Material handling areas
MO	Manufacturing order
MRP	Material requirement planning
QC	Quality control
RFQ	Requests for quotations
SCO	Sales customer order (in selling company in ICT)
SEPA	Single euro payments area
SID	Security identifier
SKU	Stock keeping unit
SRU	Standardized account statement
SSRS	SQL server reporting services
TU	Transport Unit
WIP	Work in progress
WMS	Warehouse management system

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2 Jeeves ERP's User interface

Jeeves ERP has a modern and attractive user interface. For new installations and upgrades, the Desktop client is the standard and only supported client.

Existing Classic client installations remain supported under the support policy applicable to their release, which is why Classic client functionality is still described in this document.

Jeeves ERP also offers a mobile client, a framework for designing mobile routines, and a range of standard mobile routines.

Jeeves' role based user interface can hide irrelevant information and enhance features that are important to a particular role. This helps you as a user to:

- understand the information presented
- focus on the right things and prioritize
- reduce time spent on searching for relevant and correct information

2.1 Desktop client

Jeeves Desktop client is the standard and only supported client for new installations and upgrades. It is an installed Windows client built on Windows Presentation Foundation (WPF), a .NET-based framework for building user interfaces. In supported existing installations, it can run in parallel with the Classic client.

Jeeves Desktop client is an improvement on the Classic client, adding a modern, flat user interface and an enhanced user experience, including productivity features such as a dashboard and role-based designs. It allows automatic rendering of your classic program layout. With the separate Design studio, you can further improve the designs, and publish them for specific roles. As it is a Desktop client, there are few limitations regarding integration and interaction with other Windows applications.

The Desktop client's dashboard allows you to always have an overview of the work that needs to be done. You can discover and solve problems, and do so in a way that can be personalized for your own routines and needs. The dashboard is task-oriented and makes it easy to access and manage data related to your tasks. It also makes it easy to collaborate and share information with your colleagues.

The program menu, forms, and lists presented can also be role-specific and, if you have several roles in the company, you can switch roles on-the-fly.

You can have any number of programs open at the same time. The open programs' icons are color-coded to show their current mode (locked or opened for new entry or change). The navigation panel shows exactly where you are in the system. It is also clickable, which means you can revert to any previous level.

There are filters for filtering programs in the menu, and both quick and advanced filters for filtering data in program lists.

You can print reports connected to the open program from the Desktop client, including cloud compliant reports, both Jeeves ERP standard reports and reports designed using the report designer in Design studio.

The Jeeves ERP standard dashboard has pre-built dashboards that expand the available ways for users to present and interact with information. The data in the Jeeves ERP standard dashboard is retrieved from the new ERP Server (Cloud) and can be viewed on the following on the Desktop client:

- Sales dashboard provides a visual representation of sales data such as open invoices, sales against forecast, sales by item group, etc.
- CFO dashboard provides a visual representation of financial data such as due customer and supplier invoices, performance overview, profit and loss, balance sheet, revenue, cash flow, etc.

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The Desktop client's forms and lists are designed using the Design studio as described in the next section. There are, however, some settings you can make directly from the user client, such as:

- Sort a program list or embedded list by clicking on one or more column header(s) and save the sort order
- Save search conditions for re-use and translate them
- Clone programs
- Define intellisense fields
- Define language, format, and print format settings
- Set a list as a tree view
- Copy values from previous entries to the current record

There is also basic functionality for administrators and consultants, for example a function to update database tables from the user client.

You can flag records and attach comments to them directly from the Desktop client. The dialog for managing flagged records, called Assign flagged record, lets you assign a flagged record to another user by selecting them from a drop-down list of valid users. Assigning a record to another user resets the flag color and makes the record appear as flagged for the selected user. Comments are displayed in the assigned user's Tagged records widget on the dashboard. A menu option called View flagged record info gives access to detailed information about flagged records, including comments, sender and receiver details, timestamps, and administrative functions.

It is possible to configure different shortcuts to different servers with the same installation of the Desktop client.

Much of the functionality in the user client is common to both the Desktop client and the Classic client and is therefore described in common sections further on in this chapter.

2.2 Jeeves Launchpad

Jeeves Launchpad is a dynamic loader and launcher for the Desktop client and Design studio. Cloud customers receive automatic upgrades through Launchpad, ensuring they always run the latest version.

Jeeves Launchpad supports downloading, unpacking, and starting specific versions of the Desktop client and Design studio from remote or local sources, enabling flexible multi-environment setups.

Key capabilities include automated version management with controlled upgrades, configurable startup options, multi-environment and shortcut support, visual progress indicators, and comprehensive process management.

Platform components are delivered in a modular way, allowing certain corrections and updates to be applied through client updates without requiring full system upgrades.

2.3 Design studio for Desktop client and mobile clients

In Jeeves Design studio, you can create program designs "from scratch" or create new designs based on existing ones by saving them under new names. You can save designs to file, which means they can be distributed between users and different databases, both for review and for use.

Jeeves Design studio supports the WYSIWYG concept (what you see is what you get), which implies that the designs you create here accurately reflect the appearance of the finished design in the Desktop or Mobile client.

When creating new or changing existing designs you simply:

- drag and drop new design elements into your design

- drag and drop existing design elements within the design

- use the Delete key to delete design elements

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In Jeeves Design studio, you can see the design take form as you work on it to help you more easily create better designs.

The Design studio includes:

- Desktop client design
- Mobile client design
- Design of charts (pie, column, drill-down and line charts) including charts in program designs
- Design of dashboard widgets such as text widgets, chart widgets (including spline, stacked column and horizontal bar chart widgets), list widgets, web widgets and standard widgets such as the To-do widget, the Favorite programs widget, and the Tagged records widget
- Design of the Macros tab dashboards
- Design of the program menu tree
- Smart tools like design outline and a zoom feature
- Layout control options for forms and lists. You can set the orientation of a design to either vertical (default) or horizontal, select left or right placement of a form, and define the width of a list in pixels.
- Extended configuration options for forms, lists, widgets, and tooltips, enabling clearer and more adaptable program designs.
- Enhanced macro editing with improved suggestions and editing shortcuts.

You can also have several designs ongoing at the same time and there is a built-in XAML editor that has been added for advanced users.

Program designs originating from the Classic client can be imported into Design Studio and further developed, supporting reuse of existing configurations during modernization to the Desktop client.

You can handle field settings in forms, lists and total windows from the Design studio, for example you can change font colors, automatically update lists, create own total fields, create, modify and translate tooltips for fields in forms, etc. You can handle own, custom terms/fields too.

Several standard widgets have enhanced formatting and display options. The Text widget and Web browser widget support showing or hiding the widget title and heading, translating titles and headings to any supported language, and formatting text by font size, color, and alignment.

The Card widget supports showing or hiding the widget title, changing prompt font and color, and defining value format. The Card text widget supports showing or hiding the widget title, displaying a sum value, defining numeric format, setting background color, and applying conditional formatting rules based on operator, value, and color.

The Refresh option informs you if you have unsaved changes and gives you the option of discarding them and reloading the original design.

The global search panel lets you search, open, edit and save designs, macros and connections, as well as create charts from statements.

You can modify program menus in the tree format from the Design studio. You can add application programs and folders to the program menu tree design, move programs up or down in the program menu tree design, delete application programs and folders from the program menu tree design, set folder and application program properties, expand/collapse menu sections, save and publish the menu tree design, as well as export the menu to a program menu (.pmu) file.

The Favorite programs widget displays the programs you have marked as favorites in the Desktop client and updates in real time. You can drag and drop it onto a dashboard in Design studio. The Tagged records widget displays flagged records in a procedure-list view on the Desktop client dashboard, giving users quick access to items they have flagged and commented on. It can likewise be placed on a layout using drag and drop in Design studio.

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A new design is not applied until it is published, which implies that you can work on your design, gradually saving your changes, and publish it when it is ready. New designs must be saved before they can be published, but once published you do not need to republish them. When you publish the design, you can specify to whom and what the setting shall apply:

- individual companies, company groups or all companies
- individual users, user groups or all users
- individual programs, all programs that share a window, or globally
- specific client, either Desktop or Mobile, or both

Any changes take effect automatically the next time the user(s) restart(s) or refreshes the user client.

2.3.1 Template editing in Design studio

You can manage Jeeves report templates from Design studio. The Reports tab in the Developer menu contains a link to the Template manager dialog that allows you to view templates belonging to a selected report and company, create and edit report templates, add buttons and shortcut keys for printing a report with a certain template and delete report templates.

An integrated report designer facilitates the replacement of Crystal Reports and SSRS reports with a cloud compliant format. The Report designer enables you to create new reports from scratch, bind them to data, and fully customize them. You can create new reports from a design context, open a report from site or file, save a report to site or to file (for export to another system), and publish/unpublish a report to a program or node/window context.

2.3.2 Macro editing in Design studio

From the Developer tab in the Design studio you can develop new features in program designs, for example create new macros and create buttons to execute macros, which you can then place in your design. You can link/unlink from standard program events or field-specific events to macros, delete macros and create accelerator keys for functions. When saving or publishing a macro from the Macros tab, the macro code is displayed so that you can preview it before saving and publishing.

The Macros tab shows a list of all program specific and global macros. The publish dimensions are indicated by icons. You can double-click on a macro to open it in a separate tab in the design area. You can select a macro and click the desired button on the Developer tab to perform the corresponding action from the program design/XAML tabs.

The Macro links tab shows all connections related to the selected program and linked to global macros in Desktop and Mobile designs, linked events, buttons and hotkeys. A Field column displays the name of the field to which each macro is linked. The connections can be enabled/disabled via right-click. Here you can also view information about the program, node and window context publish dimensions.

There are also tabs for viewing and modifying macro code and for removing connections. In the Macro Properties tab you can enter or modify the macro description and information.

You can also initiate the Desktop client from the Design studio in order to run a macro for verification, which gives a seamless integration experience between the two.

2.4 Mobile clients

Jeeves ERP includes a framework for setting up mobile devices and users to be able to connect to the Jeeves ERP app and for designing mobile routines and processes to run from the app. Jeeves provides a suite of standard mobile routines as described further on.

Mobile routines give users working out in the field, or off-site, access to relevant information in Jeeves, for example it gives service technicians access to their scheduled work, and makes it

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possible for them to report back when work is completed. The connectivity to Jeeves ERP, while out in the field, or off-site, is complemented by typical mobile features, for example:

- data entry using buttons, icons, drop-down lists, date/time spinners and switches
- e-mails, text messages, phone calls, contacts and calendar access directly from Jeeves apps
- maps and navigation, for example to be able to locate customer sites
- the possibility to take photographs or videos and upload them to the server in order to record activities, meter readings and states, or to upload signatures to sign off work orders, etc.
- the possibility to scan bar codes and QR (quick response) codes

The Mobile clients are available for both iOS and Android platforms. You can configure your Jeeves ERP mobile app in an MDM (Mobile device management) provider like Microsoft InTune/MobileIron to configure REST URLs and profile names to deploy configurations using MDM.

2.4.1 Features

Because there are special needs with regard to the size of the display and the behavior in mobile devices, there are many features that are specific to - or especially useful for - mobile clients. The following sections list out some of these features.

2.4.1.1 Forms

The basis of your design is a form with columns and tabs etc. (as in other clients). The difference is that in the mobile client, the form adapts to the available space, also when rotating the device.

Detail records, for example, service order rows are displayed in the form of cards, giving a modern, compact look suitable for smaller displays.

Certain controls, such as tabs, are collapsible and expandable.

Program connections, report printouts, links to documents, saved searches, etc. can be accessed via a pop-menu:

e-Mails, text messages, phone calls, contacts and the calendar can be accessed from the apps' forms. You can connect these actions to buttons or automatically trigger them on events, for example when closing a service order. You can have embedded browsers in the form for easy access to information.

You can also build dialogs with multiple data entry fields, for example to enter or remove personal data.

2.4.1.2 Lists

When working with lists, you can click on the plus (+) icon to open a record in the Add mode. After making your changes, you can tap on the Done option to save your changes or on the Undo option to discard the changes.

You can select/deselect multiple records from an embedded list in the Mobile client by long pressing and tapping on records to select. Once you have selected a records or records from the list, you can tap on an icon to perform the action associated with it. For example, to delete records, you can select the records you want to remove and tap on the floating Trash icon.

For example, in an iPad you can have the search list to the left and form with embedded list etc. to the right.

2.4.1.3 Check-off lists

Many programs in Jeeves ERP are based on a special technique that we call checking off, which means that you select a set of transactions/records, for example search for all supplier invoices that are planned to be paid on a certain date, and then check off individual invoices to be included in the payment batch.

In the mobile client, the windows that are designed for checking off have the following structure:

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The top list corresponds to the left list in the classic client and presents the records that fulfill the criteria for processing (e.g. if you are checking off invoices for payment, all supplier invoices that are ready for payment) or a selected portion of them.

You use the bottom list, which corresponds to the right list in the classic client, for selecting records for processing. You can then make any necessary adjustments to the records in the bottom list. When you are satisfied with the result in the bottom list you save it and, in many cases, run the next step in the process (e.g. journal, update or the like).

A number of options are represented by icons in both the top and bottom list. Long press on a row to enable these options.

2.4.1.4 Searches

You can make quick, local searches within the current list, instead of searches being carried out on the server. Searches are run on all fields in the routine and not just on the values in one or a few specific field(s). Mobile client users can switch to Search mode to search in any field in the form as in the Classic and Desktop clients. You can also set up predefined searches, for example, as a technician using Jeeves' Mobile Service routine automatically you can see a list of relevant service orders, without having to search.

2.4.1.5 Intellisense

You can use intellisense in fields while searching for data in related tables without having to jump to the table. You can enter the initial character(s) and click the Intellisense icon. A list of values which begin with the entered character(s) are displayed. You can use intellisense for fields like Item descriptions, company names, account names, etc.

2.4.1.6 Notifications

Notifications can be sent through a cloud based notification service with information and with a pre-defined action. For example, if a customer's address is changed, the server can send a notification and on clicking on the notification, the app can show the updated address along with the map.

2.4.1.7 Buttons and icons

You can connect buttons and icons to Jeeves' standard functions or own macros, and with special icons suitable for mobile interfaces.

2.4.1.8 Map view

You can use the map view, a mobile client specific design feature that displays a map, to for example show the customer's location.

2.4.1.9 Date and time spinners

You can use the date spinner connected to date fields or the time spinner connected to time fields to select dates and times.

2.4.1.10 Diary entry

You can make a diary entry when time reporting, for example a free text to be copied to the customer invoice.

2.4.1.11 Images and videos

You can pick an image or video from the gallery or take one with the device's camera, or use the Signature control to sign off a record, and upload the image or video to the server. Images can be handled in various formats, such as .gif, .png, jpeg, or .bmp, and videos in .mov format.

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2.4.1.12 File sharing

You can share a file (such as a document or an image) from a supported application with the Jeeves mobile app and then link that file to a record, e.g. to supplier invoices, customer orders, items etc. The Upload icon is visible when you open a record and one of more attachments are available for upload.

2.4.1.13 Open attachments

Attachments to records in Jeeves ERP can be opened in an app based on the document type (PDF, Excel, Word etc.). This works according to Jeeves' standard logic but can be used via macros too. When attempting to open an attachment, a menu is presented with the apps that are installed on the device and that support the relevant document format. (If no such apps are available no menu is displayed).

2.4.1.14 Reports

Reports in Jeeves ERP can be opened in the routine from the Reports option in the Pop-up menu. The report is displayed in the format based on the settings in the Template editor. To view SSRS reports in the Mobile client, it must be set up and the Report services option must be selected.

2.4.1.15 Enter ID by scanning bar code

If it has been set up, you can use the device's camera to scan a bar code, or a QR (quick response) code, instead of entering an ID manually or using search functionality. The mobile client automatically searches for the returned code (alphanumeric string). If a record in the list matches the scanned code, it is displayed.

2.4.2 Control session length

It is possible to terminate the session after a certain period of inactivity, i.e. when the mobile client is closed without using the logout option, by configuring the idle time-out settings in the application pool.

2.4.3 Check logs

Exceptions during calls made from the Mobile client/Jeeves server and error conditions are stored to a log file. The log file (for example, JvsRestLog 20160922 103426.txt, where 210160922 indicates the date on which the log file is created in the yyyymmdd format and 103426 to the time in the hhmmss format) is stored within the Logs folder on the REST machine.

The logs are created when the client connects and disconnects from the client, server, or REST service.

You can enable or disable the logging feature based on the app settings.

2.4.4 Macro functions and objects

A number of macro functions and object methods have been created to support the mobile client.

The SetXAMLProperty macro feature is available on the form level to customize components of the program (form and list). Currently, it is a mobile-specific function that only supports customizing UI elements. You can use it for many different purposes, for example to dynamically show/hide and change names on buttons, to keep the form in Add mode in programs used for only entering new records, to avoid having to pass via the program list, customize the program title, open/close sub nodes, customize a field, set the mode of the form, etc.

The CreateSubTransaction macro feature is available on the form level to add/update/remove rows in an embedded list (child node).

The LocationSubscription macro object provides location information of the client within the watch area.

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The Navigate command allows you to change the source and destination properties of the MapViewer object. The Mobile UI changes to reflect the new route. If the source is empty, then mobile client will consider the current location as the source.

The Request macro object allows you to call the REST API.

The RestObject macro object allows you to construct a REST service request.

2.4.5 Standard routines in Mobile client

Jeeves offers standard routines for some of the most common work flow steps that are suitable to be managed from a mobile device. These standard routines are intended to be maintained in Jeeves standard. They are delivered in a standard package of settings connected to a standard role and site groups created and maintained by Jeeves. Standard routines also use functionality developed as standard features, i.e. features developed to fulfill the requirements of a mobile routine that can also be used in Jeeves Desktop and Classic clients.

The following standard mobile routines are available:

- Jeeves Mobile order - for salespeople visiting customers to enter and confirm orders, also to search among and work on existing orders
- Jeeves Mobile shopfloor - for users on the shop floor to report on own operations
- Jeeves Mobile warehouse - for warehouse staff to report their work directly from where the work is performed
- Jeeves Mobile field service – for service technicians to view and update service orders

2.5 Classic client

The Classic client is retained for existing installations covered by the support policy applicable to their release. It is no longer delivered for new installations or upgrades.

The Classic client is Jeeves' long-standing, Windows-based client. The designs in the newer Desktop client can be automatically generated with the same look as in the Classic client, if desired.

2.5.1 Data entry

You can set up the first entry field in each form, copy selected values from the previous record. You can also choose between data entry in the form or the list.

Specified values are validated directly when entering data.

2.5.2 Design

You can install role-based tabs and frames but there is also a possibility to adapt them or create your own design. There are tools for :

- changing headers in forms and lists
- changing fonts and colors of fields
- changing background color
- changing size and position of forms and lists
- hiding or showing programs, nodes, reports and terms
- creating styles
- defining formats for input and display of dates, time, numeric and monetary values

and other functions.

2.5.3 Quick settings via right-click

In addition to the tools on the **Tools** menu, there are a number of settings that can quickly be made by right-clicking on fields in forms. These settings are field-related and require that you focus or check the current field(s) before selecting a setting option. The settings that are available are:

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- Hide all flagged fields
- Hide current field
- Add help for current field
- Settings for current field (add filter condition, activate automatic wildcards, change texts, fonts, colors of fields and make conditional formatting settings)
- Show current settings
- Tabs
- Create total field from current list field

Via right-click you can also create bookmarks regarding records and transactions in Jeeves ERP, for example, regarding a sales order, a customer, a general ledger account etc. This can be helpful if you need to make a note or if you need to ask or inform a colleague about something in the database. You can:

- save bookmark to file
- send bookmark as email
- send bookmark with current conversation

2.5.4 Advanced settings

Furthermore there are functions for advanced settings such as user-defined sorting and selections in reports, as well as user-defined right-clicks to programs.

2.5.5 Program clones

Jeeves' function for cloning programs lets you create customized entry forms while the standard program form remains intact for reference.

2.5.6 Program menus

You can have any number of program menus and either make one or more menus visible or select menus from a list. A default menu can be set up per user or group, if required.

2.5.7 Role menus

If Jeeves' roles are used there are role-based menus according to role-based processes for distribution, manufacturing, supporting processes etc.

2.5.8 Program menu tools

Jeeves ERP has built-in tools to adapt pre-defined role menus or create own program menus per user, group, language etc. Programs on a menu can have standard icons that represent the program type, or you can use own icons.

2.5.9 Favorites and last used programs, program aliases

There is a Last used folder, in which Jeeves ERP automatically stores the last used programs. This can be useful if you need to retrace your steps for any reason. The folder is cleared when you log out. There is also an option for adding programs that you use regularly to a personal Favorites folder displayed above Jeeves' program menu.

There is an option for creating own program aliases or short-names for programs in Jeeves ERP. With the Run program... option you can run a program without searching for it on the menu if you know the program's internal data name in the system, or if you use an own alias, if you are authorized to do so.

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2.5.10 User defined program links

Jeeves ERP standard has many built-in program links to make it easier to access and maintain related information without having to close the program you are using. These links are accessed via embedded lists, right-clicks or nodes. If you often need to access a certain program from another program but there is no link available in Jeeves, you can create your own links between programs.

2.5.11 Graphical tree structure

To create an easier navigable interface, you can display the content in list forms as a tree structure. You can display all list forms in this way, but it is especially useful for BOM, packages, documents, etc., that comprise of several levels.

2.5.12 Business graphics

In Jeeves ERP, there is a built-in function with which you can quickly create and display business graphics, such as diagrams and charts, from a number of selected records in the list and fields in the form. The function includes a guide for creating and adjusting the graphics. You can choose between three types of charts, line, bar or pie chart.

You can print and copy graphics and save them to a number of different graphics formats or in PDF or HTML format. You can create macros to link graphics to forms for re-display.

2.6 Setting dimensions and storage

2.6.1 Settings dimensions

Settings can apply to:

- individual companies, company groups or all companies
- individual users, user groups or all users
- a specific client (device) or all types of clients
- individual programs, all programs that share a window or globally

2.6.2 Storage of settings

Separate storage of settings ensures that they are not affected during upgrade. This also implies that several Jeeves ERP installations can share the same settings and that you can test installations with and without settings. You can even set up several different site repositories against the same Jeeves ERP installation, for example, if you want to test settings without affecting the installation.

Note

Most of the remaining sections in this chapter apply to the Desktop client and, for supported existing installations, the Classic client.

2.7 Important, general features

There are a few general features in Jeeves' user interface that are important to learn as quickly as possible:

- All programs in Jeeves ERP have at least two windows, a form window and a list window. Each form window has a list window with the same title directly linked to it. The list window displays a selection of records or all existing records in the table, while the form window displays one record at a time.
- For the majority of business transactions Jeeves ERP creates a header and one or more details, or rows. Example: If you enter a customer order you create a header and one or more order details, i.e. order rows. The header contains data common to the whole order,

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for example, the customer company, delivery and payment terms etc. The rows specify the items and services etc. to be delivered on that order and any deviations from the conditions stated on the header. This structure means that the majority of programs for the entry of business transactions contain a form for the header and separate forms (or an embedded window) for the rows. If you exit the form or list for the rows you are normally automatically returned to the header.

2.8 User friendliness

2.8.1 Keyboard shortcut commands

Shortcut commands allow users to work with the keyboard instead of the mouse in many operations.

2.8.2 Check-off lists

Many programs have lists that simplify data entry, for example, a list of order rows to be picked, invoices to be paid etc.

2.8.3 Buttons and shortcuts

Many functions are run via buttons or shortcuts. You can also create your own buttons for recurrent searches, functions, and printouts.

2.8.4 Embed graphs and browsers in forms

You can embed objects, such as PDF, web browsers, and graphs in Jeeves' forms. The function is primarily for displaying supplier invoices but is general and can be used for displaying other files.

2.8.5 Parallel programs

You can run several programs in parallel. You can then toggle between them, copy data, etc.

2.8.6 Support for touch-enabled devices

The Jeeves ERP UI supports single point touches and swipe gestures on touch-enabled devices. The touch keyboard is shown automatically when a text input field is in focus. The swipe gestures are available in the intellisense fields.

2.8.7 User profiles

User profiles are available in Jeeves mobile routine to switch between different REST servers. Each user can have one or more unique profiles. The profile contains role, company and other user related details.

2.9 Searching and retrieving information

2.9.1 Unlimited search possibilities

One of Jeeves' greatest strengths is that you can search on almost any field, with or without values. Examples of use: search transactions with no voucher number to find unposted transactions, or transactions with your signature, entered on a certain date and with a certain status.

You can search on any part of the field content, for example, all companies whose names start with or contain one or more specific character(s). You can also use wildcards for unknown/unspecified characters. If there are several companies in the database you can search in multiple companies. There is a possibility to save an unlimited number of search filters.

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2.9.2 Links to related programs

In all programs you can view related information without passing via the program menu, for example, view customer or item data from orders. There is built-in intellisense in fields linked to other tables, for example, from order entry you can directly get a list of customers whose names start with a specific character without jumping to the customer program. You can even search against several fields (columns in the intellisense grid) simultaneously. You can define several fields (columns) to be displayed in the result list, for example, when searching with intellisense in the Company name field, you might also want the company ID and city/state to be displayed. There are also other features you will recognize from Jeeves' program lists and drop-down lists, such as the following:

- You can type more initial characters to jump directly to certain records in the list (only in the first column).
- You retrieve a row to the form with Space or with double-click.
- You can zoom in and out with Ctrl and the scroll button on the mouse.

2.9.3 Drill-down

You can drill-down to included or related transactions, for example, from an invoicing journal view the included invoices, from an invoice view the included order(s), from an order view picked order rows etc.

2.9.4 Save search in overlying level

There is a function on the **Search** menu for saving a search from an underlying level to an overlying level, i.e. a search made in an embedded list, in a right-click program or in an underlying program node, to be saved for use in the main form. Example of usage: You display an order/purchase order, drill down to the item rows and search for a specific item. If at this time you want to search for all order/purchase order rows for the same item, you can use this function.

2.9.5 Retrieve previous search condition

You can retrieve the last search condition used, for example, if you want to edit or add to it.

2.10 Documentation

2.10.1 Context sensitive documentation and integrated manuals

Jeeves ERP has comprehensive and context sensitive field, table and program documentation with hyperlinks to related chapters describing the setup and work flow(s), and other functionality.

The documentation is web-based and public, which implies that you can use it immediately, without any installation or file copying. This also makes it possible for Jeeves to publish new and corrected documentation more directly.

The web-based documentation has a fresh and modern interface with powerful searchability and filtering. Version management was introduced from Update 13, so you can select which Jeeves ERP version and Update you want to read about. You can also switch between English and Swedish on-the-fly.

The documentation presents lists in the form of a grid, in which you can sort on and arrange the columns, as well as filter in them. You can search on one or more individual word(s) or on an exact phrase. You can use a wildcard for any number of unknown characters. You can also search and filter on the titles in the table of contents or make a free-text search.

If you know the number of a chapter or the name of a field, program or table, you can enter it directly in the address (URL).

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2.10.2 Customer-specific tool tips

There is also a built-in function to quickly enter own tool tips for fields and to translate tooltips if users run the system in different languages.

2.11 Language versions

2.11.1 Language versions of the product

Jeeves ERP can currently be provided in the following, maintained language versions:

- Swedish
- English
- Polish
- French
- German
- Norwegian
- Finnish
- Danish
- Dutch
- Simplified Chinese

Besides these, the following language versions exist in part or for certain, earlier Jeeves versions:

- Turkish
- Slovak
- Czech
- Russian
- Estonian
- Spanish
- Italian

2.11.2 Default language and language selection

You can set up the client's default language per user, group etc. If desired, users can switch between available languages on-the-fly. This can even be achieved for data such as payment terms, chart of accounts and the like.

2.12 Integration between applications

If required, you can use all the integration between applications. This, together with the relational database technology, eliminates the need for duplicate entries. It is also easy to retrieve data from external systems or manually enter records in applications that are not included in the installation.

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3 Print design and output management

Jeeves ERP includes a large number of predefined documents and reports, such as delivery notes, order confirmations and payment reminders, as well as different types of statistic reports and journals. The documents and reports are linked to the programs to which they belong, i.e. delivery notes, order acknowledgements and invoices are found on the report menu for the various order entry and invoicing programs.

The same document/report can have different appearances depending on the printout template. If there are several templates, they are shown in a submenu to the document/report.

Because it is necessary to be able to customize printouts, Jeeves ERP also includes a design tool called the Template editor. The Template editor can be used to create one or more templates for a document or report and edit the appearance in different ways, such as:

- move fields or other parts of the report
- add and edit fixed texts
- change size, font etc. for fields, parts of the report or the entire report
- edit copies with different layouts
- hide fields or parts of the report and
- set up which printer is to be used for the template in question

Jeeves ERP also offers several alternatives for producing your own reports, for example, an in-house developed report generator as well as an external report generator.

See also separate chapter on Jeeves Document management (EDM) describing, for example, that you can also print directly to EDM.

3.1 Languages

Template proposals are provided in several languages and Jeeves templates are provided in English and Swedish.

3.2 Output management

You print a report simply by searching for one or more records, for example, one or more orders, and then selecting the report program from the report menu and the desired print template.

3.2.1 Own report menus

With the **Show selected reports**/**Show all reports** option you specify whether you only want to see the reports you have added to the report menu from the template editor or if you want to see all reports connected to the program.

3.2.2 Print format

If desired, you can print to screen instead of to a printer. (You can also print to a printer from the Print to screen dialog.) You can also export the report to file in various formats, such as Excel, HTML, PDF, etc. and either save the file to disk or send it directly with an e-mail. You can define printout formats, separate from display formats, for dates, base currency, foreign currency and numeric terms. You can use the export functionality to store documents created in Jeeves, for example, to store customer invoices in PDF format. You can then use Jeeves Document management to link the invoice printout to the invoice record in the database. In this way, it is possible to meet the legal requirements that certain countries have regarding the electronic storage of original documents.

The following are the setting priorities for printout formats:

Priority for format settings of numeric values: 1) on the term and 2) in General settings, Print format.

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Priority for format settings of dates: 1) on the term, 2) in General settings, Print format and 3) in the system defaults.

Priority for currencies: 1) Currency format setting on base or foreign currency fields, the next priority is same as the priority settings for dates.

3.2.3 Report generators

For information about the built-in report generator please refer to the chapter on *Jeeves General ledger*.

3.2.4 Dynamic reports

Jeeves has a built-in function for dynamically creating reports on a number of selected records in the list and fields in the form. You simply search for the desired records, check the required fields in the form and start the report. The dynamic report is printed to screen.

You can change the order and the width of the columns by dragging and dropping the columns and the lines between them. You can also right-click on a column, which opens a pop-up menu with modification options such as sum, alignment and change of font.

You can export reports from the preview in a number of different formats, such as PDF, HTML, RTF, Excel, Text format, JPG, etc. You can e-mail reports directly from the preview. You can also save reports and add them to the report menu in the appropriate program.

3.2.5 Link report to button/shortcut

You can create a button for printing with a certain template. Users can then start printing simply by pressing the button or entering a shortcut command.

3.2.6 Schedule recurring printouts/updates etc.

There is a possibility to schedule print and update jobs. If a company has very large transaction volumes it can be an advantage to schedule certain processes so that they are run automatically, for example, at a set time each night. There are also processes that must be run at predetermined time intervals, and for these it is an advantage to be able to schedule the process.

3.2.7 Cloud printing service

Jeeves Cloud printing service lets you print to physical printers as well as to virtual printers such as PDF and Excel. A Cloud agents program provides an internal registry of the cloud agents used in cloud printing. The Physical devices program supports physical and virtual printer mapping, device availability, and logical device configuration. You can redirect direct print jobs to a different printer and reassign print jobs to other cloud agents if one is unavailable. Jeeves returns full printer and job metadata in JSON format, including queue assignment, printers, report and publication IDs, parameters, and file path for virtual printer output. When a print job completes, its parameters are stored in a Job parameter history table, where you can view the parameters used for each job in an embedded list.

3.2.8 XML adapter (powered by StreamServe)

Jeeves ERP includes functionality for printing XML files based on reports in Jeeves. All fields that are available in the report template are also available in the XML file. The option to print XML files can be set for any combination of report, template and user.

Jeeves' StreamServe template package includes the most used documents for an ordinary distribution or manufacturing company. The templates are delivered in Swedish and English as a part of the StreamServe base package for Jeeves ERP. There is also a tool for converting your Jeeves templates to StreamServe templates.

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3.2.9 Separate report execution from printout

Report execution can be separated from printout if required. When you run a report, you have the option to have it execute in the background and not generate a printout. For example, you can create invoices and print them out later.

For all reports that are associated with a template, you have a no printout option that allows you to run the report without generating a printout.

3.2.10 Reporting Services (SSRS)

SQL server reporting services (SSRS), which is part of our technology stack, has matured enough to be an alternative for Crystal reports.

SQL Server Reporting Services (SSRS) is a generic part of SQL Server, with a distributable development tool included in the license. SSRS reports can be viewed in an embedded web browser thereby not demanding client installations. SSRS also adds distribution and subscription possibilities.

You can use SSRS as a tool for creating new reports separate from the current Jeeves reports.

You can manage SSRS reports from the Classic, Desktop and Mobile clients.

3.2.11 Print to EDM

A selection of reports can be printed directly to EDM. On selecting the report from Jeeves, the report is generated in the desired format and automatically linked to the record. You can then view and print the report as and when needed.

3.3 Print design

You can have any number of templates per report/document, for example, different invoice templates linked to different customers.

3.3.1 Company adaptation of templates

So-called overlaid templates can be used for global information, i.e. information that is to be included in several templates. One such example is own company's address data to be added as a footer to all external documents. You can paste images such as company logos, and also add fields.

3.3.2 Layout

You can select font, size, frames etc. for fields, for a part of a report or for a whole report. There are also functions for:

- Set up printer with the possibility to link different report templates to different printers
- Write a fixed text
- Add a picture/copy
- Snap to grid
- Hide/show edit marks and details

3.3.3 Link a template to a company, country or language

A print template can be linked to a company, a country or a language etc. in the system. This is limited to certain external documents and reports. The template is then automatically selected according to the print rule.

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4 Jeeves ERP CRM

Jeeves CRM is a highly functional CRM system including contact, sales and campaign management. It is fully integrated with all other applications in Jeeves ERP, which means it has direct access to all transaction data in the ERP system, such as open orders, accounts receivable, service orders etc. Its integration with Jeeves Document management gives direct access to all relevant documents regarding customer relations, like e-mails, contracts or quotations. All in all, this gives a complete picture of your customer relations.

4.1 Grouping and classification

The module provides many options for individual groupings and classifications of companies, business opportunities and contacts. This may include geographical groupings, relationships, sectors, product information, distribution data, etc.

4.2 Companies and prospective customers

In Jeeves ERP you have a common Company table for company information regardless of the type of company – prospective customers and business opportunities, customers, carriers and suppliers. This eliminates the need for duplicate entries and maintenance of company information.

Additional, CRM-specific information is managed in related tables. Here you can handle all types of customers - domestic and international, customers with another payer, with factoring, with several points of delivery etc.

4.2.1 Language management

You can manage the language of item names, units, payment terms, delivery terms, delivery methods etc. and print external documents such as quotations in the customer's language.

4.3 Relationship management

Relationship management includes functions for maintenance of information on companies, their contact persons and contacts and relationships with these.

4.3.1 Company relations

There are extensive possibilities for own segmentation/classification of companies, own keywords, gradual build-up of information about business relationships, planning and reporting of activities, customer cards etc.

4.3.2 Contact relations

You can enter data about contacts in a company in connection with a new entry of a company or do so afterwards.

There are extensive possibilities for own classification of contacts at companies, own keywords, contact information, direct mail via button etc.

4.3.3 Planned and reported activities

You can plan and report activities at both company and contact level. Activities entered at contact level will automatically be available at company level.

4.4 Sales support

Here there are functions for entering and processing business opportunities (for instance created in contacts with companies in connection with different marketing activities) according to an own predefined sales process. By changing the status code the probability and sales phase can be

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automatically changed, planned activities can be generated or the opportunity can be set to won or lost. In addition, there are functions for processing data about possible and probable sales and margins, quotation management, pipeline analysis and forecasting.

4.4.1 Business opportunities

There are functions in Jeeves CRM for entering and following up on various types of business opportunities. The functions can be used to quickly enter all business opportunities that have been created, for example, during marketing activities such as fairs, telemarketing campaigns, etc. in order to divide them up among sales personnel to be followed up. They can also be used for entering and following up on negotiations with new or existing customers.

You can enter one or more business opportunities for a company. If you want to group a company's business opportunities, you do so by entering a project and linking the company's business opportunities to the project. Linking to a project also enables cost follow-up for the sales process.

It is possible to enter business opportunities without specifying a company since the information may be sensitive at this early stage.

All historical data for closed business opportunities is available and can be searched and analyzed.

4.4.2 Probable sales

Here there is functionality to make it easier to enter and produce possible or probable sales figures and margins, both in total and your own share (your own company's share or the salesperson's share) and also in base currency or foreign currency according to the specified exchange rate. You can search in the fields for fiscal year and/or period and view totals for the selected period in the list.

4.4.3 Sales process

There is also an option for automation of certain status changes, for example, when changing to status Refused, the probability can be changed to 0 and the business opportunity can be changed to Lost, whereupon it is automatically moved to the Business opportunities, history program. In addition, the deviation from the original or estimated decision date is calculated.

4.4.4 Quotations

In the Quotations node, you can see all quotations linked to the business opportunity. There is also a field for a quotation ID in the main form where you can specify the relevant quotation, if required.

4.4.5 Analysis

In the Pipeline analysis node, you can see a graphical pipeline analysis showing the number of records and sum of possible sales value for each sales phase. The graph shows all active business opportunities, i.e. connected to a sales phase.

In the Future income node you can see future income, one year ahead, based on possible sales value for your business opportunities, grouped by either period or quarter. The graph includes all active business opportunities, i.e. connected to a sales phase.

4.4.6 Activities

There are many alternatives you can choose between when you plan and report marketing activities. You can choose between various programs. You can select the level to which you want to link an activity, i.e. to a company, to a business opportunity or to a contact person. The activities follow the company independent of the current business relationship.

If you want a detailed follow-up on times and costs, use the integration with the Time, projects, assignments module and, for example, copy programs from that module. If you want costs/revenues to affect the reporting, the project must also be invoiced in that module.

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4.4.7 To-do list with Outlook link

There is also a program for searching among own planned activities, for re-planning and reporting, which is linked to Jeeves' internal calendar. The program also has a function for copying planned activities to the personal Outlook calendar including details about the customer, quotation, opportunity and contact person the activity is related to. The personal Outlook calendar is also embedded in the program in order to facilitate planning.

4.4.8 Statistics, credit information and other customer information

As the CRM module is integrated with other modules in Jeeves ERP, you can bring up information created in other modules, for example, information on sales, ledger balances, etc. to get a better overall picture of your customers.

4.5 Marketing support

There are functions for setting up campaigns and distribution lists and you can use Jeeves' unlimited search capabilities to make selections of the contacts to be included in the campaign/distribution list. There are also functions to facilitate entry of campaign responses and to follow up the outcome.

Campaigns can also be linked to projects for follow-up on times, incomes and costs.

4.5.1 Campaigns and mailings

There is a program for definition of distribution lists for use in campaigns, etc. with information about the companies and any associated contacts the campaign is aimed at.

A campaign (a distribution list) can be created in many different ways:

- You can create a campaign for a number of companies in the Company marketing program.
- You can create a campaign for a number of contacts in the Contacts program.
- You can copy a previous campaign using the Create campaign button in the Campaigns program.
- You can copy a previous campaign based on the entered campaign responses using the Create campaign based on responses button in the Campaigns program.
- You can create a campaign manually by manually selecting companies or contacts to be included.

There is a function to send an e-mail to a campaign recipient. The e-mail address is retrieved from the Contact details table. If there is a contact specified on the campaign row, the individual's e-mail address is retrieved, otherwise the company's e-mail address is retrieved. The e-mail address can be directed to either the address field or the BCC field. If there are documents linked to the campaign in Jeeves Document management, these will be attached to the e-mail.

4.5.2 Generate activities

There is a function to create planned activities for the salesperson responsible for a campaign recipient. The function warns if there are recipients not connected to a salesperson.

4.5.3 Reporting of responses

There is a function that simplifies reporting of responses through a checklist and standard replies, and with a link to reported activities.

4.5.4 Campaign follow-up

When a campaign has been released you can follow up the outcome in the form of number of replies, reported hours, number of opportunities, number of quotations, number of orders and

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order value. You can also see the corresponding ratios in percentage compared to the number of respondents.

4.6 Competitor monitoring

There are functions for entering information and for monitoring the company's competitors. This refers foremost to information that makes it possible to compare your own company with competitors regarding product assortment, segmentation and financial measurements. Here you also can enter details about previous occasions when you have won or lost a procurement process against a competitor and the key factors that led to the won or lost deal.

In order to simplify the maintenance of the data there is a function to mark when key performance indicators and SWOT categorizations are no longer pertinent. If you check the Obsolete box in any of these rows, the row will not be displayed for the competitor company in the future.

4.7 Integration with other applications

As the module is fully integrated with other applications in Jeeves ERP, you can produce data on customer orders, orders, manufacturing orders, invoices, ledger balances. There are ready-made links that provide a selection of data. You can also create your own program links if you need more detailed information.

There is a global link to Jeeves Document management. You can use the link to connect the documents, e-mail etc. to companies, business opportunities and contacts and thus make information accessible and easy to find.

You can also store templates for your frequently used business documents in Jeeves Document management, to enable automatic creation and storage of documents, like letters or quotations, linked to the current company, or contact. The module can be further linked to external programs such as MS Word and Excel to obtain data from the database and create compilations and letters.

To facilitate field sales there is functionality for replicating contact information between Jeeves ERP and Outlook. This can either be performed via a function on the contact record, copying the current contact to the personal contacts folder in Outlook, or as a service copying all contacts to a public folder in Outlook.

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5 Jeeves ERP Purchases

Jeeves ERP's purchasing module can help business enterprises, organizations and authorities improve and fine-tune their purchasing process, irrespective of industry or field.

With Jeeves Purchases you manage the entire purchasing process – from requirement calculations and requisitions, to matching deliveries with supplier invoices and producing supplier statistics and evaluations – within the same application. Communication with external applications, for example, with services for handling eInvoice s from suppliers, is fully integrated and handled from within the Jeeves application.

The main purchasing process and common sub processes are packaged with pre-defined roles, role-specific menus and forms as well as user guides and training material. Besides these processes, you can activate other, wider functionality for specific company needs.

5.1 Companies and suppliers

In Jeeves ERP, you have a common Company table for company information regardless of the type of company – prospective customers and business opportunities, customers, carriers and suppliers. This eliminates the need for duplicate entries and maintenance of company information.

Additional, supplier-specific information is managed in related supplier tables. Here you can handle all types of suppliers, domestic and international, suppliers with another payee, with factoring etc.

5.1.1 Supplier categorization

You can handle any number of supplier categories to control approval and posting templates as well as statistics etc. You can have any number of supplier types for supplier evaluations.

5.1.2 Language management

You can manage the language of item names, units, payment terms, delivery terms, delivery methods, currency clause texts, etc. and print external documents in the supplier's language.

5.2 Purchased products and services

Jeeves ERP also has a common Items table for information regardless of the type of products and items – finished products, semi-manufactured, components, raw materials, packaging, consumables, stocked or non-stocked.

5.2.1 Product/item classification

There are many possibilities for categorizing and classifying items and products, for example:

- product classes – can control pricing and be used for statistics
- product accounts – control inventory management and valuation, and can control inventory postings
- item groups – can control pricing and be used for statistics and budget allocation
- item classes – can be used for purchase planning, statistics and budget allocation
- item categories – can be used for statistics, for example, for collections of item groups
- item types– can be used for statistics

5.2.2 Product variants

There is functionality for handling product variants, useful for companies in the fashion industry for example, for defining applicable combinations of say, models, colors, sizes etc. of a product with the help of a matrix. The matrix has various layers that you can switch between for viewing current and planned inventory balances, item numbers, alternative item descriptions, delivery dates, minimum purchase quantities - or if there are none, economic order quantities, prices etc. for each

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combination. The matrix is used in the whole process flow, from creating purchase orders from requisitions to receiving deliveries.

5.2.3 Seasonal products

You can set up seasonal items, orders and purchase orders, which can be used together with, or without, the product variant matrix. If used together with the matrix, fields for items that are out of season are inactivated.

If you specify a season on a purchase order header, you can enter purchase order rows for both seasonal and non-seasonal items. When entering purchase order rows for seasonal items the following checks are made:

- That the season complies with the season on the purchase order header
- That the purchase order type is allowed for this season
- That the delivery date complies with this season

5.2.4 Item status

There are other alternatives for handling the status of items in sales and purchases, i.e. whether an item may be purchased or sold. In Jeeves purchases, discontinued products cannot be ordered.

Besides these, you can use Jeeves ERP's batch functionality to control the status of items. See the description of the batch functionality under the Inventory module.

5.2.5 Acquisition methods

Purchased items have a default acquisition method – purchase to order point, to plan or to requirement – which also controls the work process. This can be changed on individual purchases. It is also possible to have different acquisition methods for different inventory locations (warehouses).

5.3 Costing

Please refer to the section on costing in the chapter on *Jeeves Inventory*.

5.4 Purchase prices and agreements

Purchase prices can be entered manually on requisitions and purchase orders, in which case the manually specified price applies. Otherwise the purchase price is retrieved as follows:

- One or more purchase agreements with one or more suppliers, including tiered prices and discounts if required (if there are several agreements the main agreement or the one with highest priority applies)
- Purchase price lists per supplier, including tiered prices and discounts if required, as well as any campaign price lists
- Fixed purchase prices for items

Purchase prices can also be set up to include VAT/tax.

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5.4.1 Purchase agreements

There are several ways to enter purchase agreements with prices agreed upon with the company's suppliers. The most common methods are as follows:

- Manual entry of purchase agreements with an agreement ID and header in the Purchase agreement/RFQ, supplier program. In this case you enter an ID and general information on the header and associated rows in the embedded list window (the Alternative suppliers table).

Note

In Jeeves, purchase agreements and RFQs are stored in the same table, but with different statuses.

- Manual entry of purchase prices without an agreement ID and header directly in the Alternative suppliers table, which is accessed via the menu or a node in the Items program.
- Automatic entry/update of purchase prices without an agreement ID for new suppliers or prices, in the Alternative suppliers table. This is done automatically if you check in the Update agreement price checkbox for the supplier, purchase order or at invoice matching. If checked, new combinations of items and suppliers generate new records, and new prices for existing combinations of items and suppliers update existing records in the Alternative suppliers table.

These methods can be combined.

You can also create purchase agreements or RFQs automatically based on existing records in other tables:

- Entry of purchase agreements/RFQs (with header) based on requisitions in the RFQ from requisition program. In this case you enter an ID and general information on a header and select rows from existing requisitions.
- Automatic entry/update of purchase agreements/prices (with header) based on customer quotations/agreements in the Quotations/agreements program. In this case a header is created automatically with associated details, which is linked to the customer quotation/agreement.

You can use the agreement status to control if a purchase order can be entered against the agreement. You can also set up purchase agreements with tiered prices.

5.4.2 New purchase prices on agreements

There is a function for recalculating purchase prices in an agreement. You calculate new, preliminary purchase prices and analyze the results before updating the current purchase prices. The purchase price changes are logged.

5.4.3 Purchase price lists

If you have not specified a purchase price on an agreement, Jeeves ERP looks for a purchase price on a price list.

You can enter purchase price lists per supplier and product class or per supplier and item. You can also combine these alternatives, in which case price lists per item have higher priority.

If a supplier advertises campaign prices, you can connect a campaign price list to the supplier’s normal price list.

5.4.4 Fixed purchase price on item

Normally, the purchase price is retrieved from the item/supplier record or the purchase price list. If there are none it can be retrieved from the item, on condition that you enter a purchase price

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currency for the item. Purchase orders in this currency will automatically retrieve the fixed purchase price on the item.

5.4.5 Early settlement discounts

You can set up early settlement discounts from suppliers with one level or tiered levels.

5.4.6 Posting of accrued royalties

You can define royalties per item, item group, or supplier. Accrued royalties are posted according to posting rules when you create the receiving journal and voucher.

5.5 Requests for quotations (RFQs)

Requests for quotations can

- be entered manually for distribution to an individual supplier or
- originate from a requisition for distribution to several suppliers via a distribution list

Requests for quotations can be converted to purchase agreements simply by changing status and making any adjustments to validity, prices etc.

5.6 Requisitions

Requisitions are a means of getting an overview of your purchase or manufacturing requirements. The basis for the requirements can be created via

- one of the programs for manual entry of requisitions
- one of the reports for creating requisitions against order points or against planned inventory activities
- one of the reports for MRP
- a function button on customer orders
- a function button on manufacturing orders
- purchase forecasts

In the requisitions routine the requirements are analyzed, a supplier chosen and quantities and time for deliveries adjusted. Based on the requisition you can then automatically create purchase orders by running a report in one of the programs for requisitions.

Jeeves ERP checks against any minimum/maximum inventory balances, order points, order quantities etc.

5.6.1 Replication of requisitions

Some companies that use Jeeves Master scheduling and forecasting need to replicate relevant data, such as item forecasts and requisitions, between sales units and manufacturing units. There is a report with which you approve selected requisition rows for replication to another company.

5.6.2 Fixed delivery days from supplier

You can set up fixed delivery days from suppliers to automatically get suggestions for delivery dates on purchase requisitions and purchase orders. When entering and creating requisitions, transforming requisitions to purchase orders and when manually entering purchase orders, a delivery date is retrieved from the calendar connected to the supplier.

5.6.3 Forecast purchase orders

You can create preliminary purchase orders, forecast purchase orders, to send to suppliers as advance information on what, according to forecasts, will be ordered in the future in a longer term than can be calculated from lead times, etc.

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You can then convert the forecast purchase orders into definite purchase orders after making any necessary adjustments.

5.6.4 Alternative unit on requisitions

Jeeves supports alternative units on requisitions. You can specify an alternative unit code and add the required quantity in the alternative unit. Note that you can only do this for manual entry of requisitions as a first step.

5.7 Purchase orders

Purchases can be created in many ways. You can either enter purchase orders manually or generate them automatically from

- purchase agreements
- requisitions
- customer orders
- service orders
- net requirements
- material reservations on manufacturing orders

5.7.1 Purchase order types

The work process is controlled through the choice of purchase order type. Purchase order types are predefined and determine how the relevant type of purchase order is created, how it is numbered, if the goods should be received to inventory, how part deliveries should be processed, and so on.

5.7.2 Purchase value

Purchase values on purchase orders are updated upon receipt of incoming deliveries. You can view the actual value of the purchase order, irrespective of if any deliveries have been received in fields to retain the values:

- Purchase order value in base currency for the whole purchase order
- Purchase order value in currency for the whole purchase order

5.7.3 Inventory order

Used when purchasing items to own inventory, i.e. the inventory balance etc., is updated. Also used to order goods that are not stocked, for example, items with the consumables inventory stock type.

5.7.4 Drop shipment

Used when goods are sold but NOT from own inventory. The supplier sends the goods directly to the customer. Inventory balances are not affected. Purchase orders that are created automatically from customer orders with the Drop shipment order type are automatically assigned this type.

5.7.5 Transit

A variant of drop shipment in which a purchase order is created from a customer order, but the goods are shipped via own inventory. Processing is more like normal orders except that picking never needs to be done. The received delivery automatically reports dispatch of the customer order.

5.7.6 Order specific inventory

The goods are placed in an inventory linked to the customer/manufacturing order. The items are picked from this order specific inventory. Via a parameter you can control that any goods remaining

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in the order specific inventory when the order is released should be picked automatically. An order specific inventory is useful when the goods have a high value and remain in stock a while before being shipped to the customer. There are programs and lists that show the value of the order unique stock.

5.7.7 Subcontract

Refers to purchasing of labor for an operation in a manufacturing order. Instead of performing the labor on the shop floor, the labor is purchased.

5.7.8 Advance purchase for manufacturing

Used when you know in advance that certain material with a long delivery time must be used in a manufacturing order. Preparation/design may not be ready, but you know that certain critical material/components will be used (order specific). The purchase order row or rows must be flagged with manufacturing order number (suborder number and operation number should be blank, and are set to 0 when the purchase order row is saved). Later, when you create a manufacturing order with corresponding material reservations and press the button to create order specific purchase orders, the material is linked to the advance purchase. If no such purchase exists, purchase orders or purchase requisitions are created.

5.7.9 Transfer between inventory locations

Used to create customer orders for transfer between inventory locations in the order routine, which means that you can create picking lists and delivery notes in the same way as for normal customer orders. The receiving inventory location sees the planned delivery in the planning list and reports the delivery when it arrives.

5.7.10 Payment terms

You can store an unlimited number of payment terms, per supplier, per agreement or general. Credit periods and any early settlement discounts can be set up in numerous ways.

There are several methods for calculating due dates:

- Calculate period from arrival or invoice date or from beginning of following month(s)
- Extend period to end of month

5.7.11 VAT

In Jeeves Purchases/Accounts payable you set up suppliers and items for VAT. Your company can apply any number of VAT rates.

See also separate sections describing VAT in the *Sales* module, *VAT declarations*, as well as *VAT reporting within the European Union*.

5.7.12 Fixed delivery days from supplier

You can set up fixed delivery days from suppliers to automatically get suggestions for delivery dates on purchase requisitions and purchase orders.

5.7.13 Lead times

Suggested delivery dates, whether on fixed days or not, always take lead times into consideration. The purchase agreement shows to the total lead time, i.e.,:

Administrative lead time

+ Lead time from supplier

+ Transport time

+ Possible extra lead time for, for example, checks, inspection, etc. (LedTidExtra1)

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The lead time on the item per inventory location record is updated automatically from the purchase agreement that is flagged as the main agreement.

See also the section on Jeeves' lead time matrix in the chapter on the *Sales module*.

5.7.14 Buffer days on order-specific purchases and requisitions

If no buffer days are set up, the requirement date is based on the customer order's estimated dispatch date. The standard functionality for determining the number of buffer days between the order row's estimated dispatch date and the preparation date is determined via Dispatch methods, sales, the Internal lead time days field.

Alternatively, you can use the Preparation date field on the customer order row to receive the requirement date for linked purchase and manufacturing orders or requisitions. This means the preparation date for the customer order is taken into consideration in the programs for creating order-specific purchase orders and manufacturing orders and in the MRP program. The preparation date can in turn be calculated based on lead times for the item/item group.

5.7.15 Alternative units

Alternative units can be used for automatic or manual unit conversion in the Inventory, Sales and Purchasing modules. If the function is activated, items can be purchased, stocked, sold, and returned in different units. Costing is, however, based on your inventory stock unit.

You can enter inventory transactions, order, requisition and purchase order rows, dispatches and received deliveries in your own inventory unit or in the alternative unit. You can set the function up to allow entry of quantities in both units, if the relationship between stock units and alternative units vary. This is useful in the food industry, for example, where a stock unit can vary in weight and you need to specify both the number of units and the weight. This setup also accumulates inventory balances in alternative units in Inventory balance records.

If you set the function up for automatic conversion according to a fixed conversion rate, the unit you enter is automatically converted into the other unit based on the conversion rate. In other words, if you enter a quantity in an alternative unit, it is automatically converted to your inventory unit, and vice versa.

Example: An item is stocked per piece. It can be purchased and sold per piece but can also be purchased and sold in the alternative units of 6 pack, 12 pack or 24 pack. If you use 6 pack as alternative unit when entering an item on an order or purchase order and specify a quantity of 3, the quantity is automatically converted to 18 in your stock unit.

If you set the function up for manual conversion, you enter quantities in both units and Jeeves ERP checks that the relationship between them is reasonable according to the given conversion factor and tolerance percentages. When using manual conversion rates, Jeeves ERP suggests the quantity in the alternative unit but you can change it. This applies only when entering new order rows, not when changing existing rows.

You can search for the price from a price list that is set for an alternative unit.

You can also deactivate alternative units that are no longer to be used.

5.7.16 Own or supplier's item number

You can enter purchase order rows using your company's own item numbers or the supplier's item numbers, when using the supplier's number Jeeves ERP automatically retrieves your own numbers.

5.7.17 External texts

General texts can be used to print standard texts on documents. For example, if you want to add a New Year's greeting or the like to all external documents, you can enter the greeting as a general text.

You can decide on which documents the text is printed, for example, on purchase orders, delivery notes, invoices, or all external documents. On the other hand, you cannot control that the text

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should only be printed on certain purchase orders, delivery notes or invoices. If you specify that the text is to be printed on a document, it will be printed on all printouts of that document.

Texts can be controlled by date using the Start date and End date fields. If you have specified several texts within the same date interval on the same document, the most recently entered text will be used.

In the same way, you can create supplier-specific standard texts and control which documents these can be printed on.

5.7.18 Surcharges

Automatic item surcharges can be applied on both sales and purchases. On sales you get automatic entry and invoicing of deposits, environmental charges, etc., which by law must be reported and paid in connection with sales of certain types of products, for example, DVDs. On purchases you can control and reconcile charges on supplier invoices for such products.

A surcharge is entered as a non-stocked item connected to a (normal) stocked item. It is priced according to the price given in the item table, and a customer/purchase order row is generated with the same quantity as the basic order row.

5.7.19 Joint purchases

Joint purchases can be used to decrease the number of purchases when purchase orders are created automatically from customer orders, and there are several order rows for items to be delivered from the same supplier within a specific time.

5.7.20 Subcontract

Subcontract purchase orders are for purchasing labor operations to be carried out at another workshop/factory. This can be necessary, for example, if the workload at your own workshop is too high or for a specific type of work.

5.7.21 Order confirmation from supplier

There is a function for entering delivery confirmations from suppliers. With this function, purchase order rows can be assigned a status that shows they have been confirmed by the supplier. At the same time, the promised and confirmed delivery dates are updated on the row.

Purchase order rows are automatically split if the supplier does not confirm the whole quantity, one row for each part delivery confirmation.

5.8 Received deliveries

The work process for receiving deliveries can be set up depending on the company’s routines and the products’ need for inspection or quality control etc. It can be anything between extremely simple and very thorough, and steps can be automated or manual. Different methods can be also be used within the same company.

You can use purchase order reports as a basis for checking received deliveries. Some of these reports can be set up to include information about customer orders waiting to be dispatched, in order to save time by not having to unpack and put away the goods.

5.8.1 Goods arrivals and receipt

In Jeeves ERP, you can choose from three alternative ways of handling goods arrivals and received deliveries against purchase orders. The alternative you choose depends on the elements included in the company’s goods arrivals routines; arrival entry, arrival check and received deliveries, and who does them. The choice can be made per supplier and/or item.

Receive directly without checking is the simplest process and means that updating inventory balances is done at arrival entry.

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Arrival check means that after arrival entry the goods are sent to a special table for goods that need to be checked. From this control table, you can print reports or find the items waiting to be checked. It is from the control table that you enter the approved quantity either for further processing by delivery personnel or for delivery directly into inventory.

At arrival entry, you can enter remarks and also create return delivery notes directly if something or everything is to be sent back to the supplier.

5.8.2 Receive shipment

There is a program for finding arrival entries per shipment ID in order to report received deliveries per shipment.

5.8.3 Part deliveries/backorders

You can choose between manual or automatic numbering of part deliveries against purchase orders. You can enter the supplier's delivery note number on the order header, which will be automatically copied onto the deliveries you enter.

5.8.4 Automatic bin location

Jeeves ERP can automatically assign a bin for received goods depending on bin type and quantity.

5.8.5 Blocked inventory

You can block a quantity in stock from being used or dispatched directly during goods receipt or at a later stage. This can be necessary if special inspection is required, for example, or if you receive many complaints.

5.8.6 Returns

Returns can be entered for any items that are to be returned to the supplier. A return delivery note can be created automatically. If there is an existing, unprinted delivery note for the supplier in question you can add to it.

5.8.7 Forward cover

There is a simple function for managing forward cover of foreign currency. It calculates the value of the received delivery at the purchase order’s exchange rate.

5.9 Supplier invoices

5.9.1 Matching supplier invoices against received deliveries

Supplier invoices can be linked to purchase orders for checking before final approval and posting. You can control whether only received rows may be matched and whether goods arrivals and matching can be done simultaneously.

There is also an option to automatically link and match supplier invoices and purchase orders as far as possible. If the data in the invoice agrees with the data in the received purchase order, and if the specified limits for differences have not been exceeded, then matching and final approval will occur completely automatically and the invoice will be marked as ready for payment.

5.9.2 Purchasing costs on supplier invoices

There are several alternatives for handling purchasing costs on supplier invoices in connection with invoice matching. The work method you choose depends on whether you apply FIFO (First In First Out), if you work with average weighted costs, etc., as well as if the overheads should be added to or should replace any markup on the items, suppliers, etc.

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5.10 Contract matching

Jeeves ERP supports contract matching for supplier invoices. With contract matching, organizations can automatically match invoices that they receive at regular periods and for predictable payments to an underlying contract, without the need for any supporting documents; for example, purchase orders.

You can enter preapproved contracts along with valid from-valid to dates, posting, and periodicity information. You can set reminders for when a contract time is about to run out, match incoming invoices automatically against existing contracts and set a matching tolerance on your supplier.

You can define how difference amount will be handled during invoice matching, have different suppliers on invoice and contract, multiple supplier invoices for the same contract period and specify percentage instead of a fixed amount when defining the contract posting template.

5.11 Supplier statistics and evaluations

Thanks to Jeeves ERP's superior search possibilities, and to the fact that you can customize forms and lists by adding fields, columns, totals etc., you can compile information on many dimensions in more or less any table. You can also build your own reports and business graphics based on own search criteria and layouts. Besides this, there are a number of standard forms for viewing, and reports for printing, statistics regarding purchases and supplier performance.

5.11.1 Statistics

Jeeves ERP includes forms and reports for received delivery and invoicing statistics per supplier. There are also connections between suppliers, purchase agreements, purchase orders etc. with received delivery statistics, for example, to check previous prices and matched invoices, delivery performance etc.

5.11.2 Evaluations

You can award points to suppliers based on any concept, standard or own to perform the required evaluations. The standard evaluation includes different aspects of delivery precision, such as the number of deliveries between ordered and delivered quantity, the number of late and early deliveries, as well as the number of returns.

5.12 Cost calculation templates/Detailed purchase calculation

You can set up cost calculation templates to split costs and control their postings. This is currently based on the former Import module functionality, which means it requires that the product account is set up for Lot handling and that the relevant items are stocked in batches.

A unique template ID is defined along with a template description when you create a new template for purchase/inventory cost templates. You can connect a Template ID to alternative suppliers, items, and suppliers. The purchase/inventory cost calculation template is applied when entering a purchase order for such an item, and the value is calculated on the average or FIFO value of the item.

5.13 Posting and integration with general ledger

You create inventory postings upon final approval after invoice matching. One final approval record is created and posted per invoice. Each final approval record generates a voucher in the general ledger.

You post deliveries that have not yet been invoiced and matched at desired intervals, for example, during monthly or more regular reconciliations. One delivery journal is created and posted per delivery date. Each journal generates a voucher in the general ledger.

Posting is done according to posting rules for inventory transactions that control to which account and account segments (cost center, cost unit, etc.) the delivery should be posted.

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5.13.1 Manual posting of purchases

It is possible to manually specify account segment 1, 2, 3 and 4, when purchasing for own use (consumables), as an alternative to automatic posting through the product account.

5.13.2 Reconciliations

During period reconciliations you can use Jeeves ERP's function for reconciliation of delivered but non-invoice matched quantities and values.

There are also functions for reconciliation and analysis of differences in quantities and/or amounts, as well as for analysis of postings, on customer orders with associated purchase orders, for example, Drop shipments and Transit orders.

5.14 Claims

You can enter and log purchase claims, i.e. remarks and complaints about items, purchase orders and supplier invoices, and record how they are to be settled as well as actions to be taken.

You can define different work flows for different types of claims, specify base data for scheduling, and control how activities should be reported.

You can also access any claims records from the purchase order rows.

5.15 Industry-specifics

Most functions in Jeeves Purchases are common to all industries. There are some, however, that are specific to certain industries, as described below.

5.15.1 Product variant matrix

The product variant matrix mentioned earlier was developed with the fashion industry in mind, but can be used for any products with variants, i.e. combinations of models, colors, sizes etc., or the like.

5.15.2 Alloy surcharges

You can set up different surcharges per kilo or percent, connected to purchase agreements for items. A surcharge can occur when purchasing metal plate as an addition to the base price. Note that alloy surcharges cannot be combined with purchase price lists.

You can also set up future surcharges. If a future alloy surcharge has been set up and the estimated dispatch date on the purchase order row is the same or later than the date when the new surcharge is to take effect, the new alloy surcharge is applied instead.

5.15.3 Import module/Detailed purchase calculation

Jeeves ERP includes a special function for cost estimate and actual cost on purchase order rows and complete traceability through lot numbers that follow the purchased items throughout the system.

You determine which items should be handled in this way using the inventory stock type and the product account (the Lot handling checkbox). This means that a project with an associated purchase cost record for the inventory location in question is automatically created when you order such items, which is assigned a project number from the purchase order row

A batch is created in the purchasing routine when the items in question are received, which is automatically given the same project number (lot number).

If you use allocations, during an allocation you can retrieve the project/lot number automatically from the batch number and the number is copied when picking. Otherwise, a batch ID is required when picking items that are stocked in batches, and you must enter it manually. In this way, you obtain a link and follow-up of a lot through the project/lot number from the batch.

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5.16 Localizations/internationalization

Most functions in Jeeves Purchases are common to all countries. There are some, however, that are specific to certain countries or trading areas, as described below.

5.16.1 Cost posting and/or inventory posting of received deliveries

Some countries, for example, France and Belgium, apply cost posting at received delivery instead of inventory posting. The received delivery value is posted to an interim cost account which is then reposted to a final cost account at supplier invoice matching. It is sometimes necessary to combine the two and in Jeeves ERP it is possible to do so.

The system does not continuously post inventory accounts, nor is Cost of sold goods posted when posting customer invoices. The cost is taken directly in connection with received deliveries of goods or in connection with arrival or approval of supplier invoices. Invoice matching against purchase orders in Jeeves ERP only functions if the amounts are posted to an interim account.

5.16.2 Intrastat reporting (European union)

Jeeves ERP includes functions for producing the required declarations of dispatches and arrivals between the company's own country and other countries within the EU, including drill down to the invoices included in the dispatches and arrivals totals. This is to make it possible to reconcile the data in the report.

If you choose to use the functions, you can either use the information generated as a basis to manually fill out the Intrastat form or you can compile the Intrastat report in an external report generator. You can also export the data to MS Excel using Jeeves ERP's Dynamic reports. You can then use Intrastat Data Entry Package (IDEP), which is free software that is available on the web, to automatically import your Intrastat reports into the tax authority's web site.

See the chapter on *Jeeves General ledger* for further information.

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6 Jeeves ERP Sales

Jeeves ERP's sales module can help business enterprises, organizations and authorities improve and fine-tune their sales process, irrespective of industry or field.

With Jeeves Sales you manage the entire sales process – from quotations and agreements, to invoicing customer orders and generating sales statistics – all within the same application. Communication with external applications, for example, with e-sales for webshops, is fully integrated and handled from within the Jeeves ERP application.

The main sales process and common sub processes are packaged with pre-defined process charts, roles, role-specific menus and forms as well as user guides and training material. Besides these processes, you can activate other, wider functionality for specific company needs.

6.1 Companies and customers

In Jeeves ERP, you have a common Company table for company information regardless of the type of company – prospective customers and business opportunities, customers, carriers and suppliers. This eliminates the need for duplicate entries and maintenance of company information.

Additional, customer-specific information is managed in related customer tables. Here you can handle all types of customers, domestic and international, customers with another payer, with factoring, with several points of delivery etc.

You can click a button to automatically create an order (header) for the selected customer, based on default values for example from the customer. The order rows form is opened in Add mode so that you can directly start entering order rows for the customer.

6.1.1 Customer categorization

You can handle any number of customer categories for posting control and statistics etc., any number of customer discount classes for discount management, also customer classes for own classification purposes.

6.1.2 Language management

You can manage the language of item names, units, payment terms, delivery terms, delivery methods, currency clause texts, etc. and print external documents in the customer's language.

6.2 Products and services

Jeeves ERP also has a common Items table for information regardless of the type of products and items – finished products, semi-manufactured, components, raw materials, packaging, consumables, stocked or non-stocked.

6.2.1 Product/item classification

There are many possibilities for categorizing and classifying items and products, for example:

- product classes – can control pricing and be used for statistics
- product accounts – control inventory management and valuation, and can control inventory postings
- item groups – can control pricing and be used for statistics and budget allocation
- item classes – can be used for purchase planning, statistics and budget allocation
- item categories – can be used for statistics, for example, for collections of item groups
- item types– can be used for statistics

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6.2.2 Product variants

There is functionality for handling product variants, useful for companies in the fashion industry, for example, for defining applicable combinations of say, models, colors, sizes etc. of a product with the help of a matrix. The matrix has various layers that you can switch between for viewing current and planned inventory balances, item numbers, alternative item descriptions, delivery dates, prices etc. for each combination. Allocation and picking can be reported per item (master item) and season or assortment, if desired. The matrix is used in the whole process flow, from placing orders to printing invoices.

6.2.3 Seasonal products

You can set up seasonal items, orders and purchase orders, which can be used together with, or without, the product variant matrix. If used together with the matrix, fields for items that are out of season are inactivated.

If you specify a season on a customer order header, you can enter order rows for both seasonal and non-seasonal items. When entering order rows for seasonal items the following checks are made:

- That the season complies with the season on the order header
- That the order type is allowed for this season
- That the delivery date complies with this season

6.2.4 Item status

There are other alternatives for handling the status of items in sales and purchases, i.e. whether an item may be purchased or sold. Statuses in the sales module:

- Item OK to use – whether the item is available for use at all (in sales, service and production)
- Sellable item – whether the item is available for sales and service orders
- Item OK to quote – possibility to override the above statuses for quotation only
- Discontinued – the item is available for sale until the inventory balance reaches 0

Besides these, you can use Jeeves ERP's batch functionality to control the status of items. See the description of the batch functionality under the Inventory module.

6.3 Sales prices and discounts

You can set up sales prices in a number of ways. The most common is to work with price lists and one or more discount matrices, or customer and order value discounts. You can also combine the different pricing alternatives, for example, link a customer to a price list and a tiered order value discount.

You can control which date should be used when retrieving prices to quotation and order rows, order/quotation date or today's date.

As standard, Jeeves ERP searches for the relevant price in the order given below. Your company can, however, apply a different pricing method and change priority or disregard one or more of the pricing alternatives.

Jeeves ERP automatically sets the pricing method that has been used on each order row.

6.3.1 Quoted/agreed prices

If there is a quotation or agreement, the quoted/agreed prices apply.

6.3.2 Customer-specific price list

If you set up a customer-specific price list, i.e. with a company ID in the price header, and the function is activated in the system data, these prices apply for that customer in combination with any discount rates.

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6.3.3 Price lists

If you set up a price list, connect customers to the price list in the Customers table, and if the function is activated in the system data these prices apply for these customers in combination with any discount rates.

You can set up price lists per item or item group. Price lists can have set prices or be based on markups or reductions on cost/list prices. Validity can be controlled against order entry or dispatch date. You can combine price lists with discounts with or without quantity limits. You can link a customer to one or more price lists.

6.3.4 Campaign price lists

You can set up date controlled campaign prices. When entering orders, Jeeves ERP compares these with the customer's normal price list and applies the lowest price.

6.3.5 Supplementary price lists

Supplementary price lists can be used for items that are not included in a customer's normal price list. The idea is that you use supplementary price lists together with customer-specific price lists, to hold the prices that are common to several or all customers. They only work in one step.

6.3.6 Fixed prices for items

You can enter a fixed sales price in the item’s list price field, which applies if agreement or price list exists.

6.3.7 Prices per product class

You can use the item’s product class to either set up fixed or tiered prices, or to calculate sales prices based on cost prices per product class. This will override any calculated price per product class. Pricing method on order rows is automatically set to the method that has been used on each order row.

6.3.8 Net prices

Prices can be set up as net prices on price lists and quotations/agreements.

6.3.9 Prices including value added tax

Prices can also be set up to include VAT/tax when selling to consumers.

6.3.10 General customer discounts

You can give general customer discounts, or code controlled customer discounts i.e. controlled by codes on the customer and item.

6.3.11 Discount matrix

The discount matrix enables you to automatically determine sales discounts depending on which item the customer purchases and, if so desired, in what quantity. You can also link customers to price lists via the discount matrix.

A discount is retrieved from the matrix to the order row when you enter the order row quantity, if a discount has been specified for the combination of customer discount class, item discount class and quantity.

You can set the availability for a discount matrix based on date. The addition of the date fields allows you to preconfigure the discount matrix and simplifies the process for future changes.

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6.3.12 Order value discounts

You can set up discounts to be given on customer orders based on the order's total discountable net/gross value. These can be general order value discounts, according to agreements with customers, according to a discount matrix or determined by codes on customers and items.

You can combine the various pricing and discount alternatives, for example, a customer can be granted both a price list and a tiered order value discount.

6.3.13 Early Settlement Discount

You can set up early settlement discounts on one level or on tiered levels. You can also calculate these discounts on the invoice total or on the value of goods. If calculating on the value of goods, you can also determine which goods are to be included in the calculation.

6.3.14 Manual discount on order

Users can, of course, manually enter discounts on order rows, for all types of discounts except for order value discounts. A discount can be specified on the order header and copied to existing order rows.

6.3.15 Combined costing and pricing

Most functions for costing are gathered in Jeeves Inventory, this applies to both purchased and manufactured items. There is, however, a function in Jeeves Sales in which you can calculate both cost prices and list prices from the same program. Via this function you can also import price lists from suppliers via Microsoft Excel. The import function is based on Jeeves Message Integrator (JMI).

Jeeves ERP provides a standard XML mapping for import of price lists, based on the columns that are possible to update in the costing and pricing programs. You can adjust the mapping, for example, if the columns names in the file from the supplier are different from Jeeves ERP's column names. You can also add any additional calculations to be done in conjunction with the import.

6.4 Quotations and agreements

You can either enter quotations

- in the quotations/agreements program and, when the customer makes an order, convert the quotation into an order (the quotation is then stored in the quotation system's tables) or
- in the order entry programs but with order status Quotation (the quotation is then stored in the order system's tables).

6.4.1 Quotation templates

You can set up one or more standard templates for quotations and simply change the customer number when creating orders from the quotation templates. For this, the customer number specified in the template should be coded as an internal customer in the company table.

6.4.2 Quotations with tiered prices

You can specify tiered prices on quotations to customers where the price is determined by either a calculated dispatch date or the order row quantity.

6.4.3 Quotations connected to mailing lists/campaigns

When you connect a quotation to a mailing list or campaign, the customers in the campaign are copied to the Customers on quotation node. When you print the quotation, it is printed out for each customer and the print date is updated on these records.

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6.5 Sales orders – process/work flow

Order processing is controlled by:

- The order program you use, i.e. Order from quotation/agreement or Order processing, standard, complete or package.
- The order type you choose on each order or order row. The order type controls which activities the order is to go through. By this we mean, for example, order acknowledgement, allocating against inventory or purchase/manufacturing orders, picking, release and invoicing, as well as printing documents corresponding to these activities. The order type also determines the order and invoice number series to be used.

Customer orders and order changes can also be received and processed automatically through standardized APIs. For more information, see *Digital business documents* later in this chapter.

6.5.1 Order types

Any number of order types can be applied. A set of pre-defined order types are included but can be adjusted according to own requirements:

- Normal - Used when items are sold to customers from own inventory. Order acknowledgements, picking lists, pack lists, freight bills, invoices, etc. can be printed. Picking can be done by button or via the picking program/node.
- Internet - Used when you want a separate number series for Internet orders (necessary for replication). Works the same as Normal order in all other respects.
- Direct issue - A variant of Normal order but where picking is automatically reported when an order row is entered. No picking lists are printed. Used when entering orders after dispatch or when selling over the counter.
- Automatic release - A variant of Direct issue, but picking is automatically reported when the order is released. This means that order rows that are changed or canceled before release do not generate inventory transactions (since no picking is done). When you exit order row entry you are asked if you wish to release it.
- Drop shipment (supplier->customer) - Here the items are not sold from own inventory. A purchase order is created from the sales order and is sent to the supplier for direct delivery to the customer. No picking lists, delivery notes or freight bills are printed. The order is invoiced when the supplier confirms shipment to the customer, or when the supplier invoice arrives. Inventory balances are not affected.
- Transit (supplier->inventory->customer) - A variant of Drop shipment, but the goods pass through own inventory. The work flow is much like Normal order, except that you do not need to pick the items; they are automatically shipped upon receipt.
- Order-specific inventory - A variant of Transit via own inventory but the goods are not automatically picked on receipt. They are placed in a separate inventory linked to the customer order (order-specific inventory). The items are then picked from this order-specific inventory. Via a parameter you determine how any remaining, unpicked items are to be handled. Order-specific inventory is useful when the goods have a high value and remain in stock a while before being shipped to the customer. There are programs and lists that show the value of the order-specific stock.
- Stock replenishment ICT - Used for internal stock replenishment from own, delivering business unit to other, selling business unit (internal company/inventory location) via Inter Company Transactions (ICT). A purchase order is created at the business unit where the stock is to be replenished, and a customer order is automatically created from the purchase order at the business unit to supply the goods for replenishment. Customer/purchase orders of this type require a company ID that is flagged as an internal company, when you enter the supplier ID on the replenishment purchase order at the own business unit, and also for the customer ID used on the customer order that is automatically created at the supplying business unit. For further information, please see the chapter on *Jeeves ICT*.
- Credit order - A Credit order is a negative Normal order. The quantity on the order row must be entered as a negative quantity for return to inventory. (Note: The order type does not control the return to inventory; the quantity field does this.) Jeeves ERP does not actually

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use the order type to create a credit note/invoice; it is the invoice total that determines if the invoice is a debit or a credit invoice.

- Price adjustment - Used to adjust incorrect prices on invoices issued. The cost price is set to zero on the invoice. No inventory balances are affected.
- Sample - A variant of Normal order where the sales price on the invoice is set to zero. If you have also flagged the Temporary inventory transfer field on the order type, the items are automatically moved to the inventory location specified in the To inventory location field, i.e. the inventory location for loans/temporary deliveries. Order type flagged in this way are automatically blocked from invoicing.
- Transfer between inventory locations - Used for transferring goods between inventory locations to allow order and inventory personnel to plan orders, generate picking lists and delivery notes as with normal orders. This order type does not generate an invoice; the inventory value is instead transferred between the inventory units via inventory postings.

6.5.2 Order entry templates

You can create templates to make it quicker and easier to enter orders with the same/similar content. When entering an order based on a template, the order rows can be generated all at once or one at a time.

6.5.3 Parameter controlled work flows

Standard order statuses control the roles that can access the order and the activities that can be performed, as well as giving information about how far it has come in the order process.

There are also parameters to control the work flow according to own processes, for example, which status changes are automatic and which are manual, which status changes/entries are allowed in which programs, which documents can be printed for a certain status etc.

6.5.4 Available to promise

Available to promise (ATP) functionality gives an accurate picture of the available quantities of a product on an estimated dispatch date. These calculations/queries can be run from customer orders and/or from the Assembly analysis program.

6.6 Sales orders – terms and credit control

6.6.1 Payment terms

You can store an unlimited number of payment terms, per customer or payer, per agreement or general. Credit periods and any early settlement discounts can be set up in numerous ways.

There is an array of methods for calculating due dates:

- Credit period in days or in months
- Calculate period from delivery date or from beginning of following month(s)
- Extend period to end of month
- Set dates, for example 10th, 20th and last day of month
- Cash on delivery
- Advance payment before delivery

6.6.2 Orders for payment by installments

You can handle customer orders whose invoices shall be paid by installments. For each payment installment you specify the credit period in days and the percentage of the total amount to be paid in the installment. The sum of the percentages must be 100.

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For example, you could define:

- two equal installments with 30 days between them; enter one row for 30 days and one for 60 days with 50 percent on each or
- three installments; a first installment of 20 percent after 10 days, a second of 30 percent after 20 days and the remaining 50 percent after 30 days.

On the payment terms you can also enter a minimum limit for splitting invoices into installments.

You can adjust the installments, to create order specific installments, if desired but the sum of the percentages must be 100.

When invoicing, Jeeves ERP creates one invoice with installment records. There is a special window in the invoice report template that you can make visible in order to print information about the installments.

6.6.3 Credit limit check

Credit limit checks can be performed for customers and/or payers. Aggregated sales order and service order value will be included in the credit check. If the combined value exceeds the credit limit set for the customer, an appropriate message will be displayed.

There are two ways of working with credit limits in the order routine:

- Basic function with only a manual credit block

In order for the automatic credit limit check on orders to take effect you must define a credit limit for the customer. If you set a limit, the customer's credit will always be automatically checked and the user will receive a message, when entering order headers or rows, if the specified limit is exceeded by the total of the current A/R balance and order balance.

This type of credit limit check does not provide automatic credit blocking of orders during entry, but you can manually block an order by changing the credit block status field on the order header.

- Extended function with manual or automatic credit block

As an alternative to the basic function, you can specify which type of credit limit check is to be applied for the customer. This type of credit limit check allows you to manually or automatically block an order being entered if the customer's credit limit has already been exceeded or is exceeded during order entry. You can continue to add to and change a blocked order but the order cannot be dispatched.

You can turn off Jeeves credit control for individual users, for example, for warehouse personnel.

There is also a function in the accounts receivable for automatic credit check of customers based on the customers' due invoices.

6.6.4 Addresses

You can store an unlimited number of invoicing and delivery addresses connected to customers and enter a one-time address manually on orders. You can also specify additional information such as building name, GPS coordinates or other references regarding a delivery address.

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6.7 Sales orders – inventory control

6.7.1 Availability check

You can set up automatic availability check at order row entry as required, for example:

- simple availability check – only current inventory balance minus order rows – or
- more complex availability check – current inventory balance minus planned receipts and dispatches
- whether or not to include inventory allocations
- whether or not to include purchases and manufacturing orders
- only within the item's lead time or up to the order row's estimated dispatch time

You can receive proposals for new dispatch dates. On part deliveries Jeeves ERP performs a new availability check and calculates the delivery date for the backorder.

6.7.2 Lead time matrix

Jeeves ERP also offers more advanced functionality for describing different items' distribution via the company's warehouses and the lead times involved, in the form of a matrix. The advantage of the lead time matrix is that it can describe a distribution chain without the need for alternative supplier records and allows availability checks in multiple levels.

You enter the various types of lead times, for example, lead time from supplier, transport time and administrative lead time, between each To-From company and connected inventory location. The information displayed by the availability check will be based on the balances at each warehouse and the lead times between them.

When you enter an order row for an item recognized by the matrix, the matrix displays the available balance at each consecutive inventory location that is needed in order to fulfill the order quantity. It also shows the lead time and estimated delivery date for each step in the distribution. You are then asked if you wish to change the estimated delivery date according to this information.

You can include net requirement planning for each step in the distribution chain, which is an efficient way of working with the lead time matrix, but it is not mandatory. External business units can also be part of the distribution setup, in which case you will need to read about the setup and use of Jeeves ICT.

6.7.3 Shortage check

Jeeves ERP can check for shortages automatically each time you open a customer order and/or by running a report, with or without a printout, in a special order planning program. You determine which alternative you want to use with an application parameter. If you run the shortage check via report, order rows and the order header are updated with status codes which can then be used for searching.

6.7.4 Allocations routine

Allocating customer orders ensures that the items are in stock, so that you do not generate picking lists unnecessarily. Jeeves ERP offers several ways of working with allocating items in inventory:

- There is an allocation routine where you manually allocate orders by running a report (or pressing a button) according to rules for how your own company and your customers allow partial allocations and deliveries. You can also manually allocate individual order rows to set priorities or to make exceptions to the rules when necessary. There is also an option to delete and re-run allocations that have been made.
- In addition, there is a direct allocating function that can be useful for certain types of commerce, for example, Web commerce, which means that if there are items in inventory when placing an order they are allocated directly. This function provides the only possibility for re-allocate if other items become available, for example, through cancelations or the like.

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You can also allocate items against purchase/manufacturing orders in the case of inventory shortages.

6.7.5 Requisition/purchase order/manufacturing order from order

Jeeves ERP informs you if you need to order items to fulfill an order. This is the case for certain order types (for example, drop shipments and transit orders) and for items with certain acquisition methods (for example, purchase or manufacture to order).

You create the purchase order/requisition/manufacturing order simply by pressing a function button on the order header.

For orders that generate requisitions for material from net requirements, purchase or manufacturing requisitions, you can have an automatic pre-allocation when the requisition becomes an actual purchase or a manufacturing order.

6.7.6 Allocation and picking batches

You can allocate and pick items per batch and control the order in which the batches are to be proposed for allocation:

- Use/Best before date – against order or delivery date
- Sell before date – against order or delivery date
- Batch number order

If allowed, you can allocate a customer order row against several batches.

You can also set up any number of status controls, OK/not OK to allocate, to pick, include/exclude from available balance.

6.7.7 First in first out (FIFO) batches

Jeeves ERP can be set up for automatic allocation and picking of the oldest batch.

6.7.8 Allocating and picking serial number items

When allocating and picking you can enter an item's serial number. The Use/Best before date for the individual serial number item can be controlled in the same manner as for batches. The sale and delivery of the item, as well as any other activity, are possible to trace. For example, if service or reparation is to be performed on a serial number item in the service module, the service planner automatically gets information about the customer to whom the item was sold, as well as connections to the sales order and inventory transaction.

6.8 Sales orders – additional functions

6.8.1 Package sales

You can set up fixed and configurable packages for sales order entry. You can run availability checks to see if a whole package can be delivered within the requested dispatch date. There are numerous ways of controlling how partial deliveries, price calculations and postings should be handled. You can also set up how package details should be presented on internal and external documents.

6.8.1.1 Price calculation types, package items

You can define alternatives that can be used to control package price calculations. For example, the price can be calculated from either the package header down to details or totaled from details up to the package header.

When you enter packages, this is retrieved first from the item and then from the system table for your own company.

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You can also specify if the list price on the package items shall be calculated by adding the cost price of the included package details.

The following price calculation types are available for package items:

- 0, Price->details, post det (sales price): Price is calculated and set for the components based on current quantity and sales price for components and sales price for the package header. Posting of revenue and cost in connection with invoicing is performed based on the components' respective product accounts.
- 1, Price->details, post pkg (sales price): Price is calculated in the same as for Price calc meth, pkg = 0, i.e. Price->details, post det (sales price). The difference is that posting of revenue and cost in connection with invoicing is done based on the package item's product account.
- 2, Price->details, post det (cost price): Price is calculated and set for the components based on current quantity and cost price for components and sales price for the package header. Posting of revenue and cost in connection with invoicing is performed based on the components' respective product accounts.
- 3, Price->details, post pkg (cost price): Price calculation for details is the same as for Price calc meth, pkg = 2, i.e. Price->details, post det (cost price). The difference is that posting of revenue and cost in connection with invoicing is done based on the package item's product account.
- 5, Price=details, post details: Sale price of the package item is calculated and set by totaling the sale prices of the included details. Posting of revenue and cost in connection with invoicing is performed based on the components' respective product accounts.
- 6, Price=details, post package: Price calculation for package item is the same as for Price calc meth, pkg = 5, i.e. Price=details, post details. The difference is that posting of revenue and cost in connection with invoicing is done based on the package item's product account.

A number of application parameters are available for handling package items including an alternative price calculation method that can be activated for components.

6.8.2 Surcharges

Automatic surcharges can be used on both sales and purchases. On the sales side this gives automatic entry and invoicing of deposits, environmental charges, etc. that by law must be reported and paid in connection with sales of certain types of products, for example, DVDs. On purchases, you can control and reconcile charges on supplier invoices for such products.

6.8.3 Options/accessories

The function for options/accessories can be used during quotation and order entry to automatically propose available options/accessories for an item. For example, when selling a laser printer you could also offer an extra paper tray, color cartridges, etc. This can help generate additional sales. You can apply either fixed or automatic pricing based on the "main" item. You can have different options, a different validity period or a different price depending on the customer group.

6.8.4 Alternative/replacement items

Alternative/replacement items is a function that allows you to specify a replacement item if the item in the customer order is not available in stock during order entry. A checkbox on the customer controls if you are allowed to sell alternative/replacement items to the customer.

6.8.5 Weight and volume

If you specify of weight and volume on items, Jeeves ERP automatically calculates the total weight and volume on dispatch orders and invoices. There is also functionality for calculating the volume of stacked goods, where the volume is not always the same as the individual item's volume multiplied by the order row quantity. This is useful for furniture for example, and other goods that can be stacked.

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6.8.6 Units of measure

You can define any number of units of measure, such as piece, dozen, meter, m^3 etc. Each item is connected to a default sales unit, which can differ from stock and purchase units.

6.8.7 Alternative units

Alternative units can be used for automatic or manual unit conversion in the Inventory, Sales and Purchasing modules. If the function is activated, items can be purchased, stocked and sold in different units. Costing is, however, based on your inventory stock unit.

You can enter inventory transactions, order, requisition and purchase order rows, dispatches and received deliveries in your own inventory unit or in the alternative unit. You can set the function up to allow entry of quantities in both units, if the relationship between stock units and alternative units vary. This is useful in the food industry, for example, where a stock unit can vary in weight and you need to specify both the number of units and the weight. This setup also accumulates inventory balances in alternative units in Inventory balance records.

If you set the function up for automatic conversion according to a fixed conversion rate, the unit you enter is automatically converted into the other unit based on the conversion rate. In other words, if you enter a quantity in an alternative unit, it is automatically converted to your inventory unit, and vice versa.

Example: An item is stocked per piece. It can be purchased and sold per piece, but can also be purchased and sold in the alternative units of 6 pack, 12 pack or 24 pack. If you use 6 pack as alternative unit when entering an item on an order or purchase order and specify a quantity of 3, the quantity is automatically converted to 18 in your stock unit.

If you set the function up for manual conversion, you enter quantities in both units and Jeeves ERP checks that the relationship between them is reasonable according to the given conversion factor and tolerance percentages.

6.8.8 Forward cover

There is a simple function for managing forward cover of foreign currency, which implies invoicing of orders at the orders' exchange rate.

6.8.9 External texts

General texts can be used to print standard texts on documents. For example, if you want to add a New Year's greeting or the like to all external documents, you can enter the greeting as a general text.

You can decide on which documents the text is printed, for example, on purchase orders, delivery notes, invoices, or all external documents. On the other hand, you cannot control that the text should only be printed on certain customer orders, delivery notes or invoices. If you specify that the text is to be printed on a document, it will be printed on all printouts of that document.

Texts can be controlled by date using the Start date and End date fields. If you have specified several texts within the same date interval on the same document, the most recently entered text will be used.

In the same way, you can create order-specific texts and/or customer-specific texts and control which documents these can be printed on.

6.8.10 Order linking

With the order linking function you can automatically or manually link existing open orders or backorders for a customer to the order currently being entered for that customer. This can be used to gather backorders into a new order. It can also be used if a customer places many small orders each day and you wish to gather them to one order.

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6.8.11 Create credit order from invoice

You can use this function to create a credit order linked to an original order and thus get exactly the same data, such as price data, on the credit note as on the original order. The function also allows you to create a return of the same items that were shipped against the original order. You can create single or multiple credit orders from collective invoice and specify the desired functionality.

6.8.12 Sales against advance payment

You can handle advance payments from customers from different modules. In the sales module you can handle sales orders against advance payment in the following ways:

- You can generate an advance payment from an order
- You can link an order to one or more existing advance payment(s)
- You can enter an advance invoice and payment and manually link the advance invoice to an order or invoice

6.8.13 Delivery confirmations

You can enter delivery confirmations to customers. This implies that when a part delivery is confirmed the order row is automatically split, one row for each part delivery, and the order row is updated with a confirmed delivery date.

6.8.14 Order analyses and order entry statistics

Order statistics and analyses can be produced at any time and for any period:

- Order intake
- Open order analyses
- Order analyses, customer/item
- Cross-analyses, customer/supplier

6.8.15 Order cancellation

You can cancel orders using the button on the order header or by changing the status to Canceled. You can also cancel order rows using the button or Status field on the individual order rows.

6.8.16 Order row change log

You can log certain changes on customer order rows in the Change log, order rows table. If activated, a record is created here when a change is made to a logged field directly on an order row, or indirectly from the order header. There are fields to show the current values and corresponding fields to show the previous values before the change.

6.9 Jeeves Mobile order routine

A standard mobile routine for handling sales orders is developed and maintained in Jeeves standard.

Jeeves Mobile order routine is used for the most basic needs of salespeople out in the field. It provides a simple and efficient way of entering and managing orders for customers, and to create and send order confirmations to them.

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6.10 Dispatches

6.10.1 Shipment planning

There are two sets of programs for shipment planning; one set for use with Jeeves Dispatch planning and one set for use with Jeeves integration with Astro. You use the functionality to group and allocate dispatch orders to shipments, and to monitor that the whole shipment is picked, as opposed to the individual orders or dispatches to individual customers.

Jeeves shipment planning is based on:

- fixed departure days and times
- transportation resources and their capacity
- lead times for processing customer orders

The function also gives you the possibility to plan ahead to fill capacity.

The setup is described in more detail under the section on Jeeves Sales-WMS integration.

6.10.2 Dispatch planning

You can use Jeeves ERP's dispatch planning to coordinate a customer's orders to one or more dispatches, with collective invoicing or invoicing per order.

6.10.3 Transport automation

Jeeves ERP can be set up to automatically propose a carrier as well as fixed dispatch dates depending on transport route, carrier and dispatch method.

6.10.4 Pick lists

There is a choice of single or multiple pick list(s) per dispatch order, which gives you the possibility to have several pickers on large orders.

6.10.5 Picking

You can also choose between different picking programs and functions depending on information needs, work method etc. For example, there is a function for automatic picking of all order rows that have not been picked on an order and its release for invoicing. There is also a function for scanning the barcode on a product package, automatically matching records in the left list with the scanned barcode, and moving these records to the right list. This function only works if Serial No., qty=1.0 field is set to N on the relevant inventory stock type.

Once you scan the barcode and click OK, Jeeves ERP initiates the process of sending the information to the EMVS system for validation. However, if EMVS is not enabled for the inventory stock type, the item is moved for picking to the right list as normal. EMVS is described further on, under Industry-specifics.

6.10.6 Joint picking

With Jeeves ERP's function for joint picking you can coordinate picking per item, inventory location, dispatch date and picker/packer. The objective is to increase efficiency for packers so they do not need to retrace their steps if an item can be picked for several orders at the same time.

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6.10.7 Split item packages and miscellaneous inventory

You can set up a markup percentage or amount on split item packages. You can also store split item packages in a separate bin for miscellaneous inventory and automatically allocate inventory from this bin if the order row quantity is less than the item's package size.

6.10.8 Blocked balance/quantity

You can block a whole inventory balance, or a certain quantity within the balance, from being used or dispatched directly during goods receipt or at a later stage. There is a possibility to open up for users to pick from blocked balance, if necessary.

6.10.9 Part deliveries and backorders

Via parameters you can control if part delivery is allowed and if backorders should be created and shown on external documents.

6.10.10 Labels

You can print labels with item numbers in the desired format, such as bar code format.

6.10.11 Automatic freight calculation

You can set the order routine up to automatically calculate freight on customer orders. You can have any number of matrices to calculate freight by weight, volume or number of pallets, as well as by distance or area (County, District, Region or ZIP/Post code). Which matrix is used is controlled by dispatch method per inventory location. Jeeves ERP can handle fixed and variable costs and bulk goods.

6.10.12 Freight documents

Jeeves ERP includes a number of standard freight bills and address labels (SIS, Posten, STE etc.) for the Nordic countries. You can set up freight document and freight bill texts, as well as automatic sequence numbering, for example, as per dispatch method and freight document type. There is also integration with third party software for continuous updates of the standard freight documents in Nordic countries.

6.10.13 Packaging/packing routine

Jeeves ERP can propose the type of packaging and the number of packages required for the order row or you can manually pack orders. One packing slip is printed for each package.

You search for the order to be packed using one of the programs. If a packaging type has already been specified for the items, a packing record should have already been created and you can click on the Auto packing button to pack the order. The Create package function only works if there is a link between the packaging type and the item or items that you wish to pack. The function can be used to recreate a packaging proposal, for example, if you have accidentally deleted it.

If there is no link between the item and a packaging type (or if you want to pack the shipment differently), you must enter package contents and pack manually.

6.10.14 Transport administration

Jeeves ERP:s generic Transport administration (TA) API enables integration with multiple TA service providers' systems, for handling standardized transport documentation from Jeeves ERP:s dispatch programs.

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This includes provisions for:

- defining system data or loading the data from the external TA system, such as general/provider/user-specific settings to control which information is to be sent to TA, if and how delivery notifications are to be sent, if addresses are to be validated, etc.
- defining or loading base data, such as the service providers, carriers, transport products, package types and add-on services used
- setting additional, TA-specific data on customers, inventory locations, dispatch methods and dispatch terms
- validating addresses via the external TA system
- setting up notification methods
- setting up items for dangerous goods
- specifying shipment data and parameters on orders, dispatch orders and delivery notes at dispatch
- viewing the shipments that are created and requesting price quotes for them
- printing, releasing, tracking, recreating and deleting shipments
- viewing the message log between Jeeves ERP and the external TA system

The connector to the external TA system is set up by Jeeves Professional Services using Jeeves API platform.

6.10.15 Shipment planning for WMS (to be discontinued)

If you use Jeeves ERP's integration with Astro WMS, you use a dedicated function for shipment planning. See the section on the Sales-WMS integration.

6.11 Sales invoices

6.11.1 Invoicing

Invoicing can be done from a number of order or picking programs or from a separate invoicing program (depending on the setup of your company's roles), and can be done at any time. It is possible to postpone invoicing of individual order rows if necessary.

In the Sales module, you can invoice reported service orders from the Service module.

6.11.2 eInvoicing

Jeeves ERP supports electronic invoicing through third-party services for distribution, sender and receiver notification, storage, and online tracking.

Invoice information can also be made available through standardized APIs for transfer to external EDI and eInvoicing service providers, supporting automated electronic exchange.

6.11.3 Invoice number series

You can set up different number series for different order types and divide them into sub-number series linked to users/groups or salespersons.

6.11.4 Invoicing freight, insurance, handling charge etc.

When closing orders, besides entering the number of packages, weight, volume, etc., handling charges and other costs that the customer is to be invoiced can be specified. The charges can be entered manually or created automatically.

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6.11.5 VAT

In Jeeves Sales you set up customers, customers' points of delivery and items for VAT. Your company can apply any number of VAT rates. There is an option to create VAT specifications if there are items with different VAT rates on an invoice. You can set up any deviations from and additions to default VAT rules, for example:

- if the customer controls the VAT code (If any order row/item has a deviating VAT code it is ignored)
- if tax amount and VAT basis amount shall be calculated including or excluding VAT
- if the VAT code from sales rows is to be stored on voucher rows
- if VAT accounting date, VAT basis amount, etc. are to be stored on voucher rows
- if the VAT matrix is to be used to retrieve the VAT code

If you use the VAT matrix the VAT code on order rows (and the equivalent business transactions) considers the following:

- Ship to (country code and trading area on the company and order)
- Ship from (country code and trading area from the delivering inventory location or from the supplier in case of Triangulation)
- valid VAT number on the customer (if not a warning is displayed)
- VAT code on item if different from default domestic VAT code
- for 3rd party trade, Order type (a drop shipment indicates triangulation)
- customer's VAT code, if the VAT code from customer field is checked

See also separate sections describing Sales tax for the US and Canada and VAT/tax declarations.

6.11.6 Collective invoicing

Jeeves ERP offers collective invoicing of several released orders per payer, per delivery note, or delivery customer. You can create normal invoices and collective invoices in the same program or you can create collective invoices in a separate program.

6.11.7 Automatic matching of credit notes against debit invoices

If you create a credit note from an invoiced order, you are asked if you want to create a credit note to be automatically matched against the relevant invoice. Otherwise you can enter another invoice number to match against or leave the field blank (if no automatic matching is to take place). If you create a credit note from an order that has not yet been invoiced, you also get the possibility to enter an invoice number to match against, if desired.

The credit note cannot be matched against more than one debit invoice. Jeeves ERP also checks that the company ID is correct. Both invoices are cross-referenced. You then release and invoice the credit note.

The credit note is posted in the normal manner, during printout of the invoice journal, using templates connected to the dedicated payment method called CTD, Credit to debit. One payment journal is automatically created per invoice journal. If the debit invoice is already paid the credit note remains open. If the credit note only covers part of the invoice amount, the remaining amount remains open. If it covers more than the invoice amount, the debit invoice is fully paid and the remaining amount on the credit note remains open.

6.11.8 Currency clause

Currency clauses can be applied as protection against large changes in exchange rates, both increased and decreased rates. The currency clause makes it possible to automatically adjust the sale price during invoicing if there is a large difference in the exchange rate between the time of ordering and the time of invoicing. On the currency clause, you specify how big the difference must be for a price adjustment to be made.

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There are a number of cases when a currency clause can come into effect:

1) If the item is purchased in foreign currency and the exchange rate for the foreign currency changes during the period between the time of ordering and invoicing.

2) If the item is purchased in base currency and you invoice the customer in foreign currency and the exchange rate for the currency you invoice in changes between the time of ordering and invoicing.

3) If the item is purchased in a foreign currency and the customer is invoiced in another foreign currency.

6.11.9 Internal sales

This function makes it possible to account for revenues, costs and margins regarding internal sales, both for the department that purchases the product from supplier to inventory, and the department that purchases the product internally and sells to the customer. In "normal sales" you only account for the department that sells the product to the customer. Using this function you also account for the department that purchases the product from supplier to inventory in connection with sales.

There are three ways of activating internal sales:

- Through the combination of internal sales type, retrieved via the customer category, and item.
- If the cost centers on the salesperson and the stock location are not the same.
- If the cost centers on the salesperson and the product account are not the same.

The activation of internal sales gives two results; 1) cost, sales price and margin for internal sales are calculated and presented on quotations and orders and 2) additional postings and eliminations are created on the customer invoice journal.

Via a parameter you can select between the internal sales price and the cost price to be used while creating postings from the invoicing and journal voucher report.

6.11.10 Invoice history

In many countries it is legally required that you store historical data regarding invoices so that an invoice duplicate can be reproduced if necessary.

You can print duplicate invoices with historical data, i.e. even if company information for the customer or payer has been changed since the original invoice was printed, the change does not affect the reprint. There is also functionality to ensure that the same template is used as when printing the original invoice.

6.12 Posting and integration with ledgers

Invoices are normally automatically entered in the accounts receivable ledger upon creation.

Customer invoices are normally posted automatically in a separate routine/process after one or more batch of invoices have been produced. Invoicing can be done several times a day and posting at the end of the day or first thing the next day. One invoice journal is created and posted per invoice date. Each journal generates a voucher in the general ledger.

Posting is done based on posting rules for invoice totals and details that control all accounts and account segments (cost centers, cost units etc.) to which receivables, revenues, costs, VAT/tax etc. are posted.

You can post delivered orders/dispatch orders that have not yet been invoiced and posted at any interval, for example, during period or more regular reconciliations. If an order is picked in one period but invoiced and an invoice journal created in another, the picking journal must be created before the invoicing journal. One picking journal is created and posted per dispatch date. Each journal generates a voucher in the general ledger. Posting is performed according to posting rules

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for inventory transactions that control to which account and account segments (cost centers, cost units, etc.) the uninvoiced delivery is posted.

6.12.1 Exceptions to posting rules

You can set up which cost centers, cost units, etc. can manually be entered on order rows as exceptions to the posting rules that have been defined.

6.12.2 Reconciliations

There is a program for reconciliation of dispatched but not yet invoiced orders, which can be used in connection with period reconciliations.

There are also functions for reconciliation and analysis of differences in quantities and/or amounts, as well as for analysis of postings, on customer orders with associated purchase orders, for example, Drop shipments and Transit orders.

6.13 Sales statistics and budget

Thanks to Jeeves ERP's superior search possibilities, and to the fact that you can customize forms and lists by adding fields, columns, totals etc., you can compile information on many dimensions in more or less any table. You can also build your own reports and business graphics based on own search criteria and layouts. Besides this, there are a number of standard forms for viewing, and reports for printing sales statistics.

6.13.1 Invoicing statistics

Statistics can be produced for any period, by customer (delivery customer or payer), customer category, item, item group, item class, salesperson and project.

6.13.2 Sales budget

You can enter sales budget for comparisons against actual sales. There are programs for entering total budget, budget per customer, per item group and/or per item class. There is also a functionality for allocating sales budget per sales manager.

You can either enter each period budget manually or distribute the annual budget automatically over the periods by specifying an allocation key and pressing the Allocate budget button.

6.13.3 Intrastat statistics and European sales lists

Jeeves ERP includes functions for producing the required declarations of dispatches and arrivals between the company's own country and other countries within the EU, including drill down to the invoices included in the dispatches and arrivals totals. This is to make it possible to reconcile the data in the report.

If you choose to use the functions, you can either use the information generated as a basis to manually fill out the Intrastat form or you can compile the Intrastat report in an external report generator.

See the chapter on Jeeves *General ledger* for further information. You can also export the data to MS Excel using Jeeves' Dynamic reports. You can then use Intrastat Data Entry Package (IDEP), which is free software that is available on the web, to automatically import your Intrastat reports into the tax authority's web site.

During Intrastat reporting, you can choose to group invoices only on TARIC code or on TARIC code and delivery term. Note that the Intrastat report in Sweden is grouped on TARIC code only, whereas in Poland, Belgium and a few other countries, it is grouped both on TARIC code and delivery term.

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6.14 Claims

6.14.1 Customer claims

You can enter and log customer claims, i.e. remarks and complaints about items, orders and invoices, and record how they are to be settled as well as actions to be taken.

You can define different work flows for different types of claims, specify base data for scheduling, and control how activities should be reported.

6.15 Sales-WMS integration (to be discontinued)

If you use Jeeves ERP's integration with Astro WMS, you use a dedicated function for shipment planning as described under the section about dispatches. Here we give an overview of the integration.

6.15.1 Activation of integration

The integration can be activated per order type, inventory location and/or item.

6.15.2 Fixed departures

You set up fixed departures with lead times for when orders need to be entered, picked, packed and loaded in order to be included in a departure. These criteria can be set up per dispatch method or dispatch method and inventory location.

6.15.3 Transport resources

You also set up the company's transport resources, i.e. trucks, containers etc., with capacity information. The shipment planning function gives you the possibility to plan ahead to fill capacity.

6.15.4 Shipment plans

There is a special function for dispatches if Astro WMS is used, for:

- creation of shipment plans for fixed departures
- allocating orders on shipment plans
- information about transport resource capacity and usage, with the possibility to plan ahead to fill capacity
- release of shipment plans for picking in WMS

6.15.5 Information from WMS

Picking status is automatically updated from WMS back to the Sales module when the goods have been picked and dispatched. Balances are automatically updated on received deliveries and on returns from credited customer orders.

6.16 Digital business documents

Jeeves ERP supports automated, API-based exchange of digital business documents with customers and other trading partners. Standardized APIs can receive customer orders, order changes, and delivery schedules, provide order responses, and make invoice information available to external systems.

Automated exchange reduces manual processing and the risk of data-entry errors. Incoming orders use prioritized delivery-address selection, while administration functions support resetting and reprocessing failed transactions.

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6.17 e-Applications

6.17.1 Jeeves e-Sales

There is an optional e-application for B2B trade via the web. The solution is built on Liferay open source portal and interacts with Jeeves ERP through Jeeves Service Gateway and Jeeves Service builder (JeevesAPI). e-Sales is based on Jeeves standard business logic and is a fully functional webshop application. For more information please refer to the chapter on *Jeeves e-Applications*.

The application can be summarized as follows:

- Setup of the web interface, which is comprised of so-called classic or web-based product tree, shopping cart, currencies etc.
- Order entry with search functions and selection from product tree
- View deliveries, accounts receivable and order history
- Entry/update of company data

Please note that Jeeves e-Sales is to be discontinued and replaced by another offering. The chapters in Jeeves documentation remain for the information of any companies still using its functionality, but will be removed in a future version.

6.17.2 Integration with PEPPOL

Jeeves ERP supports the exchange of electronic business documents according to Pan-European Public Procurement On-Line (PEPPOL). PEPPOL is an infrastructure for e-commerce and e-procurement within the EU.

Jeeves solution is a cloud-based integration solution to support companies that have to provide information in the PEPPOL 3.0, SEPA, and other electronically supported formats as per legal requirements.

The eInvoicing functionality harmonizes with Jeeves ERP standard, so that if a payer ID exists on the order, the payer's customer details such as VAT number, address and the like are displayed on the invoice. If the payer ID does not exist, the details are fetched from the customer field and displayed on the invoice.

While additional setup is required to enable invoicing in the PEPPOL format using the eInvoice program, Jeeves ERP standard logic regarding order management and invoicing is applied in all other aspects.

You can include detailed package information along with package items on the PEPPOL invoices.

When an invoice or advance invoice is generated, the information is posted to a queue and then picked up by JSB for further processing. The data is validated and converted to PEPPOL format, and sent to a third party (like Pagero eInvoice service) or via HTTP (Hypertext transfer protocol)/ FTP (File transfer protocol)/ SFTP (Secure file transfer protocol).

The generated XML contains the invoice as an embedded PDF document in binary format, provided the Include PDF check box is selected and the PDM configuration is enabled.

To send a credit invoice in the PEPPOL format, the order must be released for invoicing, and the status must be changed to Invoiced. This updates the queue with the data which is then sent for further processing. The data is then validated, converted to the PEPPOL format, and sent to Pagero.

Once the order is invoiced and the invoice number is generated, the status can be checked in the Attribute queue reference program.

You can also import a PEPPOL XML invoice with a PDF file embedded in it. Once the file is imported, it is available in the Review imported supplier invoices program against the corresponding supplier invoice number. Additional details such as the PDF file name and its location are also displayed.

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The information about the invoice agreement is available on the invoice header. If the EInvoice check box is selected, the report will be eInvoice d and transferred to the Rules for printout program when the eInvoice agreement is activated.

If the Include PDF checkbox is selected, you can provide information about whether or not to include a PDF for the invoices with the eInvoicing agreement.

Jeeves e-Invoicing also includes support for Business response message handling. PUF invoice responses can be exchanged between buyer/receiver and supplier/issuer to inform the issuer of an invoice about business decisions. The delivered invoice may be technically correct and delivered to the recipient/buyer but there might be the need for a buyer to inform the issuer if the invoice is accepted or if there is a reason for a rejection.

6.17.3 Integration documents

The functionality to handle integration documents allows you to track progress, send and receive documents, validate documents, etc. with ease. You can browse the contents of the integration document and save it as a file in a chosen folder.

6.18 Industry-specifics

Most functions in Jeeves Sales are common to all industries. There are some, however, that are specific to certain industries, as described below.

6.18.1 Product variant matrix

The product variant matrix mentioned earlier was developed with the fashion industry in mind, but can be used for any products with variants, i.e. combinations of models, colors, sizes etc., or the like.

6.18.2 Sales configurator (powered by Tacton)

Jeeves ERP's Sales Configurator supports salespersons in the creation of order unique bills of material, and if required routing, from customer specific demands, and from these create customer orders in Jeeves to generate customer specific manufacturing or purchase orders. The configurator not only helps the user avoid invalid or unsuitable configuration alternatives, it can also help find the optimal configuration.

Jeeves ERP's configurator is constraint-based, which means you specify general attributes and rules for groups of items/parts, rather than for individual items. This keeps maintenance of configuration rules down to minimum. The rules are set up in Tacton studio.

In Jeeves ERP, the integration is comprised of

- Basic data for verifying and mapping models created in Tacton studio. Models are saved in Jeeves EDM for version management purposes.
- A batch handler and batch queue that enables parallel processing of the result from Tacton in order to optimize performance.
- A special order program. The configurator window is accessed from the order row and the result can either be displayed as a package with several BOMs and operations, or the BOMs and the operations just presented in an embedded window on the order row. The items are treated as any other items in Jeeves ERP.
- Functionality for copying existing order configurations to new ones in Jeeves and Tacton, to create new order rows and/or new standard items.
- A log for viewing imports, statuses and any error messages, also buttons for updating and deleting. The log is also accessed via a node in the order program.
- Auxiliary properties, which make it possible to customize the Jeeves interface, for example, insert graphics.

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6.18.3 Retail sales/Electronic point of sales (EPOS)

There is a special program for entering EPOS orders. Elements that are specific to EPOS are

- that search items and order entry are performed in a single screen where all the necessary forms and lists are available,
- that much of the functionality is linked to shortcut keys so that most things can be done from the keyboard,
- that the salesperson's authorization is checked via an acknowledgement code and
- that for closed orders picking, release and receipt printout are performed automatically. At the same time a command is sent to open the POS.

6.18.4 Retail/pallet discounts

There is a special solution for retail/pallet discounts that allows any number of discount types. The solution offers two alternatives:

- a separate discount matrix based on the standard item and customer discount classes
- pallet discounts per customer and item

With both alternatives you can specify discounts in percent and/or as amounts in currency.

6.18.5 Trading

There is a special order program for trading, with automatic creation of purchase orders for drop shipments. The cost price and sales price are automatically calculated based on a markup on the purchase price.

6.18.6 Order shuffler

The order shuffler gives the possibility to create amendment orders from ordinary orders and to credit invoices and re-invoice them, with retained traceability throughout the whole process from the order to the general ledger. You can even shuffle order rows to several different customers, for example, to different business units within your company, simply by entering a different ID for order shuffle for each customer on the order rows.

6.18.7 Estimator

The Estimator supports the estimate to quote process. Estimates can be prepared for products, services (services from both projects and service orders) as well as manufacturing operations and material. Users can create and compare different estimates and see a summarized view of different calculations, follow the process and analyze won-not won quotations and/or estimates.

The Estimator program can be used for any of the following:

- preparing price estimates for standard products
- modifying standard products, package items and product levels
- configuring products comprised of standard modules, package items and product levels
- creating new order specific products, package items and product levels including costing
- configuring services/maintenance, both during and after implementation of the products
- configuring project services, for example, for construction/design

The header includes a large number of fields for total costs and revenues summed up from the underlying details. With the Estimator you can produce several estimates/versions from which you, or your customer, can choose the most suitable solution. You can also transform order specific products into standard products and replace them on existing orders.

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6.19 Localizations/internationalization

Most functions in Jeeves Sales are common to all countries. There are some, however, that are specific to certain countries or trading areas, as described below.

6.19.1 Transport administration integration (Nordic countries only)

There is an optional integration with a third party product for transport administration (LogTrade or KSD Edicom), i.e. for handling standardized transport documentation from Jeeves' dispatch programs. This complements the standard freight documents provided in Jeeves ERP.

6.19.2 VAT refunding

Originally developed for the UK, but available to all users

In some countries, such as the UK, the VAT/tax value is based on the discounted amount even if the customer does not take advantage of the early settlement discount. In order to be able to reconcile the VAT at the end of the period, Jeeves ERP creates additional, preliminary postings on the customer invoice journal, which are subsequently, automatically reversed on the payment journal. You can adjust your invoice templates to show the required information, for example, the discount terms and the VAT amount based on the discounted amount.

6.19.3 Amendment invoicing

Originally developed for Poland, but available to all users

Amendment invoicing is a function that lets you create a so-called amendment order from an existing (normal) invoice in the database, containing corrections of quantity, price, discounts, VAT etc. against one or more invoice rows on the original invoice. The difference with amendment orders is that Jeeves ERP automatically creates a credit record with the original invoice data that corresponds to each corrected row. These credit rows cannot be changed by the user. This is handled completely automatically from the amendment order, via picking and release to invoicing.

Amendment orders are invoiced via a special invoicing program, which produces an invoice printout that contains both rows with invoice data from the original invoice and the new, corrected invoice rows from the amendment order. The invoice produced contains subtotals from both the original and the new, corrective invoice data, as well as totals for the whole invoice, which show the difference between the original and the corrective invoice data.

During packaging and delivery of goods to customers, the dispatch can potentially contain several batches and/or serial numbers. Sometimes the dispatched goods may be returned to the supplier, either entirely or just encompassing specific sets of batches or serial numbers, or part of the quantity originally dispatched. You can specify which transactions can be picked to be returned if the original order included several transactions with different serial numbers and/or batch numbers in the Temp table, changes on invoice embedded list (right list) in the Create credit order from invoice program.

For countries where it is required that the currency rate on the amendment invoice is the same as on the original invoice, Jeeves ERP includes functionality to fulfill this requirement.

6.19.4 Advance for full amount

Originally developed for Poland, but available to all users

In certain countries, a special type of invoice document is required by law if an advance invoice is for the full amount. By this we mean that the advance payment implies full deduction of the final invoice amount as well as the VAT amount.

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The invoice type is handled in the same way as normal invoices apart from:

- it uses a unique number series only applicable to invoice type 100, i.e. invoices from customer orders,
- it displays a special header on the printout and
- it not being transferred to the VAT register

6.19.5 Tax accounting on advance invoices

Originally developed for France, but available to all users

The tax authorities in some countries require tax accounting on advance invoices even before payment. Jeeves ERP includes functionality to fulfill this requirement.

6.19.6 Reminder charge and interest at invoicing

Originally developed for Norway, but available to all users

You can invoice reminder charges and interest in connection with normal invoicing of orders in the Sales module, instead of in a separate routine in the Accounts receivable. This is a common way of working in certain countries.

6.19.7 Sales taxes (USA and Canada)

Jeeves ERP includes programs for setup and collection of sales taxes according to legal requirements in the USA and Canada, comprised of the following:

- Tax jurisdictions for all tax areas in up to five levels (State/PST, County/GST, City/HST etc.)
- Tax codes (rates) connected to customers/points of delivery and items

6.19.8 Excise Tax (USA and Canada)

Jeeves ERP also includes functionality to handle excise tax where applicable, for example on sales of certain medical devices. When you enter an order, the excise tax is calculated according to the specified percentage of either the net amount or the amount including sales tax. The resulting excise tax is displayed in the same embedded list as the sales tax. When you invoice the order, the same information is displayed in a corresponding list in the invoice form.

6.19.9 Invoice numbering

Originally developed for Hungary, but available to all users

In certain countries, original invoices (i.e., when the invoice is printed the first time) and invoice reprints (when making a reprint of the original invoice) must be numbered in a specific way. Jeeves ERP includes functionality to fulfill this requirement.

6.19.10 European Medicines Verification System (EMVS)

Following the introduction of the European Medicines Verification System (EMVS) for the implementation of a functioning, secure, interoperable and cost effective system across Europe that ensures only prescription medicines which bear safety features may be put into circulation, it makes it mandatory for distributors of medical packages and pharmacies to ensure that each medical product package is qualified for consumption before sending it forward in the supply chain.

Jeeves ERP provides functionality that allows companies in the life science industry, including distributors, wholesales companies and pharmacies, to comply with the regulations set down to ensure the validity of each medical product package that is dispensed through each stage of the supply chain till it reaches the consumer. The product packages are validated against the data made available to the EMVS by the manufacturer or the parallel distributor to ensure that they can be dispensed to consumers for usage.

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Jeeves ERP functionality for validation of dispensed medical product packaging covers the following areas:

- Verification of single or multiple packages of a product
- Dispense package management
- Decommission package management
- Reactivation of packages that were previously decommissioned or dispensed
- Product verification and dispense during order picking

You can specify whether medical product verification should be enabled for the inventory stock type and select valid actions for each EMVS state to reduce external requests and attain better performance while triggering a state change for a product. You can also specify if a product package that is assigned to a particular state is allowed to be forwarded to the next stage of the supply chain or if it is valid for decommissioning. Further, you can also define if the EMVS state is applicable for pharmacies or wholesalers and if the EMVS functionality should be activated for a customer.

6.19.10.1 Initiate EMVS connections by selecting request type and method for validation

All requests from Jeeves ERP to the EMVS system are initiated from the Serial number items program.

A request type can be one of the following:

1 - Authenticate: Send an authentication request to the EMVS system. The EMVS system responds by providing the bearer token which can be used to access any service in the EMVS verification process. The token has a lifespan of 24 hours.

2 - Verify: Send a request for verification of a package.

3 - Supply: Send a request to confirm if a package can be supplied further into the supply chain.

4 - Decommission: Send a request for removing a package from the supply chain.

5 - Reactivate: Send a request for reactivating a previously decommissioned package. If you select an option other than authenticate or decommission, you can proceed by selecting an input method by which the product package information is captured and sent to the EMVS system for validation. If you select Decommission, you can proceed by selecting a Decommission state describing the reason for requesting the removal of a package. The EMVO has defined different decommission states for pharmacists and wholesalers and these may be subject to change, so we recommend you refer to the EMVS website for the latest information.

6.19.10.2 Verify package information during order picking/dispatch order picking

During order picking/dispatch order picking, product package information is captured and sent to the EMVS system for verification if the EMVS functionality is enabled for the inventory stock type/customer. The EMVS system responds by sending a bearer token which can be used to access any service in the EMVS verification process. The token has a lifespan of 24 hours. Once you have received the token, you can proceed with the picking as usual.

In case of returns, reactivation must be done manually in the Serial number items program.

6.19.11 OVT number for Finland

The Finnish Party Identification number (OVT-tunnus) is a unique 12-17 digit number linked to a business ID, which is usually used in Finland as an e-ID. You can enter the OVT number in the System data, own company and Companies tables.

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6.19.12 KSeF eInvoicing — sales invoices (Poland)

Jeeves ERP supports compliant electronic sales invoicing through Poland's National e-Invoice System (KSeF) using the FA(3) format.

The functionality supports advance and settlement invoices, third-party information, and customer-specific item information. Automated processing and validation help reduce manual handling and submission errors.

For details, see *15.11.12 KSeF eInvoicing — outgoing customer invoices (Poland)* in the *Jeeves ERP Accounts receivable* chapter.

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7 Jeeves ERP Inventory

Jeeves ERP's inventory module can help business enterprises, organizations and authorities improve their inventory performance, irrespective of industry or field.

With Jeeves Inventory you manage inventory processes – such as costing, inventory control, managing balances, counting and inventory planning including net requirement calculations – within the same application.

The main inventory process and common sub processes are packaged with pre-defined roles, role-specific menus and forms as well as user guides and training material. Besides these processes, you can activate other, wider functionality for specific company needs.

7.1 Products and services

Jeeves ERP has a common Items table for information regardless of the type of products and items – finished products, semi-manufactured, components, raw materials, packaging, consumables, stocked or non-stocked.

7.1.1 Product/item classification

There are many possibilities for categorizing and classifying items and products, for example:

- product classes – can control pricing and be used for statistics
- product accounts – control inventory management and valuation, and can control inventory postings
- item groups – can control pricing and be used for statistics and budget allocation
- item classes – can be used for purchase planning, statistics and budget allocation
- item categories – can be used for statistics, for example for collections of item groups
- item types– can be used for statistics

7.1.2 Product variants

There is functionality for handling product variants, useful for companies in the fashion industry for example, for defining applicable combinations of say, models, colors, sizes etc. of a product with the help of a matrix. The matrix has various layers that you can switch between for viewing current and planned inventory balances, item numbers, alternative descriptions, delivery dates, prices etc. for each combination. The matrix is used in the whole process flow, from automatic creation of variants and pre-packed assortments to physical inventory counts.

7.1.3 Seasonal products

You can set up seasonal items, orders and purchase orders, which can be used together with, or without, the product variant matrix. If used together with the matrix, fields for items that are out of season are inactivated.

If you specify a season on a sales/purchase order header, you can enter order rows for both seasonal and non-seasonal items. When entering order rows for seasonal items the following checks are made:

- That the season complies with the season on the sales/purchase order header
- That the sales/purchase order type is allowed for this season
- That the delivery date complies with this season

7.2 Costing

In Jeeves ERP, there are three ways of handling cost prices on purchased items:

1) Standard cost price, which you calculate using buttons or reports designed for this purpose

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2) Average weighted price, which is calculated automatically, i.e. the cost price is recalculated automatically when receiving deliveries or when approving supplier invoices

3) FIFO, also calculated automatically when receiving deliveries

You control this from a checkbox on the product account.

The following markups can be made on purchase prices:

- Freight cost
- Delivery cost
- Various markups in both percents and/or amounts
- Material overhead in percent
- Setup purchase price
- Start cost per unit
- Customs duties

The markups are totaled and can be retrieved from the:

- item
- item per inventory location*
- TARIC code for the item
- markup template for the item
- item data per supplier
- item data per supplier and inventory location*
- supplier
- product class
- product account
- delivery method and/or
- delivery terms

**) This implies that you can apply different cost markups for different suppliers and warehouses, if required. This applies regardless of whether you use standard or average cost price or FIFO.*

Costs for order specific operations and material per order row can be entered manually and taken into consideration when running costing.

You can add delivery costs as percentages or amounts, by entering the shipment cost on purchase order rows. This cost is posted with the value of goods when receiving the delivery.

In the Inventory locations program, you can set up if extra costs (refinement) should be calculated and added to the receiving value during transfers to the inventory location.

7.2.1 Standard cost prices

Cost prices are calculated based on the purchase price specified in the item data per supplier, or if there is none, the purchase price on the item. Overheads and markups, which can be entered on many different levels, are then added to the purchase price and the results are shown in the various cost price fields, either on the item or item per inventory location.

If you have several suppliers the agreement with the main supplier is used, i.e. the agreement that is flagged as the main agreement. If there is no main agreement, the supplier with the highest priority is used, i.e. the agreement with the lowest priority code (= highest priority).

Costs are also totaled upward in the structure of manufactured items and package items.

Depending on which work method you choose, the calculated cost price can either update the current cost price directly or be stored in a field for new, preliminary cost price along with a start date from which the cost price is valid. In the latter case, you transfer the preliminary price to the current price field when you are ready.

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7.2.2 Average cost prices

You use average weighted cost prices if you want the inventory to be valued according to the value of received deliveries. The function means that when new deliveries are received, the inventory is revalued with regard to the purchase price for the delivery. The delivery is "weighed in" according to the calculation. The calculation also takes into account any overheads such as delivery, freight, customs duties, startup costs, etc.

You can control whether average cost prices should be recalculated when receiving deliveries or when matching supplier invoices.

7.2.3 Distinction between costing from items and from Estimator

Companies who run costing from items, item revisions, orders and from the Estimator need to be able to distinguish between these costing records. Costing type is defined per calculation version, to determine whether calculations are of the type Item, Estimator or Item revision. You can only see versions for the relevant purpose in the different programs where you can run costing, for example, in the Estimator program you can only see calculation versions for the Estimator.

7.2.4 Cost price changes

Every cost price change is logged. If there is an inventory balance for the item, transactions are also created for integration with the general ledger in order to update the inventory value.

7.3 Revision control

7.3.1 Item master data

Jeeves Revision control is a set of programs that allows you to modify item master related data without affecting the currently approved revision. You can compare current calculations to see what effect the changes will have on cost prices. There is also a traceability perspective when using revision control, and Jeeves Revision control always saves all revisions of basic data.

The item master related basic data tables and programs include:

- documents
- routings
- BOMs
- attributes
- supplier/item records
- item balance records and
- item record related fields

You specify which item master related data, i.e. which tables and fields, are to be revision controlled. Changing one or more revision controlled field(s) activates color coding at record or field level:

- Yellow means one or more fields are changed
- Green means you have added a record.
- Red means you have deleted a record.

7.3.2 Revision process

Jeeves Revision control helps you visually track and compare current and new revisions. Your changes are also summarized, with color coding. When all changes have been made you can run a cost price estimate or check the assembly analysis tool, or start the approvals process. The final step is to manually update the revision status to Approved, ok to use in the item revision header, whereby the changes are incorporated in the item master. Any earlier version is automatically closed and automatically updated fields, such as Revision date, are color coded on the changed records.

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7.4 Quality control

Jeeves ERP includes extensive functionality for quality control. Companies within the Process industry will benefit from using these quality control tools, but they are a strong member of the ERP support for discrete manufacturers and other industries too.

7.4.1 Attribute management

You define attributes or characteristics as standalone objects and then assign them to products at item, component or routing level. Often, a product's quality varies over time and these characteristics or attributes need to be part of the product's inventory definition.

The attributes can describe:

- an item's specific characteristics
- how and what to measure or
- what and how to visually inspect, describe appearance or
- act as a checklist or
- describe what to do at different stages in a product process and
- make it possible to report on these objects, make decisions whether or not to approve steps/objects, trace and see the status and quantified target these objects had when reported on.

Jeeves ERP also supports specific types of attributes that can represent the following:

- An active component, for example, potency, stated as the power, concentration, affect, force, impact of a specific ingredient. Ingredients or materials that have an active component that is a percentage, content of the overall weight or volume of the material.
- A specific gravity when we need to calculate how much water is required to dilute the solution to the recommended grade or concentration, described in its form of occurrence in the raw material.
- Moiety can be used when the above two attributes cannot cover the required concentration demands. Moiety relates to parts of “something”.

When defining a quality attribute you also link the object to the sales, manufacturing, and purchasing or customer claims process. This means you can define different quality attributes depending on the process.

7.4.2 Quality control processes

If Jeeves Quality control (QC) is activated, QC records are automatically created when order or delivery rows and/or claims are entered for quality controlled items in the following processes:

- sales orders
- customer claims
- purchase orders
- goods arrivals, checking goods and received deliveries from purchases
- purchase claims
- manufacturing/work orders
- received deliveries from manufacturing
- internal claims

There are alternative programs for QC reporting in the different processes. This is to allow you to choose the alternative the suits your company's processes.

7.4.3 Traceability and recall management

Many companies, especially in the Process industry, face the risk of product recalls and the consequential requirements regarding traceability. Jeeves ERP's traceability functionality fulfills

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the requirements of both forwards and backwards lot traceability, with information pertaining to customer, supplier, supplier batch/lot number, work order, work order batch/lot number, etc.

Recall management often relates to product safety, regulatory compliance and demand for efficient support when communicating with the affected customers. Our recall functionality is built into the program for handling customer claims and uses our campaign and traceability functionality to find all customers who have received a specific product or a specific ingredient in the product.

When you enter a recall claim for an item and batch/lot between certain dates, a list is displayed showing all customers who have received the item within the period. You can then send out e-mails using the built-in e-mail functionality, or produce various types of documents.

You use Jeeves ERP's campaign functionality in this customer communication phase. You can handle information in different ways; enter text to be displayed in the message field of the e-mail and/or attach documents to e-mails in the recall campaign using the document management system.

You can register and follow up on the recall campaign responses. Select a response alternative from a drop-down list of standard responses and/or type in a brief note. You mark responses when the planned activity is complete and no further action is required, also report time, if required.

7.4.4 Quality control manager

As the QC manager, you can use a special program for replanning QC activities by moving records to other signatures. There is also a program for viewing the results from quality reporting in sales, purchase and manufacturing orders. Here you can select and compile QC data in order to prepare reports, certificates etc. according to your company's own work flow and quality requirements.

7.5 Item analysis and planning

7.5.1 ABC analysis

You can set up rules for and generate ABC classification of stocked items. This can be done for purchased and manufactured items, either jointly or separately. You can perform ABC analyses per

- item/item group/supplier number interval or the like
- inventory location or
- item type

The classification is then used in the planning of inventory levels to avoid shortages of the most sold items and also to avoid unnecessarily large inventory, thereby limiting inventory costs etc.

7.5.2 Safety stock

You can set safety stock levels for any selection of items and in four different ways:

1) Calculation based current consumption. You can use this method to calculate on actual shipped quantities and desired service levels by setting up inventory strategies to handle stock items based on ABC volume values and pick frequencies. The inventory strategy defines the service level (the probability of not being out of stock at the time of any demand), which in turn gives security factors. The safety stock is then calculated based on the security factor and the actual consumption (picked quantity per week and month).

2) Manually, and block automatic calculation. You can use this method for items that are new or to be discontinued for example.

3) Calculation according to safety stock in number of days to cover dispatches above the average usage. The safety stock level is calculated based on the given number of days, which is compared to annual forecast and usage.

4) Calculation according to ABC rule. The safety stock level is, in principle, calculated as in 3) but takes into consideration number of days and safety stock lead time in percent from the combination of item type and ABC class.

Items can be updated directly or via interim work tables.

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7.5.3 Order point

You can enter order points manually or calculate them automatically based on usage forecast and lead time for any selection of items. Items can be updated directly or via interim work tables. The order points are then used in purchasing and/or manufacturing to order point.

7.5.4 Inventory turnover

The items' rate of turnover is calculated automatically based on the usage statistics.

7.5.5 Economic order quantity

You can enter the items' economic order quantity manually or calculate it automatically according to Wilson's formula for any selection of items. Items can be updated directly or via interim work tables. The economic order quantities can then be used to control the calculation of proposed quantities to purchase and/or manufacture.

7.5.6 Inventory control

You can set up inventory controls in the different routines that generate inventory transactions according to own needs, for example against available balances, allocated balances, lead times, minimum/maximum balances and order points for stocked items, and against lead times for non-stocked items.

7.5.7 Available to promise (ATP)

Jeeves ERP's Available to promise (ATP) functionality provides a more accurate picture of the available quantities of a product on a requested or estimated dispatch date. ATP queries apply to package items (i.e. items with an inventory stock type set up for package items) and manufactured items (i.e. items with an acquisition method regarding manufacturing).

The ATP functionality takes into consideration requirements from customer orders where the availability has been checked but the production has not yet been planned. When using the ATP functionality, Jeeves ERP takes many more parameters and factors into consideration, and the resulting underlying requirements from previous queries are used the next time someone queries the same item.

Jeeves ERP offers alternatives for running ATP queries from sales and from assembly analysis. Via parameters, you can control if ATP calculations should be done based on planned inventory transactions, instead of the base tables, in order to speed up the calculations.

Jeeves ATP enables you to do the following:

- Enter a horizon for future requirements in ATP queries.
- Create temporary manufacturing orders, i.e. "soft" reservations from ATP queries.
- Delete all or individual temporary manufacturing orders that have been created for items by the ATP functionality.
- View the results of ATP queries on items with temporary manufacturing orders or temporary material reservations.

7.5.8 Inventory valuation

In Jeeves ERP, the following alternatives apply for stock valuation:

- Stock valuation at current inventory value, i.e. current inventory balances * current cost prices
- Stock valuation per specified date, i.e. inventory balance as of the specified date * the relevant cost price as of the specified date
- Stock valuation per period, i.e. inventory balance per specified period end * the relevant cost price per specified period end

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The latter two reports calculate values according to standard and average cost price, as well as FIFO where applicable.

You can do an automatic search of slow moving items in an inventory. The search is based on input about the number of months backwards in time from today's date that you want to search.

You can also change the current cost price of multiple selected items in an inventory location and update item data and inventory location records with the revalued cost price. This includes the current cost price of selected FIFO batch(es) in an inventory location.

7.5.9 Inventory revaluation

Jeeves Inventory revaluation lets you change the current cost price of selected item(s) or FIFO batches in an inventory location and update item data and inventory location records with the revalued cost price.

Standard Jeeves ERP logic for cost price changes automatically handles the update of revalued cost prices from the Inventory revaluation program, which means that the following types of transactions are automatically created:

- Cost price change log, Items
- Cost price change, inventory value
- Inventory history transactions

7.5.9.1 Integration source and inventory transaction types

You can specify if the functionality for cost price per inventory location should be active or not in the system and set the integration source and inventory transaction type for inventory revaluation transactions. You add details to your posting templates using the dedicated inventory transaction types.

7.5.9.2 Inventory revaluation reason

You can define the applicable reasons for inventory revaluation by selecting the required option.

By default, the Invtry rev reason field has the value obsolescence when you open a new Inventory revaluation journal record. You can change this value by selecting the appropriate option from the drop-down list.

7.5.9.3 Revise cost price of item and update item data/inventory location

You can revise the current cost price of selected item(s) or FIFO batches in an inventory location and update the item data/inventory location with the revalued price(s). If you select an inventory location in the journal header but have the same cost price for the items on several inventory locations, Jeeves ERP will create transactions for all these inventory locations.

When you have created a journal header with an inventory location and revaluation percentage, you can search for the items to be included for revaluation in a specific inventory location. Alternatively, you can do an automatic search of slow moving items in an inventory. The search is based on input about the number of months backwards in time from today's date that you want to search.

The items are presented in the left list window. Select the relevant items and transfer them to the right list window. The current value and the revised value of the contents of the right list window are summarized and shown in a total window below the Inventory revaluation, rows on the form. Double-click a record in the right list window to view the revised current cost price, average cost price, and calculated cost price of the selected item in the inventory. Note that if you want to reduce the current cost price, enter a negative (-) percentage value.

After the item values are revised, you can update the inventory revaluation values in the inventory revaluation journal, create a voucher from the inventory revaluation, and integrate with the general ledger. Note that the voucher is created only if the inventory cost price has been updated.

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You can also indicate if the current transaction for the selected item(s) in the inventory location is an inventory deposition, i.e. impacts only the voucher and not the inventory or cost price. If not, the inventory values are updated as per the revised cost price along with the update to the voucher.

7.6 Master scheduling and forecasting

7.6.1 Master scheduling

Jeeves Master scheduling makes it possible for wholesalers and manufacturing companies to use the forecasts produced by the marketing department to efficiently plan future material and resource requirements.

The function consists of programs for entry and revision of forecasts for different market segments and product families or individual items, which are then compiled into a master schedule. The master schedule is then transferred to the regular tables for item forecasts used as a basis for net requirement planning. This also includes functions to retrieve actual figures from invoicing statistics for comparison against the forecast.

There is also a function to see, on an aggregate level, how the capacities of different manufacturing resources are impacted by the master schedule.

Companies that use both master scheduling and package items can set up forecasts to be created on the underlying, stocked items in their packages.

Some companies that use Jeeves Master scheduling and forecasting need to replicate relevant data, such as item forecasts and requisitions, between sales units and manufacturing units. There is a report with which you approve selected requisition rows for replication to another company.

Jeeves Master scheduling can also be used with Product variant management. Since different models can have different sizes (or other attributes) on different markets, you can connect the same forecast family to several forecast markets. Since forecasts usually vary between SKUs for master items, you can create forecast families per master item and allocate forecast shares to their variants items.

7.6.2 Item forecasts

If your company does not work with advanced forecasts or master scheduling, you can create forecasts at item level for use as a basis for MRP.

You can enter a forecast quantity manually in the highest forecast period and then automatically distribute the forecast over the underlying forecast periods. You can define any time units as well as any number of forecast versions for these forecasts.

You can also calculate forecasts for items based on item statistics using weighting rules. When creating forecasts from history, Jeeves ERP informs if there are item statistics for the selected year and period that have not yet been updated. If this is the case you are given the possibility to update the statistics first, before creating the forecast.

7.6.3 Advanced forecasting

Jeeves Advanced forecasting consists of two tools, Stock wizard for target management (what if-analysis) and Amase for managing forecasts and alarms. In Stock wizard you classify your products and set your objectives. In Amase you manage new and changed forecasts as well as alarms. The alarms suggest corrections in order to stay on target with your objectives.

In general, the advanced forecasting tool helps you maintain the following objectives:

- satisfyingly high inventory turnover rate
- small standard deviations in service level per item

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and in production planning maintain:

- more accurate forecasts
- short/accurate delivery times

With this tool you can enter forecasts on four different levels; Item, Item group, Item class and Total. The tool supports both amount and quantity forecasts.

Amase has support for the following forecast models

- Mean
- Trend
- Season
- Crostonts
- Brown's method
- Own forecasts
- Campaigns

The forecasting models also monitor the products' lifecycle and support the following three phases:

- Growth
- Stationary
- Negative trend or phase out.

It helps you decide what to have in stock and enables you to keep inventory levels in line with forecasts.

Amase suggests a forecast based on the items' order history and it is updated monthly. Amase lets you know for which items the forecast deviates from the requirements. It also calculates the safety stock, order point and the economic order quantity (EOQ). Each items forecast is presented in a graph.

7.7 Inventory management

7.7.1 Inventory locations

Jeeves ERP can handle an unlimited number of inventory locations and related, inventory location specific item data.

7.7.2 Inventory stock types

You can define any number of inventory stock types for detailed control of different types of items' inventory management, printing to internal and external documents etc., for example:

- stocked items
- non-stocked items
- batch items
- package items
- serial number items
- consumables
- phantom items

and other stock types.

Through a number of parameters on the inventory stock type you control exactly how each type is to be handled.

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7.7.3 Blocked balance/quantity

In Jeeves ERP you can block a whole inventory balance, or a certain quantity within the balance, from being used or dispatched. You can do so directly during goods receipt or at a later stage. This can be necessary if special inspection is required, for example, or if you receive many complaints.

There is also a possibility to open up for users to pick from blocked balance, if necessary.

7.7.4 Blocked inventory location

There is also a checkbox to disable an inventory location from receiving deliveries against manufacturing or purchase orders. The value defined for this checkbox in the Item balances program overrides the value set for the checkbox in the Inventory locations program.

7.7.5 Miscellaneous item

You can define a collective item number for handling miscellaneous non-stocked items that do not require individual item numbers. If this item number is used during order entry, the order row is flagged for order-specific cost price.

7.7.6 Batch management and FIFO

Batch management means that every received delivery of goods is allocated a unique identification number. The function is used for certain types of goods that, for example, require a use by, best before or sell by date (for example: perishable goods and medicines, etc.) or the manufacturing batch (for example: wallpaper).

When receiving deliveries:

When receiving deliveries to inventory the batch ID is either set automatically based on the purchase order number, row number and date, or is specified manually (for example according to the supplier's manufacturing order, delivery note or purchase order number). The information is used to follow up received deliveries, for example, in order to follow up the use/best before date, or to trace lots with defects, at allocation and picking and in FIFO valuation.

When allocating and picking:

You can allocate and pick items per batch and control the order in which the batches are to be proposed for allocation:

- Use/Best before date – against order or delivery date
- Sell before date – against order or delivery date
- Batch number order

You can also set up any number of status controls, OK/not OK to allocate, to pick, include/exclude from available balance. You can also specify a default number of days for auto allocation for any batches that are missing a Use/best before date. Batches with a Use/best before date should normally be considered first, which implies that the default number of days should be set to a high value, if specified.

Some companies order items with a fixed batch ID from a catalogue, and that same batch ID needs to be used also when the delivery is received. If a batch ID is specified on a purchase order row it will be inherited to the received delivery row.

7.7.7 Serial number management

Serial numbers items are item individuals identified by a combination of item number and serial number. The function allows you to trace item individuals and fixed assets in all inventory movements wherever required, such as received deliveries, production, quotations, dispatches, allocating, picking, service etc. The purpose is to quickly identify where an individual is, both physically and in the relevant workflow, to which customer it has been sold, whether it is still under guarantee, etc. Serial numbers are usually entered when receiving deliveries against

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purchase orders, but can be entered in a separate program for manual entry. You can also apply serial numbers only at dispatch without affecting the inventory.

7.7.8 Bin management

You use Jeeves bin management if you need to track the location of items within an inventory or inventory location. This can be, for example, to separate different types or sizes of an item or for customer specific bins. Bin management can apply to physical inventory counts, received deliveries, allocating and picking as well as in manufacturing.

You can also associate bins with packaging types, to indicate the physical properties of the bin. This does not restrict the bin to the given packaging type; it only serves as a help to the user/planner. In addition, when creating load carriers, you can link a packaging type with an item, item balances, or inventory locations. This is then used as the default packaging type during load carrier creation.

You can add checksums (alphanumeric characters) to each bin to enable Jeeves WMS to validate that the user is at the correct physical bin when completing WMS assignments.

The purpose of load carriers/master load carriers is to hold a collection of items to be able to receive, move or pick them all at the same time, making inventory management quicker and more efficient. Considering this - and using the example of a pallet (master load carrier) containing boxes (load carriers) - it is only logical that the collection of items should always be in the same bin. The contents cannot be spread out over several bins, as that would inhibit you from receiving, moving or picking them all in the same transaction.

Several controls exist in the transactions for arrivals, received deliveries, returns, etc. including order-specific inventory:

- If bin is not to be entered, it cannot be entered for child load carriers or for the master load carrier to which they belong.
- A child load carrier cannot be in a different bin than the master load carrier to which it belongs. The existing bin is defaulted on all transactions and updates the Bin field in the Load carrier program.
- If a master load carrier does not have any bin, and you put a load carrier into it and enter a bin, the master load carrier inherits the child load carrier's bin.
- If you want to change the existing bin for a load carrier, you first need to unmap from the master load carrier and assign a new bin (requires that the balance is 0).

When receiving deliveries:

You determine whether a bin must be entered when receiving deliveries on the inventory location. If so, you set up whether the bin should be entered manually or whether Jeeves ERP should attempt to place the item automatically. You can also have Jeeves ERP check the quantity, and the maximum weight and volume that can be received and stored in the bin.

Jeeves ERP also checks if several items, i.e. with different item numbers, can be received to the same bin.

If you apply automatic bin allocation, and no suitable bin is found, i.e. with the right bin type and where received delivery quantity does not exceed the specified maximum quantity, a message displays that there is no suitable bin for the received delivery.

In the order routine:

When allocating items for customer orders, a suitable bin is checked and suggested according to a number of parameter settings but you can change the default information (bins, quantity) manually for each order row. Example of the parameter settings are customer specific bins, if it is allowed to allocate order rows against several batches etc.

There are also settings to control the bin information to be printed on picking lists and the desired sort order.

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In manufacturing:

Manufacturing companies can apply bin management for material reservations in much the same way as in the order routine described above.

7.7.9 Load carrier management

A load carrier is typically a container or pallet (with or without collars) where you load goods to store in the inventory or that may be received from the supplier. You can use the Load carrier functionality if you need more specific tracking of items within an inventory location, for example, to separate different types or sizes of an item or for a specific customer. You can activate/inactivate the Load carrier functionality per company in the database. If the load carrier functionality is activated, the priority for setting whether load carrier is required between the various levels is:

- bins per item – highest
- items per inventory location
- inventory locations
- items – lowest

Any exceptions to this order of priority can be handled by creating separate bins per item- You can set the control on specific bins and inventory locations. You can also set it as default per inventory stock type, which is inherited on items with the stock type.

You can also specify a load carrier to store work in progress (WIP) items, or to move them between work order operations. When you place your WIP in a load carrier, you cannot have inventory balance in the same load carrier on several bins in the specified inventory location at the same time i.e. the load carrier can only exist in one bin in an inventory location at a time. One work order operation may have several bins, but only one load carrier can be specified on one bin in an inventory location. The option to specify load carrier is available in the programs for labor reporting, where you manually report work order operations. Once you transfer the partially/fully completed labor operation from the left-hand embedded window to the right, you can include the load carrier details in the specific fields i.e. Inventory location, Load carrier ID, Batch No., Item serial No. and save the record.

Load carriers/master load carriers are supported in inventory allocations, both for customer orders and for material reservations on manufacturing orders, and in both manual and automatic allocations.

A load carrier can be moved around in the inventory to different areas/bins and the goods or items stored in the load carrier should reflect this in their inventory balance. Items can also be moved between load carriers and a load carrier can be delivered to the customer complete or in parts, i.e. one or more specific items in the load carrier can also be extracted and delivered to a customer.

You can use the Load carrier program if you need to track the location of items within an inventory location. This can be, for example, to separate different types or sizes of an item or for a specific customer.

You can also do inventory transfers of load carriers/master load carriers, or content within them, including order specific inventory. You also have the possibility to automate transfers when the whole content of the load carrier/master load carrier is to be transferred to a different warehouse or bin.

You can apply automatic sequence numbering to load carriers by defining one or more sequence number series connected to the number series type 8, Load carrier and/or packaging type.

Several controls and rules can be applied on different levels to refine how load carriers are used in the inventory, and what details load carriers should carry and update in the process.

Load carriers are primarily intended for use in sales/inventory/purchases/manufacturing. Item balances stored on specific load carriers or master load carriers can be allocated to customer orders both manually and automatically. Automatic allocation of inventory stored on load carriers is supported for both order fulfillment and manufacturing processes, streamlining picking and material reservation workflows.

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7.7.9.1 Master load carrier

You can use the Master load carrier functionality to make inventory management even more efficient. Any updates to a master load carrier affect all the load carriers within the master. If you use it, you can search for master load carriers using the Master load carrier checkbox.

A master load carrier can contain any number of load carriers, for example a pallet (master load carrier) containing boxes (load carriers), and can also contain other master load carriers.

If a master load carrier is moved/picked, all the load carriers within the master are also moved/picked. The same applies if you change the status on a master load carrier; the status is also changed on the load carriers within it.

You can select a load carrier to be a master load carrier and map details between master load carriers and the load carriers that they contain. In case another load carrier is connected to the load carrier, it is automatically marked as a Master load carrier.

Placing load carriers and items into master load carriers is usually done through the relevant inventory transactions. However, manual mapping can only be done if there are no balances in the Load carrier details table by selecting the load carrier to which you want to connect another load carrier and creating new mapping rows. There, enter the ID of the load carrier to be connected and adjust its status if required. Save and return to the main form to view the mapping in the embedded list.

If the load carrier in the program header was not already marked as a master load carrier, it is automatically marked as soon as there is a mapping in the embedded list. Also, the Master load carrier ID and description are automatically filled in from the Load carrier ID and description fields.

You can calculate/recalculate the gross weight, net weight and volume of master load carriers and the connected child load carriers.

7.7.9.2 Create/change load carrier ID

The change load carrier button allows you to create a load carrier ID when entering arrivals and receiving deliveries into an inventory location.

- From program header when entering arrival and receiving delivery to inventory location:
  You can create a new load carrier ID based on selected package type or manually entered load carrier ID and inventory location specified in the arrival entry/receive deliveries header for the selected purchase order number.
  When entering the arrival if you have selected more than one package type or manually entered more than one load carrier ID for a purchase order, multiple load carrier IDs are created.
- From rows in right list when entering arrival and receiving delivery to inventory location:
  You can also specify or change the load carrier ID after opening a row in the right list in the arrival entry and/or received deliveries programs.
  If you have linked a package type with an item, item balances, or inventory locations, it is considered the default package type when assigning a load carrier at the row level. The priority is as follows:
  a. Item/balance record
  b. Item record
  c. Inventory record
  d. Packaging types

  Similarly, if you have linked a load carrier rule with an item, item balances, or packaging type, it is considered the default rule when assigning a load carrier at the row level. The priority is as follows:
  a. Item/balance record
  b. Item record
  c. Packaging types

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- In manufacturing:
  You can create a load carrier ID while manually receiving deliveries from manufacturing orders/suborders based on selected package type or manually entered load carrier ID, inventory location, and load carrier rule specified for the selected MO rows.
  If you have linked a package type with an item, item balances, or inventory locations or a load carrier rule with an item, item balances, or packaging type, the priority remains the same as when entering arrival/receiving delivery as mentioned above.
- In the physical inventory programs:
  You can update the physical inventory bin for an item with the existing load carrier ID in the Physical inventory journal program as follows: After you have created a journal header with a journal number and inventory location, you can search for items to be included in the physical inventory bin by choosing the inventory location and searching for the specific item with the relevant load carrier ID. These items are presented in the left list of the journal. When you select the relevant items and transfer them to the right list window, the existing load carrier ID is copied to the right list.
  You can also update the physical inventory results record with the existing or a new load carrier ID by selecting the physical inventory bins within the journal and transfer them to the right list window. If a load carrier ID is associated with a physical inventory bin, it is copied to the right list. Alternatively, you can add a new row to the existing journal directly in the right list with the same inventory location as in the journal header and an existing or new load carrier ID.

7.7.9.3 Make load carrier mandatory

You can indicate if load carrier is mandatory when changes are made to inventory when

- reporting arrivals in the Arrival entry
- receiving deliveries to inventory
- receiving deliveries from manufacturing orders
- updating the physical inventory balance in the inventory bin for an
- transferring load carriers between inventory
- manually entering inventory transactions
- receiving order-specific deliveries or adjusting order-specific balances
- picking customer orders in the picking programs.

That load carrier is mandatory can be set on different levels and the priority between the levels is set as follows:

- bins per item – highest
- items per inventory location
- inventory locations
- items – lowest

Any exceptions to this order of priority can be handled by creating separate bins per item.

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7.7.9.4 Load carrier ID on returned goods

- Load carrier ID on reversed arrivals: The load carrier ID and description assigned to an item in Arrivals is automatically transferred to the reverse quantity transaction that has been created automatically in the right window when you click the Reverse arrival button. Load carrier ID on returns at arrival check: In the case of returns, when you enter the Returned qty and save the record, the return transaction is saved with the load carrier ID inherited from Arrivals.
- Load carrier ID when returning goods received directly:
  - From arrival entry: The load carrier ID associated with an item in the Arrivals node is assigned to the return transaction record in the Returns program when you enter the return quantity for the item and save the record.
  - From received deliveries: When you enter the return quantity on goods in the Returns node in the Received deliveries program and save the record, the load carrier ID inherited from Received deliveries and associated with the record is transferred to the new received delivery row that has been created automatically in the right window with a negative quantity according to the entered return.

7.7.9.5 Calculate load carrier weight/volume if changed in Items program

You can (re)calculate the load carrier weight and/or volume based on the selected LC calculation method if the gross/net weight or volume of any item associated with the load carrier ID is modified in the Items program.

7.7.10 Manual inventory transactions

There are several ways of reporting inventory movements that are not handled via integrated routines:

7.7.10.1 Direct entry

There is a simple transaction entry form in which you choose a transaction type and inventory location, enter an item number and, where appropriate, a transaction specific price. If required, you fill in a bin, batch number and/or serial number. This form can also be accessed from master items and the Item balances form.

7.7.10.2 Entry journals

There is also an optional, overlying form for transaction entry via a journal. This is a legal requirement in certain countries. Entry journals give greater control over which users are allowed to enter the various types of inventory transactions, and in which inventory locations. The entry journal must be authorized and its status set to Ready before a posting journal can be created and integrated with the general ledger.

Manual inventory transactions is also the normal way of entering inventory balances at implementation, that is to say, if the balances are not transferred automatically.

7.7.11 Alternative units

Alternative units can be used for automatic or manual unit conversion in the Inventory, Sales and Purchasing modules. If the function is activated, items can be purchased, stocked and sold in different units. Costing is, however, based on your inventory stock unit.

You can enter inventory transactions, including physical inventory counts, order, requisition and purchase order rows, dispatches and received deliveries in your own inventory unit or in the alternative unit. You can set the function up to allow entry of quantities in both units, if the relationship between stock units and alternative units vary. This is useful in the food industry, for example, in which you need to specify both the number of units and the weight. This setup also accumulates inventory balances in alternative units in Inventory balance records.

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If you set the function up for automatic conversion according to a fixed conversion rate, the unit you enter is automatically converted into the other unit based on the conversion rate. In other words, if you enter a quantity in an alternative unit, it is automatically converted to your inventory unit, and vice versa.

Example: An item is stocked per piece. It can be purchased and sold per piece, but can also be purchased and sold in the alternative units of 6 pack, 12 pack or 24 pack. If you use 6 pack as alternative unit when entering an item on an order or purchase order and specify a quantity of 3, the quantity is automatically converted to 18 in your stock unit.

If you set the function up for manual conversion, you enter quantities in both units and Jeeves checks that the relationship between them is reasonable according to the given conversion factor and tolerance percentages.

You can now plan a manufacturing order with alternative units in the Manufacturing suborders program and report manufactured quantity for alternative units in the Show recd delivs from MO node in the Material issues, allocation (suborders) and Mtrl issues, labor and rec deliveries programs. For more details, see the chapter on *Jeeves production*.

7.7.12 Assembly

There is a function for quick and simple reporting of received deliveries and picked components for packages and assembled items, without manufacturing orders. Because no manufacturing order is created, the program can be used for simple items that are manufactured as required or when there is free capacity.

7.7.13 Transfers between/within inventory locations

In Jeeves ERP there are three different ways to transfer items between inventory locations, either via inventory transactions or via a customer or purchase order:

- Using a special program for transferring items in the Inventory module. When you perform inventory transfers in this program you enter the item number and the FROM and TO inventory location in the same transaction. Delivery notes, picking lists and the like cannot be created in this program. The program can also be used for transfers within inventory locations, i.e. to another bin or batch within the same inventory location.
- Using an order type for transfer between inventory locations in the order routine, which means that you can create picking lists and delivery notes in the same way as for normal customer orders. The delivery is received when the customer order is released.
- Using a purchase order type that creates customer orders for transfer between inventory locations in the order routine, which means that you can create picking lists and delivery notes in the same way as for normal customer orders. The receiving inventory location sees the planned delivery in the planning list and reports the delivery when it arrives.

For all three methods you can set up how allocated inventory, statistics etc. should be updated.

7.7.14 Vendor managed/Call-off stock

To reduce the delivery time, suppliers often transfer goods to a warehouse of a customer in another EU member state. The goods remain the property of the supplier until the customer takes the goods out of the stock. This is called vendor managed or call-off stock. As per the VAT rules, effective from 1 January 2020, when the customer takes the goods out of the stock, the supplier performs a direct intra-community supply of goods to the customer. The supplier does not have to register for VAT purposes in the EU member state of arrival of the goods.

However, if the goods are not sold within 12 months to the customer or in case of theft/loss of the goods, the aforementioned measures are not applicable.

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7.8 Physical inventory counts

7.8.1 Normal physical inventory

Physical inventory counts can be performed continuously during the year. When you make a physical inventory, you select the items to be counted, usually based on search criteria such as when they were last counted, physical inventory code, inventory balance and the signature of the person responsible.

The inventory balance of the selected items is copied to a physical inventory journal, which you print out. The physical inventory date and accounting data are set to the date indicated in the journal header. The inventory is counted, the result is entered and the balance is updated. The inventory balance is adjusted (increased/decreased) with any difference between actual and registered quantity.

Since the stock is adjusted with the difference, normal inventory transactions can be made as soon as the stock has been counted, i.e. even before the results have been entered and the balances updated.

You can view physical inventory differences in both quantity and value, in the stock unit and any alternative unit. The value is updated when the physical inventory result is transferred to inventory (updates the inventory balance).

7.8.2 Quick counts

Jeeves ERP also has a function for quick physical counts. This means that you can enter the actual inventory balance directly in a quick and easy way, without the need to create an inventory journal etc.

You do so by entering a manual inventory transaction of the type Quick count. When you specify the physical count quantity, any difference between the current inventory balance and the physical count quantity is calculated, which is then entered in the field for inventory transaction quantity. The inventory balance is updated immediately, which is why the transaction type should be used with caution.

The inventory difference is posted and integrated with the general ledger when printing the inventory journal.

7.8.3 Jeeves Warehouse mobile routine

A standard mobile routine has been developed for Warehouse operations. Jeeves standard mobile routines are developed and maintained in Jeeves standard.

Jeeves Mobile warehouse routine is used by warehouse staff to report their work directly from where the work is performed. They can report inventory transfers, any inventory that is scrapped and also report quick physical counts.

7.8.4 Posting and integration with general ledger

The posting of inventory transactions is controlled by inventory posting templates. The templates are normally defined per product account and inventory location, though there can also be general templates that apply to several/all product accounts and inventory locations. The templates consist of account numbers, and other accounts segments such as cost centers where applicable, connected to the relevant inventory transaction type, such as received delivery or dispatch, and split into integration details such as Inventory account, Contra account, Price difference account etc.

Inventory transactions and associated postings are normally created in other modules. For sales from stock, inventory issues are posted according to posting templates when printing invoice journals and integrating with the general ledger. The same applies to purchases to stock; postings are created when printing received delivery journals and integrating with the general ledger. For

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manual inventory transactions that are entered directly in the inventory module there are separate programs you can use to create journals and integrate with the general ledger.

7.8.5 Reconciliations

During period reconciliations you can use Jeeves ERP's functions for inventory valuation and reconciliation of inventory against the general ledger. The latter compares postings on the relevant inventory transactions with the corresponding postings in the general ledger, and shows any differences. If there are differences, you repeat the procedure, each time with an earlier reconciliation date, until the accounts reconcile. In this way you can find out the date on which the difference arose and correct it.

7.9 Project manufacturing

Project manufacturing supports the business transactions of companies that supply customer-specific products or solutions that are meant for use in a specific project. It covers the processes of a manufacturing process from quotation to payment using a project ID. This section provides an overview of the components that are specific to inventory. Other components are described in the chapter on Jeeves Time, projects, assignments.

7.9.1 Inventory locations setup

You can define inventory locations to be used in Project manufacturing. You can also create a fictitious project to represent the normal inventory location when transferring stock to and from Project manufacturing inventory locations.

7.9.2 Requests for inventory transfers

7.9.2.1 From material reservations

If the required material is not available in the project inventory, the shop floor personnel can create a request for that material to the inventory personnel. They can do this from the material reservations in a new node in the manufacturing order programs.

7.9.2.2 From inventory request

You can search for requests created by shop floor personnel requesting inventory transfers. These requests are created in a node in the manufacturing order programs when the required material is not available in the project inventory.

It is also possible to print the requests for inventory transfers corresponding to a picking list. The report shows where the material is available to pick, and includes the items, quantities, inventory location, and the suggested bin to which the item is to be transferred for a particular requisition.

You can also manually enter requisitions for inventory transfers.

7.9.3 Movement of items between project inventories

You can move items between inventory locations with or without project ID, between bins within an inventory location, and back to an inventory location.

The inventory transfer creates manual inventory transactions in the normal manner, with the addition of the project ID.

7.9.4 Allocation of items from project bin

It is possible to allocate items for sales orders or material reservations to a prioritized bin.
For instance, when an allocation is made for a sales order or for a material reservation associated with an inventory location that is linked to a project ID, the allocation logic first searches the item balances for the bin associated with the project ID and the allocation for the order will be made against the specified bin.

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7.9.5 Material requirements planning

You can run material requirements planning (MRP) for a selected project and one or all inventory locations from the Projects programs. This helps you plan for the project inventory and gain an overview and control over the records created to supply the project with goods.

You can run MRP, either according to a pre-defined template or manually specified parameters. You can select an inventory location to run the MRP for the combination of project ID and inventory location, or leave the field blank to run it for the selected project ID and all inventory locations.

7.10 EU sustainability regulations

Jeeves ERP includes functionality to support companies in the EU to comply with sustainability regulations:

7.10.1 Carbon Border Adjustment Mechanism

Importers bringing Carbon Border Adjustment Mechanism (CBAM) goods into the EU from other countries must pay a fair price on the carbon emitted during the production of those CBAM goods. CBAM generates a cost on certain items (TARIC codes) imported from specific countries where the carbon credit requirements differ from those imposed on European producers. When determining the purchase cost of an item, CBAM related costs are included in the calculation and taken into consideration when receiving deliveries.

7.10.2 Corporate Sustainability Reporting Directive

Corporate Sustainability Reporting Directive (CSRD) is a regulatory framework that helps companies to determine the accurate carbon dioxide equivalent (Co2e) footprint of their products and overall operations, by following the Greenhouse Gas (GHG) Protocol. This is essential not only for regulatory compliance but also for identifying opportunities for emissions reduction, enhancing sustainability, and improving the company’s competitive advantage in increasingly environmentally-conscious markets.

7.10.3 SCIP management

The Substances of Concern in Products (SCIP) Management capabilities supports you with your obligation to report where there are substances of concern. It supports how you manage related products efficiently in Jeeves ERP, both in terms of structuring the relationships between substances and items as well as extracting information that needs to be reported to the ECHA system.

7.10.4 EU Deforestation Regulation (EUDR)

Jeeves ERP includes functionality to support compliance with the EU Deforestation Regulation (EUDR), which requires that products sold in or exported from the EU market do not contribute to deforestation or forest degradation. The regulation applies to products that contain, are made from, or are produced using cattle, cocoa, coffee, oil palm, rubber, soy, or wood.

A dedicated program called EUDR DDS Information stores metadata and content related to Due Diligence Statements (DDS), which are legal declarations required for placing regulated commodities on the EU market. A EUDR checkbox is available on TARIC codes and Trading areas records to indicate that a Due Diligence Statement is required for

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incoming or outgoing shipments. An application parameter controls the default batch status code for batches of items marked as EUDR-compliant.

Support for managing and distributing Due Diligence Statement information across the supply chain strengthens audit readiness and traceability for organizations subject to the regulation.

7.11 Industry-specifics

Most functions in Jeeves Inventory are common to all industries. There is one, however, that is specific to certain industries, as described above in the section on products and services.

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8 EazyStock adapter

Jeeves offers integration with EazyStock (ES) inventory optimization software (powered by Syncron). Jeeves ERP's EazyStock adapter is intended for exporting the relevant data from Jeeves ERP and for importing inventory optimization data from ES according to standardized file formats defined for ES.

Once you set up an item to be handled by EazyStock, you need to define rules for the import of data from EazyStock to Jeeves ERP. Here you define the items whose data is to be imported and the actual data that is to be updated. If you define more than one rule you should also specify their relative order of priority.

JIS uses Jeeves Service builder (JSB) in the integration between Jeeves ERP and EazyStock. JSB is an information gateway which provides a single point of access to information resources. Information can be retrieved, created, modified or deleted depending on what is supported by a so-called resource handler. It is not an integration server but a dynamic endpoint that can be used by integration servers to connect to other systems.

As part of the integration between Jeeves ERP and EazyStock (ES), you can enter a combination of own company ID and inventory location that should be used when exporting to and importing from EazyStock.

When exporting using the following procedures, the values from these two Jeeves fields are kept together as one value as required in EazyStock.

- JEEVES_EAZYSTOCK_ReadTransactionalDemands_SelectionID
- JEEVES_EAZYSTOCK_ReadPurchaseOrders_SelectionID
- JEEVES_EAZYSTOCK_ReadItems_SelectionID
- JEEVES_EAZYSTOCK_SrprForecast_SelectionID
- JEEVES_EAZYSTOCK_PneForecast_SelectionID

This ensures that the exported records can be correctly identified and processed in EazyStock.

When importing using the following procedures, they are separated as required in Jeeves:

- JEEVES_Eazystock_Insert_Tmp
- Jeeves_Init_Insert_EsInvtryParameters

This ensures that the record is imported to the correct company (foretagkod) and set to the correct inventory location (LagStalle).

8.1 Export considerations

8.1.1 Export of relevant item data

Jeeves ERP automatically exports the relevant item related data, for example item information and stock balances and sends them to ES. Much of this data depends on whether the item is a purchased item or a manufactured item.

8.1.2 Export of transaction demand data

Jeeves ERP also exports so-called transactional demand, i.e. requirements from manufacturing orders and customer orders, and sends the data to EazyStock.

8.1.3 Export of open purchase and manufacturing orders

In addition, Jeeves exports certain data from open purchase orders, i.e. purchases with a remaining quantity to be delivered, as well as open manufacturing orders, i.e. manufacturing orders with a remaining quantity to be manufactured, for items set up for export to EazyStock.

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8.1.4 Export of key forecast data

Jeeves Service Builder (JSB) checks for items that are flagged for export to EazyStock and Jeeves exports the key forecast data for these from Jeeves Forecast/net requirements, Forecast, broken down per BOM, and related tables.

8.2 Inventory optimization data

You can view inventory optimization data from EazyStock if you open an item balance record either directly or from the Items program and open the ES inventory parameters node. You can see the result of the latest import from EazyStock and any updates made for the item and inventory location in question. The data is cleared and updated every time a new import is done to Jeeves ERP. The import is run automatically by Jeeves Service Builder (JSB). Jeeves ERP then updates the relevant item balance records according to the rules you have defined.

8.3 Import of purchase and requisition proposals

Data imported from ES files can be used to create requisitions or purchase orders as follows:

1) JSB reads each row in the ES file with proposals for requisitions or purchase orders. The rows are flagged in ES for one or the other. Corresponding rows are created in the temporary table with information about item number, inventory location, supplier ID, purchase quantity, requested delivery date and any internal text.

2) JSB initiates the creation of purchase orders from the rows in the temporary table that are flagged as purchases. The purchases are grouped by supplier, i.e. one purchase order is created for each supplier. When this is complete, the corresponding rows are flagged as processed. The next time the temporary table is loaded these are cleared from the table.

3) JSB initiates the creation of requisitions from the rows in the temporary table that are flagged as requisitions. Requisitions are not grouped. When this is complete, the corresponding rows are flagged as processed. The next time the temporary table is loaded these are cleared from the table.

8.4 Export of BOM

When an item consists of a BOM, EazyStock can more accurately calculate and optimize the inventory balances for all items with BOMs if these are exported as well in a structured manner.

Jeeves ERP exports bill of materials to EazyStock based on certain conditions set up in the Items program as listed below:

- A BOM must be connected to the item in the BOMs/Package items table
- The top item must be marked for export to EasyStock

In addition, if there is a BOM revision, the BOM rev. field should specify which BOM records under an item are to be used.

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9 Jeeves ERP's Astro integration

Jeeves ERP can be run integrated with Astro warehouse management. This means that Jeeves ERP offers different levels of inventory management:

- Jeeves' own basic inventory module/processes, for companies with low volumes and a low level of complexity
- Jeeves Astro integration, for companies with high volumes and a high level of complexity

There are also other, customized solutions such as EasyWMS.

Jeeves ERP's Astro integration is comprised of the following components:

- An external integration component called Transaction handler together with a Jeeves program for the configuration of the Transaction handler.
- A number of static system tables for the definition of statuses, number series and underlying configuration data.
- A mapping tool comprised of a number of programs in Jeeves ERP for mapping outgoing and incoming XML messages.
- A few Astro specific basic data programs and fields.
- A number of Jeeves programs for monitoring XML messages, both confirmations and error messages.

9.1 Supported processes

Jeeves' Astro integration, with Jeeves' standard XML mapping, supports the processes listed below.

9.1.1 Maintenance of basic data

- Entry and maintenance of item data
- Entry and maintenance of shipment routes

Jeeves ERP "owns" all item and route data, which means new entries and maintenance can only be performed from within Jeeves ERP. Astro stores item data on the item and inventory location level.

9.1.2 The customer sales order process

- Order entry, shipment planning, dispatch and closing
- Returns
- Reverse picking

The integration between the sales process and Astro is comprised of the Shipment planning routine, which is described in the chapter on the *Sales* module.

9.1.3 The purchase order process

- Purchase order entry, received delivery against purchase order
- Update purchase order
- Cancel purchase order
- Returns
- Backorders

Purchase orders are entered as usual in Jeeves purchasing module but with a stock location prepared for Astro. When the delivery is received in Astro, an XML message is sent to Jeeves ERP that creates received delivery rows. When all rows have been fully delivered the purchase order status is automatically set to Delivered. Any returns to suppliers must be handled within Astro before the received delivery is reported back to Jeeves ERP.

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9.1.4 Basic inventory management processes

- Manual inventory transactions, including scrapped items
- Physical inventory counts

Astro "owns" all inventory balances, which means these inventory management processes are handled from within Astro. These trigger XML messages to be sent to Jeeves ERP, which create inventory balance records.

9.2 Transaction handler

The Transaction handler is an external integration component between Jeeves ERP and Astro WMS.

The Transaction handler must be configured to transform incoming messages to a format that Jeeves ERP understands, and vice versa. Messages from Astro WMS are in XML format and they are transformed into a format that Jeeves can interpret and generate SQL code from.

There is a program for setting up a standard configuration of the Transaction handler. The configuration program is distributed as part of the Transaction handler installation. All configuration is done in Jeeves ERP in order to make it as dynamic as possible for each individual installation.

9.3 Mapping XML files between Jeeves and Astro

Jeeves' Astro integrator is delivered with Jeeves' base Astro mappings. Jeeves Information Systems also distributes any new or changed templates. Partners can import these templates simply by pressing an import button and entering the name of the message in question. The template can then be copied to a mapping in the XML mapping to Jeeves program. This implies that the main task is to make adjustments to these, for example specify the name of your Astro installation, create any additional mappings or change the provided base mappings. To be able to do so you will need training in the mapping technique.

There is a base integration transaction for each process/sub process. Example: the BalanceChange transaction for updating item balances after making manual inventory transactions and/or physical counts in Astro. Balance changes are sent from Astro to Jeeves ERP and affect the inventory balance records in Jeeves ERP. The relevant columns in the inventory balance table are defined in the mapping, and they are grouped, sorted and assigned XML tags, with or without conditions etc. When a balance update is sent via XML from Astro, the XML message is interpreted according to the mapping, and results in a transaction affecting the inventory balance.

9.4 Monitoring and tracing transactions

There is a number of programs for day-to-day monitoring of transactions to and from Astro WMS. There are programs for monitoring the incoming and outbound transactions queues. Here you can monitor and search on the status of transactions. When a transaction has been executed and the import/export has been completed it is removed from the queue, after which the transaction or message can be traced via the Log - search node or the incoming/outgoing transactions log program.

9.4.1 Error handling and data synchronization

The majority of errors occur during the implementation phase and are caused by incorrect base data or balances converted from legacy systems, or by mapping problems. When these initial problems have been resolved, the only errors that will occur will be incorrect transactions, i.e. transactions containing data that does not comply with business rules or database constraints.

Errors in Jeeves' Astro integration can occur both in Jeeves and in the Transaction handler and the errors are logged in Jeeves. The Transaction handler also logs errors to its own log file which can be useful if it cannot log to Jeeves. The Transaction handler can be configured to send e-mails when an error occurs. When you receive error information you can, for example, use the transaction ID

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given in the e-mail to search for the transaction in question in the error handling programs. The programs include functions for:

- re-running messages
- deleting messages
- editing and re-running individual rows in messages
- viewing error messages
- viewing the SQL code

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10 Jeeves ERP ICT

ICT, Inter Company Transactions, is a module that can be used to manage logistics and associated financial transactions in more complex company structures, i.e. between companies or within a company with multiple sales units, distribution units/central warehouses, and perhaps manufacturing units.

The module comprises functions for:

- automatic generation of delivery customer orders at the delivering unit from purchase orders generated from customer orders in the selling unit
- item planning list for delivering unit accessible from the selling unit
- automatic creation of received deliveries against purchase orders, if required, as well as automatic picking of customer orders in selling unit, when picking delivery customer order in delivering unit
- generation and arrival posting of supplier invoices in selling unit when invoicing delivery customer orders in the delivering unit
- manual final approval or automatic matching and final approval posting of supplier invoice in the selling unit

Explanation of terminology used in this chapter:

- The 'selling unit' is the company or unit that makes the sale and takes the order from the customer.
- The 'delivering unit' is the company/warehouse/manufacturing unit that fulfills the order. Delivery to the customer can either be made by drop shipment or by transit via the selling unit.

10.1 General features

10.1.1 Intercompany relationships and connection types

Jeeves ICT can handle transactions between several warehouses in the same company, or between several companies in the same database or in different databases, even if using different servers.

Jeeves ERP makes defining internal and external business units and the relationships between them quick and simple. You set up the ICT process for each relationship, for example,

- when customer orders should automatically generate purchase orders at the delivering unit
- if received quantities should be automatically updated based on what has been delivered between the units
- when the status on purchase orders should be automatically updated based on events in the delivering unit
- when customer orders in the selling unit should be released
- how supplier invoices should be generated when the delivering unit invoices the selling unit and
- whether users are allowed to change the order quantity and sales price on customer order rows in the delivering business unit
- whether customer orders can be canceled in the delivering unit, or only in the selling unit

10.1.2 Activation of ICT functionality

The ICT functionality can easily be activated for all or for individual users.

10.1.3 Translation of item numbers between units

If the different units use different item numbers, you can specify the delivering unit's item numbers and via a parameter control that these numbers are used on the orders that are created in the delivering unit.

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10.1.4 Pricing between units

At the delivering unit you enter a price list for managing the prices between business units, for example a price list with a certain markup on the cost price. A price list can be connected to one or more business units.

You can retrieve the price based on the priority by mapping the pricing type priority parameters. You can also set the price adjustment factor for each priority type as desired in the respective Adjust factor % field. Following are the priority types that are used to retrieve the price based on the priority:

- 0 - Sales price from alternative supplier/price list + adjustment factor in percent for this pricing priority. Only used if the net price acc. to higher pricing priorities is zero.
- 1 - Sales price from delivering sales order + adjustment factor in percent for this pricing priority. Only used if the net price acc. to higher pricing priorities is zero.
- 2 - Cost price from delivering sales order + adjustment factor in percent for this pricing priority. Only used if the net price acc. to higher pricing priorities is zero.
- 3 - Net price from delivering sales order + adjustment factor in percent for this pricing priority. Only used if the net price acc. to higher pricing priorities is zero.

The price is always retrieved based on the hierarchy in the pricing priority (in descending order with prio 1 having the highest priority). A pricing priority that is lower is only considered if the net price in a higher priority is zero.

If no mapping is done, the current price is determined through a relationship between the Alternative supplier record for Items in the selling business unit and a specific price list designated for the ICT-flow in the delivering business unit.

In the underlying exchange rate tables for currencies, you can maintain agreed and fixed exchange rates per currency for trading between different business units in the ICT-flow.

10.2 Sales and purchase order entry

The customer order is entered in the normal way in any order program. A sales order at the selling unit automatically creates a purchase order, which in turn automatically creates a sales order at the delivering unit. Via parameters you can decide whether values must be retrieved from the selling unit customer order to the delivering unit customer order.

You can set up the desired functionality for the alternative supplier record to be optional regarding entry of Items for the Sales customer order (SCO) in the ICT-flow. If you set up ICT to make the Alternative supplier (AL) record mandatory for the ICT-flow, you need to provide the following information.

In the selling unit, you must enter the delivering unit as supplier of the items in the Alternative supplier program or node. This record must also point to the fictitious inventory location for the delivering unit.

Note that, if you use the ICT replenishment functionality you need to have two alternative supplier records:

- one for the ICT supplier, connected to the inventory location with Stock replenishment ICT = N
- one for the ICT supplier, connected to the replenishment inventory location with Stock replenishment ICT = Y

If the different units use different item numbers, you can specify the delivering unit's item number. You can also specify if the delivering unit's item number should be used on the orders that are created in the delivering unit.

If you set up ICT so the Alternative supplier (AL) record is not mandatory for the ICT-flow, the values regarding lead times will be retrieved from the internal company defined as ICT supplier in the current ICT connection.

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10.2.1 Automatically create manufacturing and/or purchase order

You can automatically create a manufacturing and/or purchase order if required for items in the delivering business unit set up with acquisition method Purchase to requirement in the selling unit as well as in the delivering unit. In the case of automatic manufacturing orders, the acquisition method must be set to Purchase to order point in the selling unit and Manufacture to requirement in the delivering unit.

10.2.2 Dispatch date management

You can take both the selling and the delivering unit's fixed delivery days and various lead times into consideration. This is useful for transit and order specific order types where the goods are delivered to the selling unit for further delivery to the customer.

10.2.3 Traceability Sales order->Purchase order->Delivery sales order

On the purchase order at the selling unit you can see the ID of the delivering unit and the number and status of the sales order at the delivering unit. You have the corresponding traceability on the resulting inventory transactions.

On the sales order at the delivering unit you can see the ID of the selling unit and the number of both the sales and the purchase order at the selling unit. You have the corresponding traceability on resulting inventory transactions.

10.2.4 Availability and status information at selling unit

There is an ICT specific planning list that makes it possible for the selling unit to see the current inventory balance and planned stock movements at the delivering unit.

In many steps in the work process, when the reservation status of the order rows in the customer order at the delivering unit is set, the corresponding status is automatically set on the customer order at the selling unit. The information is shown on customer order and purchase order rows at the selling unit.

10.3 Delivery reporting

You can choose between manual pick reporting at the selling unit and automatic reporting when the goods are picked, or when the order is released for invoicing, at the delivering unit. You can also automate the release of the order in the selling unit, if desired.

10.4 Stock replenishments

You can initiate internal stock replenishments, i.e. requirements that are not specifically connected to customer orders. The functionality can also be used in combination with requirement calculations. This means that a purchase order is entered or created at the business unit where the stock is to be replenished, and a customer order is automatically created from the purchase order at the business unit to supply the goods for replenishment. Jeeves has functionality to specify whether or not you can change the supplier after the stock replenishment order has been created.

10.5 View inventory movements and balances

You can view item balances at different inventory locations across companies. You can view inventory movements and the resulting balances in ICT inventory locations

An ICT-specific planning list allows you to analyze the inventory activities planned for sales, purchasing, production, and service. You can also view information about batches and item balances. In addition, you can view current inventory balance and shortage check statuses in the selling business unit for sales order rows in either the delivering or the selling business units. If the inventory location is an ICT delivering unit, the planned activities in the list and the shortage statuses are retrieved from the relevant inventory location in the delivering unit.

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10.6 Change log

You can enable/disable the log for ICT order row changes. If enabled, specific changes regarding ICT order rows are logged, including changes to order row status, inventory location, estimated dispatch date, row quantity, price currency, volume discount %, and discount%. The logs can be used to trace and view specific changes that are performed regarding Customer order rows

Note that normal changes of order row status are not logged, except when the status of the order row changes to Canceled. This is because the log of all status changes results in a heavy load to this log function. The status of customer order rows is changed during many phases of processing a customer order row.

10.7 Product configurator setup in an ICT environment

Any configurable item can be used as a configuration model in the product configurator. In an ICT environment, the configuration model is defined in both the delivering and selling units while the configuration is done in the selling unit.

You can create a stocked item with a new Configuration unique ID in the delivering unit in an ICT environment. The configuration unique ID is connected to the model in the Configuration user friendly names program. The BOM, package items and operations records will be created for the new stocked item in the ICT/non-ICT environment as required.

10.8 Invoicing between units

You can choose between manual entry of supplier invoices at the selling unit or automatic creation when creating customer invoices at the delivering unit. If you choose to automatically create supplier invoices, Jeeves ERP can create arrival postings only, or automatically match against purchase orders and create final postings.

You can specify the invoicing type for customer invoices (in the delivering unit) on ICT transactions in the Auto invoicing types ICT program and control whether drop shipment invoices and transit/order specific invoices should be created automatically from the ICT delivery unit without printing the invoice in the Order program.

If you select No auto invoicing in the Link between business units ICT program, invoices need to be created and printed in the delivering unit, else if you select Auto invoicing delivering dispatch order, invoices are generated automatically (without creating and printing the invoice in the Order program).

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11 Jeeves ERP Production

Jeeves Production includes functions for production preparation, costing, material planning, production control, shop floor reporting and attendance, as well as for configuration and net requirements planning. There are also reports and programs for following up actual costs and other forms of financial follow-up.

Jeeves ERP also offers an integration with a fully functional graphical planning tool called EQ Plan that is developed and maintained by a chosen solutions provider.

11.1 Production preparation - items/material

11.1.1 Acquisition methods

Items have a default acquisition method – purchase or manufacture to order point, to plan or to requirement.

11.1.2 Direct status control

There is a parameter to control whether a check should be performed directly when a manufactured item is specified that the component material is OK to be used. You can check at manufacturing order entry, and if required at customer order entry when the item BOM is manufactured to order.

11.1.3 Replacement items

Production planning may have specified an item that Production replaces with another item, which is used in other places in the BOM or which is easier to acquire. If this field is filled in on an item in a BOM, the item in question is automatically replaced in the list of material on manufacturing orders.

11.1.4 Bills of materials

Jeeves ERP supports:

- any number of BOM levels
- order unique BOMs
- BOM revisions
- explosion and implosion analyses
- fixed quantity in BOM

11.1.5 Change item in BOM

There is a program in which you can quickly and easily replace items in bills of materials. In the main form you enter the item to replace one or more items in bills of material. In the left embedded window you search all BOMs in which the old item is included. From these you select the BOM rows in which the component is to be replaced, transfer them to the right window and save. The relevant BOM rows are changed. The right window also shows any previous changes to the item specified in the main form.

11.1.6 Copy BOM

You can copy a BOM to another item in the Items program. There are two alternatives:

Simple copy:

Select the item whose BOM you want to copy. Press the Copy item button. In the dialog that is opened, enter a new or existing item number and check the Copy BOM box. There is a corresponding box for copying operations.

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Copy with simultaneous adjustment:

Select the item you wish to alter and select the Copy BOM node. In left window, enter the item whose BOM rows you want to copy. Transfer the desired BOM rows to the right window. This can also be used to compare BOMs of different items.

11.1.7 Item variants (earlier, simple solution)

There is a simple solution for handling different variants in connection with order-specific manufacturing, such as sizes and colors of an item in a BOM, with retrieval of components for manufacturing orders via a variant code. See also the later solution described in the section on product variants.

11.1.8 Workshop design

Jeeves ERP includes functionality for definition of:

- manufacturing operations
- manufacturing groups
- flow groups
- tools and tool lists
- specification of costs
- scrap reasons and
- instructions and linked documents on both item and operation level

…also for:

- handling design changes
- calculation of design changes and
- calculation of lead times for manufactured items based on operation times

11.1.9 Alternative routings

There is a possibility to define alternative routings for items, which can be controlled per plant/line (via the inventory location) to handle varying costs.

You can manually change to another alternative routing on a manufacturing order when manufacturing is overloaded, when manufacturing smaller quantities, etc., and then replan the order.

11.1.10 Alternative routing per inventory location

You use the Inventory location field to connect alternative routings to inventory locations where necessary. An item can have different routings within the same inventory location or depending on inventory location.

Routing can be pre-defined or can be chosen after the manufacturing order has been created. This is determined by how you set up combinations of Inventory location and Main alternative routing. If the routing is pre-defined, when manufacturing orders are created from net requirements, they are automatically assigned the correct routing and the relevant operations are automatically created.

11.1.11 Manufacturing orders with alternative units

You can now plan a manufacturing order with alternative units in the Manufacturing suborders program and report manufactured quantity for alternative units in the Show recd delivs from MO node in the Material issues, allocation (suborders) and Mtrl issues, labor and rec deliveries programs.

If the Fixed conversion factor checkbox is checked, the manufacturing order quantity will be updated as a multiple of the conversion factor of that alternative unit from the Alternative units per item program when the alternative unit is selected and the quantity of the alternative unit is

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defined. If it is not available in the Alternative units per item program, the conversion factor will be retrieved from the Alternative units program.

If you enter a conversion factor, leave the Fixed conversion factor checkbox unchecked, specify tolerance percentages, and enter quantities in both the stock unit and an alternative unit. Jeeves ERP then uses the conversion factor and the tolerance percentages to check that the relationship between the quantities specified is reasonable.

11.1.12 Overlapping operations

With this function you can plan the start of a subsequent labor operation before the preceding operation is completely finished. Overlapping can be either on quantity or on time. By doing so the manufacturing order's throughput time is reduced.

11.1.13 Subcontract

There is a possibility to define alternative data on operations used only when operations are performed by a subcontractor.

11.1.14 Product variants

There is functionality for handling product variants, useful for companies in the fashion industry for example, for defining applicable combinations of say, models, colors, sizes etc. of a product with the help of a matrix. The matrix has various layers that you can switch between for viewing current and planned inventory balances, item numbers, alternative item descriptions, delivery dates, minimum purchase quantities - or if there are none, economic order quantities, prices etc. for each combination. The matrix is used in production preparation, entry of manufacturing orders, material issues and ordering subcontract labor.

11.1.15 Revision control

See the section on *Jeeves Revision control* in the chapter on *Jeeves Inventory*.

11.2 Costing

See also the section on costing in the chapter about *Jeeves Inventory*.

You can set up manufacturing costs and overheads in the following programs:

- Flow groups
- Manufacturing groups
- Monetary factors
- Tools
- Operations

On operations you define the time to perform the operation, for example setup time, run time, and on the manufacturing groups and flow groups you define the costs of performing the operations for example fixed and variable labor, material etc. You can have different monetary factors for different manufacturing groups. Tools can also add to the costs on operations.

11.2.1 Subcontract in currency and discounts

In the Supplier/item node you can connect an alternative supplier record (al) to labor operations. This means that you can enter unit price, currency, start and setup costs here and use the standard functionality for price calculation in the Alternative supplier program. If you do not enter an alternative supplier record, the information is retrieved from the operation in the usual manner. If you need to enter several records for the same item and subcontractor, you use the Priority field to differentiate between them. There is, however, no logic for retrieving the records in this case.

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11.2.2 Costing per inventory location

As mentioned above, alternative routings for items can be controlled per plant/line (via the inventory location) to handle varying costs. If costing is to be done per inventory location you should ensure that inventory locations are defined for each plant/production line or equivalent, i.e. each manufacturing unit that has varying manufacturing and material costs. For the relevant items and routings, you define purchase agreements with their costs per inventory location (plant/line).

11.2.3 Costing with or without calculation version

Jeeves ERP offers several methods of costing, and storing the results:

- Without calculation version means that the results are stored totaled on the item.
- With calculation version means that the results are stored on a calculation version. These results can be stored for any period of time and calculation versions can later be used to update the field for new cost price on items. This is a way to calculate future costs and, if required, transfer the result to the new cost price field, which is in turn transferred to the current cost price at a certain date.

11.2.4 Simple or detailed costing

The simpler costing method saves less information in the database.

The detailed method saves the entire calculation per selected item to see the result of all components down to the lowest level and all corresponding operations. This allows you to estimate costs for individual items and add on a simulation percentage for future costs.

The detailed method always works against a calculation version.

In detailed costing, logs are also created at a detail level as to how costs have arisen and been totaled upwards in BOMs. The log is created per calculated item.

The results of cost price calculations can be analyzed in a special program before you update the item with the new cost price.

11.3 Master scheduling/forecasting

Please refer to the section on Master scheduling/forecasting in the chapter on *Jeeves Inventory*.

11.4 Material planning

11.4.1 Net requirements planning

We support a number of ways to control how net requirements should work, i.e. if to include orders already started, safety stock, allocations, finite capacity etc.

You can define templates for MRP with one or more sets of rules for how MRP is to work at your company. You can set up shortage tolerances, replanning horizons, excess inventory tolerances etc. You can also define rules for items to be included in MRP. Items can, for example, be filtered per item group or planning manager.

You can choose between:

- Material requirements planning from inventory location. This processes all items with balance records at the inventory location in question.
- Material requirements planning from item balances. You can select the items that are to receive requisitions.

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The result of the requirements calculation can be viewed in the following programs:

- Requisition, details – in a program node you can create purchase orders from the requisition details
- Manufacturing orders from net requirements
- Labor operations from net requirements
- Material reservations from net requirements

You transfer the requisitions to manufacturing orders in the following programs:

- Multiple MOs from requisition. Each requisition creates a manufacturing order header with a suborder.
- Single MO from requisition. One manufacturing order header is created under which all transferred requisitions are entered as suborders.
- Planning mgr, requisitions->MO

Via parameters, you can monitor the location of retrieval of planned inventory transactions during net requirements calculations

11.4.2 Shots management

Shots management is a functionality used in the molding industry, specifically in injection molding companies, to manage and monitor the number of shots required to produce an item.

Jeeves ERP provides support to allow you to configure the tool used for manufacturing orders (MO) to provide additional information to support shots management and to configure operations for shots management. It also allows you to calculate the time consumed (unit time) for manufacturing when entering manufacturing orders for shots management and update the values in case of any changes.

You can enter shots information in the shop floor program. Also, when reporting the number of shots in the Shop floor program, the tool is updated with the accumulated number of shots. This information is helpful to keep a track of the maintenance/overhaul of the tool.

11.4.3 Lead time control and buffer days

You can choose between manual entry of start dates for manufacturing or automatic calculation based on manufacturing lead times and buffer days.

You can define whether the calculated start date and ready date are to be based on the manufacturing order’s calendar connection. This makes it possible to take shifts that do not work the same hours every day into consideration. Jeeves ERP can be set up to check the feasibility of the calculated start date.

11.4.4 Lead time control for order specific manufacturing

If no buffer days are set up, the requirement date is based on the customer order's estimated dispatch date. The standard functionality for determining the number of buffer days between the order row's estimated dispatch date and the preparation date is determined via Dispatch methods, sales, the Internal lead time days field.

Alternatively, you can use the Preparation date field on the customer order row to receive the requirement date for linked purchase and manufacturing orders or requisitions. This means the preparation date for the customer order is taken into consideration in the programs for creating order-specific purchase orders and manufacturing orders and in the MRP program. This is determined by a parameter. The preparation date can in turn be calculated based on lead times for the item/item group. This is also determined by a parameter. Note, however, that you cannot use this together with the standard functionality described above.

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11.4.5 Net requirement pegging

Jeeves ERP includes a program for analysis of net requirements and order fulfillment. The program shows the relationship between demand and supply, order dependencies and potential shortage dates.

You can analyze the relationship between orders on different BOM levels. For example you can see which sales orders depend on a specific semi-manufactured order, a number of levels below, being completed in time.

11.4.6 Replanning proposals

In Jeeves ERP you can receive proposals for replanning of existing orders. Replanning proposals are received when the system sees that a requirement can be covered by an already planned order, but it has the wrong quantity or date (shortage situation) and therefore needs to be replanned. Replanning proposals can also be received when the system sees that there is a planned order without a corresponding requirement (excess inventory situation).

You can bring forward or postpone production, increase or reduce quantities, delete or suggest new orders, divide orders when maximum quantity is exceeded.

There is also functionality to automatically execute the proposed re-planning.

11.4.7 Manufacturing orders

Manufacturing orders can be entered manually or generated automatically from customer orders, or semi automatically via purchase/manufacturing requisitions or via MRP. You can control whether a manufacturing order header should be created per flow group that will manufacture the items. When you create a customer order-specific manufacturing order you can control whether a manufacturing order header should be created per delivery date, or per sales order row.

11.4.8 Suborders

In Jeeves, there are two levels in a manufacturing order, a manufacturing order header and a suborder. A manufacturing order header always consists of one or more suborders.

- The manufacturing order header contains only one manufacturing order number and can be considered a collection of manufacturing orders under the same number.
- The suborder contains information about the item to be manufactured, quantity, planned date, and the current costs. The suborder is also linked to a material list and an operation list.

You can plan/replan from the manufacturing order header, with sequential or parallel planning of included suborders.

11.4.9 Availability check

Once you have created a manufacturing order, you can do an availability check of materials. This check is done by running one of the availability check reports (several orders can be selected and processed in one batch) or using the buttons on the window for manufacturing orders (can only process one order at a time).

There are two types of availability checks:

1) Against planned balance (current inventory balance plus or minus planned deliveries and dispatches up to the picking date of the manufacturing order).

2) Against available balance (current inventory balance minus planned dispatches).

The result is marked on each material reservation (OK or Not OK to pick) and the connected operation, manufacturing order and MO header (all material OK or shortage in material). In this way you can see whether there is a shortage of material for any operation. If all operations are OK, the manufacturing order is also marked as OK. If all manufacturing orders are OK, the manufacturing order header is also marked as OK.

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11.4.10 Allocation of material

You can allocate material in stock to manufacturing orders. The function is used to ensure that there are items available to pick for manufacturing orders.

If a manufacturing order is fully allocated, it means that all material is available in stock and has been allocated to the manufacturing order in question. If, on the other hand, it is only partially allocated, it means that some of the material is not available. If partial allocation is not allowed and material is not available, the manufacturing order will be assigned the status: Could not allocate acc to rule.

Some types of items that are used as raw material require storage in special environments and the supplier will only guarantee the sustainability/best before date for these items under acceptable conditions. However, when the items are delivered these conditions might not be fulfilled and the best before date will deteriorate. In such scenarios, it is important to use the raw material that was delivered first rather than by the best before date.

You can prioritize the usage of items by received delivery date when making allocation to material reservations for items that are delivered in batches. Note that the best before date is also checked since the batch might be unusable.

Allocations are done as follows:

- Received delivery date
- Best before date
- Last Entry date

11.4.11 Subcontract

You can use subcontract purchase orders to purchase labor operations to be carried out at another workshop/factory. This can be necessary, for example, if the workload at your own workshop is too high or for a specific type of work. There is a special purchase order type with a separate number series for this purpose.

You can produce a delivery note, via subcontract operations, with any material which belongs to the work and which will be sent to the supplier. You can do this using a function that automatically starts the operation.

11.4.12 Forecast orders

Besides the normal manufacturing order type, there are two order types for forecast orders:

- Forecast including material can be used when a material reservation is not to be made exactly according to the BOM or if material reservations should always be shown in the planning list.
- Forecast excluding material makes it easier and quicker to enter, change and delete the order, but on the other hand you will need to create material reservations each time you run MRP.

Forecast orders cannot be reported.

11.4.13 Vacation planning

Jeeves' function for stock build-up is based on using the safety stock internally in MRP, in a special way before the vacation period.

The quantity to deliver during the vacation period is calculated for manufactured items from existing customer and manufacturing order rows. The safety stock is increased linearly from the build-up horizon's start date to the vacation period's start date. The result is that the requirement during the vacation period is produced in the form of a safety stock. The manufacturing requisition includes information about what is directly connected to orders and what is connected to the safety stock build-up.

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11.5 Alternative units for manufacturing and material handling

You can specify if the alternative unit is valid to be used on a material when creating a bill of material and in the material reservation that is added manually on manufacturing orders. Similarly, you can also specify if the alternative unit is used as the standard unit in the Bill of materials and Material reservations programs.

In addition, you can specify alternative unit quantity at parent level when creating manufacturing orders.

Note that only one alternative unit can be defined as standard and valid to use per material unit and per item number.

Jeeves also has functionality for sales, purchasing, and manufacturing to handle alternative units to your own stock unit and to handle alternative units when making material reservations. You can use the alternative units in two ways:

- Fixed relationship between own stock unit and alternative unit: When entering order, purchase order and manufacturing order quantities in alternative units, these are automatically converted to your own stock unit.
- Varying relationship between own stock unit and alternative unit: There is no automatic conversion to your own inventory stock unit. You enter order, purchase order, manufacturing order, and physical count quantities in both units. Jeeves checks that specified quantities are reasonable in relation to each other.

You can also view alternative unit used in item explosion and implosion in the Multilevel explosion embedded list.

11.6 Production control

11.6.1 Workload control

In Production control you analyze and, where applicable, replan the workload generated from manufacturing orders. Jeeves has the following levels for workload analysis:

- Total
- Product area
- Foreman area
- Work group
- Manufacturing group
- Flow group

Jeeves includes functionality for workload control against finite capacity, i.e. a workload ceiling limited by available capacity regarding machine time (manufacturing group) or man hours (work group). Note, however, that finite capacity is not supported in order-specific manufacturing.

There are functions for definition of resources in manufacturing with links to current work schedule and any exceptions. This can be for machines, workshop personnel, or other resources. There are also functions for the display of workload details and links between manufacturing orders (labor operations) and workload details.

When a manufacturing order is created, Jeeves attempts to find available time in the time pockets for the relevant manufacturing resources to cover the capacity requirement for each operation. For manufacturing orders planned by Ready date, if there is not sufficient time available up to and including today's date, you will receive a message informing that the completion date has been moved forward. If the completion date for the manufacturing order has been locked, the work overload will instead be “dumped” on today's date.

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11.6.2 Jeeves Graphical planning

You can work with workload control in Jeeves and in Jeeves Graphical planning.

11.6.3 Graphical planning system integration

Jeeves offers an integration with a fully functional graphical planning tool called EQ Plan that is developed and maintained by a chosen solutions provider. EQ Plan facilitates the management of manufacturing orders in Jeeves ERP and helps customers achieve transparency and flexibility by providing a way to interact with their manufacturing related data graphically. Note that this integration is currently only implemented with the production module.

As part of the integration, Jeeves allows you to define the different planning levels that can be set for a resource group. Based on the selected planning level, the system will load the relevant manufacturing related information from Jeeves ERP to EQ Plan (via the integration endpoint). It also lets you define resource groups and their properties for the graphical planning system. Once you have defined a resource group, you can link it to one or more manufacturing groups from the GPS resource group row embedded list. In addition, you can define settings (for e.g. prefixes, colors, fonts) and skills (for e.g. sewing) and link them to a manufacturing group. Skills can be used to assign operations to manufacturing groups. These are read by EQ Plan when loading data from Jeeves ERP. Further, admin users can trigger a locking mechanism to make sure that other users running EQ plan do not interfere with the ongoing planning.

11.7 Reporting

11.7.1 Reporting completed labor

There is a choice of reporting methods such as one single or separate programs for labor, material and received deliveries. There is also a choice between manual entry via forms or checklists, or by button, as well as automatic reporting (backflushing).

You can also set up reporting so that subsequent operations (including the material reservations linked to them) and the manufacturing order itself shall be re-planned in accordance with the reported approved quantity of the current operation.

11.7.2 Reporting material issues

You can manually report material issues:

per suborder or

per operation in the operation queue

Alternatively, material can be issued automatically (backflushing) at:

- Operation start. Material is issued automatically on startup of an operation. All material reservations belonging to the operation are picked from inventory.
- Operation reporting. Material is issued automatically according to reported labor based on the manufactured quantity of the operation. The material is picked from inventory based on planned manufacturing and on the BOM of the manufacturing order (i.e. not based on the BOM register).
- Closing of manufacturing orders

11.7.3 Scrap management

You can report scrap when receiving deliveries. The scrapped quantity and reason can be entered when reporting received deliveries of finished products. There are two, parameter-controlled alternatives; the remaining quantity is calculated either as only corrected quantity or both corrected and scrapped quantity.

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11.7.4 Delivery of manufactured items to inventory

You can manually report received deliveries from own production from the programs for reporting picked material. Alternatively, received deliveries can be reported automatically based on the approved manufactured quantity on the operation. You can set up tolerance percentages for deliveries to be considered fully delivered.

11.7.5 Integration with general ledger

Posting of material issues, labor and received deliveries from production is normally done automatically via a separate batch, for example, every day or week. One material journal, one labor reporting journal and one received delivery journal are created per reporting date. Each journal generates a voucher in the general ledger.

Posting is done according to posting rules per product account and/or flow group. The posting rules control the accounts and account segments (cost center, cost unit, etc.) on which inventory transactions, WIP, labor and production costs, etc. are to be posted.

11.7.6 Closing manufacturing orders

There is a function button for flagging manufacturing orders as completed/closed. When you do so, all material reservations and planned labor operations are checked to ensure they have been reported. The system also checks that the planned delivery has been received. It also checks against rules for cost differences.

This function changes the status of the selected manufacturing order and underlying suborders to Completed. When this has been done the manufacturing order is closed, which implies that final, actual costing can be done. The manufacturing order also disappears from the WIP list.

As long as the manufacturing order is not finally posted (Manufacturing order, profit/loss), you can open it again by going into a suborder and changing the status to Partially delivered.

11.7.7 Quick-reporting

There is a program that can be used for quick-reporting of completed manufacturing orders, in which each flow group can deal with their own orders. The function processes the manufacturing orders of the current flow group in the work table for orders to be closed.

For the order to be closed the following must have been reported:

- material issues
- labor
- received delivery from production

The following is also checked:

- there is no order specific inventory
- any manufacturing cost difference lies within the limits specified on the flow group

11.8 Attendance and Shopfloor control

Jeeves Attendance and Shop floor control are optional sub modules to the Production module that give extended, but at the same time, quick and simple-to-use functionality regarding attendance reporting and shop floor control.

The unique ID of job cards can be generated based on Jeeves Automatic sequence numbering functionality.

There is functionality to filter inactive flow groups, manufacturing groups and employees, as well as to see who is currently clocked in to work and, if applicable, what jobs they are currently working on.

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You also have the possibility to restrict the display of inactive manufacturing groups in Shopfloor/Mfg operation queue programs through the Active setting in the flow groups and/or Manufacturing groups programs

Reporting is made easy by lists that are updated in real-time with logical color-coding, and buttons for quick-reporting.

11.8.1 Attendance

The attendance module is a clock-in - clock-out system that results in calculated time per salary type. The system allows you to see who is currently clocked in to work and, if applicable, what jobs they are currently working on. The calculation can be transferred to payroll systems.

Reporting can be done by entering the employee ID number manually, or by means of barcodes or magnetic cards. The latter approach provides improved security. The employees can view their most recent clock-ins and clock-outs directly on the screen by striking a function key. This makes it easy to check that the actual hours worked are correct.

Reason codes are managed automatically by means of clock-in and clock-out rules on the schedule. If the employee clocks in and out within scheduled time, only reasons for absence are displayed, while outside the schedule only reasons for overtime are displayed.

The system allows the employees to switch schedules without having to involve the payroll administrator. This can be useful, for instance in cases where it is necessary to change schedule on days when the payroll administrator is out of the office.

11.8.2 Flexi-time and work metrics

The system has support for flexi-time, work metrics and work time reduction, including the possibility to handle work time reductions even on non-working days.

11.8.3 Integration with payroll systems

Time is calculated for each employee per selected period; this information can be transferred to payroll systems (in Sweden, for example: ADGA, Kontek, Hogia, Visma and SPCS).

11.8.4 Shop floor control

Jeeves Shop floor control is designed for manufacturing companies with intensive production reporting. The sub module consists of a real-time operation queue, where the operator can see what is to be done, in which order, and any shortages in materials. It provides an overview and a basis for decision-making for the manufacturing groups and flow groups.

Additional added value is derived from improved item balance reliability, costing and follow-up of scrapping.

The Shop floor control sub module is integrated with the Attendance sub module, which means that the production reports consider the work schedule for the employee and the employees' clocking in and out.

11.8.5 Operation card

The operation card contains detailed information about the operation, such as planned, reported and remaining quantity. You can see the previous and the next operation, and if there are drawings, the drawings can be displayed from the operation card.

The operation card also contains information about different tools that you can use. Reporting (start, sub-reporting, completion, interruption, etc.) is done from the operation card. There are shortcut commands for all buttons to finish setup, start production etc. The buttons can be enabled depending on the operation’s status.

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11.8.6 Time dividing (machine, employee)

Machine time: If there is more than one job on a manufacturing group at the same time you can choose to divide time for every job.

- Do not divide machine time
- Divide time evenly
- Divide time according to planned time

Man time: If there is more than one job on the employee at the same time you can choose to divide time for every job.

- Do not divide labor time
- Divide time evenly
- Divide time based on planned personnel/operation on operation
- Divide time according to planned time

11.8.7 Team reporting

This is used to reduce the number of reports prepared in the situation when a work team collectively loads a job.

It is only possible to be a member of one team at time.

11.8.8 Indirect job

Indirect job is used in situations where the time cannot be loaded on specific operations, example cleaning, union work and the like.

11.8.9 Assembly list

It is possible to group your operations and start them simultaneously. The reported time will be divided among the different operations.

11.8.10 Jeeves Mobile shopfloor routine

A standard mobile routine for individuals in manufacturing to report operations.

Jeeves Mobile shopfloor routine provides a productive and efficient way of reporting time and deliveries along with the statuses of operations in the manufacturing process.

11.8.11 Filter inactive employees

Shopfloor program

The 'or choose from list' drop-down field in the Start labor dialog is shown when you open a record in the Operation queue and click Start Prod . It is validated on whether the Cancel and Not active checkboxes are checked as well as on valid Active from and Active to dates in the Personnel and Shop floor personnel programs. Following the validation, only active personnel are displayed in the drop-down list.

Attendance report program:

Validation is done to check if the Cancel and Not active checkboxes are checked as well as on valid Active from and Active to dates in the Personnel and Shop floor personnel programs when;

- displaying the Employee is not active message when searching for employees in the Attendance program
- displaying the Employee is not active error message,

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11.8.12 Validation rules

Jeeves provides for maintenance of the rules in Shop floor control.

You can add records on different levels; company, flow group, manufacturing group and employee. Observe that not all keys are applicable on all levels. Company is the highest level, and there must be a record for each company. When a key value is requested, Jeeves first searches at employee, then manufacturing group, flow group and finally at company level.

If no value has been entered the key is not applicable and the system goes up one level to retrieve the key.

You can specify starting an operation only if the immediately previous operation or any previous operations are NOT finished, if there is a rule that does not allow you to start an operation before completion of the previous operation. In addition, you can also choose to divide labor time if a manufacturing group has more than one active job at the same time.

To improve the understanding of the value option of each setting when setting up the shop floor controls, fields used to control many settings are displayed as drop-down lists with the description of each value.

11.9 Internal claims

Jeeves contains a function for entering and logging internal claims (manufacturing claims), i.e. remarks and complaints about items and manufacturing orders, and for recording how they are to be settled as well as actions to be taken.

You can define different work flows for different types of claims, specify base data for scheduling, and control how activities should be reported.

11.10 Follow up and queries

11.10.1 Production information

There are programs for displaying/printout of completed material issues, reported labor and received deliveries and related transactions and postings. These display accumulated time, setup time, as well as manufactured and scrapped quantity per flow and production group.

Financial follow-up

All reporting on manufacturing orders has financial consequences. The costs are of different types:

- Material costs (all material included)
- Labor costs:
  - Wage costs (including social contributions)
  - Machine costs
  - Tool costs
- Cost for any subcontracting

As soon as anything is reported on the manufacturing order, it becomes part of Work in Progress (WIP). Any partially received deliveries of finished goods reduce the value for the order in the WIP list. When you close a manufacturing order with the Flag order completed button, the order disappears from the WIP list. You print WIP lists using, for example, the report programs:

- WIP list, manufacturing order
- WIP list, suborder details

You can print actual costs using, for example, the report programs:

- Actual cost, summary
- Actual cost, transactions

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The report shows the value of all transactions on a manufacturing order. The actual cost is listed beside the estimated costs for comparison. You can print the report to as a preliminary cost at any time.

You create the profit/loss journal for completed manufacturing orders, as a part of the final costing and financial follow-up. Postings are created for all selected, completed manufacturing orders (suborders) that have not previously been posted on any journal. One journal is created per order completion date.

11.11 Reconciliations

During period reconciliations you can use Jeeves' function for reconciliation of work in progress.

11.12 Industry-specifics

Most functions in Jeeves Production are common to all industries. There is one, however, that is specific to certain industries, as described below.

11.12.1 Product variant matrix

The product variant matrix mentioned earlier was developed with the fashion industry in mind, but can be used for any products with variants, i.e. combinations of models, colors, sizes etc., or the like.

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12 Jeeves ERP Product configurator

Jeeves Product configurator is a proprietary module in Jeeves ERP, which means that configuration rules for products are established when products are defined in Jeeves ERP and not managed as a separate system or integrated application.

The product configurator is used to create an order specific configuration of an item and is meant for companies that want to configure individual products/machines. A product configurator model is created to represent a generic product structure. A model can have a set of standard material and additional material that is model specific. Once the model is set up, items with unique bills of material and unique routings can be created from a master bill of material.

Instead of entering material and operations manually for the configured item, the model sets up the item for you based on the answers (regarding features, options, etc.) that you have entered for the configuration. For example, when configuring lighting, a certain socket would be picked based on the type of light bulb. A base item could be a set of screws that is always part of the product, or an assembly instruction, while the rest of the configuration is selected.

The definitions created from your selections are used in order entry and quotations and make the process of responding to a customer request for a quotation and order confirmation easier and faster.

When configuring, you can work with Jeeves Product configurator in an interactive question and answer dialog. All input is validated against predefined rules for each product ensuring that the resulting configuration is valid. Cost prices are calculated dynamically and lead times are displayed. By classifying questions, work flow can be applied to have multiple departments respond or validate customers' specifications and enter comments to each question.

With Jeeves Product configurator, you can manage configuration scenarios in any of the following sourcing strategies:

- Make-to-stock
- Assemble-to-order
- Engineer-to-order
- Pick-to-order
- Buy-to-order

The product configurator can be used with most applications and routines, for example the following:

- Sales order
- Quotation
- Quotation Estimator
- Production order
- Consist of (explosion)
- ICT flow
- Price simulation
- Cost price simulation
- Create item numbers with standard configuration

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12.1 Configuration feature types

The following feature types are available for product configuration:

- Option: A dynamic configuration feature; it can be a question about measurements or color, a question to which the answer is Yes or No or it could be other types of options.
- Material: Configuration feature connected to an item in the Items table. This can be any type of item, such as a package item, a phantom item, or an individual component of material.
- Numeric: Configuration feature for answering with any numeric value, with the only limitations being the number of characters and decimals allowed in the field, for example 110 cm.
- Discrete numeric: Configuration feature for generating pre-defined numeric values, where you enter a range of values and a factor, for example from 100 cm to 300 cm and a factor of 10. This would generate possible values 100 cm, 110 cm, 120 cm etc. up to 300 cm.
- Text: Configuration feature for entering or generating text, for example to enter a customer or purchase order specific item description or any other text to be included in the model. For example, you can set up a condition where if the user does not answer a question about an item's description, its standard description is retrieved from the Items table. However, if an answer is given, the user is prompted to enter an order specific item description.

12.2 Configuration answer types

User inputs is used to define the answer type. Currently, only three values are available

- User Input: The feature is generated according to the answer from the user
- Formula: The feature is generated according to a pre-defined formula
- Default: The feature is generated according to a pre-defined default value that can be overridden by the user if required.

The specified values are displayed as drop-down list options when you have answered a question to identify if a feature has been set up using a formula, user input, or a default value.

12.3 Configuration formula types

You can use formulas to define a feature by returning values that determine how a feature should be answered. A formula must be defined as one of the following types: formula returning numeric, alphanumeric (string), or an item number or formula calling a procedure.

Assigning these different formula types to formulas will help you when using and referring to formulas.

12.4 Configuration status

The applicable statuses of configurations in the product configurator are as follows:

- 10, Under entry - Configuration status when you open the configuration in the product configurator for the very first time.
- 20, Entered/changed - Configuration status when reconfiguration is done for even a single feature.
- 30, Invalid configuration - Configuration status when you press Done but the configuration is not valid or an error occurs when creating BOM and Labor.
- 40, Activated - Configuration status when you press Done and the configuration is valid.
- 90, Canceled - Configuration is canceled; this status sets manually.

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12.5 Configuration model items

You must set up an item as a model item. Unless this is done, key configuration functionality will not be available for the item as various programs and functions in the product configurator validate that the model item is set up as a configurable item.

12.6 Configuration feature groups

You can logically group features by feature groups. This is useful when you want to group related configuration questions; for example, when configuring the features for manufacturing a vehicle you might group questions relating to exterior, interior, motor, accessories, etc. Or when configuring the features of a bicycle, you may choose to group questions by main components, optional components, alternative components, and accessories. Each feature group is displayed as a tab in the Configurator, with its features on the rows.

In the product configurator, some feature groups may be invisible to the user depending on the condition set in the formula. The feature group is set to be visible or not only when the condition is met; i.e. if the value returned is True, the feature group is hidden, else if the value returned is False, the feature group is visible.

An example of when a feature group may be hidden could be when two people are responsible for the configuration of the product and the first step is performed by person 1 and additional steps by person 2. Then the second person who opens the same configuration can view additional information that person 1 does not have permission to view.

12.7 Configuration features

You can define all features that the order entry personnel must specify to transform requirements into an order specific product/resulting bill of material. Features characterize a product and each feature defines a question that is prompted during configuration or closed as a formula or matrix result.

You can enter a unique user-friendly name (alphanumeric) for referring to the feature. This will make it simpler to refer to the feature when you create a formula or condition involving the feature. If quick reference is not defined or found, the feature number is used as normal.

You can create features and link them to different feature types, user input options, and feature groups. Features can be linked to options or materials or used for user input of text, values, etc. Features can also be linked to several models (re-used), which means that a change in a feature will affect all the models to which it is linked.

You can define various combinations of configuration features, for example:

- Feature type Text and Answer type User input linked to Feature group
- Feature type Text and Answer type Formula linked to Feature group
- Feature type Option and Answer type User input linked to Feature group
- Feature type Option and Answer type Formula linked to Feature group
- Feature type Numeric and Answer type Formula linked to Feature group
- Feature type Material and Answer type Default linked to Feature group

12.7.1 Value limits on numeric features

When setting up a numeric feature, you can assign minimum, maximum, and factor values to the feature depending on the selections made for other features.

You can do this by specifying values, or by creating formulas to define these values for a configuration feature or configuration model feature.

These values are automatically updated when you use the tab key to exit an input cell while configuring an item in the Product configurator. If the value is not within the defined range, an appropriate error message is displayed.

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12.8 Configuration model features

You can define Configuration model features with a set of standard options/materials in order to reuse questions and additional options/material that are model specific.

You can select a feature to add to the list of model features linked to the model and specify if item significance key (ISK) values (explained in detail later in this chapter) are defined for the configuration model features linked to the model and if this model is the main model or a sub model. You can also add options/materials specific to a model feature. When configuring a model, a configuration option/material linked to a configuration model feature has a higher priority than one that is not linked to the configuration model feature.

12.8.1 Configuration model features matrix

You can define applicable options and/or material for model features using the model features matrix. All options and material connected to the feature (as rows) and all models connected to the feature (as columns) are displayed in the form of a matrix.

This is particularly useful when all options, for example colors, might not be applicable to all models. Likewise, all material might not be applicable to a certain model.

Once you mark the valid options and material and save your choices, the results are saved in separate tables for options and for material.

When configuring an item, only the valid options and material are presented if data exists in these tables for the model and feature.

12.8.2 Configuration model feature groups

You can manage the Configuration feature group that a feature belongs to from the specified model. If it does not exist in the Configuration feature groups table, it is added.

12.9 Configuration options

Options for a product or feature must be defined separately and entered for each feature or product/feature or customer/product/feature combination. Valid options can be alphanumeric or discrete numeric.

12.10 Configuration material

You can select the quantity and/or material for a line in a BOM or routing using selection matrices. A selection matrix consists of a multidimensional selection area that handles up to nine dimensions. This can contain values for features and item numbers. A selection line can also be date controlled to manage selections that change over time.

When doing the configuration, after you have answered all the questions, the model generates a unique bill of material and unique routing.

- The bill of material consists of the items in the BOM structure and the quantities.
- For the routing, the model generates the operations that should be included.

For the unique BOM and routing that have been generated, it is possible to calculate the cost of the configured item.

12.11 Configuration formulas

A formula can be any SQL select statement or function to select or process the information in the database. For example, you can concatenate a string of item numbers, or retrieve values such as costs from a selection of items and total them, or control settings of application parameters. It is used to create a query that returns a value that can be used to answer a feature when configuring a model. A formula can be a standard formula that has been created manually or can be generated from a formula matrix.

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Formulas use standard T-SQL but with a simplified user input interface. Once you have entered a formula, you can validate the SQL syntax of the code.

The following formula factor types are available in the product configurator:

- Internal result, calculated in the formula and reused
- Constant
- Option from a feature or value from a feature
- Retrieved value/basic data from the BOM, routing, or item master
- Stored procedure

Information about the resulting feature, answer and formula is stored for analysis purposes.

12.11.1 Formula matrix

The formula matrix in the Configuration formulas program guides and assists you in creating and maintaining formulas. You can use the formula matrix to define formulas, even if you have no knowledge of how to write code. You can also configure advanced formula matrices to generate code for setting up advanced conditions.

The configuration formula matrix provides valid combinations of features/dimensions that can be activated by setting the value of a cell to an arbitrary value or by specifying that the cell value should be parsed to another formula for further validation. The output is determined by a formula type that specifies if the output is numeric or alphanumeric.

The following function types can be associated with a specific cell in the formula matrix:

- Formula: Associates a value returned by a formula with the specified cell.
- Feature: Associates the value from a feature with the specified cell.
- Item: Associates an item with the specified cell.
- Constant: Associates a constant value with the specified cell. If the formula is of type Numeric, the value in this cell should also be numeric

The formula matrix guides the user to create formulas and formula output value for any given combination of feature answers and/or conditions or even with only conditions defined on columns and rows. It provides a visual overview of the results based on formulas. The output can be a constant, another formula, feature, configurator parameter, or even another formula matrix. This output can be used to update BOMs, operations, item rows, as input to another formula to answer a feature or change feature values depending on where you connect the formula. When defining a formula, you can also assign a default value that should be used when no other value is returned from the formula.

This functionality opens huge possibilities to create and administer formulas. You can use it to assign different formulas that can return answers or embed logic in the formulas to consider different variables like customer category, region, etc. in combination with other variables like trading area or materials.

12.12 Conditions

You can set conditions on a feature or model feature by entering a valid SQL statement. A condition set on a model feature overrides a condition set on a feature when configuring an order specific item.

You can also set a condition on an option or on material, if you want to choose a specific option or specific material although no condition is set on the feature, which means that the feature is always enabled.

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12.13 Comment code in formulas, conditions, etc.

You can comment code in fields that contain formulas, conditions, drop-down list, and code that can be parsed as follows:

- Comment row
  - The -- mark indicates the start position for the comment on the row.
- Comment text block
  - The /* mark indicates the start position of the comment.
  - The */ mark indicates the end position of the comment.

Note that the commented rows/text blocks will not be considered when validating formulas.

12.14 Configuration models

You can view the configuration model features and feature groups associated with a model (item) and mark the model as valid. Jeeves ERP does a number of checks before marking the model as valid.

You can also copy features and associated feature groups linked to a configuration model to another configuration model.

A model can also be marked for quick configuration which means that an order specific item can be automatically configured based on the significance key values.

You can create a hierarchy of components within a model item (model in model) by linking sub model items to a main model item. The hierarchy creates a link between the main model item, such as a car and its accessories such as radio, speaker, and GPS. In this case, a model item becomes a sub model item when you add it to a top model item in the product configurator. Similarly, you can also link sub model items with other sub model items such as a speaker within a radio.

12.15 Configuration feature default

You can define configuration feature default values and specify if default values should be retrieved and applied in the Product configurator/Product designer (in e-Sales) as follows:

- 0 = at closing when the configuration is being validated
- 1 = at opening when starting a new configuration.

If no value is entered (NULL), the default value 0 will be used.

12.16 Significance key values

You can use the Configuration item significance key to identify features whose answers can be reused in multiple configurations.

Example: When configuring, you may want to search among earlier configurations based on a set of base features, select one configuration and copy its answers to these base features. Then you can continue configuring the remaining features. For example, search among earlier configurations of children's bikes and copy one with the relevant frame and wheel size.

For a stocked (ISK_SKU) item that is defined with an item significance key (ISK), you can define that the latest information must be updated in the BOM/labor records when creating a manufacturing order. If you enable this function, the BOM, package items (sr) and operations (op) will be updated/recreated by using the CnfgrModelISK information and connected configuration, when the MO is created.

As an item significance key (ISK) comprises a set of answers to the features linked to a model, it can be a relatively long alphanumeric value that may be difficult to remember. To make it easier to remember, you can create a user-friendly name for the ISK. A user-friendly name is a short alphanumeric value that can be used instead of the corresponding significance key when placing an order.

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12.17 SQL code syntax

A validation button is provided in several programs to allow you to check the validity of the SQL code syntax entered by you.

12.18 Configuration model standard answer

You can set standard answers for a feature that are specific to an item (model) and customer. This will ensure that the model is configured as per the customer's requirements or as per the agreement with the customer.

You can define answers per customer, model, and feature. If you then choose to use the standard answers during product configuration, the answers provided are displayed as the default answers for the feature(s) when configuring an order specific item for a specific customer.

12.19 Product configuration, test

You can view product configuration records for testing and trouble-shooting purposes.

When setting the configuration of a product, the starting point should be to answer questions, either by:

- Picking an item from a list: If an item is picked as an answer to a question, it will be included in the BOM. The same could also apply to operations.
- Answering Yes or No: If a question is of the type Yes or No, the different alternatives will determine different things in the model. For example, the answer could determine whether a specific item or operation should be included in the model.
- Answering by picking an option in a list. If the answer is an option, it could determine different things in the same way as for the Yes/No question. An option could, for example, determine a specific optional item in the BOM or optional operations in the routing.

12.20 Product configurator screen

You can create/update an order specific item based on the selected configuration model item in the Product configurator screen.

The features connected to the model are displayed in an embedded tab grid, with one or more tabs and a grid with several rows. Each tab represents a feature group and the rows contain features linked to the model and grouped within the feature groups.

Data is read from XML and loaded and presented in a tabular format within columns and cells.

When setting the configuration of a configuration model, the starting point should be to answer questions, either by:

- Picking an item of material from a list: If an item is picked as an answer to a question, it will be included in the BOM. The same could also apply to operations. Note that if a feature is answered by a formula, the answer is automatically updated on the order row. You can also automatically view an image of the item when the tab key is used to exit an input cell. To enable this functionality, associate the image with the configuration option or item and then specify the display parameters, path to the image file, and allow automatic display of the image during configuration.
- Answering Yes or No: If a question is of the type Yes or No, the different alternatives will determine different things in the model. For example, the answer could determine whether a specific item or operation should be included in the model.
- Answering by picking an option in a list. If the answer is an option, it could determine different things in the same way as for the Yes/No question. An option could, for example, determine a specific optional item in the BOM or optional operations in the routing. Alternatively, it could determine the color, quality or execution of an item of material. It could specify a numeric value from a list of predefined values.

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12.20.1 Copy matching configuration to current configuration

You can search for configurations that match the current configuration and update the answers for the current configuration from an existing configuration.

Search configuration - lists the configurations in the current company that have been made for a model that match the answers provided in case of a non-ISK model or match the ISK text value in case of an ISK model.

Use configuration - copies the selected record to the current configuration. Answers are retrieved as defined in the item significance fields for an ISK model or from the answers provided for a non-ISK model.

12.20.2 Create stocked item via Product configurator

You can create a stocked item with a new Configuration unique ID in the delivering unit in an ICT environment.

The new configuration unique ID is connected to the model in the Configuration user friendly name program. The BOM, package items and operations will be created for the new stocked item by using the information in the Configuration user friendly name program and connected configuration. This function works in an ICT/non-ICT environment.

12.20.3 Auto close

If you have checked the Auto close option for any feature linked to the model, based on the answer selected for the feature which has this option checked, answers to the other features linked to the same model are displayed when you exit from the cell. Note that when only one value is valid, that value is used as the answer on the feature. The answers are displayed according to the following priority:

- Default/AutoClose
- Formula
- User input

12.20.4 Validate configuration

Once the configuration is ready for validation, you can validate if the configuration is complete, for example that it does not contain missing or conflicting answers.

If a complete and valid model with connections to related features exists, configuration unique data is created in the Configuration unique BOM and Configuration unique labor work tables. In addition, BOM and routing as defined by the model are created.

12.20.5 Work tables

You can view results in work tables before creating BOM and operation lists. These tables are updated with unique data for the complete product configuration structure/operations.

After the configuration unique data for the complete BOM/operations are created, the selected values and columns are linked to the relevant tables for BOM per item, operations, and order rows to create order unique data and implement the BOM.

You can view and change information relating to selected values/columns in the product configuration bill of material and to select values/columns in the configuration unique operations. You can also view and change information relating to models.

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12.21 Configure package details

You can configure package details as well as header information using the Jeeves ERP Product configurator. This provides more flexibility when configuring a product and can perform the following:

- supports configurable package headers
- supports configurable package details
- does not activate the main model until the sub models are configured
- when configuring a package where the main model and sub model contain the same features, the sub model inherits he feature value from the main model
- changing an upper level feature value is reflected in the lower level value but changing the value at the lower level value does not affect the value at the upper level
- updating a value at the lower level changes the status on the upper level
- if a formula is used for a sub model and the feature referred to is not available, the value is retrieved from the upper level configuration
- several levels are allowed below the main model

12.22 Replace item on customer order row

You can replace a configurable item on an order row with another configurable item or a nonconfigurable item by connecting a feature or a formula to the Item No. column in the Configuration order rows, items node that is accessed from the Configuration models program.

For an item that is marked as configurable, from the order row you can click on the Configurator button and configure an item based on the selected product configuration model by answering questions that define your requirements for the finished product.

After configuration, when you click the Done button,

- If the feature linked to the Item No. column is answered by selecting a configurable item, this item replaces the model item in the configurator and the item on the order row.
- If the feature linked to the Item No. column is answered by selecting a non-configurable item, this item replaces the item in the order row. However the model item in the configurator is retained as in the source item.
- If the feature linked to the Item No. column is answered by selecting the same item as in the configuration header, a loop error is displayed as an existing item cannot be replaced with itself.
- If the configurable item is replaced with another configurable item, the system will create order specific information using the replacement item.
- If the configurable item is replaced with a standard item and the formula called in the Use model definition field returns a value of “1”, the system will create order specific information using the original configurable item (i.e. the model BOM and labor definition will be used).
- Additionally, in an ICT environment, a configuration record with the same configuration unique ID as in the selling unit is created in the delivering unit.

12.23 ICT business unit

Any configurable item can be used as a configuration model in the product configurator. In an ICT environment, the configuration model is defined in both the delivering and selling units while the configuration is done in the selling unit. You can create a stocked item with a new Configuration unique ID in the delivering unit in an ICT environment. The configuration unique ID is connected to the model in the Configuration user friendly name program. The BOM, package items and operations records will be created for the new stocked item in the ICT/non-ICT environment as required.

In an ICT environment, you can specify if order details such as calculated cost price or manufacturing lead time should be retrieved from the delivering unit when you create a customer

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sales order in the selling unit and configure the item. The Use factory data function allows you to retrieve factory data regarding the product that is currently under configuration to update the order details on the order rows. So, when you configure the order specific item, the value from the specified field in the Delivering unit will be updated in the corresponding field on the order row in the Selling unit. You can view this information and make a decision on whether or not to proceed with creating the purchase/manufacturing order for the item.

12.24 Product designer in e-Sales

Support for designing items to order using the features and functions of product configuration is available in e-Sales. For more information on the Product designer, please refer to the chapter on *Jeeves e-Applications*.

12.24.1 Interaction artefacts

12.24.1.1 Configuration feature web display type

You can define how configuration feature selections should be displayed in the Product designer in e-Sales. The following display types are supported:

- Slider: For discrete numerical features, you can choose to represent feature value selections with a slider. The slider can be adjusted based on the minimum and maximum values, factor, and the number of decimal places the feature may have.
- Radio buttons: For options and materials, in addition to the drop-down list, you can choose to have each option or material represented in a radio button list from which the user can select their preference. If none of the options or materials are valid, the user can select None to indicate no selection. Radio buttons for options/materials that are not available or invalid are grayed out.
- Card grid: Alternatively, you can choose to have each option or material presented as cards within a grid. Each card will have an image and description of the option or material. When a card is selected, the card frame will be highlighted and a tick mark will appear on the top right of the card. You can have from 1 to a maximum of 12 columns in the configuration feature card grid. Options or materials that cannot be selected with the current configuration will be hidden from the grid.

You can also specify different ways features are presented in e-Sales. Apart from the web display type, you can also specify the preferred grid size and the web image URL as required.

12.24.1.2 Configuration information type

You can define different information types to indicate the status of the current configuration in e-Sales. Currently, information type is used to group information for presentation in the Product designer in e-Sales. This includes displaying actions that are required to be taken or other information relating to the configuration, such as if the configuration has an error or conflict.

The status of a feature is indicated by an icon displayed to the right of the answer. If the icon is absent, it means that the feature is not mandatory or that it contains a default value.

12.25 Tab grid control and macro objects

The tab grid control, which can be described as an embedded list with tabs, is available for the product configurator It is used to present data in the form of a tab grid when configuring an order specific item. Macro objects and related methods allow you to build up the tab grid control.

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12.26 Pricing models

Jeeves ERP has a rule based function to calculate the price of an existing or a non-existing item. A non-existing item can for example be a configurable item that is defined when placing a customer order or a quotation.

You can create any number of pricing models, built up of rule types and rules, to calculate or simulate prices when configuring items on customer orders. You must connect the pricing model to a configuration model before configuring the item and validating the price definition and simulating the price list calculation.

Calculation is based on a sequence of calculations with parameters and values that are defined and exist at the time when the price needs to be accessed.

For non-configurable items and when a pricing model is connected to calculation of item list price in the Items program, the rule type Feature is not available since it is not related and connected to a configuration. Also, formulas referring to Feature are not valid.

However, all rules types are considered when you open the Pricing model (prmh) program directly from the menu and configure the model.

Note that the first version of the pricing model functionality has only been tested together with the configurator functionality.

12.26.1 Pricing model rule types

You can use any of the following for building pricing models:

- 10, Base price - Retrieve calculated cost price from item or model
- 20, Markup - If the Related sequence exists, the calculation of the markup value will be performed by using the value on the referenced row, Operand, and Value. If the Related sequence does not exist, the value returned from the Configuration formula or Configuration feature will be used instead.
- 30, Feature - Retrieve value from the feature (FeatureID).
- 40, Formula - Retrieve the value from the Configuration formula by executing the formula.
- 50, Subtotal - Summarize all values up to the nearest subtotal rule type row for pricing models where the Sequence No. is less than the current subtotal row. Summarize all values from all rule type rows between the previous subtotal rule type row and the current row, excluding any Accumulated rule type rows.
- 60, Accumulated - Summarize the values from all of the previous rule type rows except the subtotal rule type rows.
- 70, Discount - If a reference to the Related sequence exists, the calculation of the discount value will be performed by using the value on the referenced row, Operand, and Value. If a reference to the Related sequence does not exists, the value returned from the Configuration formula or Configuration feature will be used instead.
- 80, All material - Summarize the price for all material at the top level in Order specific material/order row. Use the normal price calculation functionality.
- 90, Total price - Summarize the value from all previous rule type rows to the current row excluding the values in the Subtotal and Accumulated rows. Note that records after this record will not be considered as Active and hence, will not be calculated.

12.26.2 Quick costing

The product configurator offers a quick way to calculate the cost for a configured, order specific manufacturing order when running the configurator, including when using ICT. No order specific order rows/purchase order rows, manufacturing order rows/operations are created; this is just simulated in order to quickly calculate the cost. The cost is given when you conclude the configuration by pressing Done, if you have connected a formula for the quick costing to the model.

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12.26.3 Simulate price list calculation

You can validate the pricing definition and calculate the price of the item based on the currently defined pricing model.

To simulate or test a price calculation for a Configuration item, an Activated configuration must exist and be connected to the current pricing model. By using the configuration and the defined pricing model, you can perform a simulation. You can also view the results of the pricing simulation.

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13 Jeeves ERP Service

Jeeves Service includes routines for entering service agreements, entering and maintaining items (material) and service items, service order processing, interruption management and reporting with subsequent invoicing. Invoicing is performed either via the Time, projects, assignments module or via the Sales module's order routine.

There are several different methods for entering a service order and its associated rows. This is either done manually row by row or you use the service flows, for example, to automatically generate predefined rows.

Prices can be managed both from the order routine and the assignment routine with price lists, sector types and time types, and via the service routine where prices and discounts can be set for different service items and types.

You can plan and report service activities. You can also handle items, which in Jeeves can be normal items, serial number items, where required, or service items. These alternatives allow different service models with different characteristics.

If for any reason the service work has to be interrupted, interruption management can be used to keep track of where the object is during the interruption and the interruption's status.

13.1 Items

13.1.1 Serial number items

A serial number item is identified by a combination of item number and serial number. These combinations are stored in a dedicated table.

You use the function if the company carries out service activities on stocked serial number items as part of the service routine. The purpose is to quickly identify where an individual item is, both physically and in the service flow, to which customer it has been sold, whether it is still under guarantee, etc. In this case, serial number items are usually entered in the Serial number items table when receiving deliveries against purchase orders.

If the company carries out service on non-stocked items, you define them in the Service items program instead, and specify serial numbers directly on service orders.

13.1.2 Service items

Service items are primarily intended to be used when the company carries out service activities on non-stocked items such as, for example, reparation of mobile phones, cars and the like. The service item is linked to the item table, which means that it must be entered in the item table first.

13.2 Grouping/classification

13.2.1 Service orders

You can have any number of management areas (service manager/office/department or equivalent), service report types, service codes (in 3 layers) and priority codes for own classification and searching.

13.2.2 Error, interruption and action codes

You can have any number of own error, interruption and action classifications and search criteria. You can also use coding according to IRIS.

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13.3 Price and discount management

Prices can be managed both from the order routine and the assignment routine with price lists, sector types and time types, and via the service routine where prices and discounts can be set for different service items and reparation types.

13.3.1 Price lists

You can set up price lists for material that has been entered in the item table or services (labor) per personnel category, time type and, if so desired, service technician. You can also set a default price list on customers to be proposed when creating new assignments.

13.3.2 Price per service item

The pricing of service labor can be controlled by service items:

- You can link different types of reparation to each service item and specify a price for every reparation type. At service order row entry, you enter a reparation type and the price is then retrieved from the service item.
- You can also link several service items to a service agreement and there specify a discount rate per reparation type.

13.3.3 Discount agreements

If you have entered service items, you can also enter a discount agreement for one or more service items and link one or more customers to the agreement. Discount agreements are built up per service item and reparation type. The discount rate is retrieved to service orders for customers who are linked to the agreement depending on the specified repair type.

13.3.4 How the price is retrieved on the service order row

How the price is retrieved to the service order row depends on the type of row being entered, as described below. If no price is found, or if the suggested price is not applicable, you can enter a price manually. The price retrieved depends on what you specify in the following fields:

If the row refers to an item that is not a service item:

- Price list
- Quotation/Agreement, if a price list has been specified on the service agreement
- Company number, if a price list has been specified for the customer

You must enter a quantity.

If no price is found as above, the price is retrieved from the item's list price.

If the row refers to a service item:

- Reparation type, if a price has been entered per service item and reparation type
- Quotation/Agreement, if the service item is included in a discount agreement under this agreement
- Company number, if the customer is linked to the agreement (i.e. is included in the right-hand embedded window in the agreement)

You must enter a quantity.

If no price is found according to the above, the price is retrieved from the service item's list price in the item table.

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If the row refers to a service and invoicing is done via the assignment routine:

- Technician reporting the work, if pricing on the assignment is done per person
- Sector, if pricing on the assignment is done per sector
- Personnel category, if pricing on the assignment is done per category
- Time type (price lists for services are entered for combinations of time types and persons or sectors)

You must enter the number of hours.

13.4 Service order processing

13.4.1 Parameter-controlled work flow

The work flow for service order processing is parameter-controlled, for example:

- Service orders can be automatically closed when reporting the last activity
- Partial invoicing can be allowed or when closed

13.4.2 Priority for retrieval of inventory location on service order rows

The following priority has been set to retrieve the inventory location on the service order programs and display it on the service order rows:

1) Inventory location from the Service technicians.

2) Inventory location from the Service Managers connected to the service technician.

3) Inventory location from the Service Managers at service order.

4) Inventory location from System data, own company.

13.4.3 Service order entry

There are several ways in which you can enter service orders; with or without pre-defined templates, so-called service flows, with or without a bill of material as a master, or manually or automatically from serial number items.

You can have any number of Service order types for defining number series. You can also have different order types but with the same number series, in case you need to change the service order type after entry.

13.4.4 Master service order

You can create related service orders from a master service order in order to handle them as a batch. For example, you can use the Master service order number to close a whole batch of related service orders.
When a service order is connected to a master service order, the value in the Master order No. field is transferred to the corresponding field in the service order records header and row. Also, when you enter a service order in the embedded list, select fields inherit values from the service order header.

13.4.5 Service order copy

You can create a service order from a selected service order or you can create a corresponding credit note if you want to reverse quantity, hours, or time accrued on a service order.

When a new service order is created, it has the status Entered in the Service order rows. The Order number, Invoice number, and Preliminary invoice number are not copied into the new record. If you create the service order from the Estimator, the status is set to Quotation. You can also manually set the status to Quotation while entering the service order.

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Once you confirm the creation of a service order/credit service order, Jeeves ERP runs a credit control check of the aggregated sales order and service order value. If the combined value exceeds the credit limit set for the customer/payer, appropriate messages will be displayed showing the amount by which the credit limit is exceeded in the current order and in total.

Note that Service order rows that have a status of Canceled are not copied to the new record. If all rows are canceled, only the Service order header is copied. However, any Service order rows with a different status are copied when the Service order has a status of canceled on the header.

13.4.6 Generate planned activities from template on service order

You can automatically generate planned activities on projects from service orders (incident reports, support requests, development suggestions etc.). This requires that you enter a project ID on the service orders in question. The activities are generated using pre-defined activity templates. The Activity templates node is available from the service order program. The automatically generated planned activities are assigned the connected service order number.

You can accumulate planned and reported time on service activities and display the information on service order headers.

13.4.7 Create manufacturing orders from service orders

You can create requisition(s), purchase order(s), or manufacturing orders from the current service order.

The rows on the service order where the Order checkbox is checked will result in a requisition or a purchase order. Alternatively, a manufacturing order is created for items that have acquisition method Manufacture to requirement.

The service order row that has created a purchase order/manufacturing order is flagged with the purchase order/manufacturing order number, so you can jump directly to the purchase order/manufacturing order by right-clicking the field that contains the purchase order/manufacturing order number.

You can also create requisitions for manufacturing orders from service orders.

If the service order contains material that needs to be ordered from a supplier, you can create a purchase order from the service order by pressing a button. The requirements for this to work are 1) that the service order has status: Planned or Ongoing and 2) that the rows have the field: Ordered checked and a supplier specified.

13.4.8 Planning service order rows via iterations

You can define iterations to make it quicker and easier to schedule activities. Only active iterations are displayed in the list on planned activities and service order rows. If you select an iteration, the start and end date on the activity or service order row will be updated based on the start/end date of the iteration, but can be changed. The changed dates must, however, lie within the iteration's date interval, otherwise you will have to remove the selected iteration.

13.4.9 Serial number items

When entering service orders you can specify an item's serial number. The sale and delivery of the item, as well as any other activity, are possible to trace. For example, if service or reparation is to be performed on a serial number item in the service module, the service planner automatically gets information about the customer to whom the item was sold, as well as connections to the sales order and inventory transaction.

If only one serial number item is to be handled on the service order you enter the serial number on the service order header. If it is to handle multiple serial number items there is a special table, accessed from the service order rows. Here you can also enter a service order specific item description, if desired. There is a window in the reports that you can make visible in order to print information about the multiple serial number items.

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13.4.10 Credit limit check

Credit limit checks can be performed for customers and/or payers. Aggregated sales order and service order value will be included in the credit check. If the combined value exceeds the credit limit set for the customer, an appropriate message will be displayed.

In addition, you can recreate the total service order value for the customer. This function is useful if you intend to include service orders retroactively in the customer credit limit check. When recreating the service order values, the system will exclude service order rows with the billing code set to No invoicing and service order rows with status Cancelled.

For information of working with credit limits in the order routine, see the Credit limit check section in the Jeeves Sales chapter.

13.4.11 Addresses

The delivery address of the serial number item is entered from the customer order when delivered. If the serial number item is installed at another address, information can manually be entered in the installation address fields for the serial number item. In this case, the installation address is transferred to the delivery address fields in the service order header when entering a service order with serial number item. Otherwise, the delivery address is transferred.

You can also specify additional information regarding a delivery address, for example a building, a project site, GPS coordinates, or other references.

13.4.12 Guarantee

There are two fields on the service order header referring to a guarantee end date:

- The Service guarantee field only refers to service items and shows a calculated guarantee end date based on the different guarantee times that can be specified for the service item. If the end date has not been passed, the Internal guarantee field is checked to indicate that the work should be done under guarantee.
- The Individual guarantee field only refers to serial No. items. When a serial No. item is entered on a service order, a guarantee date according to information in the Serial No. items table is displayed on the service order header.

This is to indicate that you must decide in each case whether it should be assigned a debit code for guarantee jobs that does not initiate invoicing.

13.4.13 Service options

You can manage various options or executions/models of a service item that may affect the reparation price and/or discount.

The function is based on the concept of selling and/or offering service on items that have various options. The items are entered as stocked package items and the options are entered as non-stocked package details/components. The items (package items) must also be entered as service items with data about price changes per option.

When dispatching (suitably in Order processing, packages), the serial number is entered on the package item and the relevant option(s) are chosen. The dispatch creates a serial No. item for the service item (package item) as usual, but in this case, options are also created in the underlying program node called Service options, service items.

There is also the possibility of processing options for items that you do not sell. In manual entry of serial No. items, there is a function button to create options according to the package item's structure and thereafter manually delete non-relevant options.

When entering service orders on service items with options, the price is calculated based on the price of the service item row and the price changes of the option(s) (+/-).

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13.4.14 Service labels

There is a program for printing service labels with a bar code. The program prints the number of labels specified on the service order header. This function is used, for example, when an item is received in several parts.

13.4.15 Aggregated values

You can view the following aggregated values for service orders in the list window:

- Actual accrued time for the event(s).
- Total time on a service report.
- Total order value in base currency at the time of entry. Order rows and equivalent are subtracted from this total order value when the row is invoiced. This information forms the basis for order value discount. On customer orders, it helps in automatic calculation of the handling charges.
- Total order value in currency at the time of entry. Order rows and equivalent are subtracted from this total order value when the row is invoiced. The order value discount is based on this information. On customer orders, this value forms the basis for automatic calculation of handling charges.
- Total calculated cost value of the customer or service order, or row, expressed in base currency.
- Total calculated cost value of the customer or service order, or row, expressed in the chosen currency.

You can also view aggregated service order values incl VAT/tax in base currency and currency in the service order header and row.

13.4.16 Checklists

Checklists are available for service technicians to maintain complex service objects and to ease the performance and reporting of service activities. Checklists consist of multiple rows called checkpoints linked to service order rows and are populated from a template. Each service order row can have a linked checklist with many checkpoints.

13.5 Jeeves Graphical planning

You can create logic for the Calendar field if you use Jeeves Graphical planning connected to Service. There must then be a calendar for every service technician.

13.6 Reporting

You can plan and report service activities. You can also process service objects which in Jeeves can be normal items, serial number items, where required, or service items. These alternatives allow different service models with different characteristics.

On a service order, you can enter and report time without a link to any product. You can enter and report service for non-stocked products, which is normally done using service items. You can enter material from your own inventory/stores, or purchased from a supplier, that is used during service.

13.6.1 Reporting service labor and material

In Jeeves ERP you choose between the following alternatives when reporting service:

- "free" reporting
- against a default service flow
- against planned activities

Your work method determines which program you use.

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You can accumulate planned and reported time on service activities and display the information on service order headers.

13.6.2 Jeeves Mobile field service routine

A standard mobile routine for individuals in manufacturing to report operations.

In the Field service routine, own service orders are searched and the service technician's list of service orders are displayed. The user can of course customize and add to the search conditions.

13.6.3 Inventory transactions

There is a possibility to select, per debit code, which inventory transaction type is to be used for material issues.

13.6.4 Account segments on service order rows

You can control account segments on inventory transactions from service order rows. If no account number is entered on the service order row the account is retrieved from the inventory posting template, on condition that an account segment has also been entered (i.e. cost center, cost unit, segment 4-7). If an account segment is entered on the service order row this overrides any account segment entered on the template.

13.6.5 Spare parts management

There is an underlying program for spare parts entry, which is by default largely identical to the service order row form. The purpose of the node is to provide the option to create a separate form for entry of materials/spare parts that contain only the relevant fields.

13.6.6 Interruption management

If for any reason the service work has to be interrupted, interruption management can be used to keep track of where the object is during the interruption and the interruption's status.

Jeeves service module contains functions for reporting interruptions in service work with information about the type of interruption, interruption status, the possibility to link item numbers and also error descriptions. An interruption report can be printed and sent to the customer.

If the above information is not necessary and you only want to be able to signal that an interruption has occurred, you can instead enter a service order status called Interruption and simply change the status for the service order. The same applies when a cost proposal is first produced; if you do not want to use interruption management, you can simply check the Cost proposal box on the service order.

13.7 Invoicing

Service work that has been carried out and material that has been used can be invoiced via the order routine in the Sales module or via the assignment routine in the Time, Projects, Assignments module.

13.7.1 Invoice via order routine

When you close and save a service order row which has an item number and quantity specified, and the Create order box is checked, a customer order with a new order number is created automatically. You invoice the order according to the normal order-invoicing routine.

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13.7.2 Invoice via assignments routine

An assignment must be entered on the closed service order row from which the invoicing is done. If the row refers to time, a service technician, a sector and a time type must be specified. If the row refers to material, an item number must be specified. You create a preliminary assignment invoice or a preliminary service invoice and invoice according to the normal assignments routine.

13.7.3 Collective invoicing

There is a parameter to determine if collective invoices should be created per service order number in the Invoicing, collective invoice program. If the parameter is set there will be one collective invoice per service order. If the parameter is not set different service orders can be included in the same collective invoice.

13.7.4 Payer on service orders

You can specify a different company as payer on service orders in the same way as, for example, in the Sales module. At present this can only be used if you invoice service orders via the order routine.

13.7.5 Meter readings

There is a function that can be used to automatically calculate and charge the difference between the previous and newly entered meter reading. The meter reading and date of the last reading for the serial No. item are updated automatically when you close a service order row with a meter reading specified. The difference is entered in the field for the quantity on the service row. But note that currently debit code 5 must be used on the service order row for updating to be performed.

13.7.6 Integration with ledger and general ledger

The invoices are automatically entered in the accounts receivable ledger and posted according to the normal order or assignments routine. For further information about this, and about follow-up reports and statistics, please refer to the overviews of these modules.

13.8 Other functions

13.8.1 Logging service orders

Every status change of a service order can be logged in a log file in the service routine. The status code determines which changes are logged.

13.9 Localizations/internationalization

Most functions in Jeeves Service are common to all countries. There is, however, one function specific to Sweden. It is the possibility to integrate with the Kontek payroll system. This is described in the chapter on *Jeeves Time, projects, assignments*.

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14 Jeeves ERP Time, projects, assignments

Jeeves Time, projects and assignments module includes routines for planning and managing customer assignments and projects, including planning and assigning resources, reporting and invoicing time and expenses, closing activities and follow-up. There are also functions for managing price lists regarding professional services and expenses, as well as costs for internal debiting.

You can choose between fixed price and open account invoicing. You can invoice open account projects per assignment or per project at any time. Alternatively, you can set up open account invoicing per assignment to be done monthly. Invoices are automatically created in the accounts receivable when you invoice. The invoices are automatically posted and integrated with the general ledger when invoice journals and vouchers are created. There are also functions for on account invoicing and for manually entering advance invoices.

All project documentation can be stored connected to the specific project through Jeeves EDM.

14.1 Project management

Jeeves ERP has a built-in project hierarchy that facilitates project management. On the top level you have an assignment with one or more master projects and/or projects connected to it. You can use the master project functionality to create a project hierarchy in any number of levels. Each project can have any number of planned activities connected to it. This structure allows you to gather all business agreements on assignment level and all planning and execution on project level. The project must be connected to an assignment in order to be able to invoice time and expenses related to the project.

14.1.1 Assignments

The assignment contains information about the business agreement with the customer. It is on the agreement you specify the customer, price lists for service and expenses, pricing method, invoicing type, currency, payment plan if applicable, VAT/tax code etc.

You must also set an agreement type for the assignment; there are four different types:

- Open account – used for projects on which reported time and expenses are invoiced recurrently
- Periodic – used for projects that you want to invoice for example every month, quarter or year.
- Fixed price – used for invoicing fixed amounts on specific dates.
- No invoicing – used for projects that are not to be invoiced, for example internal projects.

The agreement type is inherited from the assignment to the connected projects, but can be changed. This implies that, even if an assignment is to be invoiced, you can set No invoicing on an individual project, and vice versa.

The assignment status controls whether time, expenses and material can be reported and invoiced; only assignments with status Active can be invoiced.

You use the invoicing type to control the level of detail on invoice rows as well as postings, for example per person, per category, per category and person, per activity, total time and expenses etc.

There are numerous other parameters and defaults you can set on the assignment that control reporting, pricing and invoicing.

14.1.2 Projects

All planning, reporting and follow up is done on the project. The project also contains information about how to integrate the project with the general ledger. The project must be connected to an assignment in order to be able to invoice time and expenses.

You can connect project members to your projects, if required, by entering user signatures in the Project personnel node.

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14.1.3 Master projects

You can use the master project functionality to create a project hierarchy in any number of levels. A project hierarchy can have several master project and sub project levels.

You can see all projects connected to the master project in the Consists of projects node in the Projects program. You can also make a setting in the list of this node in order to graphically display your project hierarchy in a tree structure.

The following data is aggregated to the master project from underlying master projects and sub projects:

- total estimated time
- total accrued time
- total remaining time
- total accrued project cost
- total theoretically remaining time

The master project shows these values from all connected sub projects, plus planned and reported time on the master project itself.

There is also an embedded list where you can view all planned activities on all connected projects. Note, however, that this only shows activities from the master project and one underlying level, but you can drill down to view activities on up to four levels.

14.1.4 Project customers/clients

You can specify a project customer on projects, if desired. The drop-down list only displays users in the Accounting and time report users table with the Valid to use box checked.

14.1.5 Resources

You can set up applicable resource types for example project manager, developer, tester etc. A default resource type can be entered on users, which is suggested as default when entering planned activities for the user, but can be changed.

14.2 Project planning

14.2.1 Planned activities

You can have an unlimited number of planned activities on each project. Each activity can have planned start and end dates. If the project is planned the activity’s planned start and end dates will be checked that they lie within the project's corresponding planned start and end dates. You can view planned, reported and remaining time on activities. On projects there are embedded lists to view all planned activities regardless of the activity status, or only current activities.

The project manager always has an up-to-date view of all active activities directly in the project form. The project member always has control over his/her activities through a separate to-do list. Once an activity is finished, you check the Activity finished box. The activity is then removed from the to-do list.

There are different ways you can enter planned activities on projects. You can manually enter planned activities either via the embedded list window or the node for planned activities. You can also automatically create planned activities based on the resource personnel category.

14.2.2 Activity templates

Alternatively, you can generate planned activities from a template. You define your templates in the Activity templates program. Then you connect the relevant templates to your projects and automatically generate planned activities from them.

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14.2.3 Main activities and sub activities

You can use the main activity functionality to create activity hierarchies in any number of levels, for example a Test activity might consist of Test planning, Test execution and Test reporting. Note that you cannot plan or report time on, or assign resources, to main activities. Main activities are automatically assigned to the default user defined in the Accounting and time reporting users program. Time is automatically accumulated from planned and reported time on sub activities. Main activities show both planned, allocated time and remaining allocated time, i.e. allocated time not yet planned on resources.

14.2.4 Assign resources to planned activities

Alternatively, you can use the function for assigning resources to planned activities, which automatically generates planned activities on the project for the resources. You will find the node for assigning resources via an underlying program node to Planned activities. If you assign several resources to the same activity, these will be connected to the first as sub activities in order to collect all time planned and reported on the same activity. The overlying activity will automatically be flagged as a main activity. New resources will automatically be added to the project personnel. You can then add information regarding dates and times to the automatically created project activities.

14.2.5 Iterations

You can define and maintain iterations, if required. Iteration date intervals can be used to automatically set start and end dates on planned activities. Iterations are defined manually according to own requirements.

14.2.6 Milestones

You can set planned activities as milestones. Only one planned activity with the same milestone ID can have this box checked. If you, for any reason, change the milestone ID of an activity defined as a milestone, Jeeves ERP warns you and gives you the possibility to change the milestone ID on all linked planned activities. If you use activity hierarchies the milestone ID is inherited from the main activity to the sub activities when resources are allocated, but can be changed.

When you close a planned activity defined as a milestone, Jeeves ERP warns if there are any unclosed, planned activities linked to the milestone and asks if you want to close them.

14.2.7 Requirement IDs

Requirement ID can be set on planned and reported activities, in order to be able to keep track of their related requirements. The requirement ID is copied to the reported activity when reporting time against the planned activity. Otherwise the field has no logic and the value does not need to be unique; several planned activities can have the same requirement ID. You can change a requirement ID on a planned activity, but note that any existing reported activities will not be updated with the changed ID.

14.2.8 Graphical planning

There is a possibility to integrate with Jeeves Graphical planning where you can plan resources and activities graphically. Jeeves Graphical planning offers an entirely different dimension of resource planning on projects.

14.2.9 Document management

All project documentation can be stored connected to the specific project through Jeeves EDM.

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14.2.10 Budget

You budget projects in the same way and in the same program as budgeting for other account segments in the general ledger. For further information, please refer to the chapter about *General ledger*.

14.3 Prices and discounts

14.3.1 Pricing of services and expenses

You can set up price lists for both services and expenses.

The services price list is used when reporting activities. Each services price list consists of three sections, price per

- Person
- Service
- Personnel category

Jeeves ERP automatically retrieves the correct price depending on the pricing method declared on the assignment. In each section you can set up different prices for different time types (such as normal/overtime etc.). Exceptions from the price list can be specified in an assignment specific price list.

The expenses price list is used when reporting expenses. This price list is the same as for sales orders.

Both services price lists and expenses price lists are specified on the assignment, together with the pricing method.

You can also apply general customer discounts on expenses.

14.4 Project execution

You can report time on planned activities but also directly on a project or an assignment with or without a planned activity. All reported time is aggregated on the project. You can use status reporting to report progress of an activity at different stages of the project. The statuses are user-defined.

In the project header you have an overview of the project's progress with access to:

- planned time
- reported time
- remaining time
- accrued project costs
- theoretic remaining time (planned time in hours minus reported time in hours)

On master project level you have corresponding total fields for the entire project hierarchy.

There is also a possibility for personnel to report time per week number. All reported time, irrespective of where it is reported, ends up in the weekly time report.

You can report expenses directly on a project or an assignment. The expense report is connected to a specific signature and you can set up general expenses as non-stocked items in the item table. You can also issue material from inventory against an inventory location, and a bin if used. This includes availability information and update of inventory balances.

Items that are purchased from suppliers can be entered as expenses on the project using a function called re-invoicing of supplier invoices. When re-invoicing you can add a markup either in percent or a fixed amount. The purchased item appears automatically as a project expense when posting the supplier invoice according to pre-defined rules. The expense is then included in the accrued project cost and you can invoice the project customer.

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The project status can be used to control if time and expenses can be reported and invoiced on the project.

14.5 Invoicing

You can create assignment invoices in different ways, based on reported time rows and expenses, fixed payment plans, periodic invoice templates, etc. You can also enter invoices directly in the system without using any preparatory program.

You can mark all invoiceable expenses and reported activities as ready for invoicing, either for a project or for a project manager. Additionally, you can also mark newly entered expenses or time rows under a project as ready for invoicing.

14.5.1 Manual or automatic invoicing

You can both manually enter invoices and automatically generate invoices based on reported time rows and expenses.

14.5.2 Preliminary invoicing

You can choose between preliminary invoicing and direct release of invoices for all types of assignment and project invoicing.

You can turn off automatic creation of preliminary invoice rows for manual entries on assignments. This is for companies who want to be able to invoice manually entered activities or item rows on assignments, without invoicing all other uninvoiced time and expenses on the assignment.

14.5.3 Invoicing open account projects

You can invoice open account projects per assignment or per project at any time. Alternatively, you can set up open account invoicing per assignment to be done monthly.

14.5.4 Periodic invoicing

Jeeves ERP's assignment routine has a function for periodic invoicing of, for example, service agreements, rent, etc. You define templates containing the periods when different types of periodic invoicing of assignments is to be done. Then you define a template for the invoice. You can enter an item number if, for example, you have defined an item for rent in the item table. Alternatively, you can just enter a price, salesperson and posting information. You can set up the system for periodization of revenues and costs; see the section about integration with the general ledger.

14.5.5 On-account invoicing

Invoicing can be based on a payment plan for an assignment (so-called on-account). On-account amounts entered on assignments can be set up to be automatically deducted when invoicing time and expenses.

14.5.6 Collective invoicing

Jeeves ERP offers collective invoicing of assignments with the same payer or delivery customer, terms and currency. You can choose if the assignment should be summarized or specified on the collective invoice. There is the possibility combine collective invoicing for assignment invoices with invoices from the order routine.

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14.5.7 Credit invoices and matching credit to debit

You can manually enter credit invoices or automatically generate credit invoice based on debit invoices. Credit invoices can be automatically matched against debit invoices via parameter setup.

If you create a credit invoice from an invoice, you are asked if you want this credit invoice to be automatically matched against the relevant invoice. Otherwise you can enter another invoice number to match against or leave the field blank (if no automatic matching is to take place).

The credit invoice is posted in the normal manner, during printout of the invoice journal, using templates connected to the dedicated payment method called CTD, Credit to debit. . If the debit invoice is already paid the credit invoice remains open. If the credit invoice only covers part of the invoice amount, the remaining amount remains open. If it covers more than the invoice amount, the debit invoice is fully paid and the remaining amount on the credit invoice remains open.

14.5.8 Advance invoicing

Advance invoices can be entered in the assignment routine. You simply set the invoicing type Advance invoice on the assignment and enter the invoice manually.

14.5.9 e-Invoicing

You can create electronic invoices and use a third party service for electronic distribution, notification of sender and receiver, storage and online tracking. The advantages of electronic invoicing are many; it is quicker, less expensive, better for the environment, minimizes administration etc.

14.5.10 Forward cover

There is a simple function for managing forward cover of foreign currency, which implies invoicing at the preliminary invoices' exchange rate.

14.5.11 Extra invoice texts

Via parameters you can retrieve extra texts describing reported time and expenses on invoice rows.

14.5.12 Invoice costs

You can setup Jeeves ERP to automatically handle charges on assignment invoices.

14.6 Integration with accounts receivable ledger and general ledger

Invoices are normally automatically entered in the accounts receivable ledger upon creation.

Customer invoices are normally posted automatically in a separate routine/process after one or more batch of invoices have been produced. Invoicing can be done several times a day and posting at the end of the day or first thing the next day. One invoice journal is created and posted per invoice date. Each journal generates a voucher in the general ledger.

When manually added stocked items are invoiced, a record for the manual inventory transaction is created and a matching record in the Inventory transaction history is created when integrated with the general ledger.

Posting is done according to templates or according to invoicing types that control all accounts and account segments (cost centers, cost units etc.) to which receivables, revenues, costs, VAT/tax etc. are posted.

You can set up and run automatic posting of internal costs for assignments/projects. You set up a cost per user and cost price factors per time type.

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14.6.1 Internal debiting

You can set up and run automatic posting of internal revenues and costs when lending employees to other projects. The project and staff must have different cost centers for it to be possible to internally debit another department. You enter an internal price on the relevant price list.

14.6.2 Periodization of revenues and costs

When invoicing periodic assignments, revenues and costs can be allocated to periods. This is controlled by specifying allocation templates for revenues and costs on the assignment's template for periodic invoicing.

14.6.3 Project coding

There is a function for coding projects in order to automatically change postings (cost units) when using certain combinations of account and project numbers on vouchers. This can be used, for example, to add a prefix that represents the department/business unit that the cost unit represents.

14.6.4 Balancing project costs

On projects you can specify a template for translation of income accounts to balance accounts when balancing a project. In the template you enter the intervals of income accounts and to which balance accounts these shall be translated.

If project costs are balanced, amounts are posted in the GL on the specified balance account every time vouchers are created on the accounts specified in the interval. Later, when you want to cost account the project, you simply delete the translation template from the project.

14.7 Project follow-up

14.7.1 Follow-up of revenues and costs

In a table accessed via a node in the Projects program you can see posted debit and credit amounts and accumulated revenues and costs in base currency from posted vouchers per master project ID, project ID, fiscal year, period and integration source. Only cost accounted projects are included, i.e. not balanced projects. If you have created a project hierarchy, the amounts are summarized on the master project. The window is updated on-the-fly and can take a while depending on the size of the general ledger and the project structure (level of hierarchies). The window also has an underlying node for viewing the related general ledger voucher details.

If you include expenses in project costs in base currency the project cost will be broken down into time and expenses.

There are nodes for viewing the individual customer and supplier invoices connected to the project and their invoice rows. Only customer invoices on posted invoice journals and posted supplier invoices are included.

You can calculate internal costs based on an estimated hourly cost per user and time type. You can also perform internal debiting between different cost units or cost centers based on the user's estimated hourly cost. For more information see the section about integration with the general ledger.

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14.7.2 Follow-up of accrued time

You have several types of reports to follow up reported time on projects, for example reported time per project or per person.

14.7.3 Follow-up of time and expenses not yet invoiced

From assignments you can follow up time and expenses that have not yet been invoiced by printing an overview comprised of the following parts:

- summary of included projects with totaling of times and costs
- list of any assignment-specific prices
- list of reported but uninvoiced times
- list of uninvoiced expenses
- summary of preliminary invoices and associated invoices
- payment plan, where applicable

14.7.4 Closing projects

There is logic connected to project status 8, Closed that allows you to automatically close all open planned activities on a project when closing the project.

14.8 Reconciliations

There is a program for on-screen reconciliation of balanced projects, per specified date, against the account(s) for balancing of projects in the general ledger.

14.9 Industry-specifics

Most functions in Jeeves Time, projects, assignments are common to all industries. There are some, however, that are specific to certain industries, as described below.

14.9.1 Assignments with counter parties

Jeeves ERP includes a simple solution for client agencies for managing assignments in, for example, legal proceedings. The solution is comprised of a program in which you can directly see the counter entries on an assignment. You can also enter supervisors for various tasks in the proceedings.

14.9.2 Percentage of completion accounting

Jeeves ERP includes functionality for percentage of completion accounting, mainly used by companies in the construction industry.

Percentage of completion accounting is a method for successively posting the results of long-term fixed cost projects. Postings are based on a revenue forecast and a percentage of completion that is calculated from cost forecasts and posted costs, i.e. the calculation does not take into consideration any posted revenues. By keeping track of the amount posted on the project between each period, you can create new postings for accumulated revenue, accumulated cost and accumulated result for the period.

If the project is forecasted at a loss you make an appropriation for the loss risk based on the whole forecasted loss.

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14.9.3 Project manufacturing

Project manufacturing supports the business transactions and processes of a manufacturing process from quotation to payment using a project ID. It covers the following features:

- Quotations and sales orders from quotations
- Project MRP
- Project manufacturing
- Project inventory
- Invoicing and payment plans
- Delivery
- Project analysis

This section provides an overview of the components that are specific to Time, projects, and assignments. Other components are described in the chapter on Inventory.

14.9.3.1 Inventory locations setup

You can define inventory locations to be used in Project manufacturing. You can also create a fictitious project to represent the normal inventory location when transferring stock to and from Project manufacturing inventory locations.

14.9.3.2 Agreement types for project manufacturing

You can associate an agreement type with project manufacturing. Projects connected to such quotations/agreements are invoiced according to a payment plan. They also require that quotation rows are defined as either main rows or optional rows.

14.9.3.3 Assign activity type to planned/reported activities

If you intend to use the project activity type in project analysis and follow up, you can connect the relevant activity types to the planned and reported activities:

- Project management
- Administration 1
- Administration 2
- Administration 3
- Administration 4

14.9.3.4 Cost types linked to cost account

You can select a cost type to connect it to the cost account. The following cost types are available as standard:

- Transportation
- Travel
- Sub contractor
- Miscellaneous

14.9.3.5 Quotation and payment plan

You can enter a quotation and specify whether an item is mandatory/main or optional.

You can specify different payment plan templates from which to create payment plans and then link the templates to quotations. An example of a payment plan could be a first part payment when initiating the manufacturing, a second for procuring certain specialized and/or expensive parts with long lead times, one or more payments at different milestones in the project and a final payment upon full delivery.

You can create or modify a payment plan from a quotation and then create or modify a project and associate it with a payment plan.

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14.9.3.6 On account records

You can update on account records with quotation type set to Project manufacturing from projects and color code all account records connected to the quotation/assignment associated with the project for easy identification.

After the payment plan is created, you can create milestone activities to control when the assignment is invoiced. When a planned activity associated with a project manufacturing milestone is closed, the on account amount to invoice record with the same project manufacturing ID as the planned activity is automatically updated to make it possible to invoice the record.

14.9.3.7 Project analysis: Estimations, forecasts, and follow up

As a project progresses, differences occur between the original cost and time estimates and the actual course of the project.

During and after the completion of a project, you can take stock of its progress through periodic analysis. Jeeves ERP allows you to do the following through a node in the projects programs:

- View estimations based on quotations provided by sales representatives
- View forecasts and calculations made by project manager/management team during various stages of the project
- Create/update forecasts
- Save/view old versions of forecasts
- View actual, accrued project costs and time usage
- Compare current forecast with actual numbers/calculations to know the updated cost of the project.

14.9.3.8 Sales order from quotation

When using Project manufacturing, you create sales orders from quotations in the normal manner. There is, however, additional logic for Project manufacturing.

You can specify an order type for the project manufacturing agreement type, in which case the order type is retrieved from the agreement type on the quotation when creating a sales order.

14.9.3.9 Split reports per shipment

The shipment ID field primarily supports the Project manufacturing functionality, but can be used by all. In large manufacturing projects, it can be used to identify and specify the contents of each transport vehicle, when large constructions need to be dissembled for transport.

14.10 Localizations/internationalization

Most functions in Jeeves Time, projects, assignments are common to all countries. There are some, however, that are specific to certain countries, as described below.

Integration with Kontek Payroll system (Sweden only)

Jeeves ERP offers Swedish users an integration with the Kontek payroll system. The integration is based on parts of the existing functionality for Attendance reporting in the Production module. The main components in the integration are:

- employee agreements connected to applicable time types and salary types,
- connection of employees to the relevant employee agreements,
- day schedules with applicable attendance reporting time types,
- work schedule templates based on the day schedules,
- connection of employees to the relevant work schedule templates,
- entry of any deviations and automatic creation of the current year’s work schedules,
- connection of reported activities to relevant time types (via time codes) and
- automatic creation of payroll files.

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14.10.1 VAT refunding

Originally developed for the UK, but available to all users

In some countries, such as the UK, the VAT/tax value is based on the discounted amount even if the customer does not take advantage of the early settlement discount. In order to be able to reconcile the VAT at the end of the period, Jeeves ERP creates additional, preliminary postings on the customer invoice journal, which are subsequently, automatically reversed on the payment journal. You can adjust your invoice templates to show the required information, for example the discount terms and the VAT amount based on the discounted amount.

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15 Jeeves ERP Accounts receivable

Jeeves Accounts receivable helps you optimize the process from managing invoices to receiving payments. Invoices that originate from Jeeves Sales or Jeeves Time, project, assignments modules are automatically integrated. You can also enter invoices from other sources manually.

Payments, including advance payments, can be entered manually or by importing payment files from financial institutions. For overdue invoices there is flexible functionality for managing reminders, interest invoices, debt collections, and payment injunctions.

Integration with the general ledger is automatically made via a journal, which generates a voucher in the general ledger. There are also programs and reports for reconciliation and functionality for revaluation.

15.1 Companies and customers

In Jeeves ERP you have a common Company table for company information regardless of the type of company – prospective customers and business opportunities, customers, carriers and suppliers. This eliminates the need for duplicate entries and maintenance of company information.

Additional, customer-specific information is managed in related customer tables. Here you can handle all types of customers, domestic and international, customers with another payer, with factoring, with several points of delivery etc.

15.1.1 Customer categorization

You can handle any number of customer categories for posting control and statistics etc., any number of customer discount classes for discount management, also customer classes for own classification purposes.

15.1.2 Language management

You can manage the language of payment terms, delivery terms, delivery methods, currency clause texts, reminder texts etc. and print external documents in the customer's language.

15.2 Invoices

15.2.1 Manual invoices

You can manually enter invoices and credit invoices without detail rows, for example invoices from external systems, in the Customer invoices program. Invoice totals can be entered but not invoice rows and invoices cannot be printed. You use this function for invoices that are not handled in the integrated modules, i.e. the Sales or the Time, projects, assignments modules.

15.2.2 Invoice notes/claims

There is a basic routine for recording and searching claims regarding customer invoices.

15.2.3 Credit check

There is a function in the accounts receivable for automatic credit check of customers based on the customers' due invoices. This is an additional check to the normal credit check during order entry in the sales module.

You specify a minimum number of due days on invoices to be checked and a maximum total due amount for the customer/payer. You also choose between credit block and warning. To perform the credit check you first make a selection of customers or invoices and then run a report program.

You can view the debt account for invoiced, unpaid customer advances in the Accounts receivable reconciliation program.

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15.2.4 Payment terms

You can store an unlimited number of payment terms, per customer or payer, per agreement or general. Credit periods and any early settlement discounts can be set up in numerous ways.

There is an array of methods for calculating due dates:

- Credit period in days or in months
- Calculate period from delivery date or from beginning of following month(s)
- Extend period to end of month
- Set dates, for example 10th, 20th and last day of month

15.2.5 Invoice history

In many countries it is legally required that you store historical data regarding invoices so that an invoice duplicate can be reproduced if necessary.

You can print duplicate invoices with historical data, i.e. even if company information for the customer or payer has been changed since the original invoice was printed, the change does not affect the reprint. There is also functionality to ensure that the same template is used as when printing the original invoice.

15.3 Posting and integration with general ledger

Customer invoices are normally posted automatically in a separate routine/process after one or more batch of invoices have been produced. Invoicing can be done several times a day and posting at the end of the day or first thing the next day. One invoice journal is created and posted per invoice date. Each journal generates a voucher in the general ledger.

For manual invoices the income is allocated in its entirety on a single account according to a posting template for invoice totals, comprised of an account for customer receivables, for VAT/tax, and for income. If you want to split it into several accounts or account segments, you do this manually or via allocation.

15.4 Integration with Intrum

You can manage receivables and/or payments against invoices using integration with Intrum, a leading credit management services company in Europe. You can choose customers whose invoice information you want to send to Intrum. The information is sent in XML format either on the invoice creation date (outsourcing of entire Accounts receivable) or on the due date (outsourcing of demands and collections only). Also, you can set a grace period before sending the invoice information to Intrum, considering extra days for processing of payments done on the due date.

For every invoice sent for collection, an update on the status is sent back from Intrum. It contains, amongst other things, information on the number of reminders sent to the customer and the latest date of reminder. The information is received in XML format which is then updated in the Customer invoices program.

15.5 Incoming payments

Incoming payments can be entered manually or imported from a bank or postal giro file.

When entering payments manually, you can choose between a number of methods:

- enter all today's incoming payments per payment method
- enter payments per customer
- if you know the invoice number, enter payment of an individual invoice

You can set up import of payments to automatically create payment journal rows. The import can be stopped when an incorrect payment is identified so that the incorrect payment can be adjusted and the import resumed. Alternatively, you can control that incorrect payments should not create payment journal rows and have to be handled manually.

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You can receive payments to multiple (own) companies in the database. By default, Jeeves ERP determines the company by checking the bank account number in the payment file. However, companies using the BG Max and Total In payment services should turn off this functionality and enter the receiving account number for each company in the database. Jeeves ERP checks the bank account number in the payment file to decide which company the payments are connected to.

You can view the payment information in a text field by clicking the Internal text button in the Incoming payment file detail node in the Incoming payment receipts program. In the Edit blob that opens, you can manually enter any own internal comments or information.

15.5.1 Partial payments, discounts, deductions, posting individual invoices

You can manually enter a partial payment, an early settlement discount or any other deduction, or post the incoming payment in a way other than according to the posting templates for incoming payments, simply by double-clicking on the planned payment and changing the fields in question.

You can set a parameter to automatically make proportional deductions of VAT when early settlement discounts take effect and other deductions are made on customer invoices.

When importing from file, there is a function to capture incomplete transactions and add information to achieve a higher rate of matching.

15.5.2 Early settlement discounts

Automatic calculation of early settlement discount based on discount date can be applied to invoice amount or value of applicable goods, on one level or as tiered discounts.

15.5.2.1 Early settlement discount on advance payments

You can enter early settlement discount amount in currency at the time of incoming/outgoing payment. The discount is calculated automatically based on the terms specified for the payment terms in question, either in the Early settlement discount days and Early settlement discount % fields, or the Early settlement discount terms field. It can also be specified manually, in which case the paid amount must be manually adjusted correspondingly.

15.5.3 Payments via check

There is functionality for preparation of payments made by check, integration with the general ledger, remittance and payment. There is also functionality for cancelation of checks.

You can set the check number automatically on all paid invoices in the journal. If you want the same check numbers to be automatically added to a set of invoices when you enter received payments by check, you must enter the check number on the payment journal header.

You can also specify that empty spaces in any field in the printout can be filled with a defined character, e.g. asterisk (*), using the Fill character field in the Template editor. In addition, you can reprint (print copy) of checks with same check number from the payment programs.

There are various options for printing checks with different layouts.

15.5.4 Lump sum payments

When entering incoming payments for individual customers you can handle lump sum payments, i.e. when a customer pays an amount that covers several invoices without specifying which invoices they intend to pay.

15.5.5 Miscellaneous payments

A statement from the bank may contain miscellaneous types of payments. The Miscellaneous payments functionality makes it quicker and easier to handle incoming payments that are not

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connected to customer invoices and to facilitate the reconciliation of the bank account. This can be, for example, a payment for something other than an invoice, or a payment to/from the bank.

You enter the amounts in the currency of the payment journal. The exchange rate in the journal header is used when integrating with the General ledger. You include the amount of the miscellaneous payments in the Paid amount in the payment journal header. You only enter the contra posting to the bank posting. The bank posting is created automatically.

15.5.6 Advance payments

In Jeeves ERP's accounts receivable you can manage advance payments from customers. When you enter an advance payment from a customer, an on-account invoice is created automatically. This can later be used against the normal customer invoice or incoming payment.

15.5.7 Payments in other currency

You can enter incoming payments in a currency other than the invoice's currency, if the function is activated. The work method is largely the same as for entering incoming payments in the invoice's currency. The difference in this program is that you can process invoices whose original currency differs from the currency in the journal header. If you open the form for an individual incoming payment (detail), you access program specific fields for entering received amounts, or deductions, etc., in the currency you have entered in the journal header.

15.5.8 Payment by installments

You can handle customer invoices to be paid by installments. For each payment installment you specify the credit period in days and the percentage of the total amount to be paid in the installment. The sum of the percentages must be 100.

For example, you could define:

- two equal installments with 30 days between them; enter one row for 30 days and one for 60 days with 50 percent on each or
- three installments; a first installment of 20 percent after 10 days, a second of 30 percent after 20 days and the remaining 50 percent after 30 days.

On the payment terms you can also enter a minimum limit for splitting invoices into installments.

You can adjust the installments, to create invoice specific installments, if desired but the sum of the percentages must be 100.

When invoicing, Jeeves ERP creates one invoice with installment records. There is a special window in the invoice report template that you can make visible in order to print information about the installments.

Jeeves ERP takes the payment installments into consideration when creating payment reminders. This implies that reminders only cover the amount that is currently due. There is a special window in the reminder report template that you can make visible in order to print information about the installments.

The following applies if your company uses the Payment by installments functionality:

- If you use the "traditional method" of interest calculation, i.e. if Balance method is not checked, Jeeves ERP takes the payment installments into consideration when creating interest invoices. This implies that the interest invoices only cover installments that are paid after their due dates.

If you use the balance method you can create interest invoices before installments are paid, in which case this cannot be applied. Forms, lists and reports that show amounts due, and/or amounts becoming due, take the installments' due dates and amounts into consideration.

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15.5.9 Reverse journal or individual payment(s)

There is a function for reversing a paid and posted payment journal. You use the function to reverse all the payments in the journal if you, for any reason, need to re-open one or more paid customer invoice(s).

Alternatively, you can reverse individual invoices by creating a new incoming payment journal and entering a negative incoming payment record in the right window. You enter the customer number, invoice number and a negative amount. When you post the journal the status of the invoice is reset to Invoiced.

15.5.10 Offsetting customer receivables against accounts payable

In Jeeves ERP you can transfer a customer invoice to the accounts payable ledger in order to settle one or more supplier invoices. This can be used to pay credit notes. The function can also be used for debit invoices if you wish to annul a payment that is already in the accounts payable ledger.

15.5.11 Direct debits

For certain types of sales, for example sales via mail order, or when organizations charge student or member fees, it is common to have a special procedure to gain the customer's permission to transfer the amount from their bank account.

In Jeeves ERP you can handle incoming direct debit payments within Sweden, Germany and the Single Euro Payments Area, SEPA. The function can also be localized by partners in other countries by replacing Jeeves ERP's standard procedure for creating payment files with an own procedure.

There is one work flow based on a dedicated Direct debit journal program. This assumes that you first mark the individuals/customers who have given permission. Then you run a report to create a so-called permission file. At regular intervals, for example when invoicing, you create a direct debit file that is sent to the Bank Giro. There is also an alternative work flow for creating a direct debit file directly in the normal incoming payment journal program. You choose the latter, for example, if you need access to Jeeves ERP's standard functionality for early settlement discounts and adjustments of invoice amounts.

15.6 Posting and integration with general ledger

Incoming payments are posted automatically according to posting templates, and integrated with the general ledger, by running a report program. When posting incoming payments, the amount on the accounts receivable account is counter-posted and the bank account increased. Any bank costs, exchange rate differences, early settlement discounts and deductions are also posted.

15.6.1 Manual posting of incoming payments

It is possible to manually specify account segment 1, 2, 3 and 4 for individual incoming payments.

15.7 Payment reminders, collection and interest invoicing

Jeeves ERP offers support for several different ways to urge customers to pay overdue invoices. The process supports the following routines, mainly based on how payment reminders are handled in the Scandinavian countries:

- Payment reminders
- Interest invoicing
- Collection
- Payment injunctions

All parts can be set up dynamically. You can set a minimum amount for demands and even handle reminder charges. Reminder letters can be printed in different languages.

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An alternative solution is also available to invoicing payment reminder charges as part of interest invoices. Instead of including reminder charges in payment reminders or future customer orders, you can include them directly when you run interest invoicing. The reason for this alternative solution is that a customer who pays late might be credit blocked, and/or they might never place another order with the company charging them interest and reminder charges. When the interest invoice amount is less than the payment reminder charge limit, extra invoice rows will be created. Since the charges are included as rows in the interest invoice, they are processed as claims and also update the general ledger.

It is also possible to remove specific customers or invoices from a reminder batch.

15.7.1 Payment reminders

In Jeeves ERP you can create up to four payment reminders regarding due and unpaid customer invoices and, if so desired, a collection journal. You set up the number of days between each reminder and before collection and via a parameter you control whether or not payment reminders should be printed. You can also set a parameter if you want to include invoices that are not yet due on reminders.

During the reminder batch, accrued interest is calculated using the formula: (invoice amount*interest days/365)*penalty interest (or 30 days/month is used if that is stipulated in the interest agreement). Accrued interest is calculated for each invoice that has been paid after the due date and has not already been included in an interest invoice, as well as invoices that are due but not paid. The date you specify when printing the reminder is then used as the 'simulated payment date' of these customer invoices. Thus, interest is not only calculated on the invoices printed on the reminder.

A reminder journal is created, which contains one reminder per customer and currency. This means that the reminder can consist of one or more invoices. If the reminder covers several invoices for which a different number of reminders have been sent, reminder text from the highest reminder number is used.

You can create and print reminders in one or three steps. The latter is recommended because this enables you to change the reminder text and delete individual invoices from the reminders before printing. You can even cancel the entire reminder for a certain customer.

Other features include:

- any number of reminder rules, customers are linked to a rule
- minimum amount for reminders
- payment reminders in several languages
- reminder charges

15.7.2 Interest invoicing

Interest can be invoiced in two different ways in Jeeves ERP. Either according to the "traditional" method which calculates interest when the invoice is paid, or according to the balance method which calculates interest day by day on the balance of due invoices and then also takes credit invoices into consideration. If you use the balance method, there is a parameter with which you can determine whether interest should be paid on the total balance or on every invoice separately.

There are a number of parameters that determine how interest invoices are calculated. These mainly lie in interest rules to which you link customers.

You can create and print interest invoices in one or several steps. The latter is recommended because this enables you to check which customers and invoices will be invoiced interest, and where necessary to block individual customers or invoices before printing.

Other features include:

- any number of interest rules, customers are linked to a rule
- minimum amount for interest invoices

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15.7.3 e- Invoicing interest

You can also use Jeeves ERP's eInvoicing functionality to create electronic interest invoices. This includes:

- electronic distribution
- electronic storage and online tracking
- notification of sender and receiver (services via Pagero)

15.7.4 Collections

Jeeves ERP includes a function for handling invoices that need to be sent for debt collection. The routine consists of creating a collection journal according to rules connected to each customer, i.e. number of reminders before collection etc. From the journal a collection file is created which can either be directly sent by e-mail or printed and sent by mail to the collection company.

15.7.5 Payment orders/injunctions

Jeeves ERP includes a simple function for creating and logging payment orders/injunctions, i.e. handling of invoices sent to the authorities for enforcement of payment.

15.7.6 Credit control, interest and reminders

There is a special program for making requests/searches regarding credit control, interest and reminders. The program only includes reports related to credit control.

15.8 Doubtful and bad debts

Doubtful debts are those debts that a business or individual is unlikely to be able to collect. The reasons for potential non-payment can include disputes over deliveries, the condition of received items or financial stress within a customer's operations.

Bad debts is the continuation of doubtful debts management, when you know for certain that the debt is not possible to collect, for example because the customer has gone out of business.

You can mark unpaid invoices as doubtful debts and post these values as doubtful receivables and as potential losses on the accounts receivable account.

Integration source 105, Doubtful debt is used for the voucher source.

You can set up posting rules with the following integration details for use in the vouchers created from the doubtful debt journal:

- 12, Doubtful debt (credit)
- 13, Counter posting of doubtful debt (debit)

These two integration details are also used by integration source 6, Incoming payments along with 90, Bad debt, for reversing doubtful receivables and posting bad debts.

You can also set up posting rules for the reversal of the doubtful debt when it becomes an actual loss, with the addition of integration detail 90, Bad debt.

You can search for invoices marked as doubtful or bad debts and to view the doubt or bad debt amount in the Customer invoices program.

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15.8.1 Create doubtful or bad debt journal

You can mark unpaid invoices as doubtful debts and post these values as doubtful receivables and as potential losses on the accounts receivables in the Doubtful debt journal program. You use the same program if it becomes necessary to transform these into bad debts and post the values as actual losses.

The program comprises of three parts:

- The journal header – here the Doubtful debt journal number is allocated and you enter general information such as journal date, currency code, etc.
- The left embedded list window – here you can open customer invoices with due dates earlier than or equal to the Doubtful debt journal date are displayed.
- The right embedded list window – here is an entry list to which you transfer the selected doubtful debt invoices. You can view invoice details such as outstanding amount, invoice total excluding VAT/tax, the number of days that have lapsed since the amount was due. You can also view or modify the doubtful debt amounts or change the exchange rate if required.

15.8.2 Post and integrate with general ledger

You can create postings and general ledger vouchers for a selected doubtful or bad debt journal.

If the Bad debt journal checkbox is unchecked (false) on the journal header, this is a doubtful debt journal. If the Bad debt journal checkbox is checked (true) on the journal header, this is a bad debt journal.

Note that the journal must be closed before printing the report.

15.8.3 Reverse doubtful invoice

You can reverse a doubtful invoice, for instance, if there is a mistake and the invoices are incorrectly selected as doubtful, but the journal has posted them to the General ledger. You can rectify this by creating a new reverse journal.

15.8.4 Recovery of doubtful or bad debt

If a customer should pay off a doubtful or bad debt, you can create a normal incoming payment journal for that invoice. The voucher created for the incoming payment journal will reverse the postings for the doubtful or bad debt. The doubtful receivables will be debited and the loss on the accounts receivable will be credited.

15.9 Ledger information

15.9.1 Customer solvency

There are functions for analysis of customers' solvency, i.e. average payment times, payment habits, etc. The Customers program and the program for accounts receivable inquiries display information about payments and payment times per customer. If you select several customers, totals are shown at the bottom of the list. If you want to see totals for the entire accounts receivables, select all customers.

The data is generated using the Generate customer solvency button and is calculated based on the specified starting invoice date. Note that only authorized users can run this calculation.

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15.9.2 Invoicing statistics

Statistics can be produced for any period, by customer (delivery customer or payer), customer category and project.

There is also a function for creating records for European sales lists, i.e. reports on arrivals and dispatches within the EU, including drill down to the invoices included in the dispatches and arrivals totals. This is to make it possible to reconcile the data in the report. This function can be used by all companies within the EU that purchase from and/or sell to other countries within the EU. A report can be printed from Jeeves ERP as a basis for manually filling in the official form.

15.9.3 Reconciliation of accounts receivable ledger

The accounts receivable ledger must regularly be reconciled against the general ledger, for example once a month. During reconciliation, the accounts receivable ledger is checked against the accounts receivable ledger account in the general ledger. You can also reconcile the accounts receivable per date by entering the from and to dates in the relevant program. The transactions for the specified general ledger account and the accounts receivable ledger for each of the dates in the date interval are generated. This makes it easier to identify which date has caused any difference immediately.

The accounts receivable, i.e. debt due, can be reconciled in report form or on screen, after which a report is printed. If required, you can print the entire accounts receivable ledger as of the reconciliation period.

15.10 Other functions

15.10.1 Customer number change

There is a function for transferring individual or all outstanding invoices to a new customer (payer) by changing the customer number. The function is useful when a customer changes their line of business or when liability to pay is assumed by a factoring company or bankruptcy estate. A customer number change log, which you can print from the **Report** menu, shows the requested change, which invoices were processed and the new invoice numbers. This log can also be used to view the original invoice numbers.

15.10.2 Revaluation of accounts receivable

In Jeeves ERP you can revalue the accounts receivable ledger at the current exchange rates. Exchange rate differences are posted automatically according to posting templates, and integrated with general ledger.

There are three alternative ways of handling the revaluation:

- as realized gains and losses in the accounts receivable
- as unrealized gains and losses in the accounts receivable which are reversed when the final exchange rate difference is determined, i.e. upon payment
- create a voucher in the general ledger to show unrealized gains and losses, which is immediately and automatically reversed, without affecting the accounts receivable

15.11 Localizations/internationalization

Most functions in Jeeves Accounts receivable are common to all countries. There are some, however, that are specific to certain countries, as described below.

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15.11.1 Bank details available in currency codes program and printed on invoices (Sweden only)

The SWIFT code and IBAN/SWIFT account fields are available in the Currency codes program.

If the report templates for customer invoices and preliminary invoices are configured to retrieve the bank account information, the information is retrieved and printed in the invoice footer as follows:

- Swedish customer invoiced in base currency: Customer invoice PG and Customer invoice BG retrieved from System data, own company.
- Foreign customer invoiced in other currency or Swedish Krona (SEK): IBAN/SWIFT account and Swift code retrieved first from the Currency codes table and if unavailable, from System data, own company.
- Swedish customer invoiced in other currency: IBAN/SWIFT account and Swift code retrieved first from the Currency codes table and if unavailable, from System data, own company.

15.11.2 Invoice reference numbers and bank account validations

Jeeves ERP supports localized reference numbers on customer invoices such as OCR in Sweden, FIK code in Denmark, KID reference in Norway, bank reference in Finland and Internal subscriber number in Switzerland. Bank account numbers are validated according to local standards. There is also the possibility to add own formats and validations.

15.11.3 Localized customer payments

Jeeves ERP supports import of payment files for Sweden, Denmark, Norway and Finland. There is also the possibility to add other localized formats.

15.11.4 Import payments in Credit advice ISO 20022 XML, Camt054 format

You can import supplier invoice payment files from banks in the Camt.054 format. The files are imported in the normal manner, using the Definition and import external data->work table program, and the result will be displayed in the Outgoing payment receipts program.

Note that not only the setup is identical for both supplier and customer payments, the same payment file can also contain payments from both suppliers and customers. When importing, Jeeves ERP identifies the type of payment and stores them in the relevant program.

15.11.5 Import payments from Norwegian banks

The currency code for the incoming payment receipts when importing a payment file from a Norwegian bank in Norwegian Crowns (NOK) is retrieved from the payment. This ensures that there is no currency mismatch when Jeeves ERP matches the payment file with the existing invoices in the accounts receivable ledger. The Receipt detail status shows the value as Transferred, the payment journal is created, and the invoice is paid as expected.

15.11.6 Interest invoicing in connection with normal invoicing

In some countries, such as Norway, payment reminders and interest invoices are printed in connection with normal invoicing of orders in the Sales module, i.e. not in a separate routine in the Accounts receivable. Jeeves ERP includes functionality for interest invoicing at normal invoicing, which is described in the chapter on the *Sales* module.

15.11.7 Advance invoicing

In some countries, such as Slovakia, it is legally required that postings of net amounts are suppressed on advance invoices. There is an application parameter with which you can suppress the postings.

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15.11.8 Import and automatic matching of bank statements

There is functionality for importing bank statements including customer and supplier payments and other bank transactions, matching the payments against open invoices and integrating the bank statement with the general ledger. The function is used in countries such as Germany and the Netherlands. Please refer to the chapter on the *General ledger* module for a description of the function.

15.11.9 Tax clearing on sales

With Jeeves ERP you can handle sales of items with interim posting of VAT/tax, i.e. where the outgoing VAT/tax is posted on an interim account until payment is made, at which time it is counter-posted to the normal VAT/tax account. This is used, for example, in France.

15.11.10 Split payments

VAT regulations in Poland require customers to pay their suppliers via a split payment mechanism. This applies only to domestic transactions between companies with the product and service groups specified in the Accounting Act. The split payments mechanism is obligatory only for transactions over 15,000 PLN. Payments with lower amounts can be made through traditional bank transfer or cash. It applies on both full and partial payments.

Jeeves ERP provides functionality to determine if customer payments should be made via a split payment mechanism, and to print information about split payments on customer invoices.

For partial payments, the paid amount and the VAT amount needs to be changed manually in Jeeves.

15.11.11 BOE ledger, bills received from customers

The bills of exchange ledger is a ledger for both received bills of exchange for payment from customers and issued bills of exchange for payments to suppliers. The BOE ledger in the accounts receivable includes functionality for preparation, acceptance and printout of bills of exchange received from customers. This is used, for example, in France. The process includes:

- acceptance
- collection
- payment
- integration with the general ledger

15.11.12 KSeF eInvoicing — outgoing customer invoices (Poland)

Jeeves ERP supports electronic sales invoicing through Poland's National e-Invoice System (KSeF) using the FA(3) format. Validation supports accurate processing and compliance with Polish eInvoicing requirements.

After a customer invoice has been cleared through KSeF, the cleared legal invoice, including its KSeF ID and QR code, can be retrieved automatically and attached to the invoice header in Jeeves ERP.

KSeF identifiers can be used in payment processing and reconciliation. They can also be used when processing incoming customer-payment notifications in ISO 20022 camt.054 format, supporting reconciliation and improving traceability between invoices and payments.

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16 Jeeves ERP Accounts payable

Jeeves Accounts payable reduces your administration and optimizes your payments by streamlining the management of supplier invoices, from purchase order matching at invoice arrival, via the approval process, to final payment. Much of the functionality is based on seamless integration with the following Jeeves modules/applications:

- Jeeves General ledger – for automatic posting as well as reports and programs for reconciliation
- Jeeves Purchases – for automatic matching of invoices against linked purchase orders
- Jeeves Fixed assets – for automatically creating fixed assets at supplier invoice entry
- Jeeves e-Approval web application – to let a workflow electronically route invoices for approval on a web interface
- Jeeves e-Approval mobile routine – makes the management of supplier invoices even more accessible

Payments are commonly made via payment files. There is support for many different localized formats. The invoices to be paid can be entered manually or selected automatically, from a cash optimizing point of view.

16.1 Companies and suppliers

In Jeeves ERP you have a common Company table for company information regardless of the type of company – prospective customers and business opportunities, customers, carriers and suppliers. This eliminates the need for duplicate entries and maintenance of company information.

Additional, supplier-specific information is managed in related supplier tables. Here you can handle all types of suppliers, domestic and international, suppliers with another payee, with factoring etc.

16.1.1 Supplier categorization

You can handle any number of supplier categories to control approval and posting templates as well as statistics etc. You can have any number of supplier types for supplier evaluations.

16.1.2 Suppliers' bank accounts

If you enter an account number for the supplier, a record is automatically saved in the Supplier's bank accounts. The number is validated according to the Bank acct check type field for the account, unless the validation is inactivated. The account check type can be, for example, PG or BG and numbers are checked to ensure that they are for a valid account, or valid plus giro or bank giro numbers. You can define additional bank account types and modification procedures are provided to allow other validations for the different standards that apply in different countries. You can also retrieve different bank accounts depending on the currency specified on the supplier invoice.

16.1.3 Supplier identification

There is a possibility to set an alternative supplier ID to identify a supplier when importing payment files. This can be used as an optional identity for the supplier in the absence of normal identity data such as Bank giro number, Plus giro number, Vat number etc.

16.2 Supplier invoices

Supplier invoices can either be entered directly or imported from a file to a temporary table, to be reviewed before converting to a supplier invoice record.

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16.2.1 Arrival handling of supplier invoices

You can choose between different work methods for handling supplier invoices:

- arrival entry with preliminary posting, then final approval with final posting
- arrival entry with final posting, then final approval without posting
- direct final approval and final posting

These postings are based on pre-defined templates. You can create a new posting template with similar posting controls as an existing one using a function button in the Posting templates program.

16.2.2 Approval rights

You can restrict approval rights to certain cost centers or group of cost centers.

16.2.3 Project ID linked to imported supplier invoice

When you enter a purchase order number on an imported supplier invoice and save the record, the project ID associated with the PO is automatically retrieved from the purchase order header table and can be viewed in the Review imported supplier invoices program. The project ID can also be automatically moved when you transfer the invoice to the supplier invoices (lr) table if required.

The project ID is copied to the preliminary posting, which can be viewed in the Preliminary posting node. This requires, according to Jeeves ERP's standard logic, that the Project accounting segment should be combined with the account.

16.2.4 Update imported supplier invoices with error in data

It is recommended that you regularly check imported supplier invoices for errors in data and keep the information updated using the Preliminary import/scanned suppl invoices program. From this program, you can view the PDF files with the preliminary supplier invoices. You can also check the invoices for errors in data (for example, incorrect Invoice date or invoice already exists in the Accounts payable). In case an error is found, you can correct the data and then transfer the invoice for review or you can delete it.

16.2.5 Manually transfer corrected supplier invoices

After you make corrections to imported supplier invoices in the Preliminary import/scanned suppl invoice, you can manually transfer the corrected supplier invoices for review via a push button.

When you click the Transfer to review button, a row-by-row transfer is made to the Review imported supplier invoices program and controls are performed as usual. If a row can be loaded, it is deleted from the preliminary table; otherwise the row remains. When all corrected records are successfully loaded, a message is displayed showing the number of records that have been transferred.

When you click the Transfer to AP - PDM button, the invoice reviewal documents are automatically checked, one by one. All valid documents are converted to supplier invoices and transferred to the supplier invoices table and the accounts payable ledger.

- If all documents are processed correctly, a message informing that all reviewed invoices are now transferred is shown.
- If an error is found in any document, it is not converted into an invoice and a message referencing the sequence number of the documents that are not transferred is shown.
- If no documents are found for processing, a message informing that there are no documents for processing is shown.

The transfer status is also updated in the Review of imported supp.invoice table/program. In case of errors, the related error message is also displayed. For all records that have errors, you must correct the error and repeat the transfer.

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During transfer, a document record is created in the EDM table, that in turn creates an approval chain in Jeeves Workflow.

16.2.6 Approve or reject/reply supplier invoices

You can approve a supplier invoice and send it to the next person/level in the approval chain or reject the invoice and send it back to the invoice clerk from the Jeeves client interface. This function facilitates invoice approval from within the Jeeves client particularly when using Jeeves Mobile client.

You can display the list of approvers in the selected approval chain. When you click this button, a dialog opens displaying the level, name, and email ID (if available) of the approvers for the selected approval chain.

16.2.7 Payment terms

You can store an unlimited number of payment terms, per supplier, per agreement or general. Credit periods and any early settlement discounts can be set up in numerous ways.

There are several methods for calculating due dates:

- Calculate period from arrival or invoice date or from beginning of following month(s)
- Extend period to end of month

16.2.8 VAT

In Jeeves Purchases/Accounts payable you set up suppliers and items for VAT. Your company can apply any number of VAT rates.

See also separate sections describing VAT in the Sales module, VAT declarations, as well as VAT reporting within the European Union. There is also a number of country specific VAT declarations described among the localizations for the General ledger.

16.2.9 Advance invoice credit (on account)

You can enter an invoice with this special type and pay it in advance. The invoice is automatically converted into a credit invoice and can later be used to credit the real invoice when it arrives.

16.2.10 Multiple tax amounts on supplier invoices

It is possible to handle up to five VAT/tax amounts on the same supplier invoice.

16.2.11 Link to fixed assets ledger

Fixed assets can automatically be entered when entering supplier invoices or general ledger vouchers by using an account with a dedicated account type. When you use an account of this type, you must specify whether the cost pertains to a new or an existing fixed asset.

- If it pertains to an existing fixed asset and component, a transaction of the type Addition costs from supplier invoice is created. The acquisition amount and remainder to be depreciated are updated automatically on the fixed asset.
- If it pertains to a new fixed asset or component, a transaction of the type Entry of new fixed asset is created. The component's fixed asset description is assigned the voucher number and, the supplier number is copied to the Supplier number field where applicable.

16.2.12 Canceling supplier invoices

An invoice that has been entered incorrectly, for example with an incorrect amount or supplier ID, can be canceled if there is only one approval or if the balance on vouchers associated with each posting has been updated. The original invoice is allocated a new number by adding _1. This means that the invoice can be entered again with its original number and the correct information.

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If you use e-approval of supplier invoices:

Jeeves ERP checks if there is a connected workflow instance in Jeeves Workflow that has not already been canceled. If there is a workflow instance connected, it will also be canceled. This eliminates the need for manual cancelation in Jeeves Workflow.

Cancelation of invoice in closed period

In Jeeves ERP, you can cancel a supplier invoice that has been arrival entered, or finally approved directly, even in closed periods.

The invoice is paid with payment method MAK and arrival entry is reversed. The canceled invoice is renumbered so it can be re-entered. The VAT/tax accounting for the period in which the invoice was posted is not affected; the adjustment is made to the period in which the cancelation is posted.

16.3 e-Application – e-Invoices

16.3.1 e-Invoices

You can receive electronic invoices including invoice images via a third party service. There are standard functions in Jeeves ERP for interpretation, review and transfer to the accounts payable ledger. Approval can either be done in the accounts payable ledger or via Jeeves ERP's e-Approval module. The advantages of electronic invoicing are many; it is quicker, less expensive, better for the environment, minimizes administration etc.

You can display electronic supplier invoices in a PDF browser. There are buttons for viewing invoices in an embedded browser and a button for viewing invoices in an external browser. This function requires that Adobe Acrobat Reader is installed on the client computer.

API calls can also be used to import Supplier invoices and credit notes from integration partners (currently only Pagero) in the Peppol BIS Billing 3.0 format. The invoices are received via Pagero integration services, transformed into the JSON format, and sent to the Finance API for further processing. Companies can have access to multiple accounts in Pagero Online, in which case a Pagero company ID can be passed with the Peppol invoices during submission of eInvoice s (in and out), so Pagero can identify to which company the invoice belongs.

The functionality can be set up so that only invoice headers are imported or also invoice rows. Additionally, business responses can be sent to the integration service, indicating whether the invoice is technically correct, accepted, or providing a reason if the invoice is rejected.

16.4 Posting and integration with general ledger

Supplier invoices are automatically posted and integrated with the general ledger according to general or supplier specific posting templates, and depending on the chosen method of arrival handling.

16.5 Invoice matching against one or more purchase orders

You can link a supplier invoice to one or more purchase order(s) in order to check it before final approval and posting. This can be set up to work in different ways, for example, you can control:

- whether only received purchase order rows may be matched
- whether received delivery reporting is allowed during invoice matching
- if the company uses average cost prices, that cost prices are calculated upon final approval after invoice matching against received deliveries, instead of during entry of received deliveries
- tolerances, i.e. maximum allowed differences between purchase order and invoice amounts

You can handle any additional costs on the invoice, such as freight, any deviations in prices, partial deliveries, as well as any difference in quantity that is to be credited, or that is approved.

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16.5.1 Automatic matching

There is also a function for automatic matching of supplier invoices against purchase orders. Automatic matching means that if the data in the invoice agrees with the data in the received purchase order, and if the specified limits for differences have not been exceeded, matching and final approval will occur completely automatically and the invoice will be marked as ready for payment. Automatic matching forms a part of Jeeves e-Approval application, but a part of this functionality can be applied even if the e-application is not used.

There are many requirements for automatic matching to function, including, among other things, a purchase order number must be specified on the supplier invoice, the invoice and purchase order must have the same supplier number, and the order must have been received or partially received. You can also specify if you want the delivery note number to be used in automatic matching of supplier invoices against purchase orders in order to improve the accuracy in the matching.

For manual arrival entry of supplier invoices, you specify a purchase order number and arrival post the invoice on an interim account. In addition, you can control whether or not automatic review will be allowed when loading supplier invoices. If any of the imported invoices is incorrect, the incorrect invoices are skipped and all other invoices are set as OK.

If automatic matching has been activated by a parameter, matching is carried out and a final approval record created when you save the arrival approval record. Otherwise, you activate it using the Auto-match invoice-PO button. You can also open imported supplier invoices and activate automatic matching with the Auto-match invoice-PO button if it has not been set up to occur automatically.

You can also schedule automatic matching of several invoices at the same time and creation of matching records for the supplier invoices using an SQL procedure once the integration of received deliveries is completed. If received deliveries integration is not completed for any supplier invoice linked to a purchase order, that invoice is not matched and the process will continue to the next invoice that meets this requirement.

Note that the entry date is automatically retrieved from the arrival voucher or set to today's date.

16.6 e-Application – e-Approval

16.6.1 e-Approval

Jeeves e-Approval is a function for the electronic distribution and approval of supplier invoices. This is built on the combination of a standard work flow in Jeeves Workflow and an e-application. The e-Approval work flow consists primarily of the following steps:

1) Supplier invoices are scanned and stored in the form of Tif files and then interpreted.

2) The files are imported, or, alternatively, the invoices can be opened and invoice data entered manually.

If Pagero AB's service for receiving eInvoice s is used, Pagero creates an eInvoice file in Jeeves ERP's standard format and a corresponding invoice file in Tif format, and notifies the user that there are files to be downloaded.

3) Jeeves ERP’s programs for review of supplier invoices allow the accounts payable administration to open an image of the scanned invoice, enter approval data, and then transfer the invoice to accounts payable. Invoice receives status Pending approval.

4) This creates one instance of the approval chain built according to the specified approval chain in Jeeves Workflow. At the same time, an e-mail is sent to the approver stating that an invoice has arrived for approval.

5) The approver reviews the invoice in a special web interface for e-approval, and once the invoice is approved, a final approval record and a general ledger voucher that updates the balance are usually created. Alternatively, via a parameter, you can choose to let the accounts payable administration perform final review, approval, and balance update.

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16.7 Supplier payments

Payments can be performed in two ways:

- Manually (cash payments or manual payments over the internet) immediately marking the invoice as paid and creating a payment voucher in the general ledger
- By payment file, with payments for several suppliers and payment dates, which is then sent to a financial institution. Before the file is sent, a payment proposal is created based on your payment rules and prioritizations. After matching against an imported payment file the invoices are finally flagged as paid and then posted.

16.7.1 Manual entry of individual payments

You can enter individual payments of supplier invoices manually. For individual payments, you enter the payment and create the general ledger voucher directly. The invoices are considered as paid in the accounts payable ledger and general ledger.

16.7.2 Partial payments, discounts, deductions

You can manually enter a partial payment, an early settlement discount or any other deduction, simply by double-clicking on the planned payment and changing the fields in question.

You can set a parameter to automatically make proportional deductions of VAT when early settlement discounts take effect and other deductions are made on supplier invoices.

16.7.3 Bank and postal giro payments to suppliers

You can handle bank and postal giro payments to suppliers in two ways:

- Via an invoice payment service, which means that you send invoices to a bank and the bank monitors them and pays them on the due date.
- Internal monitoring and payment. This approach means that you usually monitor your invoices yourself, select the ones you want to pay, create a bank or postal giro file and transfer it for immediate payment. This also creates a general ledger voucher. This in turn means that the invoices are considered to be paid in the accounts payable ledger and general ledger.

16.7.4 Individual supplier in invoice payment service

While selecting approved invoices to be included in the invoice payment service journal, there is a possibility to select the invoices for an individual supplier by entering the company ID on the Invoice payment service program header.

16.7.5 Preliminary outgoing payments

There is an alternative work method for the invoice payment service, which implies preliminary posting of outgoing payments.

Preliminary outgoing payment means that you create an outgoing payment journal for the date on which you plan to pay a number of invoices, and then link the relevant invoices to the outgoing payment journal. It is then the planned date of payment in the outgoing payment journal that controls the actual payment date and not the due date on the supplier invoices.

The outgoing payment journal can then be sent to the bank as an outgoing payment file. A voucher for 'Payment on the way' can be created when the payment file is sent from the outgoing payment journal. If such a voucher is created, it is counter-posted when the actual payment is acknowledged. When the acknowledgement is made by the bank, the outgoing payment journal normally only needs to be flagged as ready.

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16.7.6 Priority

You can set a priority directly on the supplier invoice, where 1 is the highest priority and 99 is the lowest priority. The priority is taken into account when a payment proposal is created.

16.7.7 Block supplier/invoice from outgoing payment

You have the possibility to block all a supplier's open and new invoices, or individual invoices, from payment.

16.7.8 Payment before final approval

Usually an invoice must be approved before it can be paid. In Jeeves ERP you can, however, specify that outgoing payments are allowed before final approval for individual suppliers or invoices.

16.7.9 Due date calculations

Due dates can be calculated from arrival or invoice dates. (For details see the section on payment terms on supplier invoices.) Payment proposals can take early settlement discount days into consideration. There is also a possibility to automatically update the invoice if the date on a payment plan is changed.

16.7.10 Multiple accounts and currency accounts for outgoing payments

If your company has multiple bank or plus giro numbers you enter them in the Own bank accounts program and link each to a payment method. When making outgoing payments you specify the relevant bank/plus giro number in the Account at bank field in the payment journal header.

You can use different currency accounts for international payments. If the bank account is linked to a specific currency, the outgoing payment journal can only contain invoices in that currency.

16.7.11 Cancel erroneous outgoing payment

It is possible to cancel an erroneous outgoing payment and reset the supplier invoice to unpaid. This is done by creating a new outgoing payment journal and entering the payment again but with a negative value. When you post this payment the status of the invoice is reset to Unpaid.

16.7.12 Reverse outgoing payment journal

You can reverse a paid and posted payment journal. You can reverse all the payments in the journal if you, for any reason, need to re-open one or more paid supplier invoice(s).

The journal can be reversed by pressing a button and you can determine how it should be done with an application parameter. The date of reversal with respect to the corresponding invoice is displayed in the Supplier invoices program. It is updated even for partial payments that are reversed and cannot be changed manually.

When the journal is reversed, a new reversal journal is created with voucher updated in the General ledger (status = 8. Balance update complete). The reversal voucher counter-posts the original voucher. The supplier invoice is reset with previous status and outstanding amount. Alternatively, when the journal is reversed, a new reversal journal is created with status=1 Entry complete. The supplier invoice is reset with previous status and outstanding amount.

You can also reverse individual invoices by creating a new outgoing payment journal and entering a negative outgoing payment record in the right window. You enter the supplier number, invoice number, and a negative amount. When you post the journal, the supplier invoice is reset with the previous status and outstanding amount.

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16.7.13 Matching of completed payments

Completed payments can be matched manually, or automatically by importing a payment file. There is also a function that can be used, if required, to capture incomplete transactions and add information to achieve a higher rate of matching.

16.7.14 Automatic matching

Payments can be matched automatically by importing completed supplier payments in the form of a Bank giro file.

Jeeves ERP creates one payment receipt journal per payment method for all invoices imported from file. The payment method is retrieved from each invoice. During import, Jeeves ERP checks that the supplier and invoice exist, that the amount is correct and that the invoice has not already been paid. Each invoice is flagged with its current status. Any incorrect invoices must be corrected manually.

Jeeves ERP checks each invoice at import to see if it is included in an outgoing payment journal with status Entered, sent, and the status on journals is changed to Ready, paid when all invoices on the journal are matched at import and paid. You must then integrate the journal with the general ledger.

16.7.15 Manual matching

There is a program for matching outgoing payments made through the invoice payment service. Only invoices that have been sent to the invoice payment service can be matched here.

You match payments by creating an outgoing payment journal header, in which you specify how the outgoing payments were made (outgoing payment method), on which date and any bank cost, currency and rate.

When you save the header, the invoice form opens in order for you to search for invoices, if necessary. You transfer the invoices back to the left list window in the journal form. Match and transfer the paid invoices to the right list window.

16.7.16 Capture of incomplete transactions

For outgoing payment transactions that are missing information, and therefore cannot be matched with an invoice, you can manually add the integration source (for example, Manual entry), an account and any other account segments on the outgoing payment details in the embedded list in this program.

You must also set the status to OK with other integration source, to indicate that the transaction has been completed manually and is ready for further processing.

When creating an outgoing payment journal by pressing the Create journal button in the Outgoing payment receipts program, vouchers for these manually completed payment transactions are created and stored separately as interim vouchers.

The interim vouchers can then be transferred to the general ledger in the normal manner, i.e. by flagging them ready and running the transfer report.

16.7.17 Miscellaneous payments

The Miscellaneous payments functionality makes it quicker and easier to handle payments that are not connected to supplier invoices. This can be, for example, a payment for something other than an invoice, or a payment to/from the bank. You enter the amounts in the currency of the payment journal. The exchange rate in the journal header is used when integrating with the General ledger. You only enter the contra posting to the bank posting. The bank posting is created automatically.

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16.7.18 Advance payments

To be able to handle advance payments when you have not received any invoice from the supplier but still want to be able to make a payment, there is an invoice type and payment method for advance credit. (This invoice type is mentioned above under Supplier invoices.) This type of invoice can also be created automatically. When a payment is entered with just a company ID and paid amount, i.e. without an invoice number, this automatically creates a supplier invoice of the type Advance invoice credit with final approval. You deduct the advance from the normal invoice when it arrives.

16.7.19 Recurring tax payments

Many companies use the supplier payment routine for payments to the company's tax account against the tax authority's bank giro account. The tax authority identifies a company's tax payments through a 13 character long reference number.

In order to be able to make recurring tax payments to the same supplier (the tax authority) and invoice number (tax reference number) you must enter a supplier invoice number more than 20 characters long and enter the tax reference number in the Invoice no OCR field. When you enter the numbers according to this method, the reference number is written to the bank giro payment file in place of the invoice number.

16.8 Posting and integration with general ledger

Outgoing payments are posted automatically according to posting templates, and integrated with the general ledger, by running a report program. When posting outgoing payments, the amount on the accounts payable account is counter-posted and the bank account decreased. Any bank overhead costs, exchange rate differences, price differences, early settlement discounts and deductions are also posted.

16.8.1 Manual posting of outgoing payments

It is possible to manually specify account segment 1, 2, 3 and 4 for individual outgoing payments.

16.8.2 Reconciliations

There is a program for on-screen reconciliation of the interim clearing account's balance in the ledger and the supplier debts in the general ledger.

All supplier invoices with balances on interim account(s) in the accounts payable ledger for the processing date specified are displayed in the list. At the bottom, a total of the amount on the interim account is shown for the specified reconciliation date.

16.9 Ledger information

16.9.1 Supplier statistics

Jeeves ERP includes forms and reports for received delivery and invoicing statistics per supplier. Invoicing statistics can be produced for any period.

There is also a function for creating records for European sales lists, i.e. reports on arrivals and dispatches within the EU, including drill down to the invoices included in the dispatches and arrivals totals. This is to make it possible to reconcile the data in the report. This function can be used by all companies within the EU that purchase from and/or sell to other countries within the EU. A report can be printed from Jeeves ERP as a basis for manually filling in the official form.

16.9.2 Reconciliation of accounts payable ledger

The accounts payable ledger must be reconciled against the general ledger regularly, for example, once a month. During reconciliation, the accounts payable ledger is checked against the accounts

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payable ledger account and the accounts payable interim account in the general ledger. The interim account can only be reconciled if invoices are arrival posted on an interim account before final posting in connection with final approval.

The accounts payable ledger, or the outstanding debt, can be reconciled by printing a report or directly on the screen, after which a report is printed. The interim account can only be reconciled in report form. If required, you can print the entire accounts payable ledger as of the reconciliation period.

You can also reconcile the accounts payable per date by entering the from and to dates in the relevant program. Transactions for the specified general ledger account and the accounts payable ledger for each of the dates in the date interval are generated.

16.10 Other functions

16.10.1 Revaluation of accounts payable

In Jeeves ERP you can revalue the accounts payable ledger at the current exchange rates. Exchange rate differences are posted automatically according to posting templates, and integrated with general ledger.

There are three alternative ways of handling the revaluation:

- as realized gains and losses in the accounts payable
- as unrealized gains and losses in the accounts payable, which are reversed when the final exchange rate difference is determined, i.e. upon payment
- create a voucher in the general ledger to show unrealized gains and losses, which is immediately and automatically reversed, without affecting the accounts payable

16.11 Localizations/internationalization

Most functions in Jeeves Accounts payable are common to all countries. There are some, however, that are specific to certain countries, as described below.

16.11.1 Invoice reference numbers and bank account validations

Jeeves ERP supports localized reference numbers on supplier invoices such as OCR in Sweden, FIK code in Denmark, KID reference in Norway, bank reference in Finland and Internal subscriber number in Switzerland. Bank account numbers are validated according to local standards. There is also the possibility to add own formats and validations.

16.11.2 Manually enter Intrastat records on supplier invoices

There is a possibility to manually enter information about purchases for reporting to Intrastat. You can manually enter invoice information without having to create a purchase order for the purchases or for the returns to suppliers.

16.11.3 Localized payments and international payments

In Jeeves ERP, outgoing payment routes are used for communicating domestic and international payments, via banks, outside Sweden. These control the format of the payment file that you create with the Create payment file report. There are different file formats for localized payments, depending on the bank that makes the outgoing payment. A number of outgoing payment routes and/or stored procedures are supplied as standard.

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Some examples are:

- Telepay – domestic and international payments via Norwegian banks
- Credit advice (ISO 20022 XML) – domestic payments and cross border euro payments within the Single Euro Payments Area (SEPA)
- DTAUS and DTAZV – domestic and international payments via German banks
- BACS – domestic payments via British banks and building societies, as well as others, such as the Danske Bank
- DTA – domestic and international payments via Swiss banks
- ACH – domestic payments via United States banks
- CFP – domestic and international payments via Nordea Corporate file payments

There is also a procedure template, which can be adapted to create a payment file for other banks.

16.11.4 Payment routines for SEPA

You can set the priority of international payments through payment services using SEPA format and set different conditions from specific banks and/or countries.

The priority of international payments via payment service using SEPA format is as follows:

- Normal = SEPA (Default)
- Urgent = URGP
- Same day value = SDVA

A change bearer for SEPA can also be set. It can have the following values: 'DEBT', 'SHAR, CRED or SLEV. For SEPA, the default is SLEV.

SEPA specific options can be used to set different conditions from specific banks and/or countries as follows:

- Pos 1-1 - If the bank allows sub totals or if the currency should be retrieved from own accounts.
- Pos 2-2 - Charge bearer for SEPA domestic payments.
- Pos 3-3 - If currency code is to be picked from own accounts (syk) when creating payment journal.
- Pos 4-4 - Charge bearer for SEPA international payments not IBAN.
- Pos 5-5 – Instruction priority
- Pos 6-6 - BANK or CUST in tag, depending on the bank.
- Pos 7-7 - Payment per invoice on domestic invoices when summary on supplier is chosen.

16.11.5 Summarize payment for SEPA

You can define if invoice information should be summarized in the SEPA payment file. If you select to summarize payments, the invoice information is grouped in the payment file for SEPA. This means that all payments on the same date and currency to each supplier are summarized into a total payment amount and information from all the invoices is specified on one row (when the file is unstructured) or each component of information from the invoices is shown in a separate row (when the file is structured). This needs to be done if you pay credit invoices because it is not allowed to send negative payment amounts in the SEPA payment file.

Else, the information from the invoices is ungrouped in the payment file for SEPA. This means that each invoice is specified separately in the SEPA payment file.

16.11.6 Calculation of non-deductible VAT based on VAT amount

Non-deductible VAT based on the VAT amount can be calculated and the results can be viewed whenever the VAT amount does not match with the VAT rate and invoice amount.

In some countries, only a part of the calculated VAT amount is deductible. The deductible part of the VAT is posted to the VAT account, the non-deductible part is posted as a cost. You can view the

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difference between the computed VAT and the actual entered VAT whenever the VAT/tax, curr amount field is changed and the VAT rate does not match.

16.11.7 Import and automatic matching of bank statements

There is functionality for importing bank statements including customer and supplier payments and other bank transactions, matching the payments against open invoices and integrating the bank statement with the general ledger. The function is used in countries such as Germany and the Netherlands. Please refer to the chapter on *General ledger* for a description of the function.

16.11.8 Payment to guarantee account

In some countries the tax authorities require by law that main contractors pay a certain percentage to a special guarantee account when using the services of subcontractors. This is in order to guarantee that taxes and social security etc. can be paid even if a contract is not fully completed. When paying a supplier invoice the paid amount is divided between the supplier's normal bank account and this guarantee bank account according to the percentage specified in the Guarantee % field.

16.11.9 Import payments in Credit advice ISO 20022 XML, Camt054 format

You can import supplier invoice payment files from banks in the Camt.054 format. The files are imported in the normal manner, using the Definition and import external data->work table program, and the result will be displayed in the Outgoing payment receipts program.

Note that not only the setup is identical for both supplier and customer payments, the same payment file can also contain payments from both suppliers and customers. When importing, Jeeves ERP identifies the type of payment and stores them in the relevant program.

Jeeves uses End-to-end ID, a unique number that is a combination of payment date and time, to identify each payment in the payment file. It is updated during the creation of the SEPA file and is used to unambiguously identify the transaction when importing the CAMT053 and CAMT054 bank files. This unique identification key is passed on unchanged throughout the entire end-to-end supplier invoice payment chain and can be viewed in the Bank statement details embedded.

16.11.10 Overpayments

In Jeeves ERP there is a possibility to allow overpayments, which can be applicable in some countries. If allowed, overpayments can be handled in two different ways:

- If you simply enter an overpayment in the Plan/paid amount field, the supplier invoice will automatically be updated with a negative amount in the Remaining to pay field. The paid amount will be posted in the normal manner. The invoice can later be used as a credit invoice.
- If you enter the overpaid amount as a negative amount in the Deducted amount field, the overpayment is handled and posted as a deduction.

16.11.11 Check payments

In Jeeves ERP, you can pay supplier invoices by check. Check payments in Jeeves ERP’s accounts payable works the same way as other outgoing payments except it is possible to reconcile checks, rewrite them, etc. You can also reprint (print a copy) of checks with the same check number from the payment programs if required.

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16.11.12 Print checks for miscellaneous payments

You can create/select miscellaneous check journals and print checks to employees or other persons not connected to the accounts payable and post the vouchers to the general ledger. This feature fulfills a requirement in the USA for printing miscellaneous checks for employees or others belonging to the employee cadre who are not regular suppliers. This is meant only to print checks and post them to the general ledger.

Select the relevant payment method and account in the Print miscellaneous checks program. Enter the amount and company ID. This implies that you will need to define one company for employees or the like, in the company table. You can enter the check recipient's name and address in the form before saving.

16.11.13 Cash supplier invoices (for EU stock replenishments)

There is functionality for handling cash supplier invoices that are paid directly and posted according to own posting rules. This is primarily for use during intra community stock replenishments within the EU. You enter the supplier invoice in the normal manner but specify the cash payment terms and relevant posting template with the cash account. This results in the following, automatic steps, without any actual payment transaction:

The paid amount is set according to the invoice amount including VAT and the outstanding amount is automatically set to zero.

If the invoice is posted in two steps, after arrival entry and posting the supplier invoice is automatically assigned the status Paid, not posted. After final approval and posting the status is changed to Paid.

If the invoice is posted in one step, after final approval and posting the supplier invoice is automatically assigned the status Paid. This requires that payment before final approval is allowed.

16.11.14 Interim posting of VAT/tax on supplier invoices

You can handle supplier invoices with interim posting of VAT/tax, i.e. where the input VAT/tax cannot be credited until outgoing payment. The VAT/tax is posted to an interim account at supplier invoice entry. The deducted VAT/tax is then counter-posted against a definitive VAT/tax account on outgoing payment. In order for this to work, a posting template must be set up to direct the deducted VAT/tax to the intended account.

16.11.15 Poland — regulatory compliance

16.11.15.1 Split payments

Polish VAT regulations require split payments for certain domestic business transactions involving specified goods and services when the transaction amount exceeds the applicable statutory threshold.

Jeeves ERP supports the Polish split-payment mechanism for supplier payments and helps determine whether a payment should be made using split payment. The required split-payment information can be included in ISO 20022 pain.001 payment files.

16.11.15.2 KSeF information in payment processing

KSeF identifiers can be included in payment references for outgoing supplier payments in ISO 20022 pain.001 format, supporting payment processing and traceability between supplier invoices and payments.

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16.11.16 BOE ledger, bills issued to suppliers

The bills of exchange ledger is a ledger for both received bills of exchange for payment from customers and issued bills of exchange for payments to suppliers. The BOE ledger in the accounts payable includes the following functionality:

- Printing of bills
- Posting of outgoing payments in two steps; 1) to the bill of exchange debt account and 2) bills of exchange paid
- Preliminary payment and payment
- Possibility to cancel a paid bill of exchange
- Possibility to issue multiple bills of exchange against one invoice (part payments) or multiple invoices against one bill of exchange and one supplier

Payment via bills of exchange is common in France, for example.

16.11.17 Global Cash Pool transfers

If debit and credit account is not part of Nordea's Global Cash Pool service, then payment will be processed as an ordinary Intercompany payment. To enable processing of the payment as a Global Cash Pool transfer that does not attract an additional bank fee, you can now enter the code SDVA in the SEPA category purpose field in the AP payment routes program. In this case, SDVA is considered as the service level while processing the payment file, and the bank fee is not levied. Else, the existing value (NURG or SEPA) is considered as the service level while processing the payment file.

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17 Jeeves ERP General ledger

Jeeves General ledger is the central application in Jeeves ERP's Financials suite, offering tools for planning, compiling business operations and analyzing the outcome in financial terms. It is completely integrated with all parts of Jeeves ERP that gives rise to financial transactions providing an overview of the entire business.

17.1 Chart of accounts

Jeeves ERP has eight accounting dimensions, called account segments. By default, segment 1 is called Account, 2 Cost center and 3 Cost unit, but this can be changed. Segments 4-7 can be used and named according to own requirements and the eighth segment is Project. Account segment IDs can be up to 8 alphanumeric characters and project IDs can be up to 10 alphanumeric characters.

17.1.1 Own account classifications

You can define your own account classifications that can be used when searching for accounts or defining reports.

17.1.2 Currency accounts

There is a possibility to limit postings to specific accounts to a certain currency.

17.1.3 Only debit or credit postings

You can define that all postings to a certain account, or integration source, should be debit or credit postings, which is achieved by displaying a negative amount where necessary.

17.1.4 Detailed or aggregated postings

You can also define whether all postings to a certain account should be detailed or aggregated. Detailed posting is used, for example, when posting and reconciling paid advance invoices from suppliers, also for bank accounts where you want to reconcile each individual invoice against a bank account statement.

17.1.5 Unique chart of accounts per fiscal year

You can define a unique chart of accounts per fiscal year. The company's chart of accounts is copied between fiscal years and any variances between the fiscal years are maintained manually.

It is possible to add and translate own accounts to the standard chart of accounts, as well as enter and translate own, say industry or country specific charts of accounts.

17.1.6 Change chart of accounts

Jeeves ERP includes unique functionality for supporting change-over to a new chart of accounts, with the possibility to automatically change the accounts used in the various posting templates with a minimum of manual activities.

17.2 Daily routines

The daily routines mainly consist of entering manual vouchers and posting integration journals from the sub systems, handling incorrect vouchers and if desired, performing daily reconciliations on account, segment or total level.

17.2.1 General ledger vouchers

The General Ledger is the destination where all manually entered and automatically generated postings from the sub systems finally end up. Through automatic allocation rules, new postings can

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be created or further details can be added to the financial transactions. Before a balance is updated several controls are made.

17.2.2 Integrated vouchers

The general ledger is integrated with the logistics, production and project modules as well as the accounts receivable, accounts payable and fixed assets ledgers. This implies that vouchers are created automatically from these modules. All account segments can be used in the integrated modules, according to the general requirements from IFRS.

The integration implies a high level of transparency and traceability. From a voucher you can directly identify and drill down to the source transaction in the relevant sub system. From a source transaction you can, in the same way, directly identify and drill down to the linked vouchers.

17.2.3 Manual voucher entry

General ledger vouchers can be entered manually, with or without posting templates. You can also use shortcut codes for default combinations of account segments. The balance can either be updated directly or by running a journal or voucher list.

You can type in a short text on vouchers and on individual rows. There are shortcut fields in several programs to simplify the entry of voucher rows, specially when the posting contains many account segments.

In the manual voucher entry program, you can copy an existing voucher with its account, account segments and amounts. The Manual entry checkbox and the Copy voucher allowed checkbox must be checked on the integration source. This is helpful in case of recurring vouchers where most of the account structure is the same and only minor adjustments are to be made.

17.2.4 Manual voucher entry with quantity

You can also specify a quantity on voucher rows, i.e. the number of pieces, packages, hours, or the like, that the invoice row or the voucher amount concerns.

17.2.5 Voucher number series

You can choose between automatic and manual numbering of vouchers, which is controlled per user. You set up different number series for different subsystems or modules. You can also divide them into sub-number series and link users and/or groups to the sub-number series.

17.2.6 Voucher controls

Only authorized users can enter data in any of the financials modules.

You can hide any account segments and other fields that are not used in your company's accounts, and design tabs with only relevant fields. If your installation is based on the Jeeves or Selected package, you will already have pre-designed tabs and frames.

You determine if and when individual accounts and other account segments can be used. A validity date interval can be specified for accounts, cost centers, cost units (account segment 1-3), segment 7 and Projects.

For each account you can specify which other account segments are allowed to be used in combination with the account. You can also specify whether an account segment is mandatory for the account.

You can set up a difference to be automatically balanced. If a difference arises between debit and credit during voucher entry, the amount is balanced if the difference falls below the amount you enter in this field.

A voucher can encompass several periods, but to be able to update the balance, the voucher must balance within each individual period.

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17.2.7 Voucher corrections

You can analyze incorrect vouchers from several different programs, both on-screen and through running a report. Incorrect vouchers can be corrected in underlying program nodes or by right-clicking to the voucher entry program.

There is a function to automatically reverse an existing voucher, for example if a voucher contains many errors. This creates a new voucher in which debit and credit have switched places. Also, there is a possibility to reverse the voucher in the same period as the original posting.

There is also a function for re-opening a closed voucher for changes. This can only be done if a number of conditions have been met. It is not allowed to unlock a voucher that contains an auto posting template. Updates to the general ledger are reversed and any flags indicating the voucher has been printed are removed. Furthermore, the voucher header is flagged to indicate that the voucher has been re-opened. Any changes to updated vouchers are logged. You can complete and change a re-opened voucher in the same way as at new entry.

17.2.8 Deleting vouchers

You can delete vouchers that are more than ten years old. The general ledger posting date is used to check if the voucher is old enough to be deleted.

17.2.9 Account reconciliation

There is a program to query on screen and/or print account balances per a specific date. In an underlying program node you can display information on associated voucher details for every balance row.

17.2.10 Flagging voucher rows

There is a help function for marking a selection of voucher rows with a text, for example for marking a selection of voucher rows that have been reconciled. The text is displayed in various programs for viewing vouchers and voucher rows.

17.2.11 Automatic posting/allocation on vouchers

You can define rules to automatically supplement or allocate postings, at the same moment as a certain posting is created. This can be used, for example, to distribute rent or the like to subsidiaries, to supplement postings with additional account segments or even with additional posting rows, etc. When an amount is posted to the account and any account segments specified for automatic allocation, postings are automatically created, either by creating new rows in the same voucher or by amending the original voucher row. The normal posting is debit if the source account basis is debit, and vice versa.

By using retrieval codes you can decide which normal and contra account segments are to be used. If the automatic postings are to be posted to the same account segment as the source row, you enter an asterisk (*). If you enter an equal to character (=), the value is instead retrieved according to what is specified in the auto posting rule.

You control the amount to be automatically posted using a percentage or by retrieving a constant.

17.2.12 Period balance queries and general ledger information

There are programs specially designed to make it quick and simple to make queries regarding period balances per account and account segment, in any currency, with the possibility to drill-down to underlying vouchers. You can use these queries for reconciliations per currency. You can also view any budget details for the selected account and account segment. Furthermore, you can print reports based on your queries.

There are also programs for requesting printouts of the general ledger, for specific accounts or account segments or in its entirety, either detailed printouts or only totals, as well as programs for printing income statements and balance sheets.

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17.3 Periodic routines

The periodic routines mainly consist of creating periodic vouchers, reconciling the integrated modules with the general ledger and creating external statutory reports. Typical periodic vouchers are automatic postings based on account balances, allocation of account balances to new accounts or segments, posting of fixed vouchers and finally generation of period closing vouchers.

17.3.1 Automatic posting of balances

Auto posting can be used to calculate, post and allocate business transactions related to a given account, based on the balance on the account and/or cost centers, cost units, etc. This can be used, for example, to calculate payroll tax or vacation pay based on the balance on the salary account. In most cases the source account's balance should not be affected.

You can select the account segments whose balances are to be included in the calculation and the accounts and any other account segments which the auto posting is to be posted to.

By using retrieval codes you can decide which normal and contra account segments are to be used.

- If the account segment(s) are to be identical to the source row, you only specify an asterisk (*) in relevant field(s) for the retrieval code.
- If you only want one specific account segment to be used as normal and/or contra account segment, you enter an equal sign (=) in relevant fields for retrieval code and the account segments in question in the next field.
- You can also enter account segments via a separate translation table. In this case you enter a greater than sign (>) in the relevant fields, which will retrieve the translations via the original account segment.

You determine the required calculation, by either using a constant or a percentage.

17.3.2 Automatic allocation of balances

Auto allocation can be used to allocate an account balance, or the total of a number balance rows, to a number of other account segments. This could be used for accumulated costs that are first posted to accounts without specifying any cost centers, which need to be periodically allocated to profit centers. It could also be used to add together a number of balance rows regarding costs, and allocate them to cost centers according to certain proportions.

You enter the accounts and any other account segments, to which the auto allocation is to be posted on.

By using retrieval codes you can decide which normal and contra account segments are to be used.

- If the account segment(s) will be identical to the source row, you only specify an asterisk (*) in relevant field(s) for the retrieval code.
- If you only want one specific account segment to be used as normal and/or contra account segment, you enter an equal sign (=) in the relevant fields for retrieval code and the account segments in question in the next field.
- You can also enter account segments via a separate translation table. In this case you enter a greater than sign (>) in the relevant fields, which will retrieve the translations via the original account segment.

17.3.3 Fixed periodic vouchers

In Jeeves ERP, you can enter vouchers in advance, interim vouchers, which are posted periodically according to a template. You can use the function to handle recurrent transactions with fixed amounts, for example insurance and other fees.

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17.3.4 Allocations over periods

Jeeves ERP has two functions for allocating transactions of an interim nature over several periods. You can use the functions to allocate prepaid costs, such as rent, insurance premiums, fees or telephone costs, which are invoiced per quarter or per year and need to be distributed over several periods in the accounts.

One of the functions is based on pre-defined allocation templates and the other, direct allocation, is based on a user dialog that poses a set of questions. The user's answers are used to create a temporary allocation template. Both alternatives can be used in parallel.

There are two alternatives for setting up pre-defined allocation templates; one does not exclude the other.

Alt 1: The whole cost is posted directly to an interim account. The allocation template contains the cost account to which the periodic allocation is to be made, and the periods in question. One template must be defined per cost and revenue type.

Alt 2: The whole cost is posted directly to the relevant cost account. The allocation template contains the relevant interim account and to which periods allocation is to be done. There is also a row for transferring the entire cost to the interim account. The same template can be used for different types of costs and revenues.

If you specify an allocation template for a voucher row, or at approval of supplier invoices, interim vouchers are automatically created based on the posting on the row when the voucher updates the general ledger balance. The basis can also be entered and created manually in a dedicated program.

If you use the button for direct allocation, when you click OK, the basis for the periodic vouchers is created. These can be viewed and changed in a node. The amounts for each period can be changed, but the total of the amounts cannot differ from the sum on the account.

17.3.5 Trial balance

Jeeves ERP includes functions for preparing trial balances for the accounts receivable and accounts payable for any given period, also for specific accounts, customers or suppliers if desired. There are reports for printing the trial balance for the accounts receivable and the accounts payable.

17.3.6 VAT/tax reporting

VAT reporting is set up using Jeeves VAT matrix, which is designed solely to meet VAT requirements from various countries. The aim is to make the grid solution as general as possible to allow all countries to build their country specific VAT/tax declarations based on the solution. This allows compilation of preliminary and final VAT/tax declarations and the possibility to split financial periods on a VAT declaration, for example if supplier invoices are received after the declaration has been submitted.

The grid is built up by defining all relevant combinations of:

- markets (represented by customer/supplier categories),
- VAT/tax codes,
- VAT/tax types (goods, services etc.),
- transaction types (country specific definitions) or standard integration sources and
- accounts

each connected to a VAT/tax box in the declaration form. You define either an account or one or more intervals of accounts to which a box in the grid applies. The grid and tax codes can be set up in various ways using VAT/tax types and VAT/tax transaction types, and can be set up for special cases, such as reverse charge on purchases.

If your company has several VAT/tax registration numbers you define them in the VAT reporting units program. These can be in the same country and/or in several different countries. The VAT reporting unit is retrieved to the business transactions and resulting voucher rows, and can be

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selected when creating VAT reports via the VAT grid, EC Sales lists, Intrastat reports or other reports used for internal and external reporting.

On each reporting unit you specify the VAT/tax registration number, the tax office to which the VAT should be reported and in which currency. In an underlying node you can add/view the (ship from) inventory location(s) that are connected to the reporting unit. All relevant inventory locations must be connected to a reporting unit. In another node you define invoice No. series per reporting unit, which is mandatory for companies registered for VAT in several countries. You can also define an item's VAT code per country, if an item is sold from two different reporting units/countries where the VAT/tax rate is differentiated. The amount in the VAT report can be viewed in the reporting unit's currency.

You can automatically set the VAT code on the order and choose the text to be printed on invoices when you are selling goods exempt of VAT. This setting also recognizes if there are areas within a country that are exempt from VAT.

When producing the VAT/tax declaration you can select VAT/tax periods from the accounting periods defined in Fiscal year and periods. Alternatively you can define the periods in your VAT/tax statements, if your VAT/tax periods differ from your accounting periods. You can create VAT/tax declarations for more than one period (for example, a quarter) before you print the report and you can use the function as many times as you want, for example if you have made corrections for a period, as long as you have not set the period to Ready.

You can have different VAT/tax periods for different reporting units. For example, you can have VAT/tax periods 1-12 for one unit and just 4 periods for another unit.

There is also a number of country specific VAT declarations, as mentioned in the section on localizations for the General ledger.

If a user makes mistakes it is sometimes necessary to have the option to re-open the VAT report. There is a function for setting the status of a VAT report back to Preliminary, which gives the possibility to re-create the VAT basis, set the report status to Final and re-print it. This can only be done in the last period for which the VAT report has been finalized.

You can reopen the VAT reports that are balanced and closed. This can be done by counter-posting the VAT balance journal.

17.3.7 One Stop Shop (OSS)

One Stop Shop (OSS) affects EU based businesses that sell to private customers (B2C) in other EU countries. OSS is a way to report tax for sales over €10,000 per year, but it can be done even for smaller amounts.

OSS implies that B2C sellers dispatching their goods from a single country are not required to register for foreign VAT and complete multiple VAT filings in countries where they are selling. Instead, they may opt to simply complete and file a new OSS filing alongside their regular domestic VAT return that will list all their pan-EU sales.

17.3.8 Standard audit file for VAT/tax (SAF-T)

Jeeves ERP includes functions to prepare VAT codes for export of financial data in order to prepare Standard audit files for VAT/tax in accordance with the legal requirements in OECD member countries.

There is also a general program for exporting financial data and creating standard audit files. The purpose is to provide a program that can be used for creating audit files in all countries.

You can enter valid from and to dates for the export definition and can control whether the export should be based on the date interval. You control this on the individual export procedures, depending on the country for which the files are being prepared, for e.g., Norway, Denmark, or Poland.

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17.3.9 Intrastat reporting

Jeeves ERP includes functions for producing the required declarations of dispatches (including two-digit codes) and arrivals between the company's own country and other countries within the EU, including drill down to the invoices included in the dispatches and arrivals totals. This is to make it possible to reconcile the data in the report.

If you choose to use the functions, you can either use the information generated as a basis to manually fill out the Intrastat form or you can compile the Intrastat report in an external report generator. You can also export the data to MS Excel using Jeeves ERP's Dynamic reports. You can then use Intrastat Data Entry Package (IDEP), which is free software that is available on the web, to automatically import your Intrastat reports into the tax authority's web site.

One record is generated per TARIC code, country, dispatch terms, transaction type, Intrastat transport method, and per country of origin for arrivals, based on customer invoice rows and invoice-matched purchase order row. The statistics include only sales and purchases from other countries (in other words where the country code <> your own country's country code) within the EU trading area.

The header includes definition of the report period, data about reporting manager as well as buttons for generation/deletion of dispatch and arrival records for the period. Generated totals are displayed in the header.

You can search in the left or right embedded list for any erroneous records, i.e. records with missing values, and correct them. You can search on "Taric code missing" in the Taric code field, on "Intrastat data missing" in the EU report message field. You can analyze the transactions in more detail by selecting the desired record in the embedded list window and then opening the log node. There are also dedicated programs to make it easier to search among details in Intrastat statistics.

You can view the summarized values from Intrastat report period and generation and European sales list programs in the VAT/tax reporting periods programs.

You can group invoice amounts by TARIC code by checking the relevant checkbox in the VAT/tax reporting units program.

Returns and replacement deliveries are reported in the Intrastat report according to the direction of the delivery, i.e. received goods as purchases and sent goods as sales, and depending on the company's duty to report purchases and/or sales.

From January 2022 when reporting intrastat export sales to SCB, you must send the reports with 2-digit codes and grouping must be done on EU VAT/tax registration number and Country of origin code.

17.3.10 Goods in transit

Jeeves ERP's includes functionality for posting goods in transit, i.e. interim accounting of items that have been shipped by the supplier, but have not yet been received by the customer. One of the two companies must add the cost of the goods in transit to the cost of the inventory that it owns. It is the delivery terms that determines which company is to report the goods in transit as its inventory.

You create the postings manually each month before closing the period.

17.3.11 Automatic clearance of VAT accounts

You can automatically clear the company's VAT accounts at the end of each period, after you have sent the VAT report to the tax authorities. The balance on the clearance account shows the VAT amount to receive or to pay to the tax authorities.

You compile the balances for all VAT accounts from voucher rows in a journal. You review and approve the result before setting the status to Ready and creating the voucher. When you create the voucher, the balances on the VAT accounts are cleared and the VAT balance journal's status is set to Closed. Finally, you can print the voucher from the VAT balance journal if desired.

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17.3.12 Period closing vouchers

There is a function that simplifies the entry of period closing vouchers through reversal, even over the year end.

You create period closing vouchers by specifying in the voucher header that you want to reverse the voucher on a date in another period. Only future periods can be entered manually in the Reversal date field. When you enter the reversal date, the integration source is replaced with Period closing voucher and the voucher number from this series is retrieved on the source voucher as well as the reversal voucher. When you approve the voucher, a reversed voucher is automatically created on the specified reversal date.

17.3.13 Reconciling and closing periods

You can reconcile the integrated modules with the general ledger on screen and/or using reports. You can close periods at any time, for all or individual modules. A period can either be closed as the last step, or as the first step, and then opened only for the user(s) involved in the reconciliations. Jeeves ERP includes dedicated programs for reconciliation work, including:

- Preparations (for example, checking journals from other modules and general ledger vouchers and creating periodic vouchers)
- Reconciliation of work in progress
- Reconciliation of inventory (including calculation of inventory value per end of period)
- Reconciliation of goods in transfer
- Reconciliation of order-specific inventory
- Reconciliation of dispatched, not yet invoiced deliveries
- Reconciliation of received, not yet invoice matched deliveries
- Reconciliation of accounts receivable ledger
- Reconciliation of accounts payable ledger
- Reconciliation of balanced projects
- Reconciliation of fixed assets ledger
- Reconciliation of interim account(s)
- Reconciliation of VAT/tax accounts
- Corrections
- Closing work (including printing VAT/tax reports)
- Closing the period

17.3.14 Liquidity forecasts

There is a liquidity planning function for analysis of planned incoming and outgoing payments already due or that will become due within given time intervals.

You can create liquidity plans for all customers and/or suppliers or for an individual customer or supplier. You choose whether the planning should also include open orders and purchase orders. You can see which orders, purchase orders and invoices are covered by the planning. You can include time and expenses from assignments/projects. You can also enter manual transactions of different types, either one-off transactions such as extraordinary payments, or recurring transactions such as salaries and company tax.

You can calculate opening cash balance on accounts marked as Cash accounts. You can automatically calculate and include VAT amounts from customer orders, purchase orders, upcoming assignment invoices and VAT accounts in the liquidity forecast. Alternatively, you can enter VAT transactions manually in the program node for manual transactions.

You can specify whether the forecast should be calculated in real credit time, i.e. based on the customer/supplier's average (actual) credit time (invoice date-payment date), instead of nominal credit time, i.e. based on the current payment terms on the invoice (invoice date-due date). It is even possible to create a retroactive forecast, i.e. analyze the liquidity for a specific date back in time.

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You can view the forecast in the Jeeves program form, print the forecast if desired, and you can display the forecast in the form of a matrix. A report definition has been prepared for the liquidity forecast in Jeeves ERP report generator and is provided as part of Jeeves ERP packaging.

There is a function for copying the currently selected liquidity forecast to a new one. When you press the button, all the parameter settings and conditions are copied to a new forecast number. You can make any changes to the settings and conditions, then run the calculation by pressing the Create forecast button. You can then delete, adjust or add any individual transactions and re-run the calculation.

17.4 Annual routines

The annual routines consist of creating a new fiscal year, importing and modifying the chart of accounts and generating opening balances.

17.4.1 Opening balances

In Jeeves ERP you can produce opening balances for a new fiscal year in three ways:

- Automatically generate an opening balance based on the previous year's outcome.
- Automatically create an opening balance voucher via Jeeves conversion routine. You can also translate accounts in this routine.
- Enter opening balances manually.

Manual entry is done in the normal voucher entry program but with period 0.

Automatic generation creates an opening balance voucher based on the balance table for the previous year. You can generate opening balances for several companies at the same time. Opening balance can be created several times for a fiscal year. The voucher contains the difference between the previous year's outcome and the current year's previously transferred opening balance, i.e. if an opening balance is created several times, the voucher only contains the changes between the batches. The opening balance voucher is not entered directly in the General Ledger, but in the interim voucher table, where you can make changes before transferring it to the General Ledger.

17.4.2 Change of fiscal year

In Jeeves ERP, you can create and work in an unlimited number of fiscal years. The year can have any number of periods and you can apply split fiscal years if required.

You run a program that copies all definitions on the accounts and other account segments (cost centers, cost units etc.) to the new year. It also copies all tables containing rules for integration with the general ledger.

You can generate fiscal years for several companies at the same time by setting a parameter in the system data for the companies and running this program from the template company to which they are linked.

Jeeves ERP includes functionality for replication between fiscal years to make it easier to manage changes in the charts of accounts and posting rules between fiscal years. This applies to new records, changes and deletions. If you would like to avoid deletion in subsequent fiscal years, check the Do not replicate to FY field before you delete the record.

17.4.3 Import and export of chart of accounts

Jeeves provides functionality for importing chart of accounts and SRU-codes from external sources.

You can also export a chart of accounts to file in order to import it into another database.

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17.5 Report generator

Jeeves ERP's report generator is primarily used to compile values from the accounting tables in the database, for example to produce balance sheets and income statements with different layouts and cash forecasts.

The report generator can handle all tables in the database. The report generator aggregates data by row and column. The report generator is best used in income statements and balance reports, where data must be totaled and presented in matrix format. Other matrix reports, such as sales statistics that are aggregated per district and customer category or item group, can also be produced with the report generator.

Jeeves ERP's in-house developed report generator works with all SQL databases as long as the database, tables and columns/terms are defined.

Besides compiling data from the database, you can also add free texts, own calculations etc.

17.5.1 Sub reports

Reports can be broken down in several levels, for example per cost center or to show certain cost units' share of the values for overlying cost centers. Sub reports are defined separately, and can therefore be reused in several reports.

17.5.2 Report specifications

You use this function when you want to print a specification of one, several or all rows included in a report or sub report. Specifications of account ranges, headings, totals, etc. are defined separately, and can therefore be re-used in several reports.

17.5.3 Formats on numeric values

You can define formats for numeric values in reports. You set the desired formats on the columns in your report definition. Any deviating formats can be determined by setting other formats on the rows in the report. From a drop-down list you can choose between the default formats from the Date and currency formats per company program, the predefined formats for the report generator as well as own defined formats.

17.5.4 Report groups

You can use report groups to increase efficiency in generating the company's reports that need to be produced at regular intervals. A report group consists of several report requests that are to be generated at the same time. For example, you can create report groups for weekly or monthly printouts.

17.5.5 Distribution lists

Jeeves ERP's report generator includes a function for creating distribution lists, i.e. lists of recipients of the reports. If you use this function you can print all reports from a generated batch per recipient and, with up to six selection codes, extract exactly the sub reports the recipient wants, for example only sub reports on the recipient's own cost center.

17.5.6 Presentation of generated reports

Besides being able to print the reports, the results can be stored in unique tables per report definition, which allows you to display them on screen and drill-down to the underlying transactions. The values are stored in their own columns that you can access using normal SQL commands. In this case, Jeeves ERP also automatically creates a program in which you can view and search among the results in the same way as in normal Jeeves programs.

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17.6 Financial statements in SQL Server Reporting Services

Jeeves ERP offers functionality to generate financial statements for a specified company or group, account segments and project, using SQL Server Reporting Services (SSRS). The reports can be based on entities such as companies, which are grouped a dedicated Jeeves program.

It is possible to show Profit and loss and Balance sheet data respectively for twelve consecutive months (rolling 12-month) regardless of which base month is selected, from the current month and twelve months backwards in time, or starting earlier than the current month. For Profit and loss, the report contains information for years/periods as specified and an additional column with the accumulated value for the 12 periods.

17.7 Budget

In Jeeves ERP you can enter budgets on all levels in the business, cost and project accounts. The budget routine uses the chart of accounts in the general ledger for each year, which means that the account's budget rules, account segment rules etc. are also applied. With the use of allocation curves the budgeted amounts can be allocated over the year’s periods according to predefined rules.

A budget can be created/entered in three ways:

1) Enter budget records for the fiscal year with simultaneous allocation over the year's periods.

2) Load/enter "preliminary" budget records for the fiscal year and then allocate over the year's periods and convert to "normal" budget.

3) Enter budget details without using allocation curves.

If you use automatic allocation you can choose between percentages per period, with automatic allocation of the remaining amount if required, or even allocation over the periods.

17.7.1 Budget revisions

You can define budget revisions according to own requirements. You can use budget revisions to identify which revision budget transactions belong to, and to update the budget balance table for a certain revision, as well as to lock a budget after updating the balances.

17.8 Import/Conversion

Jeeves ERP's conversion routine is mainly used to load data from external systems, for example to load accounting data before implementing a new Jeeves installation. You can, however, use the routine for other purposes.

Conversion is governed by tables and means that the external accounting material is changed to Jeeves record layout, after any account translations, according to the company’s posting rules.

Conversion is made by first describing the layout of the source table (the external system file) and its terms, and then describing the processing, i.e. the separate steps in the conversion, any conditions for conversion to take place as well as a description of how the terms in the source table should create terms in the Jeeves ERP record.

The conversion creates a voucher header that shows (among other things) the posting’s total balance and individual voucher rows. Compression can be made, which means transactions with the same posting are aggregated. The result of the conversion is stored in an interim table where the transactions can be accessed for correction. When any errors have been corrected you transfer the interim vouchers to general ledger.

17.8.1 Company consolidation

You can use the conversion routine for group consolidation. The counterpart term can be used to eliminate transactions between companies within the group.

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17.9 Other functions

17.9.1 Revaluation of the general ledger

There is a function for revaluation of the general ledger at the current exchange rates and posting of exchange rate differences based on posting templates.

You can run revaluation with or without reversal of the revaluation postings in the next period.

All accounts that have been flagged for revaluation in the chart of accounts are included in the revaluation, with the exception of accounts receivable and accounts payable accounts, which are revaluated separately.

17.10 Localizations/internationalization

Most functions in Jeeves General ledger are common to all countries. There are some, however, that are specific to certain countries, as described below.

17.10.1 Import/export (Sweden only)

There is a function for importing SIE files of type 4. You can either create a new company with a chart of accounts, posting templates etc. based on a template company in connection with the import, or you can import only vouchers to an existing company.

You can either import vouchers directly into the general ledger or load them into the interim voucher table first and transfer them to the general ledger later. The latter makes it possible to search for and correct erroneous vouchers before they are transferred to the general ledger.

17.10.2 Tax declaration (Sweden only)

There is a function to create and store Swedish tax declaration file in your system for upload to the Swedish tax agency. You can use the relevant button in the VAT/Tax reporting periods program to save the file in XML format and use it for electronic filing of the tax declaration. You can add any additional information in the file using the text editor.

17.10.3 Standardized account statement (SRU)

SRU coding is a localization for the Swedish market but can be used in any country for own grouping of accounts.

You can link the accounts in the chart of accounts to the account groups in the standardized account statement, either account by account or via a help table that works with intervals of accounts.

Jeeves ERP provides a standard chart of accounts and SRU codes, in both English and Swedish along with functions for import and export. The chart of accounts and SRU codes can be imported in new installations and when creating new fiscal years.

Each account can be linked to up to 3 different SRU codes. SRU code 2 and 3 can be used if an account occurs in several SRU groupings. If you have loaded the basic chart of accounts, there is a large number of standard records for the earnings statement for incorporated companies, financial institutions, etc., already filled in.

17.10.4 Account mapping

You can capture account mapping details, for example: for mapping Norwegian account classes against Swedish ones if the Swedish chart of accounts is used by Norwegian companies running SAFT export. Example: For type 1, SAFT Norway, for Swedish accounts 26xx you enter the corresponding Norwegian account class 27xx. This will be set as StandardAccountID 27xx in the Norwegian SAFT file.

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17.10.5 Non-VAT transactions

Originally developed for Norway, but available to all users.

The general ledger can be set up so that voucher rows from non-VAT related transactions are kept free from VAT related information. This is a requirement from auditors in some countries, such as Norway.

17.10.6 Standard Audit File for Tax (SAF-T) – Norway

All Norwegian VAT returns must be filed based on the SAF-T code format. The VAT return must be in XML format and is based on the Standard Audit File for Tax (SAF-T) code list. This is a legal requirement to be followed and used by all companies in Norway from 2022.

It is possible to create a SAF-T file with anonymized accounting and base data for testing with the Norwegian Tax Administration (Skatteetaten). This anonymization is required, otherwise the test file will be rejected by Skatteetaten.

Jeeves has a program where you can define VAT/tax report lines that can be used to make a one-to-one identification of which data is reported and with which VAT SAF-T code in the output XML file. These report lines are similar to the list of VAT/tax rates but may have separate lines for input and output entries.

If you have linked VAT/tax boxes for the respective Norway VAT report line and defined the VAT/tax declaration for the VAT boxes, you can filter records and view records on the Norwegian VAT report line. This is useful when reporting VAT for Norway to correct any historical VAT reporting submitted with errors.

17.10.6.1 Corrections in VAT reports (Norway)

All Norwegian VAT returns must be filed based on the SAF-T code format. The VAT return must be in XML format and is based on the Standard Audit File for Tax (SAF-T) code list. This is a legal requirement to be followed and used by all companies in Norway from 2022.

From 2022, it is not possible to submit an additional VAT return where only the details to be corrected or added are included when reporting VAT for Norway. To correct the historical VAT reporting, the taxpayer needs to do the following as required to replace the previously submitted return:
- original reports must be canceled and re-sent in full, including corrections
- VAT amount from adjustment or sale of fixed assets must be included in the VAT report.
- when requested, a customer should be able to separate parts of the original report and send it to the authorities.

Jeeves includes support for these country-specific reporting rules.

17.10.6.2 Fictitious import VAT

In the SAF-T file for Norway, fictitious import VAT is reported according to the Norwegian Tax authority based on setup in the VAT/tax program.

17.10.6.3 Norwegian income statements account codes

You can define a Norwegian Income statements account code; for example RF-1167 for Income statement 2. The value you enter in this field is mapped to the Norwegian income statement account and used as the basis to display the account code in the output SAF-T file.

Currently the following codes are supported:

- Standard for SAF-T Norway standard
- RF-1167 for SAF-T Norway RF-1167 (Income Statement 2)

The master files for SAF-T Norway contain checks to ensure that the accounts that appear in the master files are limited to accounts that are part of a transaction or that have an inbound balance and/or an outgoing balance. They include all customers with any invoice transaction during the

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selected interval, or those customers that have opening/closing balance in the selected period in the master files that are submitted to Norwegian Tax Administration (Skatteetaten).

17.10.6.4 Anonymized data for Norwegian SAF-T files

When creating the file for Norwegian Financial SAF-T with accounting data for testing with the Norwegian Tax Administration (Skatteetaten) via the Norwegian digital hub Altinn, the file must be anonymized. You can do this when defining the SAF-T data definition in the Standard audit file - tax program. Once you have done this, the file is anonymized when you export it; i.e., it will not contain any information that may identify the sender or any third parties, for instance, personal information like individual names and addresses and contact information or company names and information about companies, or account numbers, IBAN-numbers, etc.

17.10.6.5 Recovery of VAT/tax on bad debts and reversal of VAT/tax deductions

Other localized functionality for Norway is the possibility to show invoices as bad debts in the company's submission for recovery of VAT/tax payments made on the invoice from the tax authorities. Another is support for reversal of earlier VAT deductions on fixed assets.

17.10.7 Standard Audit File for Tax (SAF-T) – Denmark

The Danish Accounting Act requires accounting systems to support the use of the SAF-T international standard file format. Jeeves ERP includes functionality to extract the SAF-T file according to specifications from the Danish Business Authority.

17.10.8 e-Invoicing and business response messages – Denmark

Jeeves ERP includes support for eInvoicing and for sending and receiving business response messages in Denmark through NemHandel. The eInvoice s are transferred and converted to the relevant format via Pagero.

17.10.9 VAT refunding

Originally developed for the UK, but available to all users.

Jeeves ERP includes functionality for VAT refunding on customer invoices. See the chapters on Jeeves Sales and Jeeves Accounts receivable for details.

17.10.10 Journal groups

This function was originally a localization for the French market but is available as standard.

Integration sources can be grouped into journal groups. This group number and the associated description are stored on the voucher row and are intended for selection purposes. For example, you can enter each bank as a payment method for outgoing payments and the bank's Journal group number on the payment method. You can then select all voucher rows related to a certain bank within a voucher.

17.10.11 Journal entry

The Journal entry program is a preparatory system to the ledgers and the general ledger. You use one and the same program to enter different types of transactions, general ledger transactions, customer/supplier invoices, incoming/outgoing payments, etc., and then transfer these to the general ledger by running a journal. When doing so, records are created in each relevant module, i.e. an invoice in the accounts receivable ledger, a supplier invoice and associated approval record in the accounts payable ledger, an incoming/outgoing payment journal, etc. This is a common way of handling financial transactions in certain countries, for example in Norway.

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17.10.12 Cash book processing

There are two solutions for cash book processing:

- The cash book processing makes it possible to run several cash books for incoming and outgoing cash payments. Different cash books may, for example, pertain to different cost centers or currencies.
- The variation of cash book processing, originally developed for the Polish market, makes it possible to run several cash books for incoming and outgoing cash payments with increased functionality for definition, numeration, posting, and reconciliation of cash books.

Cash book processing is an extension of the Journal entry function.

17.10.13 Import and matching of bank statements

There is functionality for importing bank statements including customer and supplier payments and other bank transactions, matching the payments against open invoices and integrating the bank statement with the general ledger. The function is used in countries such as Germany and the Netherlands.

There are two alternative solutions; one for companies using the Journal entry program and one for "normal" integration between the accounts receivable/payable ledgers and the general ledger.

Jeeves ERP attempts to identify open customer/supplier invoices using the bank account number, invoice number, payment amount etc. You check the automatically matched rows and continue manually matching the remaining rows. You can also manually add rows for, for example, advance payments, interest or bank charge (if not imported), currency transactions and the like. If you enter an advance payment in this way, an advance invoice is automatically created from it.

When all the rows on a bank statement have been matched and the amount reconciled, the bank statement header is assigned the status Can be flagged ready. Only then can it be integrated with the general ledger.

Electronic bank statement files that comply with the CAMT.053 standard (countries who are a part of SEPA) can be imported, matched and integrated with general ledger.

17.10.14 FIFO valuation of currencies

In some countries, such as Hungary and Poland, it is a legal requirement that acquisitions of foreign currencies be valued according to FIFO (First in first out). The FIFO value is then used in outgoing payments in currency, as well as related costs. This means that outgoing payments are made at the earliest exchange rate. If you have more than one bank account for a currency, the FIFO value is calculated separately for each account.

Jeeves ERP creates a new FIFO balance or updates an existing record with the same accounting date and currency rate when integrating incoming payments with the general ledger. This applies to:

- incoming payments from customers
- manual incoming payments, for example, bank interest and
- credits from suppliers

During outgoing payment Jeeves ERP uses the earliest balance first and locks the relevant balance record(s). This applies to:

- outgoing payments to suppliers
- manual outgoing payments

There is also a program for printing information about the available currency balances on the bank account(s).

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17.10.15 1099 reporting (USA only)

In the United States, companies are required to send a 1099-MISC form to any external consultants and subcontractors who are paid more than a specific amount per year. Jeeves ERP includes functionality for setting up the 1099 reporting categories and thresholds, and for printing the report.

17.10.16 GDPdU export (Germany only)

Jeeves ERP includes a program for exporting financial data for auditing purposes in accordance with legal requirements in Germany. The data is exported in a format that is readable and processable in applications used by all financial authorities in Germany. The format is called GDPdU, which in English stands for "Principles of data access and auditing of digital documents".

17.10.17 Making tax digital (UK only)

Making Tax Digital (MTD) is a key part of the United Kingdom's plans to make the tax system easier for individuals and businesses to streamline tax administration. MTD introduces digital record-keeping for VAT-registered businesses in the UK. Customers of Her Majesty's Revenue and Customs (HMRC) will use digital record-keeping software to interact directly with government systems via the MTD APIs.

Jeeves will receive permission to access and send data digitally to HMRC. Once an account is registered with login information for HMRC online services using the VAT/tax number of the business, VAT/tax reports can be sent from Jeeves to HMRC. Relevant information is logged and saved in Jeeves to allow you to keep track of data relating to the submission of the VAT report to HMRC.

17.10.18 Reports in e-SFS format (Poland only)

Poland legally requires companies to prepare financial statements in electronic form with the specified logical structure and format (e-SFS). These reports are also available in .xml file format.

17.10.19 JPK_VAT reporting (Poland only)

From 1 October 2020, Poland introduced the JPK_V7M file for taxpayers to submit monthly VAT returns and JPK_V7K file for taxpayers to submit quarterly VAT returns. This is a periodical filing that will replace the existing JPK_VAT files and the VAT return declaration. The JPK_VAT is an extensive form which includes classifications of transaction types and codes to identify certain type of goods and services. A new version of the structure of the JPK_VAT invoice has been published for tax books and accounting documents which is effective from 1 January 2022, Jeeves supports the updated format with suitable fields and format updates.

The files will provide information on:

- split payments
- commodity groups depending on the type of item/service being sold - for both sales and purchase transactions
- procedures and transactions types

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18 Jeeves ERP Fixed assets

The fixed assets ledger in Jeeves ERP consists of routines for entering and managing fixed assets, for depreciations and for integration with the general ledger.

18.1 Fixed assets management

18.1.1 Entering fixed assets

Fixed assets can be entered manually or automatically from supplier invoices or from manual vouchers. If you use automatic entry you can activate cost accumulation for existing fixed assets. For both methods you can apply automatic numbering of fixed assets based on fixed assets types.

18.1.2 Fixed asset types and groups

You can handle all fixed asset types (buildings, machines, fixed assets). You can also divide them into any number of fixed asset groups that can be used when selling fixed assets, creating depreciations and postings etc.

18.1.3 Fixed asset areas and locations

There are programs for defining areas and locations of fixed assets according to own requirements. These can represent buildings, departments, rooms, working areas or the like.

18.1.4 Insurance management

There are fields that can be used as the basis for insurance management.

18.2 Fixed assets transactions

18.2.1 Fixed asset transactions

There is a large number of standard transaction types, which can be entered via a journal or directly:

- Entry of new fixed asset
- Change status of fixed asset
- Division of fixed asset
- Addition costs from supplier invoice
- Discard of fixed asset
- Sale of fixed asset
- Movement of fixed assets (placement only)
- Movement of fixed assets to other account and/or account segment(s)
- Change in acquisition value
- Change in residual value
- Change in depreciated amount

18.2.2 Direct entry

There is a simple transaction entry form in which you enter the fixed asset ID, choose a transaction type, enter the amount and any other information required for the transaction type in question.

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18.2.3 Entry journals

There is also an optional, overlying form for transaction entry via a journal. This is a legal requirement in certain countries. The entry journal gives better control of fixed assets transactions. Each transaction must be printed out, also the journal must be approved and its status set to Ready before a transaction journal can be created and integrated with the general ledger.

18.2.4 Integration with general ledger

Fixed assets transactions are posted automatically according to posting templates, and integrated with the general ledger, by running a report program.

18.3 Depreciations

Recorded depreciations can be made according to plan or according to an own depreciation method. They can be done from the beginning of the year, from the acquisition date or an activation date. Fixed assets are automatically depreciated if sold or discarded.

Fixed assets depreciation can be scheduled as a job considering depreciation methods and accounting date. A journal is created for each asset method in the Fixed assets depreciation -> GL table.

18.3.1 Recorded depreciations

Depreciation according to plan can be calculated and created based on the economic life of the asset (specified on the asset group). The fixed assets that do not have any calculated depreciation are:

- rented/leased fixed assets,
- assets with no residual value and
- assets that are acquired after the date to which the depreciation refers.

Depreciation is calculated from the period the asset was acquired and up to and including the period before it is sold/discarded. A fixed asset that is acquired at the end of one month will therefore be depreciated for this month but not for the month during which it was sold/discarded.

Jeeves ERP also keeps track of when the Fixed asset was last depreciated, so the next calculated depreciation will cover all periods that have elapsed since then. If the fixed asset has not been depreciated yet, the year and period of the acquisition date is used to calculate the number of periods.

18.3.2 Depreciation periodicity

For each fixed asset group you can set up the desired depreciation periodicity:

- Annually
- Biannually
- Every four months
- Quarterly
- Monthly

You can choose between degressive and linear depreciation, or accelerated depreciation (highest of degressive and linear).

18.3.3 Simulating depreciations

You can simulate future depreciations, which gives you the possibility to be more efficient in the budgeting and forecasting process. In the current version, simulations only cover forecasted depreciations for fixed assets with a linear depreciation method, according to plan and monthly depreciations.

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18.3.4 Integration with general ledger

Depreciations are posted automatically according to posting templates, and integrated with the general ledger, by running a report program.

18.4 Ledger information

18.4.1 Reconciliation of fixed assets ledger

There is a program for reconciliation of the fixed assets ledger balance against the general ledger per specified date. In the list displayed you will see the relevant fixed assets and at the bottom a number of totals, calculated values as of the specified date including fixed assets transactions that have not updated the general ledger, also the values as of the specified date that have updated the general ledger, as well as current values. The totals shown are:

- acquisition amounts
- depreciated amounts
- residual value

18.5 Localizations/internationalization

Most functions in Jeeves Fixed assets are common to all countries. There are some, however, that are specific to certain countries, as described below.

18.5.1 Local legal requirements regarding depreciations

There is a number of local legal requirements that are specific to certain countries but can be incorporated in Jeeves ERP by setting parameters. One example is in Germany where the current maximum depreciation percentage is limited to 25% per year by law, but an additional rule says that this amount is not allowed to exceed the linear depreciation times 2.5. A second example is "Zero" depreciations, used to retain records of fixed assets that are never really fully depreciated, for example Forest. A third example is depreciations ending at the end of the acquisition year. The monthly depreciation amount is calculated in proportion to the number of months remaining until the end of the acquisition year.

18.5.2 Tax depreciations (Sweden only)

In Jeeves ERP, there are functions for calculating and performing tax depreciations and integrating them with the general ledger.

Recorded depreciations are performed as planned and are integrated with the general ledger every month. At the turn of the year you can make a comparison between recorded depreciations and tax depreciations. Differences may arise between recorded depreciations and tax depreciations. If you wish to show the same value of fixed assets in your annual closing report as in your declaration, you can record over- and under-depreciations as appropriations.

This can be performed according to one of two rules and you can choose the one that gives the lowest outgoing value of the fixed assets, and thereby the greatest possible depreciation:

- Principle rule (also called 30 % rule) means an annual depreciation of maximum 30 % of the fixed assets' posted residual value.
- Supplementary rule (also called 20 % rule) means an annual depreciation of 20 % of the fixed assets' total requisition value.

The form is designed to comply with the tax authority's calculation basis for the principle rule on the left-hand side and the supplementary rule on the right-hand side.

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18.5.3 Half year straight line (linear) depreciations (US and Canada)

You can apply half year, linear depreciations, a method that is a legal requirement in the US for property other than real estate. The half year convention means that, irrespective of which month of the year you begin using the property, you must treat it as if you began using it in the middle of the year. Only half of the full-year depreciation is allowed in the first year, with the remaining balance being deducted in the final year of the depreciation schedule, or the year that the property is sold.

18.5.4 Negative acquisition amount

In some countries, negative fixed assets are used for accounting purpose. To accommodate this, you can enter fixed assets with a negative acquisition amount and save the record where required.

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19 Jeeves ERP Multi-company administration

In Jeeves you can quickly and efficiently set up multiple companies in the same database or in different databases. You can process and analyze transactions and produce reports for multiple companies within your organization.

19.1 Multiple companies in same database

All applications can be made available to any number of companies in the same database. New companies can be set up at any time, even during production.

Jeeves ERP provides a definition of tables to be copied to newly created companies. Almost all tables in the database are represented here, which means that additions are only necessary for own developed tables. All tables defined here are by default flagged for copying, including own developed tables, and for "normal" users this is only editable for own tables. System administrators can, however, edit all table definitions.

Here there is also a checkbox for flagging tables that CAN be replicated when entering, changing and deleting data in the template company. Specify which of these SHALL be replicated in the Common tables/companies program.

19.2 Company access

There is a program for administration of access rights to companies in multi-company databases. There is also a program, View user access to companies, you can use for viewing access rights across all companies. By default, all users have access to all companies, but if you grant a user or group access rights to a specific company, the user/group is blocked from all other companies.

19.3 Switching companies

It is quick and easy to switch company, either via drop-down menu or a company code. You can have different background colors for different companies in the database. Background settings are especially useful for differentiating between companies in order to prevent users from doing things in the wrong company.

19.4 Settings when switching companies

Settings can be saved to a specific company, which means that program menus, forms, lists, etc. can have different designs when you switch company. You can also use the ReadSiteSettingsOnCompanyChange kernel parameter to display date, time, numeric and monetary values in different formats when you switch company.

19.5 Multi-company searches

There is an option on the search menu that you can use to search in several companies at the same time. Each selection of companies is saved in a list under this menu. You can re-use your selection of companies in several searches simply by checking the selection and conducting a new search.

19.6 Template companies

A template company is used to simplify creation of new companies. You can have one template company per type of company in your area of business, for example: Manufacturing, Distribution, Service, Sales etc. Data that is to be common for several companies is entered into the template company, for example: chart of accounts, cost centers, etc. When a new company is created the data is copied from the template company to the new company. You can then decide if new entries, changes and deletions are to be replicated from the template company to the other companies.

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All tables in the database can be copied, which is useful for quickly creating training or test companies with realistic transactions and history. There is also an option (per table) to automatically update production companies with changes made in the template company.

19.7 Multi-company fiscal year routine

You can set up simultaneous creation of new fiscal year and opening balances for all or several companies with the same template company.

19.8 Defaults

In the system data you can set up global and company specific defaults such as default inventory location, default salesman etc. Much of the data that is entered here will be given as proposals when working with Jeeves ERP but can be changed.

19.9 Multi-company reports

There is a program for definition of the reports that can be printed/run simultaneously for several companies. The information you enter here is global, so you only need to enter the reports in one of the companies. You can run reports across several companies, for instance run summary reports such as account balance reports, and get the sum on the selected accounts for all companies. You can also save these company selections.

The user must have a right to work in the company in question. Also, a template must be set up for the report with the same name, but not necessarily the same appearance, in the relevant company. If you want the templates to have the same appearance, use the Copy report templates program to copy templates between the companies.

19.10 Report templates

You can set up company specific report templates or set up common report templates via a template company.

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20 Jeeves ERP Document management (EDM)

Jeeves (electronic) document management (EDM) includes functionality for entry, structure and classification of documents, as well as for searching, version management, authorization control and logging. You can also create, display, print and send files with e-mail directly from a Jeeves form. Through integration with other Jeeves modules, you can link a document to, and thereby make it accessible from, the business object or transaction (for example: customer, order, invoice etc.) from which the document originates. You can select a linked document directly from a drop-down menu on Jeeves ERP's menu bar.

Every document is represented by a record in the database that points to the file that constitutes the document in question. The files are stored on a central file server. Links to related business objects are specified on the document record. For each record a document type is also specified. This functions as a logical grouping of documents with the same content or function. The document type also specifies the directory where the files are located and which file types the documents may consist of. The file type specifies which programs may be used to open or change files of the specified file type.

A document can also be related to another document. In this way you can create document structures with several child documents related to a parent document. A document structure that refers to items can be converted into a BOM which can then be used in Jeeves Production module.

20.1 Basic document management

The basic functions/steps in Jeeves Document management are:

- create a document record
- create a file linked to the record or
- retrieve an existing file and link to the record or
- copy an existing document record with associated file to a new record with a new file

There are buttons for these actions. The relevant buttons are activated depending on the step being performed. You can also enter files using drag-and-drop.

You can associate programs, browsers and directories with document and/or file types. You can control which programs are used to create, edit and view documents, and for MS Word and Excel, which template is used for new files.

Besides the automatically generated information, you can set document properties such as own document ID, description, author and links to business objects or transactions such as item or item group, company, contact, project, customer order, manufacturing order, campaign, quotation, etc. If you jumped to the document management program using the Document management button, the link(s) to the current business object or transaction will automatically be filled in.

You can set up your own categories for categorization of your documents, for example to which organization or process they belong.

20.2 Print to EDM

You can define that certain reports should be printed directly to EDM. On selecting the report from Jeeves ERP, the report is generated in the desired format and automatically linked to the record. You can then view and print the report as and when needed. You can also print multiple records to PDM simultaneously.

The CheckPrinttoEDM parameter allows you to enable/disable the Print to EDM functionality. You specify the directory for the document type to be printed to EDM in the rules for printout.

The generated files are saved with the following naming convention: 'Report Template Name _ Record Identifier _ DateTime Stamp'. For example: rfaktfu_10176_290915122051.pdf.

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20.3 Document structures

You can use document structures to:

- organize multiple documents consisting of several records, for example drawings,
- gather associated documents, for example, documents which originate from a certain project, and
- build directory structures for easier navigation.

A parent document can in turn constitute a child document in relation to another document. In this way you can build structures at several levels. Copying business object data from the parent document to the newly created child document is controlled via a parameter.

There are embedded lists that show the structure relationships. The Consists of list displays all document structure records where the overlying record constitutes a parent document, while the Included in list shows all document structure records where the overlying record constitutes a child document. To create a more easily navigated interface, the Consists of list can be displayed as a graphical tree structure.

You can even create template structures and base new document structures on them. New document records are created in the PDM table where the Document ID, Document Description, Short note and File name fields are copied from the template record. No business object links are set for the new document record.

20.4 Version management

Each new version of a document corresponds to a new file. Version management is performed through reference to the unique ID of both a previous document record and the original record. Which version is valid is indicated by a check in the Valid PDM checkbox. There is a program that only displays valid document records, i.e. current revisions and editions. A second program shows only previous revisions and editions. Otherwise, both programs contain the same functionality for creating and viewing documents and structures.

A new version of a document corresponds to a cloned record where in principle only the unique ID, previous ID, version name and validity periods differ. Business links and other document data are the same because the new version fulfills the same function as the previous one.

Version increment is performed in two steps, a new database record is created and a new version of the related file is created. Any file link to the newly created record must be entered using drag-and-drop. The previous version does not automatically become invalid when a new version is created; you check the Valid PDM box manually. An alternative version increment method, with slightly different functionality, is check-out and check-in.

To make it easier to keep track of version changes there are two alphanumeric terms, version and revision, each linked to an underlying index, which facilitates gradual, serial increments. The user sets up alphanumeric or numeric series, and determines when and how these should be used. A common method is that Revision increases for every individual change while Version is increased for more extensive changes. Validity periods are linked to both terms. Date and signature are automatically set on version and revision changes.

In order to guarantee that documents are only changed by one person at a time they must be checked out. At check-out, a copy of a document (file) is created in a directory specified for the login signature. Checked out documents can only be edited by the check-out signature. On checking in, a new document record and file are entered with the document record's Unique ID, in the directory specified for the document type. Alternatively, there is a setting to receive a prompt at each check-in asking whether or not the previous file should be overwritten.

A version change can also be relevant for a document structure, on a change of an underlying document or on the addition of an underlying document. You can set that the document structure is complete, which means that included document structure records cannot be changed, added to or deleted.

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20.5 Access rights

You must set up all users or resource groups who are authorized to use the Document management module. Users' access rights can be controlled at two levels:

- document record - the right to view records in the EDM programs. This can be controlled using Restrictive views or default search terms. Suitably, you control which document types various user groups have access to.
- file - access rights to files. This is controlled at role level per document type. The following rights can be controlled:
  - create file
  - edit file
  - delete file
  - read file
  - publish file
  - approve file

20.6 Approval

Document approval is based on a file having to be approved by an authorized user before it is accessible to other users. A checked-in file whose document type is marked for approval will not become valid until it is approved. Consequently, it is not immediately viewable in the Document, valid program. It can, however, be viewed by authorized users in the Document approval program/node. There are checkboxes that indicate if the document record:

- is valid
- is under approval
- requires approval
- has a file linked

When approving, if there are several different versions of the file under approval, the approved version will be set to valid and the others will be deleted.

20.7 Logging

You can log both document access and version management. You define what is to be logged in the EDM activity types program. The documents which are logged are controlled per document type. The following standard types are supplied.

- Created
- Checked out
- Checked in
- Opened
- New version
- Printed
- By E-mail

You can also set up your own types, but then you must also define your own logging method.

You access the log via a node in the document management programs.

20.8 File management

Documents can be stored either centrally or locally (but not on local machines and disks). One of the main advantages with document management is a central storage site, which is why this is usually preferred. The file status specifies whether a file exists for the document record and if so, if it is local or central, or has been archived.

For centrally stored files you can specify directory paths for document types either with mapped letters or as UNC paths. You can specify an absolute path to the server and directory under which

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all directories that are defined for the different document types lie. In this way it is easy, for example, to move all documents to another directory or file server without needing to change the directories for the document types. Even if this parameter is set, you can specify directories with paths to mapped drives. Jeeves ERP checks first whether a document type directory refers to a mapped drive.

For locally stored files, the easiest way to enter local file links is using the Get file button or drag-and-drop. The file pointer is dropped in the Document file path field. After the field is validated, a file name and file extension are filled in automatically.

There are also functions with which you can:

- move files from local to central
- delete central files
- copy files to the desired directory
- create and delete directories directly from Jeeves ERP

All file transfer is done with Windows file management routines. Encrypted streaming is also supported as an alternative to file transfer. This allows a higher security level to be set and files become accessible via the Internet.

You can zip and archive files recorded in the document management module, in order to increase efficiency in file management and backup if your company has large numbers of files and images.

The following three functions can also handle files that have been zipped and archived:

- Show file
- Mail file
- Print file

20.9 MS Office integration

Jeeves ERP includes integration with Microsoft Word, Excel and Outlook. You can specify a template for document types linked to Microsoft Word or Microsoft Excel. When you create a new document using the Create file button, Word or Excel opens with an empty record as an instance of a specific format template.

You can save Word documents and Excel workbooks directly to Jeeves EDM to a certain document type. You can also save and send mail using MS Outlook. You can either save an e-mail in Outlook's msg format or as a text file, with sender, date, addressees, subject and content, in a specified directory. Attached appendices can be saved to other directories in their existing format. Appendices can also be entered as document structure records linked to the mail document.

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21 General functions and features in Jeeves ERP

The functions described below are available and applicable throughout the whole system.

21.1 VAT/tax management

21.1.1 VAT/tax rates and declarations

You can have any number of VAT rates, controlled by customer, supplier, item, or entered manually. Jeeves ERP automatically calculates VAT amounts with or without consideration of early settlement discounts. VAT codes can also control postings. You can also set up sales taxes for USA and Canada.

Jeeves ERP has a general solution for all countries' VAT declarations. There is also a number of country specific VAT declarations described among the localizations for the General ledger.

21.2 Currency management

21.2.1 Currencies and rates

You can have any number of currency codes and rates controlled by date, current rates which are used in invoicing, in ledgers and so on, and calculation rates for calculating cost prices and so on. This is mutual functionality for sales, purchases and service, in ledgers and the general ledger, for projects etc.

21.2.2 Switch currency

It is possible to switch currency on-the-fly, i.e. show amounts in any currency according to current rate (amounts are always saved in base currency).

21.3 Language management

21.3.1 Language management when printing external documents

You can enter item descriptions, units, payment terms, delivery terms, delivery methods, currency clause texts, general texts etc. in different languages. Jeeves ERP automatically selects translations when printing external documents depending on the customer’s/supplier’s language.

21.3.2 Language management of data

Jeeves ERP even includes unique functionality for translating standard/own drop-down list records and other data that needs to be translated such as account names, payment terms, activity descriptions, etc. This is used if, when switching language in the client, the language of drop-down lists and certain data is also to be changed.

21.4 EU sustainability regulations

Jeeves ERP includes functionality to support companies in the EU to comply with sustainability regulations. You can set a markup in percent on TARIC codes for Carbon Border Adjustment Mechanism (CBAM) and the CBAM cost will be calculated per purchase order row. You can also set a CO2 equivalent value per unit or kilogram on items for the Corporate Sustainability Reporting Directive (CSRD). Total CO2 equivalent values are shown per sales order row and purchase order row.

Jeeves ERP also includes support for the EU Deforestation Regulation (EUDR), with functionality for managing Due Diligence Statements and flagging relevant TARIC codes and trading areas.

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21.5 Time zone management

Jeeves ERP includes functionality for time zone management, which allows you to set up time zones for companies in the database, as well as for users, in a different time zone from the database server.

The purpose is to set dates and times in local time where each transaction takes place, and not according to the database server's clock. For example, if the database server is located in Sweden and a user in the USA picks a customer order, the picking transactions shall have the local time according to the relevant time zone in the USA.

21.6 Change log

You can log new entries, changes and deletions. You can also control whether these events require a reason code and/or a password. For further information see the overview of the System administration functions.

21.7 General import/export

Jeeves ERP's import function can be used for importing files of various formats, for example import of cost prices from an external application, freight costs from carriers etc. First, you create a table in the database in question to hold the imported data. You enter an import definition which, among other things, specifies the path and name of the data file to be imported, and into which table in which database, as well as which column in the table, the data should be imported. You save the file to the folder in question and check its content/format. You import the file by running the Import report.

Export is done in a similar manner. First, you enter an export definition which, among other things, specifies which table in which database data will be retrieved from, and where the work file should be saved, if you want to create one. If you are exporting in the SIE format, SRU codes must be entered or imported by installing Jeeves ERP base chart of accounts. Also, if other dimensions than 1, 2 and 6 - CC, CU and project - you must define these SIE dimensions. You are then ready to either export data to a work table and from there create a file, or you can create a file directly via the mail function.

21.8 Import external data to Jeeves on cloud without VPN connection

Jeeves ERP allows file handling to work without a VPN connection using the Definition and import external data->work table via dialog program.

This can be files to be imported via the conversion routine, usually financial data for the General ledger, or it can be files for incoming and outgoing payment receipts etc. You can also enter own procedures that handle the imported data for further processing in the system.

This program uses the Import file via dialog function to open a file dialog that enables you to navigate to the desired folder and select the import file specified in the In-/output file field to update the database.

Once you have selected the file to be loaded into the table specified in the Value term table name field, the stored procedure specified in the Procedure, after field is run to process the data.

The updated information can be viewed as follows:
- in the Incoming payment file header program for BGMAX in text format and Camt.054
- in the Bank statement header program for CAMT.053 in text and XML formats
- in the cloud environment, imported supplier invoices can be viewed in the Review imported supplier invoices program by clicking the View pdf invoice button.

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21.9 Export files to local system from Jeeves on cloud without VPN connection

When running Jeeves in cloud environments, a VPN connection is necessary to access and export files to your local system. However, for those who cannot use a VPN connection, Jeeves ERP allows file handling to work without a VPN connection using the Export Jeeves data -> file via dialog program.

When you want to start the export, click the Export file via dialog button to open a file dialog that enables you to navigate to the desired folder and then enter the new .txt file name or select the existing .txt file to identify the file to be created/updated with the exported data. Once you have entered the new .txt file name or selected the existing .txt file, the data will be updated from the table specified in the Value term table name field based on the SQL condition mentioned in the DB term/expression 2, where, and Sorting/weighting by fields. The filename and path are updated in the In-/output file field once the export is completed. Note that you can browse and select a different file than the one specified in the In-/output file field and this is then saved as the default file and file path.

21.10 Jeeves Message integrator (JMI)

There is a general integrator called Jeeves Message integrator (JMI). The aim is to make the JMI solution as general as possible to allow integration with any type of external application where the integration is based on XML messages. Besides XML files, JMI can also handle other file types, such as flat text files and Excel files, by using plug-ins. At present, JMI is used in the integration with the Sales configurator and Astro WMS, and for general export and import via Microsoft Excel. The latter can be very useful if you need to process data in Excel and then import the processed data back into the Jeeves database.

JMI is comprised of programs for:

- setting up the integration installation
- mapping incoming and outbound XML messages
- monitoring and tracing transactions
- logging errors and synchronizing data

There are also special Jeeves programs for handling sales configurations in customer orders and for sending shipment plans to Astro.

Jeeves ERP provides a standard XML mapping for import of price lists, based on the columns that are possible to update in the costing and pricing programs. You can adjust the mapping, for example if the columns names in the file from the supplier are different from Jeeves ERP's column names. You can also add any additional calculations to be done in conjunction with the import.

Some of the advantages of JMI are as follow:

- It provides one single preparatory system for all XML messages from all applications (most integrations are based on one table per message) and does not require interim storage.
- Most of the setup is done from within Jeeves.
- It enables parallel processing of messages where applicable, and can handle large volumes of transactions.
- It offers extensive monitoring and exception management, with the possibility to correct and re-run transactions that do not comply with business or integrity rules.

21.11 Excel integration

Besides being able to print standard reports, dynamic reports and SSRS reports to various formats, including Excel, you can also export data directly from any Jeeves program to Excel. You simply select the required records and then select the required columns in the list before choosing the Excel export option.

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21.12 Data replication, conversion and management rules

Some data, for example in Inter company transactions (ICT), will be common to several companies/units. Common data can be entered and maintained centrally and replicated in the other companies/units. You can set up conditions for replication, for example only items that are sellable should be replicated.

Some of the data may need to be converted between companies/units; for example, an item's acquisition method could be Manufacture to requirement at the delivering unit but Purchase to requirement at the selling unit.

If data is entered and maintained centrally, you need to define rules for data management, such as which companies/units are allowed to change and delete data.

There are also so-called modification points that can be used to define the data to be included in customer orders in delivering units and to be updated on changes from the selling unit, as well as for conversions and own calculations.

21.13 Jeeves Graphical planning

There is an optional tool for graphical resource planning integrated with the Production, Service and Time, projects, assignments modules.

Jeeves Graphical planning is used to graphically view capacities and levels of usage for various kinds of resources, and for detailed planning. Jeeves Graphical planning can be used to simulate various planning situations and, if you like, save changes to Jeeves. Re-planning is performed via drag-and-drop. In Jeeves Production you can plan setup, personnel/operation, subsequent activities, including overlapping operations, and control deviations from Jeeves ERP's standard calendar. Jeeves Graphical planning also helps to identify bottle-necks. You can view activities per resource and also a graph showing resource load and backlog.

21.14 Graphical planning system integration

Jeeves ERP also offers an integration with a fully functional graphical planning tool that is developed and maintained by a chosen solutions provider. The new tool called EQ Plan facilitates the management of manufacturing orders in Jeeves ERP. Note that this integration is currently only implemented with the production module.

Jeeves ERP enables you to define different planning levels that can be set for a resource group. Based on the selected planning level, the system will load the relevant manufacturing related information from Jeeves ERP to EQ Plan (via the integration endpoint). It also allows you to save and view information and settings for EQ Plan. The values, for example: color settings, are read by EQ Plan when loading data from Jeeves ERP.

You can define resource groups and their properties for the graphical planning system. Once you have defined a resource group, you can link it to one or more manufacturing groups.

A user with sufficient administrative rights can manage manufacturing operations using EQ Plan. When creating or modifying the plan for one or several resources at a specific point in time, this user can trigger a locking mechanism to make sure that other users running EQ plan do not interfere with the planning. When a resource has been locked, no other user will be able to alter the plan for that particular resource. Once a connection is successfully established between EQ Plan, the endpoint (JSB), and Jeeves ERP, EQ Plan will call the locking mechanism at regular intervals depending on how the management of the manufacturing plan is progressing.

You can organize your data using versioning in EQ Plan. The version mentioned in the Graphical planning value (GpsValue) field will be applicable to all GPS services. You can also define resource types for planning, restrict changes to start and end dates for labor operations using lock behavior, and enable/disable the following options in EQ Plan: lock, merge, split, delete, and copy. In addition, you can split a labor operation into another operation as per the conditions specified or merge two operations that were previously split into a single operation.

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21.15 Localizations/internationalization

Most general functions in Jeeves ERP are common to all countries. There are some, however, that are specific to certain countries, as described below.

21.15.1 General data protection regulation (GDPR) (EU countries)

Jeeves ERP includes functions for handling the General data protection regulation within EU countries.

Rename (anonymize) companies

For a company, if the Consent checkbox has been unchecked in the Companies program and if no customers' orders are in progress and no invoices exist for the past one year, as a GDPR controller, you can do the following:

- rename the company where you no longer have consent to keep their data,
- remove the contact persons associated with the company,
- remove the Points of delivery connected with the company, and
- remove the EU/VAT information pertaining to the company.

Remove contact persons

In the Contacts (kp) program if the Consent checkbox has been unchecked for a contact person, as a GDPR controller you can remove the contact person, including a web user. This can be done even if the contact person is associated with a number of companies.

21.15.2 Online credit check (Sweden only)

Jeeves ERP has a built-in function that makes it quick and easy to run a credit check on a company in the company table. You can also join the agreement that Jeeves Information Systems AB has signed with the Swedish credit information agency (UC).

21.15.3 Postens eBrev (Nordic countries)

Jeeves eBrev (which in English means e-Letter) is a function to define and create text files in accordance with the postal service's eBrev standard. The text files are sent to postal service for print out, sealing and distribution.

The function is available for a number of reports that are included in the Rules for printout program. For each report, eBrev templates are added which are obtained when running reports based on the current language, country, customer category for the company to be processed.

When reports are run, nothing is printed out. Instead, a record is created in an outbox for eBrev with reference to a certain eBrev template. By running a separate function, an eBrev file is created based on the records in the outbox, which are sent by FTP to the postal service.

Jeeves eBrev includes a program for definition of various eBrev templates and eBrev jobs, creating eBrev files and distribution of files using FTP.

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22 Jeeves ERP e-Applications

22.1 e-Approval of supplier invoices

Jeeves e-Approval is a function for the electronic distribution and approval of supplier invoices. This is built on the combination of a standard process in Jeeves Workflow and a web application.

22.1.1 Approval chains in e-Approval of supplier invoices

Jeeves e-approval is based on custom approval chains. An approval chain is a hierarchical and role-based approval flow. By hierarchical we mean that the chain controls the order in which an approval will be performed. By role-based we mean that approval takes place at a resource group level independent of the individual. You can create new approval chains easily from an existing approval chain. You can create approval chains for normal invoicing and for collective invoicing.

On the approval chain you can set up approval rules to control whether posting and manual allocation are allowed, whether it is allowed for postings to be changed, including others' postings.

22.1.2 e-Approval work flow

The e-Approval work flow consists primarily of the following steps:

- Supplier invoices are scanned and stored in the form of PDF files and then interpreted.
- The files are imported, or, alternatively, the invoices can be opened and invoice data entered manually. If Pagero AB's service for receiving eInvoice s is used, Pagero creates an eInvoice file in Jeeves ERP's standard format and a corresponding invoice file in PDF format, and notifies the user that there are files to be downloaded.
- Jeeves programs for review of supplier invoices allow the accounts payable administration to open an image of the scanned invoice, enter approval data, and then transfer the invoice to accounts payable. The invoice receives the status: Pending approval.
- This creates one instance of the approval chain built according to the specified approval chain in Jeeves Workflow. An e-mail is sent to the approver stating that an invoice has arrived for approval.
- The approver reviews the invoice in a special web interface for e-approval, and once the invoice is approved, a final approval record and a general ledger voucher that updates the balance are usually created. Alternatively, via a parameter, you can choose to let the accounts payable administration perform final review, approval, and balance update. If the web functionality for invoice matching is used the approver can match the invoice against one or more received purchase orders via the web before final approval.

22.1.3 e-Approval on the web

The web interface of e-Approval helps approvers to work more efficiently. The enhancements include the ability to:

- Match the invoice against one or more received purchase orders before final approval.
- Approve an invoice from the invoice matching page
- Filter purchase orders in the matching page
- Choose to view one or all purchase orders
- Add a posting in the list instead of a separate page
- Use Jeeves ERPs shortcut function to set pre-defined entries when posting
- Show arrival posting on demand
- Choose to display VAT/tax codes VAT-mandatory accounts in the Posting table page
- Enter the VAT/tax code when you add a custom row to an existing purchase order on the invoice
- View information on escalated invoices easily

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The Current approver column in the invoice list displays the signatures of the current approvers for the incoming invoices. The name associated with a signature is displayed when you place the cursor over the signature.

The approver can view arrival postings in a pop-up window. This information cannot be modified as it is read-only.

In the Supplier invoice program, there are two buttons to approve or reject an invoice that is distributed in an approval chain. These function buttons can be used in place of the equivalent buttons that exist in the web solution.

22.1.4 User interface

When you log into Jeeves e-Approval you are presented with a list of invoices you must approve or check. Invoices with numbers displayed in red will become due within 5 days or are already due for payment. The date by which the invoice must be approved is also shown in the Escalation column. Escalation means that after this date, the invoice will be forwarded to a manager or other colleague. The original approver can, however, still handle the invoice as usual.

The activities in the list are assigned a colored symbol in the left margin that shows their status, which also indicates the invoice's route to your approval list; if it came directly to you or if it was forwarded from another recipient for inquiry, etc.

When you choose an invoice from the list, a graphic image of the scanned invoice is displayed along with invoice information interpreted during scanning, any notes that were made during the review and postings that were created according to the posting template. There are numerous options you can choose:

- add notes to invoice
- block payment of invoice
- approve invoice
- send invoice back to administration
- send invoice to colleague for inquiry
- send invoice to colleague for information
- pause invoice
- add approver
- show approval chain
- create new posting
- copy posting
- restart approval chain (used by Workflow administrators in order to force the invoice to another approver)
- match invoice against received purchase order rows
- show documents
- add appendices
- display log

If you cancel a supplier invoice in the Jeeves client, Jeeves ERP checks if there is a connected workflow instance in Jeeves Workflow that has not already been canceled. If there is a workflow instance connected, it will also be canceled. This eliminates the need for manual cancelation in Jeeves Workflow.

22.1.5 Administration

The e-application includes a few simple administrative functions for:

- choosing one e-mail for each new invoice or one e-mail per day
- changing passwords
- entering absence periods and specify another colleague as a reserve
- editing voucher templates

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22.1.6 Workflow to-do list

There is also a program in the Jeeves client that shows a list of active activities from Jeeves Workflow. The activities are automatically added to your to-do list. The activities show when an invoice is currently in the process, for example that it has been distributed to an approver, waiting to be signed, and activities waiting to be executed by Administration.

The workflow to-do list also displays advance invoices. Two approval records are created as follows and the advance invoice is shown:

- An arrival approval record with no voucher
- A pending approval record with a voucher from the template associated with the supplier invoice type.

The Jeeves program also offers many options:

- Start activity
- Approve (close) activity
- Reject (cancel) activity
- Show entry (connected record in Jeeves ERP)
- Skip activity (and progress to next activity)
- Pause/resume activity
- Delegate activity (to another resource group or user)
- Delete activity
- Restart process (used by Workflow administrators in order to force the invoice to another approver)

22.1.7 Invoice list filtering

Invoice lists in e-Approval support predefined date ranges, reducing manual filtering during daily work. Improved comment visibility supports clearer collaboration among approvers and accounts payable staff when handling invoices.

22.2 e-Sales

22.2.1 Technical platform

e-Sales is compatible with the most recent version of the portal platform, Liferay. It includes a responsive mobile interface and access to applications and solutions of Liferay. Real-time data is loaded selectively from Jeeves database to provide improved performance and better user experience.

22.2.2 e-Sales B2B

e-Sales B2B (Business to business) is the e-application for Sales that handles products variants (for example models, sizes, colors etc.), package items and batch items. It is for customer companies to place orders via your website. Refer to the chapter on *Jeeves Sales* for more information.

22.2.3 User interface

You (read “the customer” in B2B) select an item, either via the product tree or by searching for the item. The shopping cart displays items and their prices and quantities as you add them to it by clicking on Buy. Estimated delivery date is displayed on each row. You can update the quantity and the requested delivery date on rows in the shopping cart and type in any order row texts. In B2S you can enter discounts on rows, in which case you override any automatically set discount. You can also remove rows.

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When the order is complete you click on Check out. A new dialog is opened in which you can choose a point of delivery, if there are several, or enter an order specific address if applicable. You can also complete the order details form. When you submit the order all information is updated in Jeeves ERP. An order acknowledgement is automatically displayed in PDF format and can also be forwarded by e-mail if required.

There are additional options:

- Quick orders (entry of item number and quantity directly in the cart)
- Copy orders from history list to current shopping cart
- View order details
- View invoice for specific order
- Cancel order
- Start new cart (clear without saving)
- Change your order number
- View deliveries and history
- Enter/update company data (such as customers, contact persons, points of delivery, subsidiaries)
- View service orders
- View own serial number items
- View customer claims
- Register customer claims

22.2.4 Product designer

Support for designing items to order using the features and functions of product configuration is extended to e-Sales.

If an item is defined as a configurable item by checking the Configurable item field in the Items program, it can be used as a configuration model in the Product designer. The Product designer screen can be accessed as follows:

- **From the products tab**: You can click the Design button associated with a configurable item in the List/Grid view or in the item details page to open the Product designer screen and specify requirements for the features connected to the model.
- **From the shopping cart**: You can click on the gear icon, which shows the text Design item when you point to it, on the right of the item row for a configurable item to open the Product designer screen and specify requirements for the features connected to the model.

The features connected to the model are displayed within named tabs or as cards within an accordion view that can be expanded or collapsed by clicking on the header.

The tabs/cards are sorted according to the feature group's sequence number. The rows on the tab/card are sorted according to the feature's sequence number. Each tab or card represents a feature group and the rows contain features linked to the model and grouped within the feature groups.

A number of visualization artifacts/aids are available in the Product designer for e-Sales including the accordion view, web display options, status icons, identification of mandatory fields, and range constraints displayed below numeric features.

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22.2.4.1 Validate configuration

Once the configuration is ready for validation, you can validate if the configuration is complete, for example that it does not contain missing or conflicting answers.

- If the design is complete; the configuration status is automatically changed to Activated, the Product designer screen is closed, and the item is moved to the shopping cart for further action.
- If the design is incomplete, you will receive a message asking if the item can be added to the cart although the product design is not complete. If you click OK, the Product designer screen is closed and the item is moved to the shopping cart and displayed under Product designs in progress for further action. If you click Cancel, you can continue making changes to the item in the Product designer.

22.2.4.2 Undo configuration changes

You can also undo any changes made by you, remove the order row, and close the Product designer screen. If any mandatory features are unanswered or any features answers are in conflict, an exclamation icon is displayed against the feature groups to which the features belong and an Answer required dialog box appears:

- If you have not answered one or more required features, each feature is displayed one by one along with the feature group that it belongs to until you answer all the features.
- If any answers are in conflict with answers provided to other features, the features in conflict are displayed along with the feature group that they belong to.

22.2.5 Customer claims

You can create a customer claim against an item in the e-Sales order page. You must provide the quantity, relevant information, and the reason for making the claim before submitting. The claim number can be used to track the progress.

You can also view the list of claims connected to the customer along with the claim number, item number, description, quantity, order number, row number, planned completion date, and status.

22.2.6 Impersonate function

Using the impersonate function, you can enter orders on behalf of other users. By clicking on Impersonate, you can see all the subsidiaries' users. Select the user on whose behalf you intend to enter orders and you will be logged in as that user.

22.3 e-Invoices

You can receive electronic invoices including invoice images via a third party service. There are standard functions in Jeeves ERP for interpretation, review and transfer to the accounts payable ledger. Approval can either be done in the accounts payable ledger or via Jeeves ERP's e-Approval module. The advantages of electronic invoicing are many; it is quicker, less expensive, better for the environment, minimizes administration etc.

You can display electronic supplier invoices in a PDF browser. There are buttons for viewing invoices in an embedded browser and a button for viewing invoices in an external browser. This function requires that Adobe Acrobat Reader is installed on the client computer.

Jeeves ERP supports integrated electronic invoice exchange through Pagero. Outgoing invoices can be monitored throughout delivery, providing visibility into processing and delivery status. Incoming invoices can be retrieved and processed automatically, with support for reliable processing, error handling, retry management, and user notifications. If processing must be repeated, an invoice can be retrieved again from within Jeeves ERP.

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23 System administration and own development

Jeeves ERP has a wealth of functions that support system administrators, implementation consultants, developers and the like when setting up, customizing, running and monitoring the system. They can also take advantage of Microsoft SQL's functionality for querying and manipulating the Jeeves database.

23.1 System configuration

Jeeves ERP can be set up in many different ways to satisfy different needs for accessibility, performance and security. The different configurations in the form of thin and rich clients allow access both within the own network and over the Internet via encrypted tunnels. Any number of server processes can be set up on one or more physical machines in order to manage increased load, and the built-in load balancing distributes the load evenly between processes. By default, all communication is encrypted and compressed but this can be turned off if desired.

To enable running the Jeeves Client through Direct Access, Jeeves ERP supports IPV6 addresses in the communication between server and client for some configurations. This offers an option for smoother system access as no VPN tunnel is required to connect to the Jeeves Client from an outside network.

23.1.1 System/configuration settings

There are a large number of parameters that can be set up by system administrators, or the equivalent, to control how Jeeves ERP should function in general. This includes parameters about how the database is set up, how information should be displayed (language, font, formats etc.), how the client should behave etc. The parameters can be set per company, group and/or user, depending on the parameter.

23.2 Jeeves Cloud optimizations

You can set up the Jeeves Cloud in AWS. You can also set up the system for domain controller independent user authentication and enable file sharing between the business server and the client.

See also the sections on importing external data to Jeeves Cloud and exporting files to local systems from Jeeves Cloud without a VPN connection under the chapter about General functions.

23.2.1 Security identifier for user authentication

The Security identifier (SID) enables you to set up authorization to Jeeves ERP for Windows users without a domain account. The SecurityMode parameter allows you to authorize Windows user authentication against a Security identifier (SID) stored in Jeeves ERP.

You can provide the authenticated Windows users' details, including domain and temporary password. If the SecurityMode kernel parameter is set to Windows user authentication against SID stored in Jeeves ERP, and a user attempts to log into Jeeves ERP, a check is made to verify the following:

- the Signature matches the signature in the Users table
- the Windows user exists in the table
- the SID exists for the user.

23.2.2 File sharing between business server and client

The ClientCache folder enables automatic distribution of files, for instance reports and images used in the reports, from the server to the clients, implying direct file access to a central server folder is not necessary.

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Manually create a shared folder on the server that mirrors the ClientCache folder on the client. This folder should contain all files that need to be distributed (mainly report files and images used in Jeeves standard reports) to all connected clients. Once this is done and you start the Jeeves client, Jeeves ERP checks that the ClientCache folder exists. Jeeves ERP then initiates a file comparison (size and checksum) between the files stored in the specified folder on the server and those in the ClientCache folder in the client. It then generates a list of files which are changed on the server or missing on the client and sends the list to the server, which streams the files to the ClientCache folder on the client.

23.2.3 AWS S3 bucket access

There are kernel parameters under the Cloud section to facilitate the copying of application files from the S3 bucket with the following configurations: Region, AccessKeyId, SecretAccessKey and PathForS3.

There are application parameters to enable AWS S3 bucket access for EDM.

23.3 Help routines/utilities

23.3.1 Utilities

There are system administration functions for checking processing times, periodic and yearly routines, language management utilities etc.

You can, for example, show the number of invoices per day, per hour, etc.

- number of invoices per day
- number of invoices per hour
- processing time per invoice
- details of the invoices that took longest

There are functions to move customers', suppliers' and items' turnover to the previous year.

There are also language management utilities with which you can manage system and base data, such as drop-down list data, in different languages. You can even use this to translate data texts, for example item descriptions, including data in own tables. The language of the texts is switched when you switch language in the client. Note that this does not affect printouts.

23.3.2 Scheduler

There is a possibility to schedule print and update jobs. This is described in the Output management section.

23.4 Security functions

With the SecurityMode kernel parameter you can specify which type of authorization Jeeves ERP shall use:

- 0 = Default SQL Login
- 1 = Windows user authentication against SQL Server and User (sy2), then SQL Login
- 2 = Windows user authentication against SQL Server and User (sy2)
- 3 = Windows user authentication against User (sy2), then SQL Login
- 4 = Windows user authentication against User (sy2)
- 5 = Windows user authentication against SID stored in Jeeves ERP
- 6 = Federated authentication using identity service

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23.4.1 Users and user groups

In Jeeves ERP, there are two ways in which you can create groups of users to which you can grant permissions and connect settings. There is an earlier, simpler solution, where the user groups are not stored in the site repository and a later, more advanced way of working with groups of users etc.

There are pre-defined site groups for the various roles that normally exist in distribution and manufacturing companies etc. This means that you only need to define user logins and connect them to the relevant role groups, and you only need to create company specific groups if there are any. For installations based on pre-defined roles and settings there are certain restrictions regarding own groupings.

23.4.2 Company access

There is a program for administration of access rights to companies in multi-company databases. By default, all users have access to all companies, but if you grant a user or group access rights to a company, the user/group is blocked from all other companies. Access rights can be set on the following levels:

- User
- Authorization group
- User group in the site

You can grant access rights either by creating new records in the embedded list and typing in the signature(s) or group id(s) that are to have access to the current company, or by using the program nodes and selecting signatures or groups from the left list and transferring them to the right list.

23.4.3 SQL authorization

If required there is a possibility to use MS SQL Server's authorization functions. There is a mechanism that allows Jeeves ERP to be logged in as the owner of the tables in the database, which means that the database administrator does not need to specify the rights for every user and table. The result is that every user initially has full rights to the database. It is then possible to deny certain rights to users.

23.4.4 Security manager

There is also a built-in tool for security management that enables you to set up authorization on almost all Jeeves objects. The Security manager works with the Users, Site group, Company and Company group dimensions.

One work method is to build a tree structure by importing a program menu file. From the programs in the object tree structure you can access all the related objects, such as fields and buttons etc., to which you can grant or deny users access. However, you can work without a menu if you have your own, structured method of adding Jeeves objects to the Security manager.

Jeeves ERP's Security manager is based on the SQL standard called CRUDE, which stands for:

C - Create (i.e. create record)

R - Read (for example, read record or field value)

U - Update (for example, update record or field value)

D - Delete (i.e. delete record)

E - Execute (i.e. execute program, computed field, user function, node or report)

These permissions are not based on permission levels, i.e. a user can have permission to delete but not to update. There are naturally certain relationships, for example you must have the permission to execute a program in order to have any of the other permissions in the program.

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Permissions are inherited by default from top level down through sublevels. If an object has no permission setting, i.e. if neither Grant nor Deny is checked, the Security manager reads the permissions at runtime are deduced from the overlying object.

There are auditing functions in the Security manager in order to comply with SOX requirements. You can print audit reports, you can search in the permissions data that is created and you can make even deeper analyses where required.

23.4.5 Federated authentication

With Jeeves cloud-based Identity service, you can log into Jeeves ERP using federated authentication. This offers possibilities for IT to streamline the user administration and apply additional security layers, like multi-factor authentication (MFA). This is available for the Desktop client, Classic client, and Mobile client (iOS and Android). It is based on Open-ID Compliant Services (OICS) and has been verified for Azure AD.

23.5 Log functions

23.5.1 Change log

You can log new entries, changes and deletions. Logging is done at field level and can include own-developed fields, even from own tables. You can also control whether these events require a reason code and/or a password. The purpose of the development is to make it possible for Jeeves ERP users to meet requirements from the American Food and Drug Administration (FDA) concerning logging of events in the database.

23.5.2 Log processes

There are also functions for automatic logging of:

- currently logged in users in Jeeves ERP,
- currently active and canceled processes,
- certain report runs (when and by whom as well as certain data),
- certain multi row inserts and updates,
- login history, etc.

23.5.3 Handling server and client log files

The server log files can be automatically zipped and moved to the backup folder. Each zipped log file is named with its creation date. You can also truncate the log files, i.e. remove the oldest file, after a specified count is reached.

The name of the Desktop client log, which is automatically created when you start the client, includes the process ID, signature as well as date and time, making the log file both small and easy to handle.

23.6 Database administration

There are built-in functions in Jeeves ERP for maintenance of the database, for extraction and copying of information to and from databases, etc. You can activate logging of SQL statements for trouble-shooting purposes. There is also an advanced log, in which you can see which macros and user functions are executed in the current program, execution times etc.

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23.7 Developer menu

There is a special menu that is normally only available to developers, system administrators and such users. The menu contains options to restart functions and reload windows when doing own development and settings without having to restart Jeeves ERP. There are also corresponding functions to reload when Jeeves' system parameters have been changed and after manual change of local settings (in the site repository) etc.

From here you can also access more advanced options for database administration, such as Recreate indexes and Recompile window procedures.

23.8 Own development

Jeeves ERP provides extensive functionality for adding own modifications and development. We also provide functionality for identifying these customizations in customers' installations for the purpose of Cloud readiness and for facilitating upgrades. We recommend that the Import all changes from DB function in the Added/modified objects program is used as part of this process. The recommendation applies to all installations from Jeeves ERP 6.0 Update 7 and later.

Note:

- You are not allowed to modify standard Jeeves code in the database or files.

- If you create a mobile routine you must store the source code in Jeeves' programs/tables for customization (their names start with the prefix "UsrDef" which stands for User defined). By doing so, you make information about your modifications, which is valuable to customers and partners, visible within Jeeves ERP.

- In order to avoid conflicts with Jeeves standard and other apps, you must follow Jeeves Information Systems' name convention for names of all non-standard objects (tables, terms, connections, programs, subtags, macros etc.) within all own modifications.

23.8.1 Name convention

Unique name q_A_B_C where:

q_ = Is used to differentiate it from Jeeves standard. This prefix is already used, and it is well established within the Jeeves sphere.

A = Your partner ID.

B = Branch/vertical/application, a suitable name of your choice.

C = Name of your own developed object, for example, an added term, table, macro, trigger, sub tag, database object, program etc.

The ID can only consist of combinations of A-Z , 0-9 and _.

Example: q_JIS_Service_itemclass

Partners must also make sure that their IDs are unique within their own organization.

You can read more about our name convention in the document about Jeeves ERP's apps standards and rules on our partner web.

23.8.2 Own adaptations in Jeeves

If you use Jeeves ERP’s programs to hold own programs, windows, views, tables, connections, constraints, own database SQL objects (added, replaced or subtags), this can be used as a form of documentation of your adaptations. Most of the objects can also be a part of an application.

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23.8.3 Own program functionality

- You can add your own tables, views, windows and programs to Jeeves ERP and also make fields from your own tables available in standard form windows.
- Tables and views must be created in the database along with primary key, indexes and defaults.
- Own programs can display information from Jeeves ERP's tables and views or own tables and views.
- Own programs can consist of an unlimited number of levels with embedded list windows and/or underlying program nodes. They can also be used as embedded lists and nodes in Jeeves ERP's standard programs.
- You can make own windows and check-off (left-right list) windows
- You can make total windows in your own programs
- You can choose which fields to display in the form and/or list, and how to display: drop-down lists, show fields as checkboxes
- You can choose if your program allows insert, update and or delete
- You can also add Jeeves ERP’s standard nodes to other standard nodes
- You can create right-clicks in standard Jeeves programs as well as in own programs
- You can create drop-down fields in own program based on standard or own data.
- Drop-down lists based on own tables can be translated and displayed in the relevant language when you switch language in the client.

23.8.4 Managing own modifications in database logic

- You can use subtags (subtags are tagged own code to be placed in existing location tags in our standard code) to modify standard database SQL objects such as procedures, triggers, views and database functions.
- Note that the subtag must be placed in a location tag in the object. If location tag is missing we, Jeeves Information Systems, must add it in our standard code. Otherwise it will disappear after upgrading.
- If subtags are entered in the intended program and an upgrade takes place, the subtags are re-installed in the database automatically. However, if you install fixes manually or by JvsSQLRun, you must remember to re-install subtags manually in this program. You can use the same program for own added or replaced standard procedures, triggers, views, database functions and table functions. In this program, there are nodes for history regarding whole objects and subtags.
- The program with content can be used as a documentation of own adaptations of modified standard code (in subtags) as well as own or replaced triggers, procedures views etc.
- You can add own language managed texts to be used by macros and own SQL code

Jeeves Information Systems provide templates for stored procedures, views, functions and table functions.

Jeeves ERP also provides several applications for exporting/importing own development; from a simpler export/import program to the more comprehensive Apps builder. They run the same code, but the Apps builder is recommended since it automatically provides documentation of all included customizations.

23.8.5 .NET integration

It is possible to integrate Jeeves ERP and Microsoft .NET to expand Jeeves ERP’s macro language with your own functionality written in any of .NET's programming languages, for example, integrating Jeeves ERP with your own Web services, adding your own forms in the macro, etc.

You can use Jeeves .NET integration in two ways; you can call .NET from Jeeves ERP and call Jeeves ERP from .NET. When doing so, you have full access to all objects in the Jeeves database.

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23.8.6 Define own macros

Users can add their own functionality in Jeeves ERP, such as own calculations and controls, by creating a macro using the Macro option on the **Tools** menu. The statements allowed in a user-defined macro are the same as in the built-in functions in Jeeves, User functions. These functions are defined in the Appl.ufu file. You can look at this file for tips when writing macros.

Note

Do not include complex SQL code in a macro. Move the code to a procedure and call the procedure from the macro instead. Basic rule: Business logic in the database and user interaction logic in macros.

Some examples of different ways of using macros:

- To do something that is done often. In this case you can create a button or shortcut command to activate the macro.
- To let the macro fill in a field with a calculated value during entry. In this case you link the macro so that it is activated when the user has entered a value in a certain field.
- To check what is entered in a field. You link the macro so that it is activated at validation of the entered value. In this way you can check the entered value using the macro and decide whether the value should be accepted.
- To check whether a record can be entered
- To start another Windows application
- To display own calculated values in own fields (macro terms)
- Print reports from Jeeves ERP
- Execute a procedure
- Execute database maintenance options

Macros can be executed from:

- the macro editor
- buttons
- short commands (accelerators)
- macro events
- other macros
- own fields (macro-terms)

There are several hundred functions that can be inserted into macros for handling programs, fields, forms, files, SQL expressions, mathematical functions, communication etc. All macro functions are described in Jeeves documentation.

23.8.7 Communication with external programs

It is possible to exchange information and communicate with Jeeves ERP from other programs, such as MS Word, via an ActiveX/COM interface.

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23.9 Jeeves API Platform and standard APIs

Jeeves API Platform is a standardized integration service platform that can be used for both standard API development by Jeeves and for development of customized APIs by Jeeves consultants and partners. Jeeves standard APIs are delivered as packages in addition to the annual releases.

- Jeeves ERP's Sales API is intended to make it easier for Jeeves customers to integrate with Jeeves ERP:s sales capabilities from other, external systems such as CRM and e-commerce systems, to support their sales processes.
- Jeeves ERP's Finance APIs are intended to add efficiency and compliance in the financial processes. It is part of Jeeves' strategy to create connectivity for a variety of external systems and sources.
- Jeeves ERP:s Manufacturing APIs are intended to enable interaction with external systems and sources for resource planning and management. For example, an external system may trigger start and stop of manufacturing operations, report picked and scrapped materials, etc.
- Jeeves ERP:s Inventory optimization APIs will open up the possibility to interact with multiple external Warehouse optimization systems. As in our earlier integrations, they will be used for sending information from Jeeves ERP, such as item information, forecasts, transactional demand, etc. They will also be used to create purchase requisitions and purchase orders in Jeeves ERP from external systems, as well as provide suggestions to adjust inventory parameters where needed. The initial version is only mapped against the third party software EazyStock, powered by Syncron.
- Jeeves ERP:s generic Transport administration (TA) API enables integration with multiple TA service providers' systems, for handling standardized transport documentation from Jeeves ERP:s dispatch programs. For further details, see the chapter about the Sales module.

One common component to the APIs is the API service customization program, which can be used to make adaptations to the standard interface where necessary to adapt to the external system. From this program you can publish/unpublish API endpoints by using the Valid to use checkbox. You can restrict API endpoints to a client or group of clients. By default, all endpoints are hidden unless a client or group is specified.

You can customize any of the API stored procedures. Modification tags are also provided in each section to make it easy to customize. For further information, see the chapters on the API platform and APIs in Jeeves ERP documentation.

Jeeves API platform also includes a repository where Jeeves maintains the properties that are used in Jeeves standard APIs, and where developers/consultants should maintain the properties in their custom APIs. This way, we will achieve a single, uniform definition of each property in all API response and request bodies, so that the APIs can more easily be kept consistent.

Once the properties have been defined, a request and response schema can be created by adding a record in the model definition with schema details and all related properties and sub schemas (arrays). Both on the model rows and the model, there are functions for automatically validating and generating the JSON schema, if desired.

The repository also provides a tool for documenting the properties and their mappings with Jeeves tables and columns, which is also reflected in Swagger.

Consumers of Jeeves APIs, both standard and custom APIs, who need up-to-date data or handle big data that are slow moving, can subscribe to changes/events in the Jeeves database. Subscription is based on entities defined on the dynamic API endpoint in Jeeves API platform. Through pagination, consumers can retrieve data sets in a structured manner. When an API endpoint returns a large amount of data, pagination allows the data to be divided into smaller portions.

The API platform includes functionality for exporting API endpoint definitions to external SQL files in order to submit for scanning for security, performance etc.

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23.9.1 Jeeves Outbound API Service

The Jeeves Outbound API Service enables Jeeves ERP business processes to connect securely to external web services and cloud-based services. External services can be called directly from business processes, using either synchronous calls for an immediate response or asynchronous calls for background processing.

The service provides centralized, secure management of credentials and supports multiple authentication methods, including client credentials, JWT-based authentication, and API keys. Multiple external services can be configured for each Jeeves environment, enabling organizations to extend their business processes through controlled and standardized integrations.

23.10 Jeeves Apps builder

You can build your own applications in Jeeves ERP. (Apps is an abbreviation of the word *applications*, and refers to a piece of software or a packaged solution.) Jeeves Apps builder offers many advantages; here are some examples:

- Built-in import and export functions make it quick and easy to manage solutions.
- Objects and solutions can be re-used.
- The need for technical expertise and consulting can be reduced when introducing new functionality or customizations.
- You can export your adaptations and import them into another database.

A Jeeves application can consist of one or more of the following components:

23.10.1 Jeeves application

- database objects (views, triggers, procedures, database functions, table functions own or replaced Jeeves' standard)
- sub tags (application's unique modifications to Jeeves ERP's standard procedures, triggers, views and database functions)
- customers' own programs, nodes, windows, tables, views, fields, connections
- own added nodes to Jeeves standard
- own added computed columns
- script executed when importing apps

23.10.2 Data

- template companies (all tables with a value in the Company code column)
- database tables and columns:
  - tables without the Company code column
  - tables to be excluded from the template company
  - tables in the template company filtered with a certain condition
  - tables with or without data

23.10.3 Settings – all settings saved to site repository groups

- site repository groups and group structures
- settings
- menus
- macros
- buttons etc.

The applications are defined in a program in Jeeves ERP and thereafter exported with a separate executable program, Jeeves Apps builder. The program can either be started from within Jeeves ERP, or run stand-alone or by script.

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The export results in a zip file that can be imported in any Jeeves installation of the same version as the application from which the application was exported. The zip file has a logical folder structure to make it easy to identify the individual objects on which the application is built. This makes the maintenance of the application easier, enabling changes without making a new export.

When you import an application, the contents of the application are installed in the import database (site and/or Jeeves database). Example: Database objects are compiled into the database. Tables are created with primary keys and defaults, with or without data depending on import type. Site group structures are created and all site data, such as macros, macro events, buttons, settings and menus etc. are installed and ready to use.

Jeeves Apps Builder is used when installing Jeeves ERP to install all settings in our Manufacturing and Distribution packages, as well as data in our template companies.

Note

When working with apps and own modifications it is important to follow our name convention in order to avoid conflicts with Jeeves standard and other apps. This applies to names of all non-standard objects (tables, terms, connections, programs, subtags, macros, groups, apps etc.) within all own modifications.

23.10.4 Extended name convention

Unique name q_A_B_C where:

q_ = Is used to differentiate it from Jeeves standard. This prefix is already used and it is well established within the Jeeves sphere.

A = Your partner ID.

B = Branch/vertical/application, a suitable name of your choice.

C = Name of your own developed object, for example: an added term, table, macro, trigger, sub tag, database object, program etc.

The ID can only consist of combinations of A-Z , 0-9 and _.

Example: q_JIS_Service_itemclass

Partners must also make sure that their IDs are unique within their own organization.

You can read more about our name convention in the document about Jeeves ERP's apps standards and rules on our partner web.

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